

Academic Regulations Course Structure & Detailed Syllabus

CHOICE BASED CREDIT SYSTEM

R22

COMPUTER SCIENCE & ENGINEERING

Bachelor of Technology (B.Tech)

For the batches admitted from the A.Y. 2022-23



MLR Institute of Technology

(Autonomous)

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ACADEMIC REGULATIONS (R22)

B. Tech. - Regular Four Year Degree Programme (For batches admitted from the academic year 2022-23)

1.0 For pursuing four year Under Graduate Degree Programme of study in Engineering & Technology (UGP in E&T) offered by MLR Institute of Technology under Autonomous status is herein referred to as MLRIT (Autonomous):

All the rules specified herein approved by the Academic Council will be in force and applicable to students admitted from the Academic Year 2022-23 onwards. Any reference to "Institute" or "College" in these rules and regulations shall stand for M L R Institute of Technology (Autonomous).

All the rules and regulations, specified hereafter shall be read as a whole for the purpose of interpretation as and when a doubt arises, the interpretation of the Chairman, Academic Council is final. As per the requirements of statutory bodies, the Principal, M L R Institute of Technology shall be the chairman of Academic Council.

2.0 Eligibility for Admission

- 2.1** Admission to the undergraduate(UG) programme shall be made either on the basis of the merit rank obtained by the qualified student in entrance test conducted by the Telangana State Government (EAMCET) or the College or on the basis of any other order of merit approved by the University/TSHE, subject to reservations as prescribed by the government from time to time.
- 2.2** The medium of instructions for the entire undergraduate programme in Engineering & Technology will be **English** only.

3.0 B.Tech. Programme Structure

- 3.1** A student after securing admission shall complete the B.Tech. programme in a minimum period of **four** academic years (8 semesters), and a maximum period of **eight** academic years (16 semesters) starting from the date of commencement of first year first semester, failing which student shall forfeit seat in B.Tech course. Each student shall secure 160 credits (with CGPA ≥ 5) required for the completion of the undergraduate programme and award of the B.Tech. degree.
- 3.2** **UGC/AICTE** specified definitions/descriptions are adopted appropriately for various terms and abbreviations used in these academic regulations/norms, which are listed below.

3.2.1 Semester Scheme

Each undergraduate programme is of 4 academic years (8 semesters) with the academic year divided into two semesters of 22 weeks (90 instructional days) each and in each semester - 'Continuous Internal Evaluation (CIE)' and 'Semester End Examination (SEE)' under Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) indicated by UGC, and curriculum/course structure suggested by AICTE are followed.

3.2.2 Credit Courses

All subjects/courses are to be registered by the student in a semester to earn credits which shall be assigned to each subject/course in an L: T: P: C (lecture periods: tutorial periods: practical periods: credits) structure based on the following general pattern.

- One credit for one hour/week/semester for Theory/Lecture (L) courses or Tutorials.
- One credit for two hours/week/semester for Laboratory/Practical (P) courses.

Courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab are mandatory courses. These courses will not carry any credits.

3.2.3 Subject Course Classification

All subjects/courses offered for the undergraduate programme in E&T (B.Tech. degree programmes) are broadly classified as follows. The College has followed almost all the guidelines issued by AICTE/UGC.

S. No.	Broad Course Classification	Course Group / Category	Course Description
1	Foundation Courses (FnC)	BS–Basic Sciences	Includes Mathematics, Physics and Chemistry Subjects.
2		ES–Engineering Sciences	Includes Fundamental Engineering Subjects.
3		HS–Humanities and Social Sciences	Includes subjects related to Humanities, Social Sciences and Management.
4	Core Courses (CoC)	PC–Professional Core.	Includes core subjects related to the parent discipline/department/branch of Engineering.
5	Elective Courses (E&C)	PE–Professional Electives	Includes elective subjects related to the parent discipline / department / branch of Engineering.
6		OE–Open Electives	Elective subjects which include interdisciplinary subjects or subjects in an area outside the parent discipline / department / branch of Engineering.
7	Core Courses	Project Work	B.Tech. Project or UG Project or UG Major Project or Project Stage I & II.
8		Industry Training / Internship / Industry Oriented Mini- project / Mini- Project / Skill Development Courses.	Industry Training / Internship / Industry Oriented Mini-Project / Mini-Project / Skill Development Courses.
9		Seminar	Seminar / Colloquium based on core contents related to parent discipline / department/branch of Engineering.
10	Minor Courses	-	1 or 2 Credit Courses (subset of HS)
11	Mandatory Courses (MC)	-	Mandatory Courses (non-credit)

4.0 Course Registration

- 4.1 A 'faculty advisor or counsellor' shall be assigned to a group of 20 students, who will advise the students about the undergraduate programme, its course structure and curriculum, choice/option for subjects/courses, based on their competence, progress, pre- requisites and interest.
- 4.2 The academic section of the college invites 'registration forms' from students before the beginning of the semester through 'on-line registration', ensuring 'date and time stamping'. The on-line registration requests for any 'current semester' shall be **completed before the commencement of SEEs (Semester End Examinations) of the 'preceding semester'**.
- 4.3 A student can apply for **on-line** registration, **only after** obtaining the '**written approval**' from faculty advisor/counsellor, which should be submitted to the college academic section through the Head of the Department. A copy of it shall be retained with the Head of the Department, Faculty Advisor/Counsellor and the student.
- 4.4 A student may be permitted to register for all the subjects/courses in a semester as specified in the course structure with maximum additional subject(s)/course(s) limited to 6 Credits (any 2 elective subjects), based on **progress** and SGPA/CGPA, and completion of the '**pre-requisites**' as indicated for various subjects/courses, in the department course structure and syllabus contents.
- 4.5 Choice for '**additional subjects/courses**', not more than any 2 elective subjects in any Semester, must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Mentor/HOD.
- 4.6 If the student submits ambiguous choices or multiple options or erroneous entries during **on-line** registration for the subject(s)/course(s) under a given/specified course group/category as listed in the course structure, only the first mentioned subject/course in that category will be taken into consideration.
- 4.7 Subject/course options exercised through **on-line** registration are final and **cannot** be changed or inter-changed; further, alternate choices also will not be considered. However, if the subject/course that has already been listed for registration by the Head of the Department in a semester could not be offered due to any inevitable or unexpected reasons, then the student shall be allowed to have alternate choice either for a new subject (subject to offering of such a subject), or for another existing subject (subject to availability of seats). Such alternate arrangements will be made by the Head of the Department, with due notification and time-framed schedule, within **a week** after the commencement of class-work for that semester.
- 4.8 Dropping of subjects/courses may be permitted, only after obtaining prior approval from the faculty advisor/counsellor 'within a period of 15 days' from the beginning of the current semester.
- 4.9 **Open Electives:** The students have to choose three Open Electives (OE-I, II & III) from the list of Open Electives given by other departments. However, the student can opt for an Open Elective subject offered by his own (parent) department, if the student has not registered and not studied that subject under any category (Professional Core, Professional Electives, Mandatory Courses etc.) offered by parent department in any semester. Open Elective subjects

already studied should not repeat/should not match with any category (Professional Core, Professional Electives, Mandatory Courses etc.) of subjects even in the forthcoming semesters.

4.10 Professional Electives: The students have to choose six Professional Electives (PE-I to VI) from the list of professional electives given.

5.0 Subjects/courses to be offered

5.1 A subject/course may be offered to the students, **only if** a minimum of 15 students opt for it.

5.2 More than **one faculty member** may offer the **same subject** (lab/practical may be included with the corresponding theory subject in the same semester) in any semester. However, selection of choice for students will be based on - '**first come first serve** basis and CGPA criterion' (i.e. the first focus shall be on early **on-line entry** from the student for registration in that semester, and the second focus, if needed, will be on CGPA of the student).

5.3 If more entries for registration of a subject come into picture, then the Head of the Department concerned shall decide, whether or not to offer such a subject/course for **two (or multiple) sections**.

5.4 In case of options coming from students of other departments / branches / disciplines (not considering **open electives**), first **priority** shall be given to the student of the '**parent department**'.

6.0 Attendance requirements:

6.1 A student shall be eligible to appear for the semester end examinations, if the student acquires a minimum of 75% of attendance in aggregate of all the subjects/courses (including attendance in mandatory courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab) for that semester. **Two periods** of attendance for each theory subject shall be considered, if the student appears for the mid-term examination of that subject.

6.2 Shortage of attendance in aggregate upto 10% (65% and above, and below 75%) in each semester may be condoned by the college academic committee on genuine and valid grounds, based on the student's representation with supporting evidence.

6.3 A stipulated fee shall be payable for condoning of shortage of attendance.

6.4 Shortage of attendance below 65% in aggregate shall in **NO** case be condoned.

6.5 **Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examinations of that semester. They get detained and their registration for that semester shall stand cancelled, including all academic credentials (internal marks etc.) of that semester. They will not be promoted to the next semester.** They may seek re-registration for all those subjects registered in that semester in which the student is detained, by seeking re-admission into that semester as and when offered; if there are any professional electives and/or open electives, the same may also be re-registered if offered. However, if those electives are not offered in later

semesters, then alternate electives may be chosen from the **same** set of elective subjects offered under that category.

- 6.6** A student fulfilling the attendance requirement in the present semester shall not be eligible for readmission into the same class.

7.0 Academic Requirements

The following academic requirements have to be satisfied, in addition to the attendance requirements mentioned in Item No. 6.

- 7.1** A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/course, if student secures not less than 35% (21 marks out of 60 marks) in the semester end examination, and a minimum of 40% (40 marks out of 100 marks) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together; in terms of letter grades, this implies securing 'C' grade or above in that subject/course.
- 7.2** A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship (or) Seminar, if the student secures not less than 40% marks (i.e. 40 out of 100 allotted marks) in each of them. The student is deemed to have failed, if he (i) does not submit a report on Industry Oriented Mini Project/Internship, or (ii) not make a presentation of the same before the evaluation committee as per schedule, or (iii) secures less than 40% marks in Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship evaluations.

A student may reappear once for each of the above evaluations, when they are scheduled again; if the student fails in such 'one reappearance' evaluation also, the student has to reappear for the same in the next subsequent semester, as and when it is scheduled.

7.3 Promotion Rules

S. No.	Promotion	Conditions to be fulfilled
1	First year first semester to first year second semester	Regular course of study of first year first semester.
2	First year second semester to Second year first semester	(i) Regular course of study of first year second semester. (ii) Must have secured at least 20 credits out of 40 credits i.e., 50% credits up to first year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3	Second year first semester to Second year second semester	Regular course of study of second year first semester.

4	Second year second semester to Third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Third year first semester to Third year second semester	Regular course of study of third year first semester.
6	Third year second semester to Fourth year first semester	(i) Regular course of study of third year second semester. (ii) Must have secured at least 72 credits out of 120 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
7	Fourth year first semester to Fourth year second semester	Regular course of study of fourth year first semester.

- 7.4** A student (i) shall register for all courses/subjects covering 160 credits as specified and listed in the course structure, (ii) fulfills all the attendance and academic requirements for 160 credits, (iii) earn all 160 credits by securing SGPA $\geq$ 5.0 (in each semester), and CGPA $\geq$ 5 (at the end of 8 semesters), (iv) **passes all the mandatory courses**, to successfully complete the undergraduate programme. The performance of the student in these 160 credits shall be considered for the calculation of the final CGPA (**at the end of undergraduate programme**), and shall be indicated in the grade card/marks memo of IV-year II semester.
- 7.5** If a student registers for '**extra subjects**' (in the parent department or other departments/branches of Engg.) other than those listed subjects totalling to 160 credits as specified in the course structure of his department, the performances in those '**extra subjects**' (although evaluated and graded using the same procedure as that of the required 160 credits) will not be considered while calculating the SGPA and CGPA. For such '**extra subjects**' registered, percentage of marks and letter grade alone will be indicated in the grade card/marks memo as a performance measure, subject to completion of the attendance and academic requirements as stated in regulations Items 6 and 7.1 – 7.4 above.
- 7.6** A student eligible to appear in the semester end examination for any subject/course, but absent from it or failed (thereby failing to secure '**C**' grade or above) may reappear for that subject/course in the supplementary examination as and when conducted. In such cases, internal marks (CIE) assessed earlier for that subject/course will be carried over, and added to the marks to be obtained in the SEE supplementary examination for evaluating performance in

that subject.

7.7 A student detained in a semester due to shortage of attendance may be re-admitted in the same semester in the next academic year for fulfillment of academic requirements. The academic regulations under which a student has been re-admitted shall be applicable. Further, no grade allotments or SGPA/CGPA calculations will be done for the entire semester in which the student has been detained.

7.8 A student detained due to lack of credits, shall be promoted to the next academic year only after acquiring the required number of academic credits. The academic regulations under which the student has been readmitted shall be applicable to him.

8.0 Evaluation - Distribution and Weightage of Marks

8.1 The performance of a student in every subject/course (including practicals and Project Stage – I & II) will be evaluated for 100 marks each, with 40 marks allotted for CIE (Continuous Internal Evaluation) and 60 marks for SEE (Semester End-Examination).

8.2 In CIE, for theory subjects, during a semester, there shall be two mid-term examinations. Each Mid-Term examination consists of two parts i) **Part – A** for 10 marks, ii) **Part – B** for 20 marks with a total duration of 2 hours as follows:

1. Mid Term Examination for 30 marks:
 - a. Part - A : Objective/quiz paper for 10 marks.
 - b. Part - B : Descriptive paper for 20 marks.

The objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 10 marks. The descriptive paper shall contain 6 full questions out of which, the student has to answer 4 questions, each carrying 5 marks. The **average of the two Mid Term Examinations** shall be taken as the final marks for Mid Term Examination (for 30 marks).

The remaining 10 marks of Continuous Internal Evaluation are distributed as:

2. Assignment for 5 marks. (**Average of 2 Assignments** each for 5 marks)
3. Subject Viva-Voce/PPT/Poster Presentation/Case Study on a topic in the concerned subject for 5 marks.

While the first mid-term examination shall be conducted on 50% of the syllabus, the second mid-term examination shall be conducted on the remaining 50% of the syllabus.

Five (5) marks are allocated for assignments (as specified by the subject teacher concerned). The first assignment should be submitted before the conduct of the first mid-term examination, and the second assignment should be submitted before the conduct of the second mid-term examination. The

average of the two assignments shall be taken as the final marks for assignment (for 5 marks).

Subject Viva-Voce/PPT/Poster Presentation/Case Study on a topic in the subject concerned for 5 marks before II Mid-Term Examination.

- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Overall 40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores $\geq 35\%$ (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled in spite of appearing the SEE.

There is NO Computer Based Test (CBT) for R22 regulations.

The details of the end semester question paper pattern are as follows:

8.2.1 The semester end examinations (SEE), for theory subjects, will be conducted for 60 marks consisting of two parts viz. i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- a. Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- b. Part-B consists of five questions (numbered from 2 to 6) carrying 10 marks each. Each of these questions is from each unit and may contain sub-questions. For each question there will be an “either” “or” choice, which means that there will be two questions from each unit and the student should answer either of the two questions.
- c. The duration of Semester End Examination is 3 hours.

8.3 For practical subjects there shall be a Continuous Internal Evaluation (CIE) during the semester for 40 marks and 60 marks for semester end examination. Out of the 40 marks for internal evaluation:

1. A write-up on day-to-day experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for 10 marks
2. **10 marks for viva-voce** (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
3. Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 10 marks.
4. The remaining 10 marks are for Laboratory Project, which consists of the Design (or) Software/Hardware Model Presentation (or) App Development

(or) Prototype Presentation submission which shall be evaluated after completion of laboratory course and before semester end practical examination.

The Semester End Examination shall be conducted with an external examiner and the laboratory teacher. The external examiner shall be appointed from the other colleges which will be decided by the principal of the College.

In the Semester End Examination held for 3 hours, total 60 marks are divided and allocated as shown below:

1. 10 marks for write-up
 2. 15 for experiment/program
 3. 15 for evaluation of results
 4. 10 marks for presentation on another experiment/program in the same laboratory course and
 5. 10 marks for viva-voce on concerned laboratory course.
- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Overall 40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores $\geq 35\%$ (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled in spite of appearing the SEE.

8.4 The evaluation of courses having ONLY internal marks in I Year I Semester and II Year II Semester is as follows:

1. I Year I, II Semester course (ex., *Elements of CE/ME/EEE/ECE/CSE/Seminar & etc.*): The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e. 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course. Seminar evaluation conducted through day to day evaluation of the students in presentation/abstract.
2. II Year II Semester *Real-Time (or) Field-based Research Project* course: The internal evaluation is for 50 marks and it shall take place during I Mid-

Term examination and II Mid-Term examination. The average marks of two Mid-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e. 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the internal committee as per schedule, or (ii) secures less than 40% marks in this course.

- 8.5** There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization. Students shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation/semester break & during III Year without effecting regular course work. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc., Internal Supervisor and a Senior Faculty Member of the Department. There shall be **NO internal marks** for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.
- 8.6** The UG project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Stage – I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of his Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start his project work.
- 8.7** UG project work shall be carried out in two stages: Project Stage – I for approval of project before Mid-II examinations in IV Year I Semester and Project Stage – II during IV Year II Semester. Student has to submit project work report at the end of IV Year II Semester. The project shall be evaluated for 100 marks before commencement of SEETheory examinations.
- 8.8** For Project Stage – I, the departmental committee consisting of Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he does not submit a report on Project Stage - I or does not make a presentation of the same before the evaluation committee as per schedule.

A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

- 8.9** For Project Stage – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini Project/Internship/SDC etc. and the main Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

For conducting viva-voce of project, Principal/COE selects an external examiner from the list of experts in the relevant branch submitted by the Principal of the College.

A student who has failed, may reappear once for the above evaluation, when it is scheduled again; if student fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

- 8.10** A student shall be given only one time chance to re-register for a maximum of two subjects in a semester:

- If the internal marks secured by a student in the Continuous Internal Evaluation marks for 40 (Sum of average of two mid-term examinations consisting of Objective & descriptive parts, Average of two Assignments & Subject Viva-voce/PPT/Poster presentation/Case Study on a topic in the concerned subject) are less than 35% and failed in those subjects.

A student must re-register for the failed subject(s) for 40 marks within four weeks of commencement of the classwork in next academic year.

In the event of the student taking this chance, his Continuous Internal Evaluation marks for 40 and Semester End Examination marks for 60 obtained in the previous attempt stand cancelled.

- 8.11** For mandatory non-credit Audit courses, The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Student shall have to earn 40%, i.e. 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course.

- 8.12** For Mandatory Non-Credit Courses offered in a Semester, after securing $\geq 65\%$ attendance and has secured not less than 40% of marks in the sum Total. Then the student is **PASS** and will be qualified for the award of the degree. No marks

or Letter Grade shall be allotted for these courses/activities. However, for non-credit courses '**Satisfactory**' or '**Unsatisfactory**' shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.

9.0 Grading Procedure

9.1 Grades will be awarded to indicate the performance of students in each Theory Subject, Laboratory/Practicals/Industry-Oriented Mini Project/Internship/SDC and Project Stage. Based on the percentage of marks obtained (Continuous Internal Evaluation plus Semester End Examination, both taken together) as specified in item 8 above, a corresponding letter grade shall be given.

9.2 As a measure of the performance of a student, a 10-point absolute grading system using the following letter grades (as per UGC/AICTE guidelines) and corresponding percentage of marks shall be followed:

% of Marks Secured in a Subject/Course (Class Intervals)	Letter Grade (UGC Guidelines)	Grade Points
Greater than or equal to 90%	O (Outstanding)	10
80 and less than 90%	A ⁺ (Excellent)	9
70 and less than 80%	A (Very Good)	8
60 and less than 70%	B ⁺ (Good)	7
50 and less than 60%	B (Average)	6
40 and less than 50%	C (Pass)	5
Below 40%	F (FAIL)	0
Absent	Ab	0

9.3 A student who has obtained an '**F**' grade in any subject shall be deemed to have '**failed**' and is required to reappear as a 'supplementary student' in the semester end examination, as and when offered. In such cases, internal marks in those subjects will remain the same as those obtained earlier.

9.4 To a student who has not appeared for an examination in any subject, '**Ab**' grade will be allocated in that subject, and he is deemed to have '**Failed**'. A student will be required to reappear as a 'supplementary student' in the semester end examination, as and when offered next. In this case also, the internal marks in those subjects will remain the same as those obtained earlier.

9.5 A letter grade does not indicate any specific percentage of marks secured by the student, but it indicates only the range of percentage of marks.

9.6 A student earns Grade Point (GP) in each subject/course, on the basis of the letter grade secured in that subject/course. The corresponding 'Credit Points' (CP) are computed by multiplying the grade point with credits for that particular subject/course.

Credit Points (CP) = Grade Point (GP) x Credits For a course

9.7 A student passes the subject/course only when **GP ≥ 5 ('C' grade or above)**

9.8 The Semester Grade Point Average (SGPA) is calculated by dividing the sum of credit points ($\sum CP$) secured from all subjects/courses registered in a semester,

by the total number of credits registered during that semester. SGPA is rounded off to **two** decimal places. SGPA is thus computed as

$$\text{SGPA} = \{\sum_{i=1}^N C_i G_i\} / \{\sum_{i=1}^N C_i\} \dots \text{For each Semester,}$$

where 'i' is the subject indicator index (considering all subjects in a semester), 'N' is the no. of subjects '**registered**' for the semester (as specifically required and listed under the course structure of the parent department), C_i is the no. of credits allotted to the i^{th} subject, and G_i represents the grade points (GP) corresponding to the letter grade awarded for that i^{th} subject.

- 9.9** The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student in all semesters considered for registration. The CGPA is the ratio of the total credit points secured by a student in **all** registered courses (of 160) in **all** semesters, and the total number of credits registered in **all** the semesters. CGPA is rounded off to **two** decimal places. CGPA is thus computed from the I year II semester onwards at the end of each semester as per the formula

$$\text{CGPA} = \{ \sum_{j=1}^M C_j G_j \} / \{ \sum_{j=1}^M C_j \} \dots \text{for all S Semesters registered (i.e., up to and inclusive of S Semesters, } S \geq 2),$$

where 'M' is the TOTAL no. of Subjects (as specifically required and listed under the Course Structure of the parent Department) the Student has '**REGISTERED**' from the 1st Semester onwards up to and inclusive of the Semester S (obviously $M > N$), 'j' is the Subject indicator index (takes into account all Subjects from 1 to S Semesters), C_j is the no. of Credits allotted to the j^{th} Subject, and G_j represents the Grade Points (GP) corresponding to the Letter Grade awarded for that j^{th} Subject. After registration and completion of I Year I Semester however, the SGPA of that Semester itself may be taken as the CGPA, as there are no cumulative effects.

Illustration of calculation of SGPA:

Course/Subject	Credits	Letter Grade	Grade Points	Credit Points
Course 1	4	A	8	$4 \times 8 = 32$
Course 2	4	O	10	$4 \times 10 = 40$
Course 3	4	C	5	$4 \times 5 = 20$
Course 4	3	B	6	$3 \times 6 = 18$
Course 5	3	A+	9	$3 \times 9 = 27$
Course 6	3	C	5	$3 \times 5 = 15$
	21			152

$$\text{SGPA} = 152/21 = 7.24$$

Illustration of Calculation of CGPA up to 3rd Semester:

Semester	Course/Subject Title	Credits Allotted	Letter Grade Secured	Corresponding Grade Point (GP)	Credit Points (CP)
I	Course 1	3	A	8	24
I	Course 2	3	O	10	30
I	Course 3	3	B	6	18
I	Course 4	4	A	8	32
I	Course 5	3	A+	9	27
I	Course 6	4	C	5	20
II	Course 7	4	B	6	24
II	Course 8	4	A	8	32
II	Course 9	3	C	5	15
II	Course 10	3	O	10	30
II	Course 11	3	B+	7	21
II	Course 12	4	B	6	24
II	Course 13	4	A	8	32
II	Course 14	3	O	10	30
III	Course 15	2	A	8	16
III	Course 16	1	C	5	5
III	Course 17	4	O	10	40
III	Course 18	3	B+	7	21
III	Course 19	4	B	6	24
III	Course 20	4	A	8	32
III	Course 21	3	B+	7	21
	Total Credits	69		Total Credit Points	518

$$\text{CGPA} = 518/69 = 7.51$$

The calculation process of CGPA illustrated above will be followed for each subsequent semester until 8th semester. The CGPA obtained at the end of 8th semester will become the final CGPA secured for entire B.Tech. programme.

9.10 For merit ranking or comparison purposes or any other listing, **only** the 'rounded off' values of the CGPAs will be used.

9.11 SGPA and CGPA of a semester will be mentioned in the semester Memorandum of Grades if all subjects of that semester are passed in first attempt. Otherwise the SGPA and CGPA shall be mentioned only on the Memorandum of Grades in which sitting he passed his last exam in that semester. However, mandatory courses will not be taken into consideration.

10.0 Passing Standards

10.1 A student shall be declared successful or 'passed' in a semester, if he secures a $GP \geq 5$ ('C' grade or above) in every subject/course in that semester (i.e.

when the student gets an SGPA ≥ 5.0 at the end of that particular semester); and he shall be declared successful or 'passed' in the entire undergraduate programme, only when gets a CGPA ≥ 5.00 ('C' grade or above) for the award of the degree as required.

- 10.2** After the completion of each semester, a grade card or grade sheet shall be issued to all the registered students of that semester, indicating the letter grades and credits earned. It will show the details of the courses registered (course code, title, no. of credits, grade earned, etc.) and credits earned. **There is NO exemption of credits in any case.**

11.0 Declaration of results

- 11.1** Computation of SGPA and CGPA are done using the procedure listed in 9.6 to 9.9.
- 11.2** For final percentage of marks equivalent to the computed final CGPA, the following formula may be used.

$$\% \text{ of Marks} = (\text{final CGPA} - 0.5) \times 10$$

12.0 Award of Degree

- 12.1** A student who registers for all the specified subjects/courses as listed in the course structure and secures the required number of 160 credits (with CGPA ≥ 5.0), within 8 academic years from the date of commencement of the first academic year, shall be declared to have '**qualified**' for the award of B.Tech. degree in the branch of Engineering selected at the time of admission.
- 12.2** A student who qualifies for the award of the degree as listed in item 12.1 shall be placed in the following classes.
- 12.3** A student with final CGPA (at the end of the undergraduate programme) > 8.00 , and fulfilling the following conditions - shall be placed in '**First Class with Distinction**'.
- However, he
- (i) Should have passed all the subjects/courses in '**First Appearance**' within the first 4 academic years (or 8 sequential semesters) from the date of commencement of first year first semester.
 - (ii) Should not have been detained or prevented from writing the semester end examinations in any semester due to shortage of attendance or any other reason.
- A student not fulfilling any of the above conditions with final CGPA > 8 shall be placed in '**First Class**'.
- 12.4** Students with final CGPA (at the end of the undergraduate programme) ≥ 7.0 but < 8.00 shall be placed in '**First Class**'.
- 12.5** Students with final CGPA (at the end of the undergraduate programme) ≥ 6.00 but < 7.00 , shall be placed in '**Second Class**'.
- 12.6** All other students who qualify for the award of the degree (as per item 12.1), with final CGPA (at the end of the undergraduate programme) ≥ 5.00 but < 6 , shall be placed in '**pass class**'.

12.7 A student with final CGPA (at the end of the undergraduate programme) < 5.00 will not be eligible for the award of the degree.

12.8 Students fulfilling the conditions listed under item 12.3 alone will be eligible for award of '**Gold Medal**'.

12.9 Award of 2-Year B.Tech. Diploma Certificate

1. A student is awarded 2-Year UG Diploma Certificate in the concerned engineering branch on completion of all the academic requirements and earned all the 80 credits (within 4 years from the date of admission) up to B. Tech. – II Year – II Semester, if the student want to exit the 4-Year B. Tech. program. The student **once opted and awarded for 2-Year UG Diploma Certificate, the student will not be permitted to join** in B. Tech. III Year – I Semester and continue for completion of remaining years of study for 4-Year B. Tech. Degree.
2. A student may be permitted to take one year break after completion of II Year – II Semester or B. Tech. – III Year – II Semester (with Principal permission through the HOD of the department well in advance) and can re-enter the course in **next Academic Year in the same college** and complete the course on fulfilling all the academic credentials within a stipulated duration i.e. double the duration of the course (Ex. within 8 Years for 4-Year program).

13.0 Withholding of results

13.1 If the student has not paid the fees to the College at any stage, or has dues pending due to any reason whatsoever, or if any case of indiscipline is pending, the result of the student may be withheld, and the student will not be allowed to go into the next higher semester. The award or issue of the degree may also be withheld in such cases.

14.0 Transitory Regulations

A. For students detained due to shortage of attendance:

1. A Student who has been detained in I year of R18 Regulations due to lack of attendance, shall be permitted to join I year I Semester of R22 Regulations and he is required to complete the study of B.Tech./B. Pharmacy programme within the stipulated period of eight academic years from the date of first admission in I Year.
2. A student who has been detained in any semester of II, III and IV years of R18 regulations for want of attendance, shall be permitted to join the corresponding semester of R22 Regulations and is required to complete the study of B.Tech./B.Pharmacy within the stipulated period of eight academic years from the date of first admission in I Year. The R22 Academic Regulations under which a student has been readmitted shall be applicable to that student from that semester. See rule (C) for further Transitory Regulations.

B. For students detained due to shortage of credits: A student of MLR18 Regulations who has been detained due to lack of credits, shall be promoted to the next semester of R22 Regulations only after acquiring the

required number of credits as per the corresponding regulations of his/her first admission. The total credits required are 160 including both MLR18 & R22 regulations. The student is required to complete the study of B.Tech. within the stipulated period of eight academic years from the year of first admission. The R22 Academic Regulations are applicable to a student from the year of readmission. See rule (C) for further Transitory Regulations for re-admitted students in R22 Regulations:

3. A student who has failed in any subject under any regulation has to pass those subjects in the same regulations.
4. The maximum credits that a student acquires for the award of degree, shall be the sum of the total number of credits secured in all the regulations of his/her study including R22 Regulations. **There is NO exemption of credits in any case.**
5. If a student is readmitted to R22 Regulations and has any subject with 80% of syllabus common with his/her previous regulations, that particular subject in R22 Regulations will be substituted by another subject to be suggested by the Concerned BOS.

15.0 Student Transfers

- 15.1 There shall be no branch transfers after the completion of admission process.
- 15.2 There shall be no transfers from one college/stream to another within the constituent colleges and units of Jawaharlal Nehru Technological University Hyderabad.
- 15.3 The students seeking transfer to colleges affiliated to JNTUH from various other Universities/institutions have to pass the failed subjects which are equivalent to the subjects of JNTUH, and also pass the subjects of JNTUH which the students have not studied at the earlier institution. Further, though the students have passed some of the subjects at the earlier institutions, if the same subjects are prescribed in different semesters of JNTUH, the students have to study those subjects in JNTUH in spite of the fact that those subjects are repeated.
- 15.4 The transferred students from other Universities/Institutions to College who are on rolls are to be provided one chance to write the CBT (for internal marks) in the **equivalent subject(s)** as per the clearance letter issued by the University.
- 15.5 The college have to provide one chance to write the internal examinations in the **equivalent subject(s)** to the students transferred from other universities/institutions to JNTUH autonomous affiliated colleges who are on rolls, as per the clearance (equivalence) letter issued by the University.

16.0 Malpractice Prevention Committee

A malpractice prevention committee shall be constituted to examine and punish the students who involve in malpractice/indiscipline in examinations. The committee shall consist of:

- a) Controller of examinations - Chairman
- b) Addl. Controller of examinations.- Member Convener
- c) Subject expert - member
- d) Head of the department of which the student belongs to. - Member
- e) The invigilator concerned - member

The committee shall conduct the meeting after taking explanation of the student and punishment will be awarded by following the malpractice rules meticulously.

Any action on the part of candidate at the examination like trying to get undue advantage in the performance at examinations or trying to help another, or derive the same through unfair means is punishable according to the provisions contained hereunder. The involvement of the Staff who are in charge of conducting examinations, valuing examination papers and preparing/keeping records of documents relating to the examinations, in such acts (inclusive of providing incorrect or misleading information) that infringe upon the course of natural justice to one and all concerned at the examination shall be viewed seriously and will be recommended for appropriate punishment after thorough enquiry.

17.0 Scope

- 17.1 The academic regulations should be read as a whole, for the purpose of any interpretation.
- 17.2 In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Chairman of Academic Council is final.
- 17.3 The College may change or amend the academic regulations, course structure or syllabi at any time, and the changes or amendments made shall be applicable to all students with effect from the dates notified by the college authorities.
- 17.4 Where the words “he”, “him”, “his”, occur in the regulations, they include “she”, “her”, “hers”.

ACADEMIC REGULATIONS FOR B.TECH (LATERAL ENTRY SCHEME)
FROM THE AY 2023-24

1. **Eligibility for the award of B.Tech Degree (LES)**

The LES students after securing admission shall pursue a course of study for not less than three academic years and not more than six academic years.

2. The student shall register for 120 credits and secure 120 credits with CGPA ≥ 5 from II year to IV-year B.Tech. programme (LES) for the award of B.Tech. degree.
3. The students, who fail to fulfil the requirement for the award of the degree in six academic years from the year of admission, shall forfeit their seat in B.Tech.
4. The attendance requirements of B. Tech. (Regular) shall be applicable to B.Tech. (LES).

5. **Promotion rule:**

S. No	Promotion	Conditions to be fulfilled
1	Second year first semester to second year second semester	Regular course of study of second year first semester.
2	Second year second semester to third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 24 credits out of 40 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3	Third year first semester to third year second semester	Regular course of study of third year first semester.
4	Third year second semester to fourth year first semester	i) Regular course of study of third year second semester. ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Fourth year first semester to fourth year second semester	Regular course of study of fourth year first semester.

6. All the other regulations as applicable to B. Tech. 4-year degree course (Regular) will hold good for B. Tech. (Lateral Entry Scheme).
7. LES students are not eligible for 2-Year B. Tech. Diploma Certificate.

Malpractices Rules
Disciplinary Action For/Improper Conduct in Examinations

	Nature of Malpractices/ Improper conduct	Punishment
	If the student:	
1. (a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which student is appearing but has not made use of (material shall include any marks on the body of the student which can be used as an aid in the subject of the examination).	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other student orally or by any other body language methods or communicates through cell phones with any student or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the students involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the student is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The hall ticket of the student is to be cancelled

3.	Impersonates any other student in connection with the examination.	The student who has impersonated shall be expelled from examination hall. The student is also debarred and forfeits the seat. The performance of the original student who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.
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4.	Smuggles in the answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.
5.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.
6.	Refuses to obey the orders of the chief superintendent/COE/ACoE/any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the college campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the student(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The students also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.

7.	Leaves the exam hall taking away answer script or intentionally tears off the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.
8.	Possesses any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat.
9.	If student of the college, who is not a student for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat. Person(s) who do not belong to the college will be handed over to the police and, a police case will be registered against them.
10.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared for including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the student has appeared for including practical examinations and project work of that semester/year examinations.

12	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the Principal for further action to award a suitable punishment.	
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Malpractices Identified by Squad or Special Invigilators

1. Punishments to the students as per the above guidelines.
2. Punishment for staff: (if the squad reports that the staff is also involved in encouraging malpractices)
 - a. A show-cause notice shall be issued to the staff.
 - b. Impose a suitable fine on the staff.

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COURSE STRUCTURE

COURSE STRUCTURE

B. TECH – COMPUTER SCIENCE AND ENGINEERING

REGULATIONS: R22

I B.Tech.- I Semester									
Induction program for one weeks									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal	External	Total
A6BS01	Linear Algebra and Calculus	BSC	3	1	0	4	40	60	100
A6CS02	Programming for Problem Solving	ESC	3	0	0	3	40	60	100
A6HS01	English for Skill Enhancement	HSMC	3	0	0	3	40	60	100
A6EE60	Basic Electrical and Electronics Engineering	ESC	3	0	0	3	40	60	100
A6EC03	Electronic Devices and Applications	ESC	2	0	0	2	40	60	100
A6CS03	Programming for Problem Solving Lab	ESC	0	0	3	1.5	40	60	100
A6HS02	English Language and Communication Skills Lab	HSMC	0	0	3	1.5	40	60	100
A6EC04	Introduction to Internet of Things	ESC	0	0	2	1	40	60	100
A6HS04	Seminar	HSMC	0	0	2	1	100	0	100
TOTAL			14	1	10	20	420	480	900
I B.Tech.- II Semester									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal	External	Total
A6BS02	Numerical Methods and Integral Transforms	BSC	3	1	0	4	40	60	100
A6BS07	Applied Physics	BSC	3	1	0	4	40	60	100
A6BS09	Engineering Chemistry	BSC	3	1	0	4	40	60	100
A6ME02	Engineering Drawing	ESC	1	0	3	2.5	40	60	100
A6BS08	Applied Physics Lab	BSC	0	0	3	1.5	40	60	100
A6CS04	Python Programming Lab	ESC	0	0	3	1.5	40	60	100
A6ME04	Engineering Work Shop	ESC	0	0	3	1.5	40	60	100
A6CS01	Elements of Computer Science and Engineering	HSMC	0	0	2	1	100	0	100
A6BS11	Environmental Science	MC	2	0	0	0	50	0	50
TOTAL			12	3	14	20	430	420	850

II B.Tech.- I Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6BS03	Computer Oriented Statistical Methods	BSC	3	1	-	4	40	60	100
A6CS28	Digital Electronics and Computer Organization	ESC	3	-	-	3	40	60	100
A6CS05	Data Structures	ESC	3	-	-	3	40	60	100
A6IT02	Object Oriented Programming through JAVA	PCC	3	-	-	3	40	60	100
A6CS07	Software Engineering	PCC	3	-	-	3	40	60	100
A6CS06	Data Structures Lab	ESC	-	-	3	1.5	40	60	100
A6IT03	Object Oriented Programming through JAVA Lab	PCC	-	-	3	1.5	40	60	100
A6CS52	Skill Development (Data Visualization using R)	SKL	0	0	2	1	0	100	100
A6HS05	Gender Sensitization	MC	2	-	-	0	50	0	50
Total			17	01	8	20	330	520	850

II B.Tech.- II Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6CS08	Discrete Mathematics	BSC	3	-	-	3	40	60	100
A6HS08	Business Economics and Financial Analysis	HSMC	3	-	-	3	40	60	100
A6CS09	DataBase Management Systems	PCC	3	-	-	3	40	60	100
A6CS11	Operating System	PCC	3	-	-	3	40	60	100
A6CS13	Software Testing Fundamentals	PCC	3	-	-	3	40	60	100
A6CS10	Database Management Systems Lab	PCC	-	-	3	1.5	40	60	100
A6CS12	Operating System Lab	PCC	-	-	3	1.5	40	60	100
A6CS14	Real Time Research Project/Societal Relevant Project	PWC	-	-	2	1	50	0	50
A6CS53	Skill Development (HTML, XML, Javascript, NodeJS/ Mobile Appl)	SKL	0	0	2	1	0	100	100
A6HS06	Constitution of India	MC	2	-	-	0	50	0	50
Total			17	00	10	20	380	520	900

III B.Tech.- I SEMESTER									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6CS15	Design and Analysis of Algorithms	PCC	3	-	-	3	40	60	100
A6IT13	Cloud and DevOps	PCC	3	-	-	3	40	60	100
A6CS18	Computer Networks	PCC	3	-	-	3	40	60	100
PCC	Professional Elective - I	PEC	3	-	-	3	40	60	100
PCC	Professional Elective - II	PEC	3	-	-	3	40	60	100
A6CS19	Computer Networks Lab	PCC	-	-	3	1.5	40	60	100
A6IT16	Cloud and DevOps Lab	PCC	0	0	3	1.5	40	60	100
A6HS03	Advanced Communication Skills Lab	HSMC	-	-	2	1	40	60	100
A6CS54	Independent Study/MOOCs/NPTEL	PWC	0	0	2	1	0	100	100
A6HS10	Human Values & Professional Ethics	MC	2	-	-	0	50	0	50
Total			17	0	10	20	370	580	950

III B.Tech.- II Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6IT39	Introduction to Artificial Intelligence	PCC	3	-	-	3	40	60	100
A6IT11	Automata and Compiler Design	PCC	3	-	-	3	40	60	100
A6CS16	Data Mining and Machine Learning	PCC	3	-	-	3	40	60	100
PCC	Professional Elective - III	PEC	3	-	-	3	40	60	100
OEC	Open Elective - I	OEC	3	-	-	3	40	60	100
PCC	Professional Elective III - Lab	PEC	-	-	3	1.5	40	60	100
A6CS17	Data Mining and Machine Learning Lab	PCC	-	-	3	1.5	40	60	100
A6CS21	Industry Oriented Mini Project/Internship	PWC	-	-	4	2	0	100	100
A6BS11	Environmental Science (only for Lateral Entry Students)	ES	2	-	-	-	50	0	50
Total			17	0	10	20	330	520	850

IV B.Tech.- I Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6CS22	Distributed Computing	PCC	3	-	-	3	40	60	100
A6CY06	Cryptography and Network Security	PCC	3	-	-	3	40	60	100
PCC	Professional Elective - IV	PEC	3	-	-	3	40	60	100
PCC	Professional Elective - V	PEC	3	-	-	3	40	60	100
OEC	Open Elective II	OEC	3	-	-	3	40	60	100
A6CY08	Cryptography and Network Security Lab	PCC	-	-	2	1	40	60	100
A6CS25	Research Project Phase -1	PWC	-	-	8	4	100	0	100
Total			15	0	10	20	340	360	700

IV B.Tech. - II Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
PCC	Professional Elective - VI	PEC	3	-	-	3	40	60	100
OEC	Open Elective III	OEC	3	-	-	3	40	60	100
A6CS26	Industry Specific Training/Internship/Technical Paper	PWC	-	-	4	2	40	60	100
A6HS15	Organizational Behavior	HSMC	-	-	2	1	40	60	100
A6CS27	Major Project Phase – 2	PWC	-	-	22	11	40	60	100
Total			06	-	28	20	200	300	500

SEM	I	II	III	IV	V	VI	VII	VIII	Total
Credits	20	20	20	20	20	20	20	20	160

***Environmental Studies have to be taken by LE in 2-1**

****MC: Satisfactory/Non-Satisfactory**

PROFESSIONAL ELECTIVES			
PE – I		PE – II	
A6IT23	Introduction to Data Science	A6CS43	Data Warehousing and Business Intelligence
A6IT29	Soft Computing	A6AI14	Natural Language Processing
A6CS29	Blockchain Technology and Web 3.0	A6CS30	Permissioned Block chain – Hyper ledger
A6CS36	Cyber Security& Cyber laws	A6CY10	Ethical Hacking
A6IT26	Information Retrieval Systems	A6CS37	Semantic Web
PE – III		PE – IV	
A6DS06	Big Data Technologies	A6AI18	Knowledge Representation and Reasoning
A6AI17	Deep Learning	A6AI21	Cognitive Computing
A6CS31	Smart Contract Development and Audit	A6CS33	Blockchain Taxonomy & Crypto Currencies
A6CY16	Crime Investigation & Digital Forensics	A6CY13	Vulnerability Assessment and Penetration Testing
A6CS39	Image Processing Techniques	A6CS41	Speech and Video Processing
PE – V		PE – VI	
A6CS46	Web and Social Media Analytics	A6CS47	Nature Inspired computing for Data Science
A6CS55	Perception and Computer Vision	A6IT38	Human Computer Interaction
A6CS34	Block Chain for FinTech and Non FinTech	A6CS35	Financial Risk Management for Crypto currencies
A6CY30	Web & Database Security	A6CS38	Cloud Security
A6AI22	Augmented Reality & Virtual Reality	A6CS42	IoT Cloud Processing and Analytics
PE – III LAB			
A6DS07	Big Data Technologies Lab		
A6AI19	Deep Learning Lab		
A6CS32	Smart Contract Development and Audit Lab		
A6CY17	Crime Investigation & Digital Forensics Lab		
A6CS40	Image Processing Techniques Lab		

OPEN ELECTIVE COURSES

OPEN ELECTIVE - I			
S. No.	Course Code	Course Name	Offering Department
1	A6AE19	Fundamentals of Avionics	Aeronautical Engineering
2	A6AE20	Introduction to Aerospace Technology	
3	A6CS48	Software Project Management Fundamentals	Computer Science and Engineering
4	A6CS37	Semantic Web	
5	A6EC55	Microprocessors and Interfacing	Electronics & Communication Engineering
6	A6EC56	Principles of Communications	
7	A6EE55	Electrical Wiring and Safety Measures	Electrical &Electronics Engineering
8	A6EE56	Electrical Materials	
9	A6IT28	Adhoc & Sensor Networks	Information Technology
10	A6IT35	Object Oriented Programming	
11	A6ME55	Basic Mechanical Engineering	Mechanical Engineering
12	A6ME60	Fundamentals of Engineering Materials	
13	A6HS08	Business Economics and Financial Analysis	HSM
14	A6HS09	Basics of Entrepreneurship	
OPEN ELECTIVE - II			
S. No.	Course Code	Course Name	Offering Department
1	A6AE25	Introduction to jets and rockets	Aeronautical Engineering
2	A6AE26	Non-Destructive Testing Methods	
3	A6CS09	Database Management System	Computer Science and Engineering
4	A6CS43	Data Warehousing and Business Intelligence	
5	A6EC57	Microcontrollers and Applications	Electronics & Communication Engineering
6	A6EC58	Fundamentals of Image processing	
7	A6EE57	Neural Networks and Fuzzy Logic	Electrical &Electronics Engineering
8	A6EE58	Energy Storage Systems	
9	A6IT30	Security Testing	Information Technology
10	A6IT36	Human Computer Interaction	
11	A6ME65	Fundamentals of Operations Research	Mechanical Engineering
12	A6ME70	Industrial Engineering & Management	
13	A6HS11	Advanced Entrepreneurship	HSM

OPEN ELECTIVE - III			
S. No.	Course Code	Course Name	Offering Department
1	A6AE40	Introduction to Aircraft Industry	Aeronautical Engineering
2	A6AE41	Unmanned Aerial Vehicles	
3	A6CS51	Artificial Intelligence and Applications	Computer Science and Engineering
4	A6CS36	Cyber Security& Cyber laws	
5	A6EC59	Introduction to Sensors and Actuators	Electronics & Communication Engineering
6	A6EC60	Introduction to Computer Vision	
7	A6EE59	Solar Energy and Applications	Electrical & Electronics Engineering
8	A6EE65	Instrumentation and Control	
9	A6IT37	Introduction to Computer Networks	Information Technology
10	A6IT31	High Performance Computing	
11	A6ME75	Renewable Energy Sources	Mechanical Engineering
12	A6ME80	Elements of Electric and Hybrid vehicles	
13	A6HS12	Management Science	HSM

Academic Guidelines

Professional Electives:

- a. Professional Electives are streamed to either Data Science/AIML/Block Chain/Cyber Security
- b. Students must choose the same stream in all Professional Electives.

Skill Development:

- a. Skill development courses will be conducted as practical hours and evaluated for one credit.

Summer Internship/Research Project in II-II semester

Summer Internship/Internship

- a. Student can either do internship from AICTE approved Internships or in any Industry.
- b. In case of Internship from Industry, two supervisors will be supporting the students. Supervisors: 1. One Internal supervisor and 2. One Industry Personnel (External)
- c. Evaluation will be done in III-I

Rubrics - Internship

	Internal Supervisor	External Supervisor	Total
Marks	40 Mid – 20; Final 20;	60	100

Mini Project in III-II

Student can carry out mini project based on the courses learned in II and III years.

- a. Student can carry out project in guidance of an internal supervisor.
- b. Students can carry out projects in team of three or individually.
- c. One supervisor per student/team will supervise a project and the supervisor shall be internal supervisor.
- d. Evaluation will be done in III-II itself.

Research Project:

- a. Student can carry out project with respect to research question.
- b. Students can carry out projects in team of three or individually.
- c. One supervisor per student/team will supervise a research project and the supervisor shall be internal supervisor or external supervisor from JNTU/IIIT/IIT.
- d. Evaluation will be done in III-I.

Rubrics

S.No.	Tasks	Marks	Review
1	Research Problem/Question	5	Review-I
2	Research Design	5	
3	Methodology	5	
4	Implementation	40	Review-II
5	Results and Discussion	20	
6	Deployment	10	Review-III
7	Publish/Patent	10	
8	Viva/Presentation	5	

Massive Open Online Courses (MOOCs)

- a. Student can undergo one MOOC course from NPTEL/INFOSYS Springboard/any platform which permits credit transfer as per AICTE.
- b. It is purely self-paced/self-study course.
- c. Certificate should be produced by student to the department/exam cell for considering the credit at the end of semester.
- d. Evaluation will be done in III-II itself for 100%

Global Certification

- a. Student can undergo Global Certification on their own or in support of Placement cell or department.
- b. Student can choose a course in such way that the course must have minimum 40 hours of duration and certificate must be directly from company.
- c. Ex: Computer Networks and Security [CISCO], AWS Certification [AWS Academy]
- d. Evaluation will be done in III-II itself for 100%

I B.TECH I SEMESTER SYLLABUS

LINEAR ALGEBRA AND CALCULUS

I B.TECH - I SEMESTER

I B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6BS01	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	40	60	100
Contact Classes: 44	Tutorial Classes: 08	Practical Classes: Nil				Total Classes: 52		

COURSE OBJECTIVES

The course will enable the students to:

1. Concept of Rank of a matrix, Consistency and solving system of linear equations.
2. Concept of Eigen values, Eigen vectors and diagonalization of the matrix.
3. The concept of differential equations and solve them using appropriate methods.
4. Evaluate multiple integrals and improper integrals.
5. The partial derivatives of several variable functions.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Solve the system of linear equations using rank of the matrices.
2. Find the Eigen values and Eigen vectors of a matrix.
3. Identify the different types of differential equations and solve those using appropriate methods.
4. Evaluate the improper integrals using beta and gamma functions.
5. Find the Maxima and Minima of several variable functions.

UNIT - I	MATRICES AND THEIR APPLICATIONS	CLASSES: 08
Real matrices: Symmetric-skew-symmetric and orthogonal matrices –Complex matrices: Hermitian, Skew – Hermitian and Unitary matrices –Elementary row and column transformations –Elementary matrix-Finding rank of a matrix by reducing to Echelon form and Normal form-Finding the inverse of a matrix using elementary row/column transformations (Gauss-Jordan method)-Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix –Solving m n and n n linear system of equations by Gauss Elimination, Gauss sie del Method		
UNIT - II	EIGEN VALUES, EIGEN VECTORS	CLASSES: 08
Eigen values and Eigen vectors and its properties (without proof), Cayley-Hamilton theorem (Statement and verification)-Finding inverse and powers of a matrix by Cayley-Hamilton theorem, Diagonalization of matrices. Quadratic forms and Nature of the Quadratic Forms, Reduction of Quadratic form to Canonical forms by Orthogonal Transformation.		
UNIT - III	ORDINARY DIFFERENTIAL EQUATIONS AND THEIR APPLICATIONS	CLASSES: 10
Introduction- Exact and reducible to Exact differential equations-Newton's Law of cooling-Law of Growth and Decay. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type $Q(x) = e^{ax}$, $\sin ax$, $\cos ax$, $e^{ax}v(x)$, $x^n v(x)$ - Method of variation of parameters L-C-R Circuits.		

UNIT - IV	MULTIPLE INTEGRALS, BETA AND GAMMA FUNCTIONS	CLASSES: 10
Double and triple integrals (Cartesian and polar), Change of order of integration in double integrals, Change of variables (Cartesian to polar) in double integrals. Finding the area and volume of a region using double and triple integral. Beta-Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions.		
UNIT - V	CALCULUS OF SEVERAL VARIABLES	CLASSES: 08
Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative - Chain rule, Jacobians- functional dependence & independence. Applications: Maxima and Minima of functions of two variables without constraints and Lagrange's method (with constraints).		
TEXT BOOKS		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.		
REFERENCE BOOKS		
1. G.B.Thomas, Calculus and Analytical Geometry, 9th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal, A Text of Engineering Mathematics, Laxmi publications, 2008. 3. E.L.Ince, Ordinary differential Equations, Dover publications, 1958.		
WEB REFERENCES		
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/		
E -TEXT BOOKS		
1. https://www.e-booksdirectory.com/details.php?ebook=10166		
MOOCS COURSE		
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/		

PROGRAMMING FOR PROBLEM SOLVING

I B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS02	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
Contact Classes: 64	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 64			
COURSE OBJECTIVES								
The course will enable the students to:								
1. To familiarize with the syntax and semantics of C programming language. 2. To learn the usage of structured programming approach in solving problems. 3. To use arrays, pointers, strings and structures in solving problems.								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Apply algorithmic thinking to understand, define and solve problems 2. Develop computer programs using programming constructs and control structures and to use arrays to develop C programs 3. Decompose a problem into functions to develop modular reusable code and to use pointers to solve complex problems. 4. Use Strings and structures to formulate algorithms and programs. 5. Use FILE to perform read and write operations.								
UNIT - I	INTRODUCTION - PROBLEM SOLVING AND ALGORITHMIC THINKING& INTRODUCTION TO C LANGUAGE						CLASSES: 12	
Algorithm -Definition, Characteristics of Algorithm. Constituents of algorithms: - Sequence, Selection and Repetition. Algorithm with Example: Roots Of a Quadratic Equations, Minimum and Maximum Numbers of a Given Set, Given number is prime number or not, given integer is palindrome or not, etc. Flowchart/Pseudo Code with examples. Introduction To C Language: Structure of C Program, Data Types, Input and Output statements, Operators, Precedence and Associativity of operators, Evaluation of Expressions, Type Conversions In Expressions.								
UNIT - II	CONTROL STRUCTURES AND ARRAYS						CLASSES: 15	
Control structures: Decision statements; if and switch statement; Loop control statements: while, for and do while loops, Jump statements: break, continue, goto statements. Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi-dimensional arrays								
UNIT - III	FUNCTIONS AND POINTERS						CLASSES: 17	
Functions: Function definition, Types of Functions: User defined and built-in Functions, Advantages of User Defined Functions. Parameter passing in functions: Call by value, Call by reference, Passing arrays to functions, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Towers of Hanoi etc... Storage classes. Pointers: Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, Functions returning pointers, Dynamic memory allocation.								

UNIT - IV	STRINGS AND USER DEFINED DATA TYPES	CLASSES: 10
Strings: Array of characters, variable length character strings, inputting character strings, character library functions, String Handling Functions, Arrays Of Strings Structures and Unions: Structure definition, initialization, accessing structures, nested structures, arrays of structures, Structures and functions, Self-referential structures, unions, typedef, enumerations.		
UNIT - V	FILE HANDLING,SEARCHING AND SORTING	CLASSES: 10
File Handling: Command Line Arguments, File Modes, And Basic File Operations Read, Write and Append, Example Programs. Random Access Using fseek, ftell and rewind Functions. Basic Searching And Sorting Algorithms: Linear and Binary Search, Bubble Sort, Insertion Sort, Quick Sort.		
TEXT BOOKS		
1. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) 2. Byron Gottfried, "Programming with C", Schaum's Outlines Series, McGraw Hill Education, 3rd edition, 2017. 3. Programming in C E. Balagurusamy Edition 3 Publisher Tata McGraw-Hill Publishing, 1990		
REFERENCE BOOKS		
1. W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2nd Edition, 1988. 2. Yashavant Kanetkar, "Exploring C", BPB Publishers, 2nd Edition, 2003. 3. Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4th Edition, 2014. 4. R. S. Bichkar, "Programming with C", Universities Press, 2nd Edition, 2012. 5. Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2nd Edition, 2006. 6. Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4th Edition, 2014.		
WEB REFERENCES		
1. https://en.wikipedia.org/wiki/Computational_thinking 2. https://nptel.ac.in/courses/106/104/106104128/ 3. https://en.cppreference.com/w/c/language 4. https://www.learn-c.org/		
E-TEXT BOOKS		
1. https://slidelegend.com/queue/computational-thinking-for-the-modern-problem-solver_59d6f01e1723ddb0c7a0df47.html 2. http://flowgorithm.altervista.org/#elf_l1_Lw 3. http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm		
MOOC COURSE		
1. https://www.coursera.org/learn/computational-thinking-problem-solving 2. https://onlinecourses.nptel.ac.in/noc18_cs33/preview 3. https://www.alison.com/courses/Introduction-to-Programming-in-c 4. http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-programming-in-c-and-c-january-iap-2014/index.html		

ENGLISH FOR SKILL ENHANCEMENT

I B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS01	HSMC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
Contact Classes: 64	Tutorial Classes: NIL	Practical Classes: NIL			Total Classes: 64			

COURSE OBJECTIVES

The course will enable the students to:

1. Develop language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write/compose clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.
5. Develop language components to communicate effectively in formal and informal situations.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Will be able to acquire language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write/compose clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.
5. Use language components to communicate effectively in formal and informal situations

UNIT - I	'TOASTED ENGLISH' BY R. K. NARAYAN	CLASSES: 08
Vocabulary	The Concept of Word Formation -The Use of Prefixes and Suffixes – Acquaintance with Prefixes and Suffixes from Foreign Languages to form Derivatives – Synonyms and Antonyms	
Grammar	Identifying Common Errors in Writing with Reference to Articles and Prepositions	
Reading:	Reading and Its Importance- Techniques for Effective Reading	
Writing	Sentence Structures – Use of Phrases and Clauses in Sentences – Importance of Proper Punctuation – Techniques for Writing precisely – Paragraph Writing – Types, Structures and Features of a Paragraph – Creating Coherence – Organizing Principles of Paragraphs in Documents.	
UNIT - II	'APPRO JRD' BY SUDHA MURTHY	CLASSES: 09
Vocabulary	Words Often Misspelt–Homophones, Homonyms and Homographs	
Grammar	Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement	

Reading	Sub-Skills of Reading –Skimming and Scanning– Exercises for Practice	
Writing	Nature and Style of Writing-Defining/Describing People, Objects, Places and Events – Classifying-Providing Examples or Evidence.	
UNIT - III	‘LESSONS FROM ONLINE LEARNING’ BY F.HAIDER ALVI, DEBORAH HURST ET AL	CLASSES: 09
Vocabulary	Words Often Confused– Words from Foreign Languages and their Use in English.	
Grammar	Identifying Common Errorsin Writing with Reference to Misplaced Modifiers and Tenses	
Reading	Sub-Skills of Reading – Intensive Reading and Extensive Reading –Exercises for Practice	
Writing	Format of a Formal Letter - Writing Formal Letters E.g., Letter of Complaint, Letter of Requisition, Email Etiquette, Job Application with CV / Resume	
UNIT - IV	ART AND LITERATURE	CLASSES: 08
Vocabulary	Standard Abbreviations in English	
Grammar	Redundancies and Clichés in Oral and Written Communication	
Reading	Survey, Question, Read, Recite and Review (SQ3RMethod) – Exercises for Practice	
Writing	Writing Practices- Essay Writing-Writing Introduction and Conclusion-Précis Writing	
UNIT - V	GO, KISS THE WORLD’ BY SUBROTO BAGCHI	CLASSES: 08
Vocabulary	Technical Vocabulary and their Usage	
Grammar	Common Errors in English (Covering all the other aspects of grammar which were not covered in the previous units)	
Reading	Reading Comprehension-Exercises for Practice	
Writing	Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) -Types of Reports - Writing a Report	
TEXT BOOKS		
1. “English: Language, Context and Culture” by Orient BlackSwan Pvt. Ltd, Hyderabad.2022.Print.		
REFERENCE BOOKS		
1. Effective Academic Writing by Liss and Davis(OUP) 2. Richards,JackC.(2022)InterchangeSeries.Introduction,1,2,3.CambridgeUniversityPress 3. Wood,F.T.(2007). Remedial English Grammar. Macmillan. 4. Chaudhuri, Santanu Sinha.(2018). Learn English: A Fun Book of Functional Language, Grammar and Vocabulary.(2 nd ed.,). Sage Publications India Pvt. Ltd. 5. (2019). Technical Communication. Wiley India Pvt. Ltd. 6. Vishwamohan, Aysha.(2013). English for Technical Communication for Engineering Students. McGraw-Hill Education India Pvt. Ltd. 7. Swan, Michael. (2016) . Practical English Usage. Oxford University Press. Fourth Edition		

WEB REFERENCES

1. <http://www.bbc.co.uk/learningenglish>
2. <http://learnenglish.britishcouncil.org>
3. <https://www.cambridgeenglish.org/learning-english/>
4. <https://study.com/academy/subj/english.html>

E-TEXT BOOKS

1. <https://www.pdfdrive.com/advanced-english-books.html>

MOOC COURSES

1. <http://nptel.ac.in/courses/109/106/109106067>
2. <https://www.britishcouncil.org.tr/en/english/mooc>

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING

I B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE60	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Develop fundamentals, including Ohm's law, Kirchhoff's laws and be able to solve for currents, voltages and power in electrical circuits.
2. Develop EMF equation and analyze the operation of DC Machines.
3. Analyze the working principle of Transformer.
4. Discuss the operation of AC Machines.
5. Analyze the operation of PN junction diode and rectifiers.
6. Discuss the operation and characteristics of Transistors.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Evaluate current and voltage values in resistive circuits with independent sources.
2. Explain the working of DC machines and solve the numerical problems.
3. Explain the working of AC electrical machines and solve the numerical problems.
4. Analyze the V-I characteristics of PN – junction diode and describe the operation of rectifiers.
5. Analyze the different configurations of Transistors and obtain its characteristics.

UNIT - I	ELECTRICAL CIRCUITS	CLASSES : 12
Basic definitions-Ohm's Law, types of elements, types of sources, Kirchhoff's Laws – simple problems. Series & parallel resistive networks with DC excitation, star to delta and delta to star transformations.		
UNIT - II	DC MACHINES	CLASSES : 12
Principle of Operation of DC Motor, types of DC motor, Torque equation & Losses and problems. DC Generator construction and working Principle, EMF Equation types of generators and problems.		
UNIT - III	AC MACHINES	CLASSES : 12
Working principle and Construction of transformer, Emf Equation & problems. Principle operation of 3-phase induction motor, slip and torque Equation, Torque –slip characteristics & problems.		
UNIT - IV	DIODE AND ITS CHARACTERISTICS	CLASSES : 12
PN JUNCTION DIODE: Operation of PN junction Diode: forward bias and reverse bias, Characteristics of PN Junction Diode – Zener Effect – Zener Diode and its Characteristics. Rectifiers, Half wave, Full wave and bridge Rectifiers –capacitor filters, inductor filters		
UNIT - V	TRANSISTORS	CLASSES : 10
Bipolar Junction Transistor - NPN & PNP Transistor, CB, CE, CC Configurations and Characteristics.		

TEXT BOOKS

1. Basic Electrical Engineering by *M.S.Naidu and S.Kamakshaiah* TMH
2. Electronic Devices and circuits by J.Millman, C.C.Halkias and Satyabrata Jit 2ed.,

REFERENCE BOOKS

1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, "Basic Electrical, Electronics and Computer Engineering", Tata McGraw Hill, Second Edition, (2006).
2. Nagsarkar T K and Sukhija M S, "Basics of Electrical Engineering", Oxford press (2005).
3. Mehta V K, "Principles of Electronics", S.Chand & Company Ltd, (1994).
4. Mahmood Nahvi and Joseph A. Edminister, "Electric Circuits", Schaum' Outline Series, McGraw Hill, (2002).

ELECTRONIC DEVICES AND APPLICATIONS

I B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC03	ESC	L	T	P	C	CIE	SEE	Total
		2	-	-	2	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To introduce components such as diodes, BJTs and FETs.
2. To know the applications of devices.
3. To know the switching characteristics of devices.

COURSE OUTCOMES

Upon completion of the Course, the students will be able to:

1. Acquire the knowledge of various electronic devices and their use on real life.
2. Know the applications of various devices.
3. Acquire the knowledge about the role of special purpose devices and their applications.

Course	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	-	-	1	1	-	-	-	-	1
CO2	3	2	3	-	-	2	1	-	-	-	-	1
CO3	3	3	3	-	-	2	1	-	-	-	-	1

UNIT - I	DIODES	CLASSES: 10
Diodes: Diode – Static and Dynamic resistances, Equivalent circuit, Diffusion and Transition Capacitances, V-I Characteristics, Diode as a switch-switching times.		
UNIT - II	DIODE APPLICATIONS	CLASSES: 12
Diode Applications: Rectifier - Half Wave Rectifier, Full Wave Rectifier, Bridge Rectifier, Rectifiers with Capacitive and Inductive Filters, Clippers-Clipping at two independent levels, Types of Clampers.		
UNIT - III	BIPOLAR JUNCTION TRANSISTOR (BJT)	CLASSES: 09
Bipolar Junction Transistor (BJT): Principle of Operation, Common Emitter, Common Base and Common Collector Configurations, Transistor as a switch, Need for Biasing, BJT as Amplifier.		
UNIT - IV	JUNCTION FIELD EFFECT TRANSISTOR (FET)	CLASSES: 10
Junction Field Effect Transistor (FET): Construction, Principle of Operation, Pinch-Off Voltage, Volt-Ampere Characteristic, Comparison of BJT and FET, FET as Voltage Variable Resistor, FET as amplifier.		
UNIT - V	SPECIAL PURPOSE DEVICES	CLASSES: 11
Special Purpose Devices: Zener Diode - Characteristics, Zener diode as Voltage Regulator, Operation of - SCR, UJT, Photo diode, Solar cell, LED.		

TEXT BOOKS

1. Jacob Millman - Electronic Devices and Circuits, McGraw Hill Education
2. Robert L. Boylestead, Louis Nashelsky Electronic Devices and Circuits theory, 11th Edition, 2009, Pearson.

REFERENCE BOOKS

1. Horowitz-Electronic Devices and Circuits, David A. Bell–5th Edition, Oxford.
2. Chinmoy Saha, Arindam Halder, Debaati Ganguly – Basic Electronics – Principles and Applications, Cambridge, 2018.

PROGRAMMING FOR PROBLEM SOLVING LAB

I B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS03	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
Contact Classes: Nil	Tutorial Classes: Nil	Practical Classes: 36			Total Classes: 36			
COURSE OBJECTIVES								
The course will enable the students to:								
1. To be familiarize with flowgorithm to solve simple problems. 2. To develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc. 3. To develop modular, reusable and readable C Programs using the concepts like functions, arrays, strings pointers and structures.								
COURSE OUTCOMES								
At the end of the course, student will be able to:								
1. Solve simple mathematical problems using Flowgorithm. 2. Correct syntax errors as reported by the compilers and logical errors encountered at run time Develop programs by using decision making and looping constructs. 3. Implement real time applications using the concept of array, pointers, functions and structures. 4. Solve real world problems using matrices, searching and sorting.								
LIST OF EXPERIMENTS								
WEEK - 1	INTRODUCTION TO FLOGORITHM							
1. Installation and working of Flowgorithm Software. 2. Write and implement basic arithmetic operations using Flowgorithm – sum, average, product, difference, quotient and remainder of given numbers etc. 3. Draw a flowchart to calculate area of Shapes (Square, Rectangle, Circle and Triangle). 4. Draw a flowchart to find the sum of individual digits of a 3 digit number. 5. Draw a flowchart to read input name, marks of 5 subjects of a student and display the name of the student, the total marks scored, percentage scored. 6. Draw a flowchart to find roots of a quadratic equation. 7. Draw a flowchart to find the largest and smallest among three entered numbers and also display whether the identified largest/smallest number is even or odd. 8. Draw a flowchart to check whether the triangle is equilateral, isosceles or scalene triangle. 9. Draw a flowchart to check whether a given number is palindrome or not.								
WEEK - 2	BASIC DATA TYPES							
1. Write a C program to find division of 1 integer 1 float numbers. 2. Write a C program to find division of 2 integer numbers. 3. Write a C program to find average of (1int, 1float) numbers. 4. Write a C Program to Swap Numbers without Using Temporary Variable 5. Write a C Program to Swap two Numbers Using Temporary Variable								

6.	Write a C program to read the values of x, y and z and print the results of the following expressions in one line. a. $(x+y+z)/(x-y-z)$ b. $(x+y+z)/3$ c. $(x+y) * (x-y) * (y-z)$
WEEK - 3	OPERATORS
1.	Write a C program to convert temperature from Fahrenheit to Celsius and vice versa ($c=(f-32)/1.8$)
2.	Write a C program to find area and perimeter of a circle. ($area=\pi r^2$ perimeter= $2\pi r$)
3.	Write a C program to calculate area and perimeter of a right angled triangle. a. ($Area=1/2*b*h$ perimeter= $w+h+sqrt(w*w+h*h)$)
4.	Find the sum of natural numbers 1 to n.(read n as input) (Use formula $sum=n(n+1)/2$)
5.	Write a C program to calculate Simple interest ($SI=PTR/100$)
6.	Write a C program to calculate area and perimeter of a rectangle. $Area=l*b$ Perimeter= $2*(l+b)$
7.	Write a C program to calculate the value of the third angle of a triangle if two angles are given as input. ($a+b+c=180$)
8.	Write a C program to read the consumer number and number of units consumed and the cost per unit and print the amount to be paid. ($Amt=num\ of\ units*cost$)
9.	Write a C Program to calculate area and perimeter of a triangle. a. Perimeter= $(a+b+c)$ b. $s=(a+b+c)/2$ c. $Area=sqrt(s*(s-a)*(s-b)*(s-c))$
10.	Write a C program to read five Subject marks and find the average.
11.	Write a C program to Calculate Compound interest ($CI=p(1+r/100)^n$)
WEEK - 4	CONDITIONAL STATEMENTS
1.	Write a C program to find largest and smallest of given numbers.
2.	Write a C program which takes two integer operands and one operator from the user (+, -, *, /, % use switch)
3.	Write a program to compute grade of students using if else ladder. The grades are assigned as followed: marks<50 F 50≤marks< 60 C 60≤marks<70 B 70≤marks B+ 80≤marks<90 A 90≤marks≤ 100 A+
4.	Write a C program to whether given year leap year or not.
5.	Write a C program to find whether given triangle is scalene or isosceles or equilateral.
WEEK - 5	LOOPING STATEMENTS
1.	Write a C program to find Sum of individual digits of given integer
2.	Write a C program to generate first n terms of Fibonacci series
3.	Write a C Program to find the Sum of Series $SUM=1-x^2/2! +x^4/4!-x^6/6!+x^8/8!-x^{10}/10!$
4.	Write a C program to print the Fibonacci sequence up to given value of n.
5.	Write a C program to print the multiplication table to the given value of n.
6.	Write a C program to check whether given number is palindrome or not
7.	Write a C program to check whether given number is perfect or not
WEEK - 6	NESTED LOOPING STATEMENTS
1.	Write a C program to generate prime numbers between 1 and n
2.	Write a C program to generate Pascal's triangle.

3. Write a C program to generate the following pyramid of numbers.

```

      1
    1 3 1
  1 3 5 3 1

```

4. Write a C program to generate the pattern

```

*****
 *****
  *****
   *****
    *****

```

WEEK - 7**ARRAYS**

1. Write a C Program to implement following searching methods
 - i. Binary Search
 - ii. Linear Search
2. Write a C program to find largest and smallest number in a list of integers
3. Write a C program
 - iii. To add two matrices
 - iv. To multiply two matrices
4. Write a C program to find Transpose of a given matrix

WEEK - 8**FUNCTIONS**

1. Write a C program to find the factorial of a given integer using non recursive functions
2. Write a C program to find GCD of given integers using non recursive functions
3. Write a C Program to find the power of a given number using non recursive functions
4. Write a C program to find sum of natural numbers using non recursive function.
5. Write a C program to reverse a given integer number using non recursive functions
6. Write a C Program to find binary equivalent of a given decimal number using recursive functions.
7. Write a C Program to print Fibonacci sequence using recursive functions.
8. Write a C Program to find LCM of 3 given numbers using recursive functions
9. Write a C program to find the factorial of a given integer using recursive functions
10. Write a C program to print Fibonacci series till n terms using recursion.

WEEK - 9**STRINGS**

1. Write a C program using to Insert a sub string into a given main string from a given position
2. Write a C program using to Delete n characters from a given position in a string
3. Write a C program to determine if given string is palindrome or not
4. Write C Programs to demonstrate the following string handling functions.
 - a. strcat()
 - b. strcmp()
 - c. strrev()
 - d. strcpy()
 - e. strlen()
 - f. strstr()
 - g. strncpy()
 - h. strncat()
 - i. strncmp()

WEEK - 10	POINTERS
1. Write a C program to read the elements of 1-d array using pointers and print them in reverse order using pointers. 2. Write a C Program to read two elements dynamically using malloc() function and interchange the two numbers using call by reference. 3. Write a C Program to read and print the elements of 1-D array using calloc() memory allocation function and reallocate memory for the array by increasing the size of the array, read and print the elements of reallocated array. 4. Write a C Program to print 2-D array using pointers.	
WEEK - 11	STRUCTURES
1. Write a C Program using functions to a) Reading a complex number b) Writing a complex number c) Add two complex numbers d) Multiply two complex numbers Note: represent complex number using structure 2. Write a C program to read employee details employee number, employee name, basic salary, hra and da of n employees using structures and print employee number, employee name and gross salary of n employees.	
WEEK - 12	FILES
1. Write a C program to read and print the content of a file. 2. Write a C program copy the content of one file to another file 3. Write a C program to merge two file into third file. 4. Write a C Program to find the number of lines in a text file	
TEXT BOOKS	
1. Riley DD, Hunt K.A. Computational Thinking for the Modern Problem Solver. CRC press, 2014 Mar 27. 2. B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) Yashavant Kanetkar, "Let Us C", BPB Publications, New Delhi, 13th Edition, 2012.	
REFERENCE BOOKS	
1. Ferragina P, Luccio F. Computational Thinking: First Algorithms, Then Code. Springer; 2018 2. King KN, "C Programming: A Modern Approach", Atlantic Publishers, 2nd Edition, 2015. 3. Kochan Stephen G, "Programming in C: A Complete Introduction to the C Programming Language", Sam's Publishers, 3rd Edition, 2004. 4. Linden Peter V, "Expert C Programming: Deep C Secrets", Pearson India, 1st Edition, 1994.	
WEB REFERENCES	
1. http://www.flowgorithm.org/documentation/ 2. http://www.sanfoundry.com/c-programming-examples 3. http://www.geeksforgeeks.org/c 4. http://www.cprogramming.com/tutorial/c	

ENGLISH LANGUAGE AND COMMUNICATION SKILLS LAB

I B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS02	HSMC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
Contact Classes: 00	Tutorial Classes: 00	Practical Classes: 39			Total Classes:39			

COURSE OBJECTIVES

The course should enable the students to:

1. To facilitate computer-assisted multi-media instruction enabling individualized and independent language learning
2. To sensitize the students to the nuances of English speech sounds, word accent, intonation and rhythm
3. To bring about a consistent accent and intelligibility in students' pronunciation of English by providing an opportunity for practice in speaking
4. To improve the fluency of students in spoken English and neutralize the impact of dialects.
5. To train students to use language appropriately for public speaking, group discussions and interviews

COURSE OUTCOMES

Students will be able to:

1. Acquire the skill of independent language learning
2. Overcome with the nuances of English speech sounds, word accent, intonation and rhythm.
3. Obtain the consistent accent and intelligibility in pronunciation.
4. Neutralize the impact of dialects.
5. Apply language appropriately public speaking, group discussions and interviews

English Language and Communication Skills Lab (ELCS) shall have two parts:

- a. Computer Assisted Language Learning (CALL) Lab
- b. Interactive Communication Skills (ICS) Lab

Listening Skills

Objectives

1. To enable students develop their listening skills to appreciate its role in the LSRW skills approach to language and improve their pronunciation.
2. To equip students with necessary training in listening so that they can comprehend the speech of people of different backgrounds and regions.

Students should be given practice in listening to the sounds of the language, to be able to recognize them and find the distinction between different sounds, to be able to mark stress and recognize and use the right intonation in sentences.

- Listening for general content
- Listening to fill up information
- Intensive listening
- Listening for specific information

Speaking Skills**Objectives**

1. To involve students in speaking activities in various contexts
2. To enable students express themselves fluently and appropriately in social and professional contexts
 - Oral practice
 - Describing objects/situations/people
 - Role play – Individual/Group activities
 - Just A Minute (JAM) Sessions

EXERCISE – I**CALL Lab:**

Understand: Listening Skill- Its importance – Purpose- Process- Types- Barriers- Effective Listening.

Practice: Introduction to Phonetics – Speech Sounds – Vowels and Consonants – Minimal Pairs-Consonant Clusters- Past Tense Marker and Plural Marker- *Testing Exercises*

ICS Lab:

Understand: Spoken vs. Written language- Formal and Informal English.

Practice: Ice-Breaking Activity and JAM Session- Situational Dialogues – Greetings – Taking Leave – Introducing Oneself and Others.

EXERCISE - II**CALL Lab:**

Understand: Structure of Syllables – Word Stress– Weak Forms and Strong Forms – Stress pattern in sentences – Intonation.

Practice: Basic Rules of Word Accent - Stress Shift - Weak Forms and Strong Forms- Stress pattern in sentences – Intonation - *Testing Exercises*

ICS Lab:

Understand: Features of Good Conversation – Strategies for Effective Communication.

Practice: Situational Dialogues – Role Play- Expressions in Various Situations –Making Requests and Seeking Permissions - Telephone Etiquette.

EXERCISE - III**CALL Lab:**

Understand: Errors in Pronunciation-Neutralising Mother Tongue Interference (MTI).

Practice: Common Indian Variants in Pronunciation – Differences between British and American Pronunciation - *Testing Exercises*

ICS Lab:

Understand: Descriptions- Narrations- Giving Directions and Guidelines – Blog Writing

Practice: Giving Instructions – Seeking Clarifications – Asking for and Giving Directions – Thanking and Responding – Agreeing and Disagreeing – Seeking and Giving Advice – Making Suggestions.

EXERCISE - IV**CALL Lab:**

Understand: Listening for General Details.

Practice: Listening Comprehension Tests - *Testing Exercises*

ICS Lab:

Understand: Public Speaking – Exposure to Structured Talks - Non-verbal Communication- Presentation Skills.

Practice: Making a Short Speech – Extempore- Making a Presentation.

EXERCISE - V	
CALL Lab: <i>Understand:</i> Listening for Specific Details. <i>Practice:</i> Listening Comprehension Tests - <i>Testing Exercises</i> ICS Lab: <i>Understand:</i> Group Discussion <i>Practice:</i> Group Discussion	
REFERENCE BOOKS	
1. (2022). English Language Communication Skills – Lab Manual cum Workbook. Cengage Learning India Pvt. Ltd. 2. Shobha, KN & Rayen, J. Lourdes. (2019). Communicative English – A workbook. Cambridge University Press 3. Kumar, Sanjay & Lata, Pushp. (2019). Communication Skills: A Workbook. Oxford University Press 4. Board of Editors. (2016). ELCS Lab Manual: A Workbook for CALL and ICS Lab Activities. Orient Black Swan Pvt. Ltd. 5. Mishra, Veerendra et al. (2020). English Language Skills: A Practical Approach. Cambridge University Press	
WEBSITES	
1. https://www.britishcouncil.org 2. https://www.bbc.co.uk 3. https://www.grammarly.com 4. https://www.fluentu.com 5. https://www.cambridgeenglish.org/exams-and-tests/business-preliminary 6. https://www.cambridgeenglish.org/exams-and-tests/business-vantage	

INTRODUCTION TO INTERNET OF THINGS

I B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC04	ESC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. To develop basic programming skills through graphical programming 2. To learn hardware interfacing and debugging techniques 3. To design and develop android apps								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Write the algorithms for simple problems 2. CO 1: Able to demonstrate various sensor interfacing using Visual Programming Language. 3. CO 2: Able to analyze various Physical Components. 4. CO 3: Able to demonstrate Wireless Control of Remote Devices. 5. CO 4: Able to design and develop Mobile Application which can interact with Sensors								
INTRODUCTION TO IOT 1. Introduction to basic electronic components and digital electronic 2. Introduction to sensors and Actuators 3. Introduction to microcontroller 4. Introduction to Arduino IDE								
LIST OF EXPERIMENTS								
WEEK - 1	Blinking of LED with different delays							
WEEK - 2	Digital I/O Interface [IR Sensor, PIR Sensor]							
WEEK - 3	Analog Interface [ADC, Temperature Sensor]							
WEEK - 4	Motor speed And Direction control							
WEEK - 5	Serial Communication							
WEEK - 6	Wireless Interface – Bluetooth & Wi-Fi Technologies							
WEEK - 7	Wireless Control of wheeled robot							
WEEK - 8	Smart Home Android App Development							
REFERENCES 1. Sylvia Libow Martinez, Gary S Stager, Invent To Learn: Making, Tinkering, and Engineering in the Classroom, Constructing Modern Knowledge Press, 2016 2. Michael Margolis, Arduino Cookbook, Oreilly, 2011								

SEMINAR

I B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS04	HSMC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	50	0	50

COURSE OBJECTIVES

Seminar is an important component of learning in an Engineering College, where the student gets acquainted with preparing a report & presentation on a topic.

PERIODICITY/FREQUENCY OF EVALUATION : Twice

PARAMETERS OF EVALUATION:

1. The seminar shall have topic allotted and approved by the faculty.
2. The seminar is evaluated for 50 marks for internal only.
3. The students shall be required to submit the rough drafts of the seminar outputs within one week of the commencement of the class work.
4. Faculty shall make suggestions for modification in the rough draft. The final draft shall be presented by the student within a week there after.
5. Presentation schedules will be prepared by Department in line with the academic calendar.

THE SEMINARS SHALL BE EVALUATED IN TWO STAGES AS FOLLOWS:

A. ROUGH DRAFT

In this stage, the student should collect information from various sources on the topic and collate them in a systematic manner. He/She may take the help of the concerned faculty.

The report should be typed in "MS-Word" file with "Calibri" font, with font size of 16 for main heading, 14 for sub-headings and 11 for the body text. The contents should also be arranged in Power Point Presentation with relevant diagrams, pictures and illustrations. It should normally contain 10 to 15 slides, consisting of the followings:

1.	Topic, name of the student & faculty	1 Slide
2.	List of contents	1 Slide
3.	Introduction	1 Slides
4.	Descriptions of the topic (point-wise)	6 - 10 Slides
5.	Conclusion	1 - 2 Slides
6.	References/Bibliography	1 Slide

The soft copy of the rough draft of the seminar presentation in MS Power Point format along with the draft report should be submitted to the concerned faculty, with a copy to the concerned HOD within stipulated time.

Diodes: Diode – Static and Dynamic resistances, Equivalent circuit, Diffusion and Transition Capacitances, V-I Characteristics, Diode as a switch-switching times.

The evaluation of the rough draft shall generally be based upon the following.

1	Punctuality in submission of rough draft	2
2	Dress Code	3
3	Resources from which the seminar have been based	2
4	Report , and content of Presentation	5
5	Depth of the students knowledge in the subject	5
6	Reception from Questions	5
7	Time Management, Classroom Dynamic	3
	Total Marks	25

After evaluation of the first draft the supervisor shall suggest further reading, additional work and fine tuning, to improve the quality of the seminar work. Within 7 days of the submission of the rough draft, the students are to submit the final draft incorporating the suggestions made by the faculty.

I B.TECH II SEMESTER SYLLABUS

NUMERICAL METHODS AND INTEGRAL TRANSFORMS

I B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6BS02	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	40	60	100
Contact Classes: 44	Tutorial Classes: 08	Practical Classes: Nil			Total Classes: 52			
COURSE OBJECTIVES The course will enable the students to: 1. Curve fitting and Interpolation techniques. 2. Numerical techniques. 3. Fourier series for periodic function. 4. Laplace transforms. 5. Concept and application of Fourier Transforms and Vector differentiation.								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Apply Curve fitting and Interpolation techniques. 2. Apply various numerical techniques. 3. Find the Fourier series of the periodic functions. 4. Obtain the Laplace transforms of functions. 5. Find Fourier transforms and apply vector differentiation techniques.								
UNIT - I	INTERPOLATION AND CURVE FITTING						CLASSES: 08	
INTERPOLATION: Finite differences: Forward, Backward and Central differences - Other difference operators and relations between them - Difference of a polynomial – Missing terms - Newton’s forward interpolation, Newton’s backward interpolation, Gauss’s forward and backward interpolation formulae. Interpolation with unequal intervals – Lagrange’s interpolation. CURVE FITTING: Method of least squares - Fitting a straight line, second degree parabola and non-linear curves of the form $y= a e^{b^x}$, $y= a x^b$, $y= a b^x$ by the method of least squares.								
UNIT - II	NUMERICAL TECHNIQUES						CLASSES: 08	
ROOT FINDING TECHNIQUES: Bisection method Regula-falsi method, Iteration method and Newton Raphson method. NUMERICAL INTEGRATION : Trapezoidal rule - Simpson’s one-third rule - Simpson’s three-eighth rule. NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS: Taylor’s series method – Euler’s - modified Euler’s Method – Runge-Kutta method.								
UNIT - III	FOURIER SERIES						CLASSES: 10	
Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Half range Fourier sine and cosine expansions.								
UNIT - IV	LAPLACE TRANSFORMS						CLASSES: 10	
Laplace transforms of elementary functions- First shifting theorem - Change of scale property – Multiplication by t^n - Division by t – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms- Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem – Applications of Laplace transforms to ordinary differential equations.								

UNIT - V	FOURIER TRANSFORMS AND VECTOR DIFFERENTIATION	CLASSES: 10
Fourier Integral theorem (Statement only)-Fourier Sine and Cosine Integrals, Fourier Transforms, Cosine and Sine transforms, properties, Inverse transforms. Vector functions, vector differentiation, gradient, directional derivative, divergence, curl and scalar potential.		
TEXT BOOKS		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.		
REFERENCE BOOKS		
1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics, Laxmi publications,2008. 3. E.L.Ince, Ordinary differential Equations, Dover publications,1958.		
WEB REFERENCES		
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/		
E-TEXT BOOKS		
1. https://www.e-booksdirectory.com/details.php?ebook=10166		
MOOCS COURSE		
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/		

APPLIED PHYSICS

I B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6BS07	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Learn the basic principles of quantum physics and its applications
2. Understand the formation of energy bands and atomic structure in solids for material classification
3. Understand the underlying mechanism involved in construction and working properties of different types of semiconductor devices
4. Learn the basic principles of laser and optical fiber in information technology
5. Identify the importance of Nano scale and various fabrication and characterization techniques and quantum computations in engineering applications

COURSE OUTCOMES

At the end of the course students will be able to:

1. **Analyze** the microscopic properties of materials using principles of quantum physics for engineering applications
2. **Explain** the behaviour of different electronic materials based on the concepts of band theory
3. **Apply** the knowledge of Solar PV cells for choice of materials in efficient alternate energy generation
4. **Gain** the knowledge of production of laser and usage of fibers in fiber optic communication technology
5. **Comprehend** the knowledge of quantum physics in quantum computation for secure information technology

UNIT - I	QUANTUM PHYSICS	CLASSES: 10
Black body radiation, Stefan-Boltzmann's law, Planck's radiation law (Qualitative treatment), Photoelectric effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation, Physical Significance of the wave Function, Particle in One Dimensional Potential Box.		
UNIT - II	ELECTRONIC PROPERTIES OF MATERIALS & BAND THEORY OF SOLIDS	CLASSES: 10
Electronic properties of Materials: Classical free electron theory and Quantum free electron theories of metals, success and drawbacks, Bloch theorem, Kronig-Penny model (Qualitative treatment), E-k diagram, effective mass of electron. Band theory of solids: Origin of energy band formation in solids, Fermi energy level, Fermi-Dirac distribution law, Classification of materials as conductors, insulators and semiconductors.		
UNIT - III	SEMI-CONDUCTORS & SEMICONDUCTOR DEVICES	CLASSES: 08
Semiconductors: Intrinsic and Extrinsic Semiconductors, formation of PN junction diode and its V-I characteristics, Direct and Indirect band gap semiconductors, Hall effect and its applications. Semiconductor Devices: Construction, working and V-I characteristics of PIN Diode, LED, Solar cell and their applications.		

UNIT - IV	LASER & FIBER OPTICS	CLASSES: 12
Laser: Characteristics of Laser, Absorption, Spontaneous and Stimulated emission of radiations. Lasing actions-Pumping mechanism, Meta stable state and Population inversion, Nd-YAG laser, CO2 laser, Applications of lasers in different fields. Fiber Optics: Structure of fibers, Total Internal Reflection, Acceptance angle – Numerical Aperture, Types of fibers- SI and GI fibers, Single and Multimode fibers - SMSI, MMSI, MMGI, Fiber Optic Communication system, Signal Degradation - Attenuation mechanism, Dispersion, Applications of fibers in different fields.		
UNIT - V	PHYSICS OF QUANTUM COMPUTING & QUANTUM GATES	CLASSES: 10
Physics of Quantum computing: Idea of classical bits and qubits, advantages with qubits over classical bits, Bloch vector representation of state of qubits. Quantum gates-Single qubit logic gates- Pauli X, Y, Z and Hadmard gate in matrix form, Two level gates- CNOT and SWAP gates and representation in matrix form, Comments on No cloning theorem, Entanglement, Quantum Teleportation – Basic Idea, Quantum Key distribution protocol - BB84 protocol.		
TEXT BOOKS		
1. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IVth Ed. 2. Nielsen M. A., I. L Chung, Quantum Computation & Quantum Information, Cambridge Univ. Press. 3. M. N. Avadhanulu, P.G. Kshirsagar & TVS Arun Murthy" A Text book of Engineering Physics"- S. Chand Publications, 11th Edition 2019.		
REFERENCE BOOKS		
1. Quantum Physics, H.C. Verma, TBS Publication, 2nd Edition 2012 2. Fundamentals of Physics – Halliday, Resnick and Walker, John Wiley & Sons, 11th Edition, 2018. 3. Engineering Physics by Shatendra Sharma and Jyotsna Sharma, Pearson Publication, 2019 4. B.K Pandey and Chaturvedi, Engineering Physics, Cengage Learning, 2nd Edition, 2022		

ENGINEERING CHEMISTRY**I B.TECH - II SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6BS09	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	40	60	100
Contact Classes:50	Tutorial Classes: 0	Practical Classes: 0			Total Classes: 50			

COURSE OBJECTIVES

The course will enable the students to:

1. To bring adaptability to new developments in Engineering Chemistry and to acquire the skills required to become a perfect engineer.
2. To include the importance of water in industrial usage, fundamental aspects of battery chemistry, significance of corrosion it's control to protect the structures.
3. To imbibe the basic concepts of petroleum and its products.
4. To acquire required knowledge about engineering materials like cement, smart materials and Lubricants.

COURSE OUTCOMES

At the end of the course student will be able to:

1. Understand the basic properties of water and its usage in domestic and industrial purpose.
2. Acquire the basic knowledge of electrochemical procedures related to corrosion
3. Learn the fundamentals and general properties of polymers and other engineering materials.
4. Predict potential applications of chemistry and practical utility in order to become good engineers and entrepreneurs.

UNIT - I	WATER AND ITS TREATMENT	CLASSES: 10
Introduction to hardness of water – Estimation of hardness of water by complexometric method and related numerical problems. Potable water and its specifications - Steps involved in the treatment of potable water - Disinfection of potable water by chlorination and break - point chlorination. Defluoridation- Determination of F- ion by ion- selective electrode method. Boiler troubles: Sludges, Scales and Caustic embrittlement. Internal treatment of Boiler feed water -Calgon conditioning - Phosphate conditioning - Colloidal conditioning, External treatment methods -Softening of water by ion- exchange processes. Desalination of water –Reverse osmosis.		
UNIT - II	BATTERY CHEMISTRY AND FUEL CELLS	CLASSES: 10
Introduction - Classification of batteries- primary, secondary and reserve batteries with examples. Basic requirements for commercial batteries. Construction, working and applications of: Zn-air and Lithium ion battery, Applications of Li-ion battery to electrical vehicles. Fuel Cells- Differences between battery and a fuel cell, Construction and applications of Methanol Oxygen fuel cell and Solid oxide fuel cell. Solar cells - Introduction and applications of Solar cells. Corrosion: Causes and effects of corrosion – theories of chemical and electrochemical corrosion –mechanism of electrochemical corrosion, Types of corrosion: Galvanic, water-line and pitting corrosion. Factors affecting rate of corrosion, Corrosion control methods - Cathodic protection – Sacrificial anode and impressed current methods.		

UNIT - III	POLYMERIC MATERIALS	CLASSES: 10
Definition – Classification of polymers with examples – Types of polymerization – addition (free radical addition) and condensation polymerization with examples – Nylon 6:6, Terylene Plastics: Definition and characteristics- thermoplastic and thermosetting plastics, Preparation, Properties and engineering applications of PVC and Bakelite, Teflon, Fiber reinforced plastics (FRP). Rubbers: Natural rubber and its vulcanization. Elastomers: Characteristics – preparation – properties and applications of Buna-S, Butyl and Thiokol rubber. Conducting polymers: Characteristics and Classification with examples-mechanism of conduction in trans-polyacetylene and applications of conducting polymers. Biodegradable polymers: Concept and advantages - Polylactic acid and poly vinyl alcohol and their applications.		
UNIT - IV	ENERGY SOURCES	CLASSES: 10
Introduction, Calorific value of fuel – HCV, LCV- Dulong's formula. Classification- solid fuels: coal –analysis of coal – proximate and ultimate analysis and their significance. Liquid fuels – petroleum and its refining, cracking types – moving bed catalytic cracking. Knocking – octane and cetane rating, synthetic petrol - Fischer-Tropsch's process; Gaseous fuels – composition and uses of natural gas, LPG and CNG, Biodiesel – Transesterification, advantages.		
UNIT - V	ENGINEERING MATERIALS	CLASSES: 10
Cement: Portland cement, its composition, setting and hardening. Smart materials and their engineering applications Shape memory materials- Poly L- Lactic acid. Thermo response materials- Polyacryl amides, Poly vinyl amides Lubricants: Classification of lubricants with examples-characteristics of a good lubricants – mechanism of lubrication (thick film, thin film and extreme pressure)- properties of lubricants: viscosity, cloud point, pour point, flash point and fire point		
TEXT BOOKS		
1. Engineering Chemistry by P.C. Jain and M. Jain, Dhanpatrai Publishing Company, 2010 2. Engineering Chemistry by Rama Devi, Venkata Ramana Reddy and Rath, Cengage learning, 2016 3. A text book of Engineering Chemistry by M. Thirumala Chary, E. Laxminarayana and K. Shashikala, Pearson Publications, 2021. 4. Textbook of Engineering Chemistry by Jaya Shree Anireddy, Wiley Publications		
REFERENCE BOOKS		
1. Engineering Chemistry by Shikha Agarwal, Cambridge University Press, Delhi (2015) 2. Engineering Chemistry by Shashi Chawla, Dhanpatrai and Company (P) Ltd. Delhi (2011)		

ENGINEERING DRAWING

I B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME02	ESC	L	T	P	C	CIE	SEE	Total
		1	-	3	2.5	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. Aimed to develop basic graphic skills to enable them to use for the preparation of engineering drawings, their readings and interpretations 2. To prepare the students to use technical skills and modern engineering tools necessary for the engineering practices 3. To exposure to a CAD package								
COURSE OUTCOMES At the end of the course the student should be able to: 1. Understand various commands and create drawing in AutoCAD. 2. Construct various engineering curves and know their importance. 3. Prepare orthographic projections of objects by visualizing them in different positions. 4. Solve the problem of projections of planes and solids in different positions. 5. Construct the isometric view into orthographic views and vice versa.								
UNIT - I	INTRODUCTION TO ENGINEERING DRAWING						CLASSES: 10	
Introduction to Engineering Drawing: Principles and their significance. Introduction to Computer Aided Drafting: Initial Setup Commands, Draw Commands, modify commands, 2D Drawings-Simple Exercises.								
UNIT - II	ENGINEERING CURVES						CLASSES: 10	
Engineering Curves: Ellipse, Parabola, and Hyperbola (General Method only), Involute.								
UNIT - III	ORTHOGRAPHIC PROJECTIONS OF POINTS AND LINES						CLASSES: 12	
Principles of Orthographic Projections –Projections of points. Projection of lines inclined to both planes (First angle projection only).								
UNIT - IV	PROJECTIONS OF PLANES AND SOLIDS						CLASSES: 10	
Projections of Planes: Projections of regular planes inclined to one plane. Projection of Solids: Solids inclined to one plane (Regular solids).								
UNIT - V	ISOMETRIC VIEW AND ORTHOGRAPHIC VIEWS - CONVERSION						CLASSES: 10	
Isometric view: Drawing Isometric circles, Dimensioning Isometric Objects. Conversion of Isometric view to Orthographic views and Orthographic to isometric view.								

TEXT BOOKS

1. Bhatt N.D., Panchal V.M. & Ingle P.R., Engineering Drawing, Charotar Publishing House.
2. Agrawal B. & Agrawal C. M., Engineering Graphics, TMH Publication.

REFERENCE BOOKS

1. D.M. Kulkarni, A.P.Rastogi, A.K. Sarka "Engineering Graphics with AutoCAD" PHI publications.
2. Narayana, K.L. & P Kannaiah, Text book on Engineering Drawing, Scitech Publishers.
3. Shah, M.B. & Rana B.C., Engineering Drawing and Computer Graphics, Pearson Education.

APPLIED PHYSICS LAB

I B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6BS08	BSC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the temperature and other dependent properties of semiconductor devices and their usages in different applications
2. To know the Laser and Fibre optic technologies and its applications in real time scenario
3. Understand the electromagnetic properties using experimental knowledge
4. Understand the method of least squares fitting

COURSE OUTCOMES

By the end of the course students will be able to:

1. **Analyze** the electric properties of semiconductor materials by determining energy gap of semiconductors, charge carrier concentration in Semiconductors using Hall effect and threshold voltage of LEDs, photo current in Photo diodes, solar cell, and temperature effect on resistance using thermistor.
2. **Identify** the optical properties of light such as diffraction phenomenon using grating material for calculation of the wavelength of Laser and acceptance angle, NA of optical fiber using OFC and determine the value of Plank's constant using a light source and interference by using Newton's rings
3. **Analyze** the electromagnetic properties of a current carrying coil by using Stewart Gee's experiment
4. **Analyze** the least squares fitting method for data analysis using experimental data of Tensional pendulum

APPLIED PHYSICS LABORATORY LIST OF EXPERIMENTS

Experiment - 1	ENERGY GAP OF P-N JUNCTION DIODE: To determine the energy gap of a given semiconductor diode
Experiment - 2	SOLAR CELL: To study the V-I and V-P characteristics and determine the fill factor of solar cell
Experiment - 3	LIGHT EMITTING DIODE: To study the characteristics of LED by plotting V-I graph and determine the threshold value of given LEDs
Experiment - 4	HALL EFFECT: To determine Hall co-efficient and charge carrier concentration of a given semiconductor
Experiment - 5	PIN PHOTO DIODE: To study the V-I Characteristics of Photo Diode with respect to intensity of light
Experiment - 6	OPTICAL FIBRE: To determine the numerical aperture and acceptance angle of an optical fiber
Experiment - 7	LASER: To determine the wavelength of a given laser source by using diffraction grating method
Experiment - 8	NEWTON'S RINGS: To determine the radius of curvature of a given Plano convex lens by forming Newton's rings
Experiment - 9	THERMISTOR: To study the variation of resistance with respect to temperature using thermistor

Experiment - 10	Understanding the method of least squares - Torsional pendulum as an example
Experiment – 11	PLANCK'S CONSTANT: To determine value of planck's constant by measuring radiation in fixed spectral range
Experiment - 12	STEWART GEE'S EXPERIMENT: To study the variation of magnetic field along the axis of a circular coil and calculation of magnetic flux
Note: Students have to perform any 8 experiments	
REFERENCE BOOKS	
1. Applied Physics Lab Manual"- Dr. Radhika Devi, Mr. A V Laxman Rao, N. Noel 2. S. Balasubramanian, M.N Srinivasan "A Text book of practical Physics" – S Chand Publishers, 2017.	

PYTHON PROGRAMMING LAB

I B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS04	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
COURSE OBJECTIVES The course will enable the students to: - To understand the problem-solving approaches. - To learn the basic programming constructs in Python. - To practice various computing strategies for Python-based solutions to real world problems. - To use Python data structures – lists, tuples, dictionaries. - To do input/output with files in Python								
COURSE OUTCOMES Upon successful completion of this course, student will be able to : - Write, test, and debug simple Python programs. - Implement Python programs with conditions and loops. - Develop Python programs step-wise by defining functions and calling them. - Use Python lists, tuples, dictionaries for representing compound data. - Read and write data from/to files in Python								
LIST OF PROGRAMS								
WEEK - 1								
		- Write a program to perform different Arithmetic Operations on numbers in Python - Write a Python program which accepts the radius of a circle from the user and compute the area. - Write a Python program to get the Python version you are using. - Write a Python program that accepts an integer (n) and computes the value of n+nn+nnn						
WEEK - 2								
		- Write a Python program to convert temperatures to and from Celsius, Fahrenheit. [Formula: $c/5 = f-32/9$] - Write a python script to print the current date in the following format “Sun May 29 02:26:23 IST 2017” - A library charges a fine for every book returned late. For first 6 days the fine is 50 paise, for 10-15 days fine is one rupee and above 15 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled. - Write a python program to accept the number of days the member is late to return the book and display the fine or the appropriate message.						
WEEK - 3								
		- Write a python function to find largest of three numbers. - Write a Python function that prints prime numbers in between 50 and 100. - Write a python program to find factorial of a number using Recursion. - Write a function that receives marks received by a student in 6 subjects and returns the average and percentage of these marks. Call this function from main() and print the result in main						

WEEK - 4	
Write a program to demonstrate working with Strings and string operations.	
WEEK - 5	
Write a program to demonstrate working with dictionaries in python	
WEEK - 6	
Write a program to demonstrate working with tuples and List in python.	
WEEK - 7	
a) Write a script named hellow.py. This script should prompt the user for the names of two text files. The contents of the first file should be input and written to the second file. b) Write a program that inputs a text file. The program should print all of the unique words in the file in alphabetical order	
WEEK - 8	
Write python programs to demonstrate class & object, static and instance method implementation.	
WEEK - 9	
Write python programs to demonstrate Inheritance and Polymorphism	
WEEK - 10	
Write python programs to demonstrate Exception Handling in python	
WEEK - 11	
Write python programs to demonstrate NumPy library and supporting functions.	
WEEK - 12	
a) Draw an Olympic Symbol in Python using Turtle b) Develop a simple login page using GUI Tkinter	
TEXT BOOKS	
1. Allen B. Downey, Think Python: How to Think like a Computer Scientist, 2 nd Edition, Oâ€™Reilly Publishers, 2016. 2. Karl Beecher, Computational Thinking: A Beginners Guide to Problem Solving and Programming, 1st Edition, BCS Learning and Development Limited, 2017.	

REFERENCE REFERENCES

1. Paul Deitel and Harvey Deitel, Python for Programmers, Pearson Education, 1st Edition, 2021.
2. G Venkatesh and Madhavan Mukund, Computational Thinking: A Primer for Programmers and Data Scientists, 1st Edition, Notion Press, 2021.
3. John V Guttag, Introduction to Computation and Programming Using Python: With Applications to Computational Modeling and Understanding Data Structures, Third Edition, MIT Press, 2021
4. Eric Matthes, Python Crash Course, A Hands – on Project Based Introduction to Programming, 2nd Edition, No Starch Press, 2019. <https://www.python.org/>
5. Martin C. Brown, Python: The Complete Reference, 4th Edition, Mc-Graw Hill, 2018.

ENGINEERING WORK SHOP**I B.TECH - II SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME04	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To assemble the different components.
2. To identify and apply suitable tools for different trades of engineering processes.
3. Enable to practice on manufacturing of components using workshop trades including Soldering, Carpentry, Fitting and Tin smithy & Fabricate Components with their own hands.
4. To apply basic electrical engineering knowledge for house wiring practice.
5. To learn the safety precautions for various operations in basic trades.

COURSE OUTCOMES

At the end of the course the student should be able to:

1. Assemble the different components.
2. Identify and apply suitable tools for different trades of engineering processes.
3. Practice on manufacturing of components using workshop trades including Soldering, Carpentry, Fitting and Tin smithy & Fabricate Components with their own hands.
4. Apply basic electrical engineering knowledge for house wiring practice.
5. Learn the safety precautions for various operations in basic trades.

WEEKS	TRADES FOR EXERCISES
	FITTING
WEEK 01	Filing Four Sides of MS Work piece
WEEK 02	L Fit
	CARPENTRY
WEEK 03	Half Lap Joint
WEEK 04	Dove Tail Joint
	TIN SMITHY
WEEK 05	Prepare a Rectangular Tray
WEEK 06	Prepare A Square Tin
	ELECTRICAL
WEEK 07	Series and Parallel Connection One Way Switch
WEEK 08	Two Way Switch Connection Stair Case Wiring

	ELECTRONICS
WEEK 09	Soldering - Series Connection & Parallel Connection
WEEK 10	Desoldering & Construction of Wheat stone bridge
	TRADES FOR DEMONSTRATION AND EXPOSURE
WEEK 11	Introduction to Black smithy
WEEK 12	Introduction to Plumbing
TEXT BOOKS	
1. Work shop Manual - P. Kannaiah/K.L. Narayana/Scitech. 2. Workshop Manual/K. Venugopal/Anuradha.	
REFERENCE BOOKS	
1. Hajra Choudhury S.K., Hajra Choudhury A.K.and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010. 2. Workshop Manual/Venkat Reddy/BSP	

ELEMENTS OF COMPUTER SCIENCE AND ENGINEERING

I B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS01	HSMC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	100	0	100

COURSE OBJECTIVES

The course will enable the students to:

1. To provide an overview of the subjects of computer science and engineering.
2. To Know different types of operating system, database systems.
3. To understand the significance of networks, internet, and WWW and cyber security.
4. To understand Autonomous systems, the application of artificial intelligence.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Know the working principles of functional units of a basic Computer
2. Understand program development, the use of data structures and algorithms in problem solving.
3. Know the need and types of operating system, database systems.
4. Understand the significance of networks, internet, and WWW and cyber security.
5. Understand Autonomous systems, the application of artificial intelligence.

UNIT - I	BASICS OF A COMPUTER	CLASSES: 10
Basics of a Computer – Hardware, Software, Generations of computers. Hardware - functional units, Components of CPU, Memory – hierarchy, types of memory, Input and output devices. Software –systems software, application software, packages, frameworks, IDEs.		
UNIT - II	SOFTWARE DEVELOPMENT	CLASSES: 10
Software development – waterfall model, Agile, Types of computer languages – Programming, markup, scripting Program Development – steps in program development, flowcharts, algorithms, data structures – definition, types of data structures		
UNIT - III	OPERATING SYSTEMS AND DATABASE MANAGEMENT SYSTEMS	CLASSES: 10
Operating systems: Functions of operating systems, types of operating systems, Device & Resource management Database Management Systems: Data models, RDBMS, SQL, Database Transactions, data centers, cloud services		
UNIT - IV	COMPUTER NETWORKS	CLASSES: 10
Computer Networks: Advantages of computer networks, LAN, WAN, MAN, internet, Wi-Fi, sensor networks, vehicular networks, 5G communication. World Wide Web – Basics, role of HTML, CSS, XML, Tools for web designing, Social media, Online social networks. Security – information security, cyber security, cyber laws		

UNIT - V	AUTONOMOUS SYSTEMS	CLASSES: 10
Autonomous Systems: IoT, Robotics, Drones, Artificial Intelligence – Learning, Game Development, natural language processing, image and video processing. Cloud Basics		
TEXT BOOKS		
1. Invitation to Computer Science, G. Michael Schneider, Macalester College, Judith L. Gersting University of Hawaii, Hilo, Contributing author: Keith Miller University of Illinois, Springfield. 5 th Edition, 2010		
REFERENCE BOOKS		
1. Fundamentals of Computers, Reema Thareja, Oxford Higher Education, Oxford University Press, 2 nd Edition, 2020. 2. Introduction to computers, Peter Norton, 8th Edition, Tata McGraw Hill. 3. Computer Fundamentals, Anita Goel, Pearson Education India, 2010. 4. Elements of computer science, Cengage		

ENVIRONMENTAL SCIENCE

I B.TECH - II SEMESTER								
Course Code:	Category	Hours/Week			Credits	Maximum Marks		
A6BS11	MC	L	T	P	C	CIE	SEE	Total
		2	-	-	-	50	-	50
Contact Classes: 30	Tutorial Classes: 0	Practical Classes: 0			Total Classes:30			
COURSE OBJECTIVES The course will enable the students to: - To identify the consequences of human actions on the global economy and quality of human life. - To evaluate the strategies for scientific, social, economic and legal environmental protection. - To study the impact of conservation of biodiversity. - Should be able to analyze the reasons for environmental pollution.								
COURSE OUTCOMES On Successful completion of this course, Students will be able to: - Identify the consequences of human actions on the web of life, global economy and quality of human life. - Evaluate the strategies for scientific, social, economic and legal environmental protection. - Study the impact of conservation of biodiversity. - Analyze the reasons for environmental pollution. - Assess the environmental impact of air, water, biological and socio-economic aspects and risk assessment towards sustainable future.								
UNIT - I	ECOSYSTEMS					CLASSES: 7		
Ecosystems: Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food web and ecological pyramids. Flow of energy, Biogeochemical cycles, Bio accumulation, Bio magnification, eco system value, services and carrying capacity.								
UNIT - II	NATURAL RESOURCES					CLASSES: 8		
Natural Resources: Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems-case studies Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources. Land resources: Forest resources. Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy source, case studies.								

UNIT - III	BIODIVERSITY AND BIOTIC RESOURCES	CLASSES: 5
Bio diversity and Biotic Resources: Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values. India as a mega diversity nation: Hot spots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of bio diversity: In-Situ and Ex-situ conservation. National Biodiversity Act-Case studies.		
UNIT - IV	ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES	CLASSES: 5
Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, Air Pollution, Water Pollution drinking water quality standards. Soil Pollution Impacts of modern agriculture, Noise Pollution Health hazards, standards. Concepts of bioremediation. Global Environmental Problems and Global Efforts: Ozone depletion and Ozone depleting substances (ODS) Concepts of Bioremediation International conventions/Protocols: Earth summit, Kyoto protocol and Montréal Protocol; NAPCC-Gol Initiatives, COP 24, COP25.		
UNIT - V	ENVIRONMENTAL POLICY, LEGISLATION & EIA	CLASSES: 5
Environmental Policy, Legislation & Environmental Impact Assessment (EIA): Environmental Protection act, Legal aspects Air Act1981, Water Act, Forest Act, Wildlife Act, Municipal solid waste management, biomedical waste management hazardous waste management and handling rules. Environmental Impact Assessment: EIA structure, methods of baseline data acquisition. Strategies for risk assessment, Towards Sustainable Future: Concept of Sustainable Development, Urban Sprawl, Concept of Green Building.		
TEXT BOOKS		
1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission. 2. Environmental Studies by R.Rajagopalan, Oxford University Press.		

II B.TECH I SEMESTER SYLLABUS

COMPUTER ORIENTED STATISTICAL METHODS

II B. TECH- I SEMESTER

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS03	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	40	60	100

COURSE OBJECTIVES

1. Evaluation of Probability distribution of Discrete and Continuous random variables and their moments.
2. The concept of correlation and regression, covariance and sampling distribution
3. Evaluation of the given data for appropriate test of hypothesis for large samples.
4. Evaluation of the given data for appropriate test of hypothesis for small samples and one way ANOVA
5. To learn the concept of Markov chain, transition probabilities in discrete & continuous time and Stochastic simulation techniques.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Evaluation of Probability distribution of Discrete and Continuous random variables and their moments.
2. Apply the concept of correlation and regression to find covariance and Sampling distribution
3. Evaluate the given data for appropriate test of hypothesis for Large samples.
4. Evaluate the given data for appropriate test of hypothesis for small samples and one way ANOVA
5. Recognize if a given stochastic system with finite number of states is a Markov chain or not and also identify classes of states in Markov chains and Characterize the states.

UNIT - I	RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS	CLASSES: 08
Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution, mathematical expectation,. Binomial, Poisson, Normal distributions -their Properties. Moments about origin, central moments, skewness, Kurtosis and find the mean and variance.		
UNIT - II	CORRELATION & REGRESSION AND SAMPLING DISTRIBUTIONS	CLASSES: 09
Coefficient of correlation, the rank correlation, Covariance of two random variables. Regression- Regression Coefficient, The lines of regression. Sampling: Definitions of population, sampling, statistic, parameter. Types of sampling, Expected values of Sample mean and variance, sampling distribution, Standard error, Sampling distribution of means and sampling distribution of variance. Parameter estimation- Point estimation and interval estimation		
UNIT - III	TESTING OF HYPOTHESIS - I	CLASSES: 09
Testing of hypothesis: Null hypothesis, Alternate hypothesis, Type I & Type II errors – critical region, confidence interval, Level of significance. One sided test, Two sided test. Large sample tests: (i) Test of Equality of means of two samples, equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances) (ii) Tests of significance of difference between sample S.D and population S.D. (iii) Tests of significance difference between sample proportion and population proportion, difference between two sample proportions.		
UNIT - IV	TESTING OF HYPOTHESIS-II	CLASSES: 08

Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, its properties, Chi-square test of goodness of fit. One way ANOVA

UNIT - V**STOCHASTIC PROCESSES AND MARKOV CHAINS****CLASSES: 08**

Introduction to Stochastic processes- Markov process. Transition Probability, Transition Probability Matrix, First order and Higher order Markov process, n-step transition probabilities, Markov chain, Steady state condition, Markov analysis.

TEXT BOOKS

1. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.
2. Probability and Statistics for Engineers by Richard Arnold Johnson, Irwin Miller and John E. Freund, New Delhi, Prentice Hall.
3. Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning

REFERENCE BOOKS

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. Fundamentals of Mathematical Statistics by S.C. Gupta & V.K. Kapoor, S. Chand
3. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press

WEB REFERENCES

1. https://www.efunda.com/math/math_home/math.cfm
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

E-TEXT BOOKS

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>

MOOCS COURSE

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

DIGITAL ELECTRONICS AND COMPUTER ORGANIZATION

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS28	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand different number systems and Boolean Algebra.
2. Design of combinational and sequential logic circuits
3. Understand different Computer Instructions and addressing modes
4. Understand concepts of register transfer logic and arithmetic operations.
5. Learn different types of memory hierarchy

COURSE OUTCOMES

1. Apply minimization techniques to simplify Boolean expressions.
2. Apply the principles of Digital electronics to design combinational and sequential logic circuits.
3. Understand the basics of instruction set and their impact on processor design
4. Illustrate register transfer operations
5. Analyze memory hierarchy and I/O Communication.

UNIT - I	NUMBER SYSTEMS AND BOOLEAN ALGEBRA	CLASSES: 14
Representation of numbers with different radix, Number base conversions, r-1's and r's Complements, Introduction to Binary codes. Basic Logic Gates, Universal Gates, Basic Theorems and Properties of Boolean algebra, Canonical and Standard forms, K-Map Method (upto 5-Variables)		
UNIT - II	COMBINATIONAL AND SEQUENTIAL LOGIC CIRCUITS	CLASSES: 12
Design of Half adder, full adder, half subtractor, full subtractor. Decoder, Encoder, Multiplexer, De-multiplexer and comparator. Latches and Flip-flops, truth tables, Characteristic table and excitation tables, Conversion of flip-flops, Classification of sequential circuits (synchronous and asynchronous).		
UNIT - III	DESIGN OF SEQUENTIAL CIRCUITS AND BASIC COMPUTER ORGANIZATION	CLASSES: 12
Design of Ripple counters, design of synchronous counters, Johnson counter, ring counter, shift registers, bi-directional shift register, universal shift register. Instruction codes, Computer Registers, Computer Instructions and Instruction cycle. Timing and Control Memory-Reference Instructions, Input-Output and interrupt.		
UNIT - IV	REGISTER TRANSFER AND MICRO-OPERATIONS	CLASSES: 11
Central processing unit: Stack organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Complex Instruction Set Computer (CISC), Reduced Instruction Set Computer (RISC), Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-Operations, Logic Micro-Operations, Shift Micro-Operations, Arithmetic logic shift unit. Pipeline and Parallel Processing: Parallel processing, Pipelining, Arithmetic pipeline, Instruction pipeline		
UNIT - V	MEMORY SYSTEM	CLASSES: 11

INPUT-OUTPUT ORGANIZATION: I/O interface, Programmed IO, Memory Mapped IO, Interrupt Driven IO, DMA.

MEMORY ORGANIZATION: Memory Hierarchy, Main Memory, Auxiliary memory, Associate memory, Cache memory.

TEXT BOOKS

1. Digital Design- Morris Mano, PHI, 3rd Edition.
2. Digital Principles and Applications by Leach, Paul Malvino, 5th Edition.
3. Computer System Architecture- M. Morris Mano, 3rd edition, Pearson/PHI, India.

REFERENCE BOOKS

1. Switching and Finite Automata Theory by Zvi.Kohavi, Tata McGraw Hill.
2. Fundamentals of Digital circuits, A. Anand Kumar, Third Edition, 2013, PHI.
3. Computer Organization and Architecture William Stallings Sixth Edition, Pearson/PHI
4. Structured Computer Organization Andrew S. Tanenbaum, 4th Edition PHI/Pearson

DATA STRUCTURES

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS05	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

1. Understand basic concepts about stacks, queues and their applications.
2. The student should be able to choose appropriate data structures, understand the performance, and use it to design algorithms for a specific problem.
3. Understand basic concepts of singly linked lists and singly linked lists.
4. Write programs for sorting and searching and hashing.
5. Use advanced data structures like Tress and Graphs for efficient problem solving.
6. Introduces pattern matching algorithms.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Solve problems using data structures such as stacks and Queues.
2. Solve problems using singly linked lists and double linked list.
3. Compare various types of searching and sorting techniques in terms of implementation, operation and performance.
4. Implement Tree data structure and its variants.
5. Identify the importance and application of Graph data Structure with problem solving techniques.

UNIT - I	Introduction to Data Structures, Stacks & Queues	CLASSES: 14
Introduction: Introduction to Data Structures, abstract data types, The Stack ADT: Stack operations, implementing the Stack using Array, Stack Applications: Balanced Delimiters, Evaluating Postfix Expressions, The Queue ADT: Queue operations, Implementing the Queue using Array, Circular Queue operations, Implementing the Circular Queue using Array, Priority Queues: Priority Queue operations, The Priority Queue ADT Implementation.		
UNIT - II	Linked Lists	CLASSES: 12
Linked List ADT - Singly Linked Lists (Insertion, Deletion and Traversing), Doubly Linked Lists (Insertion, Deletion and Traversing), Circular Linked Lists (Insertion, Deletion and Traversing), Stacks implementation Using Linked list, Queues Implementation Using Linked list.		
UNIT - III	Searching and Sorting	CLASSES: 8
Searching: Linear Search - Binary search. Sorting: Bubble Sort - Selection Sort - Insertion Sort - Quick Sort - Merge Sort - Heap sort, comparison of sorting methods.		
UNIT - IV	Trees	CLASSES: 14
Trees –Terminology, Representation of Trees ,Binary Tree, Binary Tree traversals, Binary search trees: operations-Insertion, Delete and Searching – AVL Trees: operations- Insertion, Delete and Searching, B-Trees, B-Tree of order m, height of a B-Tree, insertion, deletion and searching Red-Black Trees, Splay Trees, Comparison of Search Trees.		
UNIT - V	Graphs & Text Processing	CLASSES: 12
Graphs: Introduction, Definition, Terminology, Graph ADT, Graph traversals- DFS and BFS, Dijkstra's Algorithm, Bellman Ford Algorithm, Floyd Warshall Algorithm.		

Text Processing: Pattern matching algorithms-Brute force, Knuth Morris-Pratt algorithm, Tries-Standard Tries, Compressed Tries, and Suffix tries.

TEXT BOOKS

1. Fundamentals of Data Structures in C, 2 nd Edition, E. Horowitz, S. Sahni and Susan Anderson Freed, Universities Press.
2. Data Structures using C A. S.Tanenbaum, Y. Langsam, and M.J. Augenstein, PHI/Pearson Education.

REFERENCE BOOKS

1. Data Structures: A Pseudo code Approach with C, 2 nd Edition, R. F. Gilberg and B.A.Forouzan, Cengage Learning.

WEB REFERENCES

1. <https://hackr.io/tutorials/learn-data-structures-algorithms>
2. <https://www.geeksforgeeks.org/fundamentals-of-algorithms/>
3. <https://www.udemy.com/introduction-to-algorithms-and-data-structures-in-c/>
4. <https://leetcode.com>

OBJECT ORIENTED PROGRAMMING THROUGH JAVA

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT02	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the basic object oriented programming concepts and apply them in problem solving.
2. Illustrate inheritance and polymorphism concepts for reusing the program.
3. Demonstrate on the exception handling mechanism
4. Develop multi-threading and data-centric applications using JDBC.
5. Understand the basics of java collection framework

COURSE OUTCOMES

At the end of the course, student will be able to

1. Understand the basic programming concepts and apply them in problem solving
2. Implement object-oriented principles, such as encapsulation, inheritance, polymorphism, and abstraction.
3. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws and finally).
4. Develop java application to interact with database by using relevant software component (JDBC Driver).
5. Build real world applications using Collection framework

UNIT - I	JAVA BASICS	CLASSES: 12
JAVA BASICS: Review of Object oriented concepts, History of Java, Java buzzwords, JVM architecture, Data types, Variables, Scope and life time of variables, arrays, operators, control statements, type conversion and casting, simple java program, constructors, methods, Static block, Static Data, Static Method, String and String Buffer Classes, Using Java API Document.		
UNIT - II	INHERITANCE, POLYMORPHISM, PACKAGES AND INTERFACES	CLASSES: 11
INHERITANCE AND POLYMORPHISM: Basic concepts, Types of inheritance, Member access rules, Usage of this and Super key word, Method Overloading, Method overriding, Abstract classes, Encapsulation, Need for encapsulation in java, Data hiding vs Encapsulation, getter and setter methods, Dynamic method dispatch, Usage of final keyword.		
PACKAGES AND INTERFACES: Defining package, Access protection, importing packages, Defining and Implementing interfaces, and Extending interfaces		
UNIT - III	EXCEPTION HANDLING AND FILES	CLASSES: 10

EXCEPTION HANDLING: Exception types, Usage of Try, Catch, Throw, Throws and Finally keywords, Built-in Exceptions, Creating own Exception classes.

I / O STREAMS AND FILES: Concepts of streams, Stream classes- Byte and Character stream, Reading console Input and Writing Console output, IO/Serialization, File Handling,

UNIT – IV MULTITHREADING AND JDBC

CLASSES: 10

MULTI THREADING: Concepts of Thread, Thread life cycle, creating threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter Thread communication, Concurrency, Executors framework

JDBC-Connecting to Database - JDBC Type 1 to 4 drives, connecting to a database, querying a Database and processing the results, updating data with JDBC

UNIT - V COLLECTION FRAMEWORK

CLASSES: 10

COLLECTION FRAMEWORK: Introduction to Java Collections, Overview of Java Collection frame work, Generics, Commonly used Collection classes- Array List, Vector, Hash table, Stack, Enumeration, Iterator, String Tokenizer, Random, Scanner, calendar and Properties, Lambdas & Functional Interfaces

TEXT BOOKS

1. Herbert Schildt and Dale Skrien, *Java Fundamentals – A comprehensive Introduction*, McGraw Hill, 1st Edition, 2013.
2. Herbert Schildt, *Java the complete reference*, McGraw Hill, Osborne, 7th Edition, 2011.
3. T.Budd, *Understanding Object- Oriented Programming with Javall*, Pearson Education, Updated Edition (New Java 2 Coverage), 1999.

REFERENCE BOOKS

1. P.J.Dietel and H.M.Dietel, *Java How to program*, Prentice Hall, 6th Edition, 2005.
2. P.Radha Krishna, *Object Oriented programming through Javall*, CRC Press, 1st Edition, 2007.
3. S.Malhotra and S. Choudhary, *Programming in Javall*, Oxford University Press, 2nd Edition, 2014.

SOFTWARE ENGINEERING

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS07	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

1. To familiarize with software engineering methods, practices and its applications.
2. To familiarize with software requirements and the SRS documents.
3. To facilitate students in software design.
4. To familiarize with testing strategies.
5. Understanding working knowledge, quality management and risk management of large software development projects.

COURSE OUTCOMES

1. Analyze software development life cycle and select appropriate model for software application.
2. Analyze the customer's requirements for a project to be developed and formulate the software requirements document.
3. Conceptualize the system through design on architectural modelling and user interface.
4. Classify software testing strategies and recommend testing techniques during the construction of software projects.
5. Analyze the risk management, quality management and quality standards before releasing the software product.

UNIT - I	INTRODUCTION TO SOFTWARE ENGINEERING	CLASSES: 10
Introduction to Software Engineering: The evolving role of software, changing nature of software, software myths. A Generic view of process: Software engineering- a layered technology, a process framework, the capability maturity model integration (CMMI). Process models: The waterfall model, Spiral model and Agile methodology		
UNIT - II	SOFTWARE REQUIREMENTS & REQUIREMENTS ENGINEERING PROCESS	CLASSES: 10
Software Requirements: Functional and non-functional requirements, user requirements, system requirements, interface specification, the software requirements document. Requirements engineering process: Feasibility studies, requirements elicitation and analysis, requirements validation, requirements management		
UNIT - III	DESIGN ENGINEERING	CLASSES: 10
Design Engineering: Design process and design quality, design concepts, the design model. Creating an architectural design: software architecture, data design, architectural styles and patterns, architectural design, conceptual model of UML, basic structural modeling, class diagrams, sequence diagrams, collaboration diagrams, use case diagrams, component diagrams.		

UNIT – IV	TESTING STRATEGIES	CLASSES: 10
Testing Strategies: A strategic approach to software testing, test strategies for conventional software, black-box and white-box testing, validation testing, system testing, the art of debugging. Metrics for Process and Products: Software measurement, metrics for software quality.		
UNIT - V	RISK MANAGEMENT & QUALITY MANAGEMENT	CLASSES: 10
Risk management: Reactive Vs proactive risk strategies, software risks, risk identification, risk projection, risk refinement, RMMM. Quality Management: Quality concepts, software quality assurance, software reviews, formal technical reviews, statistical software quality assurance, software reliability, the ISO 9000 quality standards.		
TEXT BOOKS		
1. Software Engineering, A practitioner's Approach- Roger S. Pressman, 6th edition, McGraw Hill International Edition. 2. Software Engineering- Sommerville, 7th edition, Pearson Education.		
REFERENCE BOOKS		
1. The unified modeling language user guide Grady Booch, James Rumbaugh, Ivar Jacobson, Pearson Education. 2. Software Engineering, an Engineering approach- James F. Peters, Witold Pedrycz, John Wiley. 3. Software Engineering principles and practice- Waman S Jawadekar, The McGraw-Hill Companies. 4. Fundamentals of object-oriented design using UML Meiler page-Jones: Pearson Education.		

DATA STRUCTURES LAB

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS06	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

1. Introduce fundamental data structures, teach Stack and Queue ADTs, and explore Priority Queues.
2. Teach Linked List ADT, including various types, and demonstrate Stacks and Queues using linked lists.
3. Cover searching (linear and binary search) and sorting algorithms (bubble sort, quick sort, etc.), and compare their efficiency.
4. Explain tree concepts, binary trees, and various search trees (BST, AVL, etc.), and compare their properties.
5. Introduce graphs, their traversals, algorithms (Dijkstra's, Bellman Ford, etc.), and text processing (pattern matching, tries).

COURSE OUTCOMES

1. Students will implement Stacks, solve delimiter and postfix expression problems, implement Queues, and understand Priority Queues.
2. Students will implement linked lists and use them to create effective Stack and Queue data structures.
3. Students will apply search and sorting algorithms and make informed choices for specific problems.
4. Students will create, manipulate, and choose appropriate tree structures for different scenarios.
5. Students will work with graphs, perform traversals, apply graph algorithms, and implement text processing techniques.

LIST OF EXPERIMENTS

WEEK -1	INTRODUCTION TO STRUCTURES
1. Write a program that implement structures (its operations) using i) Structures ii) Array of Structures iii) Nested Structures iv) Pointer to Structures	
WEEK -2	STACK OPERATIONS
2. Write a program that implement Stack (its operations) using array and linked list	
WEEK -3	QUEUE OPERATIONS
3. Write a program that implement Queue (its operations) using array and linked list	
WEEK -4	SINGLY LINKED LIST DOUBLY LINKEDLIST
4. Write a program that uses functions to perform the following operations on singly linkedlist & doubly linked list i) Creation ii) Insertion iii) Deletion iv) Traversal	
WEEK -5	CIRCULAR LINKED LIST
5. Write a program that uses functions to perform the following operations on circular linkedlist. i) Creation ii) Insertion iii) Deletion iv) Traversal	

WEEK -6	SORTING TECHNIQUES – I
6. Write a program that implements the following sorting methods. i) Bubble sort ii) Insertion sort iii) Selection sort	
WEEK -7	SORTING TECHNIQUES - II
7. Write a program that implements the following sorting methods. i) Quick sort ii) Merge sort iii) Heap sort	
WEEK -8	TREE TRAVERSALS
8. Write a program to implement the tree traversal methods. i) Pre-order ii) In order iii) Post order	
WEEK -9	BINARY SEARCH TREES
9. Write a program to implement i) Binary Search tree ii) B Trees	
WEEK -10	BINARY SEARCH TREES
10. Write a program to implement i) B+ Trees ii) AVL trees iii) Red - Black trees	
WEEK -11	GRAPH TRAVERSAL
11. Write a program to implement the graph traversal methods. i) DFS ii) BFS	
WEEK -12	PATTERN MATCHING
12. Implement a Pattern matching algorithms using Brute force, Knuth-Morris-Pratt	
TEXT BOOKS	
1. Fundamentals of Data Structures in C, 2nd Edition, E. Horowitz, S. Sahni and Susan Anderson Freed, Universities Press. 2. Data Structures using C A. S. Tanenbaum, Y. Langsam, and M. J. Augenstein, PHI/Pearson Education.	

OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6IT03	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity.
2. Understand the use of different exception handling mechanisms and concept of multithreading for robust and efficient application development.
3. Understand and implement concepts on file streams and operations in java programming for a given application programs.
4. Develop java application to interact with database by using relevant software component (JDBC Driver).

LIST OF EXPERIMENTS

WEEK - 1 JAVABASICS

- a. Write a java program that prints all real solutions to the quadratic equation $ax^2+bx+c=0$. Read in a, b, c and use the quadratic formula.
- b. The Fibonacci sequence is defined by the following rule. The first two values in the sequence are 1 and 1. Every subsequent value is the sum of the two values preceding it. Write a java program that uses both recursive and non recursive functions.

WEEK - 2 ARRAYS

- a. Write a java program to sort given list of integers in ascending order.
- b. Write a java program to multiply two given matrices.

WEEK - 3 STRINGS

- a. Write a java program to check whether a given string is palindrome.
- b. Write a java program for sorting a given list of names in ascending order.

WEEK - 4 OVERLOADING & OVERRIDING

- a. Write a java program to implement method overloading and constructors overloading.
- b. Write a java program to implement method overriding.

WEEK - 5 INHERITANCE

Write a java program to create an abstract class named Shape that contains two integers and an empty method named print Area (). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contains only the method print Area() that prints the area of the given shape.

WEEK - 6	INTERFACES
a. Write a program to create interface A in this interface we have two method meth1 and meth2. Implements this interface in another class named MyClass. b. Write a program to give example for multiple inheritance in Java.	
WEEK - 7	EXCEPTION HANDLING
Write a program that reads two numbers Num1 and Num2. If Num1 and Num2 were not integers, the program would throw a Number Format Exception. If Num2 were zero, the program would throw an Arithmetic Exception Display the exception.	
WEEK - 8	I/O STREAMS
a. Write a java program that reads a file name from the user, and then displays information about whether the file exists, whether the file is readable, whether the file is writable, the type of file and the length of the file in bytes.	
WEEK - 9	MULTI THREADING
Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number	
WEEK - 10	GENERIC
a. Write a Java program to swap two different types of data using Generics. b. Write a Java program to find maximum and minimum of two different types of data using Generics.	
WEEK - 11	COLLECTIONS
Create a linked list of elements. a. Delete a given element from the above list. b. Display the contents of the list after deletion	
WEEK - 12	CONNECTING TO DATABASE
Write a java program that connects to a database using JDBC and does add, delete, modify and retrieve operations.	
TEXT BOOKS	
1. P.J.Dietel and H.M.Dietel , —Java How to programII, Prentice Hall, 6th Edition, 2005. 2. P.Radha Krishna , —Object Oriented programming through Javall, CRC Press, 1st Edition, 2007. 3. S.Malhotra and S. Choudhary, —Programming in Javall, Oxford University Press, 2nd Edition, 2014.	

SKILL DEVELOPMENT (DATA VISUALIZATION USING R)

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS52	SKL	L	T	P	C	CIE	SEE	Total
		0	0	2	1	0	100	100

Course Overview:

Data analysis is the process of data analyzing and extracting useful information which can help in making useful business decisions. As part of this course basics of data analytics are discussed and statistical analysis tools are introduced.

COURSE OBJECTIVES

The course should enable the students to:

1. To learn about the Data Visualization and Process of Visualization.
2. To learn different statistical method for Data visualization.
3. To learn Basics of R and usage of Watson studio
4. To learn Basic Visualization Techniques.
5. Learn about advanced Visualization Techniques.

COURSE OUTCOMES

1. Apply Data Preprocessing for visualization
2. Apply the Inferential statistics and Probability
3. Understand the basics of R Programming and Visualization with R
4. Understand the basic visualization Techniques.
5. Apply the techniques for spatial, Multivariate Data

WEEK - 1

1. What is R?, What is R Studio?, RObjects, RDataframes, Write a R Program to plot Student Hall ticket number with student GPD.

WEEK - 2

2. Reading Datasets using R, Import a CSV file in R and check the output.
Name, Age, Sex
Shaan, 21, M
Ritu, 24, F
Raj, 31, M

WEEK - 3

3. Outliers- a) Consider a Dataframe and remove the missing data in the dataframe.
b) Combining Data sets in R-Write a R program to merge two data frames.

WEEK - 4

4. R Functions- Write a R program to count the number of even numbers in a vector.		
WEEK - 5		
5. i) The summary of 150 flower data including Sepal Length, Sepal Width, Petal Length and Petal Width. He also wants the summary of Sepal Length vs petal length. ii) the mean Petal Length of each species. iii) Write a R Program to segregate the data of flowers having Sepal length greater than 7. iv) Write a R Program to segregate the data of flowers having Sepal length greater than 7 and Sepal width greater than 3 simultaneously. v) Write a R Program to view 1 st 7 rows of data. vi) Write a R Program to segregate the data of flowers having Sepal length greater than 7 and Sepal width greater than 3 simultaneously. vii) Write a R Program to view 1 st 7 rows of data.		
WEEK - 6		
6. Understanding the types of probability. i) Find the probability of getting 3 doublets when a pair of fair dice are thrown for 10 times. ii) Find the probability of getting 3 or lesser doublets. iii) Suppose there are twelve multiple choice questions in an English class quiz. Each question has five possible answers, and only one of them is correct. Find the probability of having four or less correct answers if a student attempts to answer every question at random.		
WEEK - 7		
7. i) If there are twelve cars crossing a bridge per minute on average, find the probability of having seventeen or more cars crossing the bridge in a particular minute. ii) If you throw a dice 20 times then what is the probability that you get following results: a. 3 sixes b. 6 sixes c. 1,2 and 3 sixes		
WEEK - 8		
8. i) Write a R Program to calculate permutation for a given values using Functions. ii) Write a R Program to calculate Sum of two numbers using function.		
WEEK - 9		
9. Write a R program to barplot movies, IMBD rating for the given Movie dataset.		
WEEK - 10		
10. i) Write a R Program to Visualize using histogram for runtime of movies. (Consider Movies dataset) ii) Write a R Program to Visualize using Pie chart for genre of movies. (Consider Movies dataset)		
TEXT BOOKS		
1. R in Action By - Robert L. Kabacoff, Second Edition - Publisher - Dreamtech Press 2. R for Data Science, By - Hadley Wickham and Garrett Gorlemund, O'Reilly		
REFERENCE BOOKS		
1. R Programming for Data Science by Roger D. Peng (References) 2. The Art of R Programming by Norman Matloff Cengage Learning India.		

GENDER SENSITIZATION

II B. TECH- I SEMESTER

Course Code	Category	Hours /Week			Credits	Maximum Marks		
A6HS05	MC	L	T	P	C	CIE	SEE	Total
		-	-	2	-	40	60	100
Contact Classes:16	Tutorial Classes: 0	Practical Classes:00			Total Classes:16			

COURSE OUTCOMES

- Students will have developed a better understanding of important issues related to gender in contemporary India.
- Students will be sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender. This will be achieved through discussion of materials derived from research, facts, everyday life, literature and film.
- Students will attain a finer grasp of how gender discrimination works in our society and how to counter it.
- Students will acquire insight into the gendered division of labour and its relation to politics and economics.
- Men and women students and professionals will be better equipped to work and live together as equals.
- Students will develop a sense of appreciation of women in all walks of life. Through providing accounts of studies and movements as well as the new laws that provide protection and relief to women, the textbook will empower students to understand and respond to gender violence.

UNIT - I	UNDERSTANDING GENDER: Gender:	
UNDERSTANDING GENDER: Gender: Why Should We Study It? (Towards a World of Equals: Unit -1) Socialization: Making Women, Making Men (Towards a World of Equals: Unit -2) Introduction. Preparing for Womanhood. Growing up Male. First lessons in Caste. Different Masculinities		
UNIT - II	GENDER AND BIOLOGY: Missing Women	
GENDER AND BIOLOGY: Missing Women: Sex Selection and Its Consequences (Towards a World of Equals: Unit - 4)Declining Sex Ratio. Demographic Consequences. Gender Spectrum: Beyond the Binary (Towards a World of Equals: Unit -10)Two or Many? Struggles with Discrimination		
UNIT - III	GENDER AND LABOUR	
GENDER AND LABOUR: Housework: the Invisible Labour (Towards a World of Equals: Unit -3) "My Mother doesn't Work." "Share the Load." Women's Work: Its Politics and Economics (Towards a World of Equals: Unit -7) Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading: Wages and Conditions of Work.		
UNIT - IV	ISSUES OF VIOLENCE	
ISSUES OF VIOLENCE: Sexual Harassment: Say No! (Towards a World of Equals: Unit -6) Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment- Further Reading:"Chupulu". Domestic Violence: Speaking Out (Towards a World of Equals: Unit -8) Is Home a Safe Place? -When Women Unite [Film]. Rebuilding Lives. Additional Reading: New Forums for Justice. Thinking about Sexual Violence (Towards a World of Equals: Unit -11) Blaming the Victim-"I Fought for my Life...." - Additional Reading: The Caste Violence		

UNIT - V	GENDER: CO – EXISTENCE: Just Relationships	
GENDER: CO – EXISTENCE: Just Relationships: Being Together as Equals (Towards a World of Equals: Unit -12) Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and Fathers Additional Reading: Rosa Parks-The Brave Heart		
TEXT BOOKS		
1. All the five Units in the Textbook, “Towards a World of Equals: A Bilingual Textbook on Gender” written by A.Suneetha, Uma Bhargubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by Telugu Akademi, Hyderabad, Telangana State in the year 2015.		
REFERENCE BOOKS		
1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012		
2. Abdulali Sohaila. “I Fought For My Life...and Won.” Available online at: http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulali/		

II B.TECH II SEMESTER SYLLABUS

DISCRETE MATHEMATICS**II B.TECH - II SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS08	BSC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To help students understand discrete and continuous mathematical structures
2. To impart basics of relations and functions
3. To facilitate students in applying principles of Recurrence Relations to calculate generating
4. Functions and solve the Recurrence relations
5. To acquire knowledge in graph theory

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Analyze and examine the validity of argument by using propositional and predicate calculus
2. Apply basic counting techniques to solve the combinatorial problems
3. Apply sets relations and digraphs to solve applied problems
4. Solve the given recurrence relation using different methods such as substitution, Generating function and characteristics roots equation.
5. Use the basic concepts of graph theory and some related theoretical problems

UNIT - I	MATHEMATICAL LOGIC	CLASSES: 11
Statements and notations, Connectives, Well-formed formulas, Truth Tables, Tautology, Equivalence implication, Normal forms, Logical Inference, Rules of inference, Direct Method, Direct Method using CP(Conditional Proof), Consistency, Proof of contradiction, Automatic Theorem Proving.		
UNIT - II	RELATIONS	CLASSES: 16
Introduction to set theory, Relations, Properties of Binary Relations, Equivalence Relation, Transitive closure, Compatibility and Partial ordering relations, Lattices, Hasse diagram. Functions: inverse Function, Composition of functions. Algebraic Systems, Semi groups and Monoids		
UNIT - III	ELEMENTARY COMBINATORICS	CLASSES: 12
Basis of counting, Combinations & Permutations, Enumeration of Combinations and Permutations, Enumeration of Combinations and Permutations With repetitions, Enumerating Permutations with Constrained repetitions, Binomial Coefficients, Binomial and Multinomial theorems, The principles of Inclusion – Exclusion, Pigeon- hole principles and its applications.		
UNIT - IV	RECURRENCE RELATION	CLASSES: 11
Generating Functions, Function of Sequences, Calculating Coefficient of generating function, Recurrence relations, Solving recurrence relation by substitution and Generating functions, The method of Characteristics roots,		

UNIT - V	GRAPH THEORY	CLASSES: 10
Basic Concepts, Isomorphism and Subgraphs, Trees and their properties, Spanning Trees- DFS,BFS, Minimal Spanning Trees- Prims, Kruskal's Algorithm, Planar Graphs, Euler's Formula, Multi graph and Euler circuits, Hamiltonian Graphs, Chromatic number.		
TEXT BOOKS		
1. Discrete Mathematical Structures with Applications to Computer Science: J.P. Tremblay, R. Manohar, McGraw-Hill, 1st ed. 2. Discrete Mathematics for Computer Scientists & Mathematicians: Joe I. Mott, Abraham Kandel, Theodore P. Baker, Prentis Hall of India, 2nd ed.		
REFERENCE BOOKS		
1. Discrete and Combinatorial Mathematics - an applied introduction: Ralph.P. Grimald, Pearson education, 5th edition. 2. Discrete Mathematical Structures: Thomas Kosy, Tata McGraw Hill publishing co.		

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

II B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS08	HSMC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES								
The course will enable the students to:								
1. To enable the student to understand and appreciate, with a particular insight, the importance of certain basic issues. 2. To get the basic understanding of the governing the business operations namely; demand and supply, production function, cost analysis, markets, forms of business organizations. 3. To get the knowledge of the capital budgeting and financial accounting and financial analysis.								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures. 2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis. 3. Analyze how capital budgeting decisions are carried out. 4. Understanding the framework for both manual and computerized accounting process. 5. Know how to analyze and interpret the financial statements through ratio analysis.								
UNIT - I	Introduction & Demand Analysis						CLASSES: 11	
Introduction & Demand Analysis: Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.								
UNIT - II	Production & Cost Analysis						CLASSES: 09	
Production & Cost Analysis: Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA) - Determination of Break-Even Point (simple problems) - Managerial Significance.								
UNIT - III	Markets & New Economic Environment						CLASSES: 12	
Markets & New Economic Environment: Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario.								

UNIT - IV	Capital Budgeting	CLASSES: 10
Capital Budgeting: Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).		
UNIT - V	Introduction to Financial Accounting & Financial Analysis	CLASSES: 08
Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, and Trial Balance - Final Accounts (Tracing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.		
TEXTBOOKS		
1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009. 2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013. 3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.		
REFERENCEBOOKS		
1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012. 2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012. 3. Lipsey & Chrystel, Economics, Oxford University Press, 2012. 4. Domnick Salvatore: Managerial Economics in a Global Economy, Thomson, 2012. 5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012. 6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012. 7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012. 8. Dwivedi: Managerial Economics, Vikas, 2012. 9. Shailaja & Usha: MEFA, University Press, 2012. 10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012. 11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011. 12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.		

DATABASE MANAGEMENT SYSTEMS

II B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS09	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the basic concepts and the applications of database systems.
2. To master the basics of SQL and construct queries using SQL.
3. Topics include data models, database design, relational model, relational algebra, transaction control, concurrency control, storage structures and access techniques.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Gain knowledge of fundamentals of DBMS, database design and normal forms
2. Master the basics of SQL for retrieval and management of data.
3. Be acquainted with the basics of transaction processing and concurrency control.
4. Familiarity with database storage structures and access techniques

UNIT - I	DATABASE SYSTEM APPLICATIONS & INTRODUCTION TO DATABASE DESIGN	CLASSES: 10
Database System Applications: A Historical Perspective, File Systems versus a DBMS, the Data Model, Levels of Abstraction in a DBMS, Data Independence, Structure of a DBMS Introduction to Database Design: Database Design and ER Diagrams, Entities, Attributes, and Entity Sets, Relationships and Relationship Sets, Additional Features of the ER Model, Conceptual Design With the ER Model		
UNIT - II	INTRODUCTION TO THE RELATIONAL MODEL	CLASSES: 10
Introduction to the Relational Model: Integrity constraint over relations, enforcing integrity constraints, querying relational data, logical database design, introduction to views, destroying/altering tables and views. Relational Algebra, Tuple relational Calculus, Domain relational calculus		
UNIT - III	SQL & SCHEMA REFINEMENT	CLASSES: 10
SQL: QUERIES, CONSTRAINTS, TRIGGERS: form of basic SQL query, UNION, INTERSECT, and EXCEPT, Nested Queries, aggregation operators, NULL values, complex integrity constraints in SQL, triggers and active databases. Schema Refinement: Problems caused by redundancy, decompositions, problems related to decomposition, reasoning about functional dependencies, First, Second, Third normal forms, BCNF, lossless join decomposition, multivalued dependencies, Fourth normal form, Fifth normal form.		
UNIT - IV	TRANSACTION CONCEPT	CLASSES: 10
Transaction Concept, Transaction State, Implementation of Atomicity and Durability, Concurrent Executions, Serializability, Recoverability, Implementation of Isolation, Testing for serializability, Lock Based Protocols, Timestamp Based Protocols, Validation- Based Protocols, Multiple Granularity, Recovery and Atomicity, Log-Based Recovery, Recovery with Concurrent Transactions.		

UNIT - V	DATA ON EXTERNAL STORAGE	CLASSES: 10
Data on External Storage, File Organization and Indexing, Cluster Indexes, Primary and Secondary Indexes, Index data Structures, Hash Based Indexing, Tree based Indexing, Comparison of File Organizations, Indexes- Intuitions for tree Indexes, Indexed Sequential Access Methods (ISAM), B+ Trees: A Dynamic Index Structure		
TEXT BOOKS		
1. Database System Concepts, Silberschatz, Korth, McGraw hill, V edition.3rd Edition 2. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, Tata Mc Graw Hill		
REFERENCE BOOKS		
1. Database Systems design, Implementation, and Management, Peter Rob & Carlos Coronel 7th Edition. 2. Fundamentals of Database Systems, Elmasri Navrate, Pearson Education 3. Introduction to Database Systems, C. J. Date, Pearson Education 4. Oracle for Professionals, The X Team, S.Shah and V. Shah, SPD. 5. Database Systems Using Oracle: A Simplified guide to SQL and PL/SQL, Shah, PHI. 6. Fundamentals of Database Management Systems, M. L. Gillenson, Wiley Student Edition		

OPERATING SYSTEMS

II B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS11	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To explain main components of OS and their working
2. To familiarize the operations performed by OS as a resource Manager
3. To familiarize with Dead Locks
4. To impart various scheduling policies of OS
5. To teach the different memory management techniques.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Analyze the different structures and services of operating system.
2. Analyze various algorithms used for OS services with respect to defined/chosen criteria.
3. Solve the resource allocation and sharing problems.
4. Assess different methods to solve OS problems.
5. Analyze the memory management approaches of operating systems.

UNIT - I	OPERATING SYSTEMS OVERVIEW	CLASSES: 14
OPERATING SYSTEMS OVERVIEW: Introduction, operating system operations, process management, memory management, storage management, protection and security, distributed systems. OPERATING SYSTEMS STRUCTURES: Operating system services and systems calls, system programs, operating system structure, operating systems generations		
UNIT - II	PROCESS MANAGEMENT	CLASSES: 16
PROCESS MANAGEMENT: Process concepts, process state, process control block, scheduling queues, process scheduling, Scheduling algorithms, and multithreaded programming. CONCURRENCY AND SYNCHRONIZATION: Process synchronization, critical section problem, Peterson's solution, synchronization hardware, semaphores, classic problems of synchronization, readers and writers problem, dining philosophers problem, monitors, synchronization examples(Solaris).		
UNIT - III	DEAD LOCKS	CLASSES: 14
DEADLOCKS: System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock banker's algorithm. MEMORY MANAGEMENT: Swapping, contiguous memory allocation, paging, structure of the page table, segmentation, virtual memory, demand paging, page-replacement algorithms, allocation of frames, thrashing.		

UNIT - IV	FILE SYSTEM	CLASSES: 12
FILE SYSTEM: Concept of a file, access methods, directory structure, file system mounting, file sharing, protection. File system implementation: file system structure, file system implementation, directory implementation, allocation methods, free-space management, efficiency and performance.		
UNIT - V	MASS STORAGE STRUCTURE	CLASSES: 14
MASS STORAGE STRUCTURE: overview of mass storage structure, disk structure, disk attachment, disk scheduling algorithms, swap space management, stable storage implementation, tertiary storage structure. I/O: Hardware, application I/O interface, kernel I/O subsystem, transforming I/O requests to hardware operations, streams, performance		
TEXT BOOKS		
1. Operating System Principles- Abraham Silberchatz, Peter B. Galvin, Greg Gagne 7th Edition, John Wiley. 2. Advanced programming in the UNIX environment, W.R. Stevens, Pearson education.		
REFERENCE BOOKS		
1. Operating Systems - Internals and Design Principles, William Stallings, Fifth Edition–2005, Pearson Education/PHI 2. Operating System A Design Approach- Crowley, TMH. 3. Modern Operating Systems, Andrew S. Tanenbaum 2nd edition, Pearson/PHI 4. UNIX programming environment, Kernighan and Pike, PHI/Pearson Education 5. UNIX Internals -The New Frontiers, U. Vahalia, Pearson Education.		

SOFTWARE TESTING FUNDAMENTALS

II B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS13	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES To learn Software Testing has different goals and objectives: 1. Finding defects which may get created by the programmer while developing the software. 2. Gaining confidence in and providing information about the level of quality. 3. To prevent defects.								
COURSE OUTCOMES Upon successful completion of the course, the student is able to: 1. Understand software testing methods 2. apply various software testing techniques 3. Design and conduct a software test process for a software testing project 4. Designing solutions for various software testing problems by selecting appropriate software test model 5. Implement various practice oriented software testing projects								
UNIT - I	INTRODUCTION						CLASSES: 08	
Basics of software testing, Testing objectives, Principles of testing, Test Life Cycle, Types of testing, Software defect tracking.								
UNIT - II	TESTING METHODOLOGIES						CLASSES: 12	
White Box And Black Box Testing, Static Testing, Static Analysis Tools, Structural Testing, Unit/Code functional, testing, Code coverage testing, Code complexity testing, Black Box testing, Requirements based testing.								
UNIT - III	INTEGRATION TESTING						CLASSES: 10	
Integration, System, and Acceptance Testing Top down and Bottom up integration, Functional versus Non-functional testing, Deployment testing, Beta testing, Scalability testing, Reliability testing, Stress testing, Acceptance testing								
UNIT - IV	TEST SELECTION & MINIMIZATION FOR REGRESSION TESTING						CLASSES: 10	
Test Selection & Minimization for Regression Testing Regression testing, Regression test process, Initial Smoke or Sanity test, Selection of regression tests, Execution Trace, Dynamic Slicing, Test Minimization, Tools for regression testing, Ad hoc Testing: Pair testing, Exploratory testing, Iterative testing, Defect seeding.								
UNIT - V	TEST MANAGEMENT AND AUTOMATION TEST PLANNING						CLASSES: 10	
Test Management and Automation Test Planning, Management, Execution and Reporting, Software Test Automation: Scope of automation, Design & Architecture for automation, Generic requirements for test tool framework, Test tool selection.								

TEXT BOOKS
1. S. Desikan and G. Ramesh, "Software Testing: Principles and Practices", Pearson Education. 2. Aditya P. Mathur, "Fundamentals of Software Testing", Pearson Education.
REFERENCE BOOKS
1. Naik and Tripathy, "Software Testing and Quality Assurance", Wiley 2. K. K. Aggarwal and Yogesh Singh, "Software Engineering", New Age International Publication.

DATABASE MANAGEMENT SYSTEMS LAB

II B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS10	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100
COURSE OBJECTIVES								
The course should enable the students to:								
1. Apply the basic concepts of Database Systems and Applications.								
2. Use the basics of SQL and construct queries using SQL in database creation and interaction								
3. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.								
4. Analyze and Select storage and recovery techniques of database system.								
COURSE OUTCOMES								
The course should enable the students to:								
1. Apply the basic concepts of Database Systems and Applications.								
2. Develop an ER model for a given database.								
3. Use the basics of SQL and construct queries using SQL in database creation and interaction.								
4. Design a commercial relational database system (Oracle, MySQL) by writing SQL using the system.								
5. Analyze and Select storage and recovery techniques of database system.								
LIST OF EXPERIMENTS								
WEEK - 1	DDL COMMANDS							
Creation of Tables using SQL-Overview of using SQL tool and Data types in SQL a) Altering Tables b) Dropping Tables								
WEEK - 2	CREATE TABLE WITH PRIMARY KEY AND FOREIGN KEY & DML COMMANDS							
Creating Tables(along with Primary and Foreign keys),Practicing DML commands- a) Insert b) Update c) Delete								
WEEK - 3	SELECTION QUERIES							
Practicing Select command using following operations a) AND,OR b) ORDERBY c) BETWEEN d) LIKE e) Apply CHECK constraint								
WEEK - 4	AGGREGATE FUNCTIONS AND VIEWS							
Practice Queries using following functions a) COUNT b) SUM c) AVG d) MAX e) MIN								
Apply constraint on aggregation using a) GROUPBY b) HAVING VIEWS Create, Modify and Drop								
WEEK - 5	NESTED QUERIES							
Practicing Nested Queries using UNION, a) INTERSECT b) CONSTRAINTS c) IN								

WEEK - 6	CO-RELATED NESTED QUERIES
Practicing Co – Related Nested Queries using a) EXISTS b) NOT EXISTS c) ANY d) ALL	
WEEK - 7	JOIN QUERIES
Practicing Join Queries using a) Inner join b) Outer join c) Equi join d) Natural join	
WEEK - 8	TRIGGERS
Practicing Triggers Creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger.	
WEEK - 9	PROCEDURES
Procedures -Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure	
WEEK - 10	CURSORS
Cursors -Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor	
WEEK - 11	PL/SQL Part 1
Practice PL/SQL a) Block structure b) Variables c) data types	
WEEK - 12	PL/SQL Part 2
Practice PL/SQL a) Operators b) Control structures	
Case study 1: College Management Case study 2: An Enterprise/Organization Case study 3 : Library Management system Case study 4: Sailors and shipment	
TEXT BOOKS	
1. Database Management Systems, Raghurama Krishnan, Johannes Gehrke, Tata Mc Graw Hill, 3rd Edition 2. Database System Concepts, Silberschatz, Korth, McGraw Hill, 5th edition.	
REFERENCE BOOKS	
1. Database Systems design, Implementation, and Management, PeterRob & Carlos Coronel 7 th Edition. 2. Fundamentals of Database Systems, ElmasriNavrate, Pearson Education 3. Introduction to Database Systems, C.J.Date, Pearson Education 4. Oracle for Professionals, The XTeam, S.Shah and V.Shah, SPD. 5. Database Systems Using Oracle: A Simplified guide to SQL and PL/SQL, Shah, PHI. 6. Fundamentals of Database Management Systems, M.L. Gillenson,Wiley Student Edition.	

OPERATING SYSTEMS LAB

II B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS12	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To provide an understanding of the design aspects of operating system concepts through simulation
2. Introduce basic Unix commands, system call interface for process management, Inter-process communication and I/O in Unix

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Simulate and implement operating system concepts such as scheduling, deadlock management, file management and memory management.
2. Able to implement C programs using Unix system calls

LIST OF EXPERIMENTS

WEEK - 1	
	Write C programs to simulate the following CPU Scheduling algorithms a) FCFS b) SJF
WEEK - 2	
	Write C programs to simulate the following CPU Scheduling algorithms a) Round Robin b) priority
WEEK - 3	
	Write programs using the I/O system calls of UNIX/LINUX operating system (open, read, write, close, fcntl, seek, stat, opendir, readdir)
WEEK - 4	
	Write a C program to simulate Bankers Algorithm for Deadlock Avoidance and Prevention
WEEK - 5	
	Write a C program to implement the Producer – Consumer problem using semaphores using UNIX/LINUX system calls.
WEEK - 6	
	Write C programs to illustrate the following IPC mechanisms a) Pipes b) FIFOs
WEEK - 7	
	Write C programs to illustrate the following IPC mechanisms a) Message Queues b) Shared Memory

WEEK - 8	
Write C programs to simulate the following memory management techniques a) Paging b) Segmentation	
WEEK - 9	
Write C programs to simulate Page replacement policies a) FCFS	
WEEK - 10	
Write C programs to simulate Page replacement policies a) LRU	
WEEK - 11	
Write C programs to simulate Page replacement policies a) Optimal	
TEXT BOOKS	
1. Operating System Principles- Abraham Silberchatz, Peter B. Galvin, Greg Gagne 7th Edition, John Wiley 2. Advanced programming in the Unix environment, W.R.Stevens, Pears	
REFERENCE BOOKS	
1. Operating Systems – Internals and Design Principles, William Stallings, Fifth Edition–2005, Pearson Education/PHI 2. Operating System - A Design Approach-Crowley, TMH. 3. Modern Operating Systems, Andrew S Tanenbaum, 2nd edition, Pearson/PHI 4. UNIX Programming Environment, Kernighan and Pike, PHI/Pearson Education 5. UNIX Internals: The New Frontiers, U. Vahalia, Pearson Education	

SKILL DEVELOPMENT COURSE (HTML,XML,JAVASCRIPT,NODEJS)

II B.TECH - II SEMESTER								
Course Code:	Category	Hours/Week			Credits	Maximum Marks		
A6CS53	SKL	L	T	P	C	CIE	SEE	Total
		-	-	2	1	0	100	100
COURSEOBJECTIVES The course will enable the students to: 1. Teach Students to build static web pages using HTML and CSS 2. Explain web application development procedures 3. Facilitate students to build dynamic, responsive web pages using Angular JS 4. Teach students the basics of server side scripting using PHP 5. Familiarize various concepts Java Framework Structs.								
COURSEOUTCOMES At the end of the course a student should be able to: 1. Use HTML and CSS to design web pages. 2. Infer the role of XML in web applications and use them. 3. Apply the concepts of JavaScript to perform client side validation and create dynamic web pages. 4. Create server side applications using PHP. 5. Create web application using AngularJS that depend on MVC architecture. 6. Understand the concept of Struts framework.								
WEEK - 1								
Note: 1. Apache, MySQL and PHP for the Lab Experiments. Though not mandatory, encourage the use of Eclipse platform wherever applicable 2. The list suggests the minimum program set. Hence, the concerned staff is requested to add more problems to the list as needed 3. Install the following on the local machine 4. Apache Web Server (if not installed) 5. Tomcat Application Server locally 6. Install MySQL (if not installed) 7. Install PHP and configure it to work with Apache web server and MySQL (if not already configured)								
WEEK - 2								
1. Write a HTML Page to create a Student registration form that collects various student information such as first name, last name, Roll No., email id, mobile number, etc. and a two buttons: submit and reset. Apply different CSS properties								
WEEK - 3								
2. Implement the following web application using XML. A user validation web application, where the user submits the login name and password to the server. The name and password are checked against the data already available in the XML file and if the data matches, a successful login page is returned. Otherwise a failure message is shown to the user.								
WEEK - 4								
3. Create an XML Http Request to retrieve data from an XML file and display the data in an HTML table.								
WEEK - 5								
4. Write an HTML page that contains a selection box with a list of 5 countries. When the user selects a country, its capital should be printed next to the list. Add CSS to customize the properties of the font of the capital (color, bold and font size).								

WEEK - 6	
5. Write an HTML page that has one input, which can take multi-line text and a submit button. Once the user clicks the submit button, it should show the number of characters, words and lines in the text entered using an alert message. Words are separated with while space and lines are separated with new line character.	
WEEK - 7	
6. Implement the following web applications using AngularJS: A simple calculator web application that takes two numbers and an operator (+, -, /, * and %) from an HTML page and returns the result page with the operation performed on the operands	
WEEK - 8	
7. Create an Angular JS Application to make a shopping list where you can add or remove items.	
WEEK - 9	
8. Implement the following web applications using AngularJS: A user validation web application, where the user submits the login name and password to the server. The name and password are checked if the data matches, a successful login page is returned. Otherwise a failure message is shown to the user.	
WEEK - 10	
9. Write an HTML page including any required PHP that takes a number from one text field in the range of 0 to 999 and shows it in range of 0 to 999 and shows it in another text field in words. If the number is out of range, it should show "out of range" and if it is not a number, it should show "not a number" message in the result box.	
WEEK - 11	
10. Implement the following web applications using PHP: A user validation web application, where the user submits the login name and password to the server. The name and password are checked against the data already available in Database and if the data matches, a successful login page is returned. Otherwise a failure message is shown to the user.	
WEEK - 12	
11. A web application that lists all cookies stored in the browser on clicking "List Cookies" button. Add cookies if necessary.	
TEXT BOOKS	
1. Web Technologies, Uttam K Roy, Oxford University Press, 2010 2. The Complete Reference PHP – Steven Holzner, Tata McGraw-Hill 3 rd Edition	
REFERENCE BOOKS	
1. Web Programming, building internet applications, Chris Bates 2 nd edition, Wiley Dremtech 2. Java Server Pages – Hans Bergsten, SPD O'Reilly 3. Java Script, D.Flanagan, O'Reilly, SPD. 4. Beginning Web Programming-Jon Duckett WROX. 5. Programming world wide web, R.W. Sebesta. Fourth Edition, Pearson. 6. Internet and World Wide Web – How to program, Dietel and Nieto, Pearson Edition, A Press.	

SKILL DEVELOPMENT COURSE (MOBILE APPLICATION DEVELOPMENT)

II B.TECH - II SEMESTER

Course Code:	Category	Hours/Week			Credits	Maximum Marks		
A6CS53	SKL	L	T	P	C	CIE	SEE	Total
		-	-	2	1	0	100	100

COURSE OBJECTIVES

The course will enable the students to:

1. To enable students to develop mobile applications

COURSE OUTCOMES

At the end of the course a student should be able to:

1. Identify various concepts of mobile programming that make it unique from programming for other platforms
2. Critique mobile applications on their design pros and cons.
3. Utilize rapid prototyping techniques to design and develop sophisticated mobile interfaces.
4. Program mobile applications for the Android operating system that use basic and advanced phone features.
5. Deploy applications to the Android marketplace for distribution.

WEEK - 1

1. Build and Run first ANDROID Hello World program in Emulator and work with Layouts and UI

WEEK - 2

2. Design Count App which Increments and Decrements Count Value by 1 in Text View

WEEK - 3

3. (a) Create an Android Application for registration form using Relative Layout and perform its validation
(b) Design Color Changer Mobile with at least three colors

WEEK - 4

4. (a) Create an Android Application for currency converter.
(b) Design Android APP to demonstrate Activity Life Cycle States

WEEK - 5

5. (a) Create an App that simulates sample ATM operations: Balance Enquiry, Deposit, Withdraw.
(b) Design Android APP using Image View and Rotate Animations

WEEK - 6

6. Create an Android Application for calculator

WEEK - 7

7. (a) Create App for visiting a web URL using implicit intent
(b) Create App for navigating from one activity to another activity using Explicit Intent

WEEK - 8	
8. (a) Develop an APP which plays Video files (b) Design Android APP for Sharing Implicit Intent	
WEEK - 9	
9. Create an App to Identify the Current Location using GOOGLE MAP API	
WEEK - 10	
10. (a) Create Slide Show of Multiple Images using Android APP (b) Create an APP for Login Authentication using Firebase	
WEEK - 11	
11. Create an APP to insert and retrieve data from SQLite database	
WEEK - 12	
12. (a) Create an APP which performs authentication of registration form using Firebase (b) Design Layouts for Online Quiz App	
TEXT BOOKS	
1. Lauren Darcey and Shane Conder, "Android Wireless Application Development", Pearson Education, 2nd ed. (2011) 2. Reto Meier, "Professional Android 2 Application Development", Wiley India Pvt Ltd	
REFERENCE BOOKS	
1. Reto Meier, "Professional Android 2 Application Development", Wiley India Pvt Ltd 2. Mark L Murphy, "Beginning Android", Wiley India Pvt Ltd 3. Android Application Development All in one for Dummies by Barry Burd, Edition: I	

CONSTITUTION OF INDIA

II B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS06	MC	L	T	P	C	CIE	SEE	Total
		-	-	2	-	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Understand the need for constitution
2. Appreciate the fundamental duties and rights of the citizens of India.
3. Explain the role and amendments of constitution in a democratic society.
4. Describe the directive principles of state policy and their significance.
5. List the key features of the constitution, union government and state government.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Create awareness about the constitutional values and objectives written in the Indian constitution.
2. List fundamental rights and fundamental duties of Indian citizens.
3. Identify the division of legislative, executive and financial powers between the union and state governments.
4. Understand the working of Indian democracy, its institutions and processes at the local, state and union levels.
5. Explain the functions and responsibilities of election commission of India and union public service commission.

UNIT - I	HISTORY OF MAKING OF THE INDIAN CONSTITUTION	
History of Making of the Indian Constitution: Introduction to the constitution of India, the making of the constitution and salient features of the constitution.		
UNIT - II	PHILOSOPHY OF THE INDIAN CONSTITUTION	
Philosophy of the Indian Constitution: Preamble Salient Features, Contours of Constitutional Rights & Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties, Amendment of the constitutional powers and procedures.		
UNIT - III	UNION GOVERNMENT	
Union Government: Union Government, Union Legislature (Parliament), Lok Sabha and Rajya Sabha (with powers and functions), president of India (with powers and functions), Prime minister of India (With powers and functions), Union judiciary (Supreme court), Jurisdiction of the supreme court.		
UNIT - IV	STATE GOVERNMENT	
State Government: State Government, State legislature (Legislative Assembly/Vidhan Sabha, Legislative council/Vidhan parishad), powers and functions of the state legislature, State executive, Governor of the state (with powers and functions), The chief Minister of the state (with powers and functions), State Judiciary (High courts)		

UNIT - V	ELECTION COMMISSION	
Election Commission: Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.		
TEXT BOOKS		
1. M.V.Pylee, Indian Constitution 2. Durga Das Basu, Human Rights in Constitutional Law, Prentice – Hall of India Pvt. Ltd .New Delhi 3. Noorani, A.G.,(South Asia Human Rights Documentation Centre),Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press, 2012 4. P.M.Bakshi, The constitution of India, Universal Law Publishing Co., 5. The Constitution of India, 1950 (Bare Act), Government Publication. 6. Dr.S.N.Busi, Dr. B.R.Ambedkar framing of Indian Constitution,1st Edition, 2015.		
REFERENCE BOOKS		
1. M.P.Jain, Indian Constitution Law, 7thEdn. LexisNexis, 2014. 2. D.D.Basu, Introduction to the Constitution of India, Lexis Nexis, 2015. 3. Indian constitution at work, NCERT 4. Subash Kashyap, Indian Constitution, National Book Trust 5. J.A.Siwach, Dynamics of Indian Government & amp; Politics 6. D.C.Gupta, Indian Government and Politics 7. H.M.Sreevai,ConstitutionalLawofIndia,4theditionin3volumes(UniversalLawPublication) 8. J.C.Johari, Indian Government and Politics Hans 9. J.RajIndian Government and Politics		
E-RESOURCES		
1. nptel.ac.in/courses/109104074/8 2. nptel.ac.in/courses/109104045/ 3. nptel.ac.in/courses/101104065/ 4. www.hss.iitb.ac.in/en/lecture-details 5. www.iitb.ac.in/en/event/2nd-lecture-institute-lecture-series-indian-constitution		

III B.TECH I SEMESTER SYLLABUS

DESIGN AND ANALYSIS OF ALGORITHMS

III B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS15	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSEOBJECTIVES The course should enable the students to: 1. Explain the asymptotic performance of algorithms. 2. Demonstrate the complexity of an algorithm in terms of time and space. 3. Help to design and implement programs in various programming paradigms. 4. Familiarize with efficient algorithms in software design and development.								
COURSEOUTCOMES Upon successful completion of this course, student will be able to: 1. Identify various Time and Space complexities of various algorithms. 2. Understand Tree Traversal method and Greedy Algorithms. 3. Apply Dynamic Programming to solve various problem. 4. Apply Backtracking, Branch and Bound concept to solve various problems. 5. Implement different performance analysis methods for non-deterministic algorithms.								
UNIT - I	INTRODUCTION TO ALGORITHMS, DIVIDE AND CONQUER						CLASSES: 12	
Algorithm specification, Performance Analysis. Divide and Conquer: The general method: Binary search, finding maximum and minimum, Merge sort, Quick sort, and Strassen's Matrix multiplication.								
UNIT - II	THE GREEDY METHOD						CLASSES: 12	
The Greedy Method: The general method, Container Loading, Knapsack problem, Job sequencing with deadlines, minimum cost spanning trees, Optimal storage on tapes, single source shortest paths.								
UNIT - III	DYNAMIC PROGRAMMING, BASIC SEARCH AND TRAVERSING TECHNIQUES						CLASSES: 16	
The general method, multistage graphs, all pair's shortest paths, optimal binary search trees, reliability design, the travelling sales person problem. Basic search and traversing techniques: Techniques for Binary trees, Techniques for Graphs, connected components and spanning trees, Bi-connected components, and depth first search.								
UNIT - IV	BACK TRACKING, BRANCH AND BOUND						CLASSES: 12	
Back Tracking: The General Method, Eight Queens problem, Sum of subsets, Graph coloring, Hamiltonian cycle. Branch and Bound: The method, traveling sales person problem, 0/1 knapsack problem, efficiency considerations.								
UNIT - V	ALGEBRAIC PROBLEMS,NP-HARD & NP-COMPLETE PROBLEMS						CLASSES: 12	
Algebraic Problems: The general method, Evaluation and Interpolation. NP-Hard & NP-Complete Problems: Basic Concepts, NP-Hard Graph Problems.								

TEXT BOOKS

1. Ellis Horowitz, S. Sahni, Fundamentals of Computer Algorithms, 2/e, University Press. 2008.
2. Thomas H. Cormen, Charles E. Leiserson, Introduction to Algorithms, et.al., 3/e, MIT Press, 2012.

REFERENCE BOOKS

1. Aho, Hopcraft, Ullman, The Design and Analysis of Computer Algorithms, 1/e, 2002.
2. Michel T. Goodrich & Roberto Tamassia, Algorithm Design: Foundations, Analysis, and Internet Examples, 1/e, John Wiley and Sons, 2001.
3. Sara Baase, Allen Van Gelder, Computer Algorithms: Introduction to Design and Analysis, 3/e, Pearson Education, , 1999.
4. Mark Allen Weiss, Data Structures and Algorithm Analysis in JAVA, 3/e, Pearson Education, 2011.
5. Jon Kleinberg, Eva Tardos, Algorithm Design, 1/e, Pearson, 2013 Pearson, 2013.

WEB LINKS

1. https://onlinecourses.swayam2.ac.in/cec20_cs03/preview
2. <https://www.coursera.org/learn/algorithms-part1>

CLOUD AND DEVOPS

III B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT13	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. To understand the fundamentals of cloud computing, including its benefits and different service models.
2. To understand the different Version control tools like Git,
3. To understand the concepts of Continuous Integration/ Continuous Testing/ Continuous Deployment)
4. To understand Configuration management using Ansible
5. Illustrate the benefits and drive the adoption of cloud-based Devops tools to solve real world problems

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Implement and manage core cloud services like IAM, EC2, S3, VPC, and RDS effectively.
2. Understand different actions performed through Version control tools like Git.
3. Perform Continuous Integration and Continuous Testing and Continuous Deployment using Jenkins by building and automating test cases using Maven & Gradle.
4. Ability to Perform Automated Continuous Deployment and to do configuration management using Ansible.
5. Build, test, and deploy a simple sample application through the CI/CD pipeline, demonstrating automation and efficiency in software delivery.

UNIT - I	CLOUD COMPUTING INTRODUCTION	CLASSES: 10
Cloud Computing, Why Cloud Computing, Advantages of Cloud Computing, Virtualization, cloud service models, Cloud deployment models, Public clouds: Amazon Web Services, Microsoft Azure, and Google Cloud Services. AWS Cloud services: IAM Roles, and Policies, EC2, S3, VPC, RDS		
UNIT - II	DEVOPS INTRODUCTION	CLASSES: 10
DevOps definition, History of DevOps, DevOps Roles, DevOps Need, DevOps Problems & Solution, DevOps Main Objectives, DevOps: Continuous Delivery and Benefits, DevOps and Software Development Life Cycle: Waterfall Model, Agile Model, DevOps on Cloud. Version control systems: Git and Github GIT Features, 3-Tier Architecture, GIT – Clone /Commit / Push, GIT Hub Projects, GIT Hub Management, GIT Rebase & Merge, GIT Stash, Reset, Checkout, GIT Clone, Fetch, Pull.		
UNIT - III	COMPILE, BUILD & CONTINUOUS INTEGRATION	CLASSES: 12
Introduction, Installation of Maven, POM files, Maven Build lifecycle, Build phases(compile build, test, package) Maven Profiles, Maven repositories(local, central, global),Maven plugins, Maven create and build Artifacts, Dependency management, Installation of Gradle, Understand build using Gradle CONTINUOUS INTEGRATION/CONTINUOUS DELIVERY USING JENKINS Definition of Jenkins, Install & Configure Jenkins, Jenkins Architecture Overview, Creating a Jenkins Job, Configuring a Jenkins job, Introduction to Plugins, Commonly used plugins (Git Plugin,). Jenkins Build Pipe Line.		

UNIT - IV	CONFIGURATION MANAGEMENT AND CONTAINERIZATION	CLASSES: 10
Containerization: Docker, What is Docker Image, Docker Installation, Working with Docker Containers, Container, Docker Engine, Crating Containers with an Image, Working with Images, Docker Command Line Interface, Docker Compose, Docker Hub, Docker Trusted Registry, Docker swarm, Docker attach, Docker File & Commands. Configuration: Ansible Introduction, Installation, Ansible master/slave configuration, YAML basics, Ansible modules, Ansible Inventory files, Ansible playbooks, Ansible Roles.		
UNIT - V	BUILDING DEVOPS PIPELINES USING AWS	CLASSES: 08
Create Github Account, Repository, Create AWS Organization, Create a new pipeline, Build a sample code, Modify AWS-pipelines.		
TEXT BOOKS		
1. Stephen Baron, "AWS: The Complete Beginner's Guide", 2020 2. "The DevOps Adoption Playbook: A Guide to Adopting DevOps in a Multi-Speed IT Enterprise" by Sanjeev Sharma 3. "DevOps. Building CI/CD Pipelines with Jenkins, Docker Container, AWS ECS, JDK 11, Git and Maven by John Edward Cooper Berg		
REFERENCE BOOKS		
1. Hands-On Azure Devops: Cidc Implementation For Mobile, Hybrid, And Web Applications Using Azure Devops And Microsoft Azure: CICD Implementation for ... DevOps and Microsoft Azure (English Edition) Paperback – 1 January 2022. by Mitesh Soni 2. Jeff Geerling, "Ansible for DevOps: Server and configuration management for humans", First Edition, 2015. 3. David Johnson, "Ansible for DevOps: Everything You Need to Know to Use Ansible for DevOps", Second Edition, 2016. 4. Mariot Tsitoara, "Ansible 6. Beginning Git and GitHub: A Comprehensive Guide to Version Control, Project Management, and Teamwork for the New Developer", Second Edition, 2019. 5. https://www.jenkins.io/user-handbook.pdf 6. https://maven.apache.org/guides/getting-started/ 7. "Cloud Computing: From Beginning to End" by Ray J. Rafaels 8. Douglas E. Comer, "The Cloud Computing Book", Edition: 1st, 2023.		

COMPUTER NETWORKS

III B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS18	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: - To introduce the fundamental s of various types of computer networks. - To demonstrate the TCP/IP and OSI models with merits and demerits. - To explore the various layers of OSI model. - To introduce UDP and TCP models.								
COURSE OUTCOMES Upon successful completion of the course, the student is able to: - Identify computer networks and its components. - Identify the different types of network topologies and protocols. - Enumerate the layers of the OSI model and TCP/IP. Explain the function(s) of each layer. - Select and use various sub netting and routing mechanisms. - Design a network diagram for a given scenario.								
UNIT - I	INTRODUCTION						CLASSES: 14	
INTRODUCTION: Network applications, network hardware, network software, reference models: OSI, TCP/IP, Internet, Connection oriented network - X.25, frame relay. THE PHYSICAL LAYER: Theoretical basis for communication, guided transmission media, wireless transmission, the public switched telephone networks, mobile telephone system.								
UNIT - II	DATA LINK LAYER						CLASSES: 14	
THE DATA LINK LAYER: Design issues, error detection and correction, elementary data link protocols, sliding window protocols. THE MEDIUM ACCESS SUBLAYER: Channel allocations problem, multiple access protocols, Ethernet, Data Link Layer switching, Wireless LAN, Broadband Wireless, Bluetooth.								
UNIT - III	NETWORK LAYER						CLASSES: 12	
THE NETWORK LAYER: Network layer design issues, routing algorithms, Congestion control algorithms, Internet working, the network layer in the internet (IPv4 and IPv6), Quality of Service.								
UNIT - IV	TRANSPORT LAYER						CLASSES: 12	
THE TRANSPORT LAYER: Transport service, elements of transport protocol, Simple Transport Protocol, Internet transport layer protocols: UDP and TCP.								
UNIT - V	APPLICATION LAYER						CLASSES: 12	
THE APPLICATION LAYER: Domain name system, electronic mail, World Wide Web: architectural overview, dynamic web document and http. APPLICATION LAYER PROTOCOLS: Simple Network Management Protocol, File Transfer Protocol, Simple Mail Transfer Protocol, Telnet.								

TEXT BOOKS
1. Computer networks-Andrew S Tanenbaum, 4th edition, Pearson education Data communication and networking-Behrouz. A. Forouzan , fifth edition, TMH, 2013
REFERENCE BOOKS
1. Behrouz A. Forouzan (2006), Data communication and Networking, 4 th Edition, Mc Graw-Hill, India. 2. Kurose, Ross (2010), Computer Networking: A top down approach, Pearson Education, India.

INTRODUCTION TO DATA SCIENCE**(PROFESSIONAL ELECTIVE – I)****III B.TECH - I SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT23	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

PRE-REQUISITES: Probability and Statistics, Database Management Systems

COURSE OBJECTIVES

The course will enable the students to:

1. Practice data analytics and machine learning approaches.
2. Adapt big data solutions in different domains including social media, transportation, finance and medicine.
3. Apply the capability and the functionality of No-SQL systems
4. Optimize business decisions and create competitive advantage with Big Data analytics.
5. Solve problems using Hadoop and introducing map reduce paradigm.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Apply mathematical logic to solve problems and identify the characteristics of datasets and compare the trivial data and big data for various applications.
2. Implement machine learning techniques and computing environments that are suitable for the applications under consideration.
3. Solve problems associated with batch learning and online learning, and the big data characteristics such as high dimensionality, dynamically growing data, and in particular scalability issues.
4. Apply scaling up machine learning techniques and associated computing techniques and technologies.
5. Recognize and implement various ways of selecting suitable model parameters for different machine learning techniques.

UNIT - I		CLASSES: 12
SQL using R & Correlation and Regression Analysis: Introduction to NoSQL, Connecting R to NoSQL databases, Excel and R integration with R connector, Regression Analysis, Assumptions of OLS Regression, Regression Modelling, Correlation, ANOVA, Forecasting, Heteroscedasticity, Autocorrelation, Introduction to Multiple Regression, etc.		
UNIT - II		CLASSES: 12
Understand the Verticals - Engineering, Financial and others: Understanding systems viz. Engineering Design, Manufacturing, Smart Utilities, Production lines, Automotive, Technology, etc., Understanding Business problems related to various businesses.		
UNIT - III		CLASSES: 12
Manage your work to meet requirements: Understanding Learning objectives, Introduction to work & meeting requirements, Time Management, Work management & prioritization, Quality & Standards Adherence. Work effectively with Colleagues, Introduction to work effectively, Team Work, Professionalism, Effective Communication skills, etc.		

UNIT - IV		CLASSES: 12
Data Management & Introduction to Big Data Tools: Design Data Architecture and manage the data for analysis, Understand various sources of Data like Sensors/signal/GPS etc. Export all the data onto Cloud ex. AWS/Rackspace etc. Introduction to Big Data tools like Hadoop, Spark, Impala, etc., Data ETL process, identify gaps in the data and follow-up for decision making.		
UNIT - V		CLASSES: 12
Big Data Analytics & Machine Learning Algorithms: Run descriptive to understand the nature of the available data, collate all the data sources to suffice business requirements, Run descriptive statistics for all the variables and observe the data ranges, Outlier detection and elimination, Hypothesis testing, and determining the multiple analytical methodologies, Train Model on 2/3 sample data using various Statistical/Machine learning algorithms, Test model on 1/3 sample for prediction, etc.		
TEXT BOOKS		
1. Bill Franks, Taming the Big Data Tidal Wave, 1st Edition, Wiley, 2012. 2. Dalgaard, Peter, —Introductory statistics with RII, Springer Science & Business Media, 2008.		
REFERENCE BOOKS		
1. Richard Cotton, —Learning RII, O Reilly, 2013. 2. Frank J. Ohlhorst, Big Data Analytics, 1st Edition, Wiley, 2012. 3. Brain S. Everitt, —A Handbook of Statistical Analysis Using RII, Second Edition, LLC, 2014. 4. T. Hastie, R. Tibshirani, and J. Friedman, —Elements of Statistical LearningII, Springer, 2009.		

SOFT COMPUTING**(PROFESSIONAL ELECTIVE – I)****III B.TECH - I SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT29	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Teach basic neural networks, fuzzy systems, and optimization algorithms concepts and their relations
2. Provide knowledge of Neuron model, and Applications of NN to discuss their work.
3. Provide the graduate the better understanding of Fuzzy Logic and Evolutionary Computations.

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Identify and describe soft computing techniques and their roles in building intelligent machines
2. Analyze fuzzy logic and reasoning to handle uncertainty and solve various engineering problems.
3. Apply neural networks and genetic algorithms to combinatorial optimization problems.
4. Evaluate and compare solutions by various soft computing approaches for a given problem.
5. Identify various Fuzzy oriented Databases and genetic algorithms.

UNIT - I	BASICS OF ARTIFICIAL NEURAL NETWORK	CLASSES: 12
Characteristics of Neural Networks, Structure and working of a biological neural network, artificial neural network: terminology, models of neurons: McCulloch Pitts model, Perceptron model, Adaline model, topology, Basic learning laws. FUNCTIONAL UNITS FOR ANN FOR PATTERN RECOGNITION TASK: Pattern recognition problem, Basic functional units, PR by functional units.		
UNIT - II	FEEDFORWARD NEURAL NETWORKS	CLASSES: 12
SUPERVISED LEARNING - I: Perceptrons - Learning and memory, Learning algorithms, Error correction and gradient decent rules, Perceptron learning algorithms. SUPERVISED LEARNING-II: Backpropagation, Multilayered network architectures, Back propagation learning algorithm, Example applications of feed forward neural networks.		
UNIT - III	FEEDBACK NEURAL NETWORKS & SELF ORGANIZING FEATURE MAP	CLASSES: 12
Introduction, Associative learning, Hopfield network, Error performance in Hopfield networks, simulated annealing, Boltzmann machine and Boltzmann learning, state transition diagram and false minima problem, stochastic update, simulated annealing, Boltzmann machine, bidirectional associative memory, bam stability analysis. Self-organization, generalized learning laws, competitive learning, vector quantization, self-organizing feature map, applications of self-organizing feature map.		
UNIT - IV	FUZZY LOGIC	CLASSES: 12
Fuzzy set theory, crisp sets, operations on crisp set, fuzzy sets, fuzzy versus crisp, operations, fuzzy relations, crisp relations, properties. Fuzzy logic Application: Fuzzy Control of Blood Pressure.		

UNIT - V	FUZZY LOGIC IN DATABASE AND INFORMATION SYSTEMS	CLASSES: 12
Fuzzy Information, Fuzzy Logic in database Systems, Fuzzy Relational data Models, operations in Fuzzy Relational data Models, Design theory for Fuzzy Relational databases, Fuzzy information Retrieval and Web search, Fuzzy Object Oriented databases. GENETIC ALGORITHMS: Introduction to Genetic Algorithms, Evolutionary Algorithms.		
TEXT BOOKS		
1. Satish Kumar (2004), Neural Networks a classroom Approach, Tata McGraw Hill Publication, New Delhi. 2. Lotfi A. Zadeh(1997), Soft computing and Fuzzy Logic, World Scientific Publishing Co., Inc. River Edge, NJ, USA.		
REFERENCE BOOKS		
1. B. Yegnanarayana (2006), Artificial Neural Networks, Prentice Hall of India, New Delhi, India. 2. John Yen, Reza Langari (2006), Fuzzy Logic, Pearson Education, New Delhi, India. 3. S. Rajasekaran, Vijaylakshmi Pari (2003), Neural networks, Fuzzy Logic and Genetic Algorithms Synthesis and Applications, Prentice Hall of India, New Delhi, India.		

BLOCKCHAIN TECHNOLOGY AND WEB 3.0**(PROFESSIONAL ELECTIVE – I)****III B.TECH - I SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS29	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To understand the basics about block chain technologies.
2. To understand the bitcoin and ethereum transactions.
3. To understand different types of Blockchain.
4. To understand different types of cryptography technologies in blockchain.
5. To understand the blockchain in web.

COURSE OUTCOMES

After completion of the course, the student should be able to:

1. Develop block chain based solutions.
2. Perform a transaction in bitcoin test nets and Develop block chain based solutions and write smart contract using Ethereum Framework.
3. Evaluate the setting where a blockchain based structure may be applied, its potential and its limitations.
4. Demonstrate the application of hashing and public key cryptography in protecting the blockchain.
5. Understand the modern currencies and its market usage

UNIT - I	BLOCKCHAIN BASICS	CLASSES: 12
Introduction to Blockchain. Blockchain terminology. Blockchain properties. Centralization vs. Decentralization. Blockchains in action. A brief history of Money. Properties of Money. How is Money created? Chaumian E-Cash and the Issue of Double Spending.		
UNIT - II	INTRODUCTION TO BITCOIN AND ETHEREUM	CLASSES: 14
Money backed by computation. Hashcash and Proof of Work. Bitcoin Blockchain. How Bitcoin solves double spend issue. Bitcoin Issuance. Introduction to Ethereum. Smart Contracts. Difference between Bitcoin and Ethereum. Ethereum Transactions.		
UNIT - III	TYPES OF BLOCKCHAIN	CLASSES: 16
Types of Blockchain: Public blockchains, Private blockchains, Semi-private blockchains, Side chains, Permissioned ledger, Distributed ledger, Shared ledger, Fully private and proprietary blockchains, Tokenized blockchains, Tokenless blockchains, CAP theorem and blockchain		
UNIT - IV	BLOCKCHAIN AND HASH FUNCTION	CLASSES: 16
Cryptography, Confidentiality, Integrity, Authentication, Cryptographic primitives, Public and private keys, RSA, Discrete logarithm problem, Hash Function: Message Digest (MD), Secure Hash Algorithms (SHAs), Design of Secure Hash Algorithms (SHA), SHA-256, Design of SHA3, Elliptic Curve Digital signature algorithm		

UNIT - V	WEB3 AND THE FUTURE OF BLOCKCHAIN	CLASSES: 16
What problems can Blockchain solve? L0, L1 and L2 Blockchains. Anonymity and confidential transactions. Zkledger. Zkrollups. Scams and Attacks. Culture. Public policy. Regulations. Blockchain governance. Crypto currencies and DeFi. DAOs. Interoperability and Scalability. NFTs. Metaverse. Decentralized computing.		
TEXT BOOKS		
1. Mastering Blockchain, Imaran Bashir, Second Edition, 2018 2. Blockchain: Blueprint for a New Economy by Melanie Swan, 2015, 6th Edition		
REFERENCE BOOKS		
1. Blockchain For Dummies®, IBM Limited Edition, John Wiley & Sons Inc. 2. Blockchain Quick Reference, Brenn Hill, Samanyu Chopra, Paul Valencourt, Packt Publishing, 2018 3. Blockchain: Blueprint for a New Economy, Melanie Swa, 2015 4. Mastering Bitcoin: Programming the Open Blockchain, Andreas M. Antonopoulos, 2017		
WEB REFERENCE LINKS		
1. https://www.simplilearn.com/tutorials/blockchain-tutorial/blockchain-technology 2. https://www.ibm.com/topics/blockchain		

CYBER SECURITY AND CYBER LAWS

(PROFESSIONAL ELECTIVE - I)

III B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS36	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To learn cybercrime cyber security and cyber law.
2. To understand the cyber attacks.
3. To understand information gathering.
4. To learn how to detect and prevent a cyber attack and tools for mitigating them.
5. Develop Competencies For Dealing With Frauds And Deceptions (Confidence Tricks, Scams) And Other Cyber Crimes

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Explain the basics of cyber security, cyber crime and cyber law.
2. Classify various types of attacks and learn the tools to launch the attacks.
3. Apply various tools to perform information gathering.
4. Understand Cyber crimes and cyber laws.
5. Explore The Legal And Policy Developments In Various Countries To Regulate Cyberspace

UNIT - I	INTRODUCTION	CLASSES: 16
Cyber Security-definition, Need for Cyber Security, Protection, Security, risk, threat, vulnerability, exploit, attack, confidentiality, integrity, availability, non-repudiation, authentication, authorization, codes, plain text, encryption, decryption, cipher text, key, ciphers, Symmetric and asymmetric cryptography, Public key, private key, Crypt analysis, Cyber forensics.		
UNIT - II	CYBER ATTACKS	CLASSES:14
CYBER ATTACKS (definitions and examples): Denial-of-service attacks, Man-in-the middle attack, Phishing, spoofing and spam attacks, Driveby attack, Password attack, SQL injection attack, Cross-site scripting attack, Eavesdropping attack, Birthday attack, Malware attacks, Social Engineering attacks		
UNIT - III	HANDLING THE ATTACKS	CLASSES: 11
Firewalls, logging and intrusion detection systems, e-mail security, security issues in operating systems, ethics of hacking and cracking.		
UNIT - IV	CYBER CRIMES	CLASSES: 12
Cyber Crime and Information Security – History of Cyber Crime, classifications of Cyber Crimes – Tools and Methods –Password Cracking, Key loggers, Spywares, SQL Injection – Network Access Control – CloudSecurity – Web Security – Wireless Security		



UNIT - V	CYBER SPACE AND CYBER LAWS	CLASSES: 16
Cyber Space- Fundamental definitions -Interface of Technology and Law – Jurisprudence and-Jurisdiction in Cyber Space - Indian Context of Jurisdiction - Enforcement agencies – Need for IT act - UNCITRAL – E-Commerce basics .Information Technology Act, 2000 - Aims and Objects — Overview of the Act – Jurisdiction Cyber Law, A Global Perspective on Cyber Crimes; Cyber Laws – The Indian IT Act – Cybercrime and Punishment.		
TEXT BOOKS		
1. Nina Godbole, Sunit Belapure, “Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives”, Wiley Publishers, 2011. 2. Cyber Law- The Indian Perspective, Pavan Duggal, 2002		
REFERENCE BOOKS		
1. David Kim, Michael G. Solomon, “Fundamentals of Information Systems Security”, Jones & Bartlett Learning Publishers, 2013 2. Patrick Engebretson, “The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made easy”, Elsevier, 2011 3. Justice Yatindra Singh: Cyber Laws, Universal Law Publishing Co., New Delhi, 2016 4. Farouq Ahmed, Cyber Law in India, New Era publications, New Delhi, 2017		

INFORMATION RETRIEVAL SYSTEMS

(PROFESSIONAL ELECTIVE – I)

III B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT26	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

PRE-REQUISITES: Relational Database Management Systems, Data Mining

COURSE OBJECTIVES

The Students will learn the following:

1. To use different information retrieval techniques in various application areas
2. To apply IR principles to locate relevant information large collections of data
3. To analyze performance of retrieval systems when dealing with unmanaged data sources
4. To implement retrieval systems for web search tasks.

COURSE OUTCOMES

After the successful completion of the course, the students would be able to:

1. Understand the basics of web search
2. Identify the data structures like Inverted Indices used in Information retrieval systems
3. Apply the different techniques on compression of index including the dictionary and its posting list
4. Understand and Apply various text search algorithms
5. Identify the evaluation metrics of information system

UNIT - I	INTRODUCTION	CLASSES: 12
Definition, Objectives, Functional Overview, Relationship to DBMS, Digital libraries and Data Warehouses. Information Retrieval System Capabilities - Search, Browse, Miscellaneous.		
UNIT - II	CATALOGING AND INDEXING	CLASSES: 12
Objectives, Indexing Process, Automatic Indexing, Information Extraction. Data Structures: Introduction, Stemming Algorithms, Inverted file structures, N - gram data structure, PAT data structure, Signature file structure, Hypertext data structure. Automatic Indexing: Classes of automatic indexing, Statistical indexing, Natural language, Concept indexing, Hypertext linkages		
UNIT - III	DOCUMENT AND TERM CLUSTERING	CLASSES: 12
Introduction, Thesaurus generation, Item clustering, Hierarchy of clusters. User Search Techniques: Search statements and binding, Similarity measures and ranking, Relevance feedback, Selective dissemination of information search, Weighted searches of Boolean systems, Multimedia Searching.		
UNIT - IV	INFORMATION PRESENTATION AND TEXT SEARCH ALGORITHMS	CLASSES: 12
Information presentation Introduction, Presentation of the hits, Display of the Item, Collaborative Filtering, Multimedia presentation, Human Perception and Presentation. Text Search Algorithms Introduction, Software text search algorithms, Hardware text search systems.		

UNIT - V	INFORMATION SYSTEM EVALUATION	CLASSES: 12
Introduction, Measures used in system evaluations, Multimedia Information retrieval Evaluation, Measurement example – TREC Evolution.		
TEXT BOOKS		
1. Gerald Kowalski, Information Retrieval Architecture and Algorithms, Springer Science, 2011. 2. Ricardo Baeza-Yates, Modern Information Retrieval, Pearson Education, 2007. 3. David A Grossman and Ophir Frieder, Information Retrieval: Algorithms and Heuristics, 2nd Edition, Springer, 2004		
REFERENCE BOOKS		
1. William B Frakes and Ricardo Baeza - Yates, Information Retrieval Data Structures and Algorithms, Pearson Education, 1992. 2. Robert Korfhage, John Wiley & Sons, Information Storage and Retrieval.2012 3. Christopher D. Manning and Prabhakar Raghavan, Introduction to Information Retrieval, Cambridge University Press, 2008		

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DATA WAREHOUSING AND BUSINESS INTELLIGENCE**(PROFESSIONAL ELECTIVE-II)****III B.TECH - I SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS43	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. This course is concerned with extracting data from the information systems that deal with the day-to-day operations and transforming it into data that can be used by businesses to drive high-level decision making
2. Students will learn how to design and create a data warehouse, and how to utilize the process of extracting, transforming, and loading (ETL) data into data warehouses.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Understand architecture of data warehouse and OLAP operations.
2. Understand Fundamental concepts of BI and Analytics
3. Application of BI Key Performance indicators and Design of Dashboards, Implementation of Web Analytics
4. Understand Utilization of Advanced BI Tools and their Implementation.
5. Implementation of BI Techniques and BI Ethics.

UNIT - I	DATA WAREHOUSE	CLASSES:12
DATA WAREHOUSE: Data Warehouse-Data Warehouse Architecture- Multidimensional Data Model-Data cube and OLAP Technology-Data Warehouse Implementation -DBMS schemas for Decision support - Efficient methods for Data cube computation.		
UNIT - II	BUSINESS INTELLIGENCE	CLASSES: 10
Business Intelligence: Introduction – Definition, Leveraging Data and Knowledge for BI, BI Components, BI Dimensions, Information Hierarchy, Business Intelligence and Business Analytics. BI Life Cycle. Data for BI - Data Issues and Data Quality for BI.		
UNIT - III	BI IMPLEMENTATION	CLASSES: 10
BI Implementation - Key Drivers, Key Performance Indicators and Performance Metrics, BI Architecture/Framework, Best Practices, Business Decision Making, Styles of BI-vent-Driven alerts – A cyclic process of Intelligence Creation. The value of Business intelligence -Value driven and Information use.		
UNIT - IV	ADVANCED BI	CLASSES:10
Advanced BI – Big Data and BI, Social Networks, Mobile BI, emerging trends, Description of different BI-Tools (Pentaho, KNIME)		
UNIT - V	BI IMPLEMENTATION	CLASSES: 12



Business intelligence implementation-Business Intelligence and integration implementation-connecting in BI systems- Issues of legality- Privacy and ethics- Social networking and BI.

TEXT BOOKS

1. Data Mining – Concepts and Techniques - JIAWEI HAN & MICHELINE KAMBER, Elsevier. 2011
2. Rajiv Sabherwal “Business Intelligence” Wiley Publications, 2012.

REFERENCE BOOKS

1. Efraim Turban, Ramesh Sharda, Jay Aronson, David King, Decision Support and Business Intelligence Systems, 9th Edition, Pearson Education, 2009.
2. David Loshin, Business Intelligence - The Savy Manager's Guide Getting Onboard with Emerging IT, Morgan Kaufmann Publishers, 2009
3. Philo Janus, Stacia Misner, Building Integrated Business Intelligence Solutions with SQL Server, R2 & Office, TMH, 2011.
4. Business Intelligence Data Mining and Optimization for decision making [Author: Carlo-Verellis] [Publication: (Wiley)], 2009
5. Data Warehousing, Data Mining & OLAP- Alex Berson and Stephen J. Smith- Tata McGrawHill Edition, Tenth reprint 2007

WEB LINKS

1. <https://www.coursera.org/learn/data-warehousing-business-intelligence>
2. https://www.vssut.ac.in/lecture_notes/lecture1428550844.pdf

NATURAL LANGUAGE PROCESSING

(PROFESSIONAL ELECTIVE - II)

III B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AI14	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

Contact Classes: 56 Tutorial Classes: Nil Practical Classes: Nil Total Classes: 56

COURSE OBJECTIVES

The course should enable the students to:

1. Provide theoretical concepts of language processing that shows how to explore interesting bodies of text.
2. Familiarize with fundamental topics in language processing that include tagging, classification, and information extraction using tiny Python programs.
3. Facilitate understanding of formal grammar to describe the structure of an unlimited set of sentences.
4. Acquaint with methods to parse a sentence, recognize its syntactic structure and construct - representations of meaning.
5. Familiarize with design of existing corpora, the typical workflow for creating a corpus and the life cycle of a corpus.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Discuss basic concepts and applications of Natural Language processing.
2. Analyze words using prefix, suffix, grammar, PoS.
3. Analyze structures of Semantic and Discourse to determine what words and phrases mean in relationship.
4. Analyze Natural Language Generation and apply machine translation.
5. Implement levels of NLP system using the Components or lexical resources to demonstrate Morphology syntax of a language.

UNIT - I	OVERVIEW AND LANGUAGE MODELING	CLASSES:12
Overview: Origins and challenges of NLP Language and Grammar-Processing Indian Languages-NLP Applications-Information Retrieval. Language Modeling: Introduction-Various Grammar-based Language Models-Statistical Language Model.		
UNIT - II	WORD LEVEL AND SYNTACTIC ANALYSIS	CLASSES:12
Word Level Analysis: Introduction Morphological Parsing-Spelling Error Detection and correction-Words and Word Classes-Part-of Speech Tagging. Parsing: Constituency Parsing - Probabilistic Parsing.		
UNIT - III	SEMANTIC ANALYSIS AND DISCOURSE PROCESSING	CLASSES:12
Semantic Analysis: Introduction- Meaning Representation-Lexical Semantics Ambiguity-Word Sense Disambiguation. Discourse Processing: Introduction- cohesion-Reference Resolution Discourse Coherence and Structure.		
UNIT - IV	NATURAL LANGUAGE GENERATION AND MACHINETRANSLATION	CLASSES:12
Natural Language Generation: Introduction-Architecture of NLG Systems Generation Tasks and Representations-Application of NLG. Problems in Machine Translation, Characteristics of Indian Languages-Machine Translation Approaches Translation involving Indian Languages.		

UNIT - V	APPLICATIONS AND LEXICAL RESOURCES	CLASSES:12
Information Extraction, Automatic Text Categorization and Text Summarization, Question-Answering System. LEXICAL RESOURCES: Introduction - WordNet- FrameNet – Stemmers - POS Tagger, Research Corpora, NLTK.		
TEXT BOOKS		
3. Tanveer Siddiqui, U.S. Tiwary, "Natural Language Processing and Information Retrieval", Oxford University Press, 2008.		
REFERENCE BOOKS		
2. Christopher Manning "Foundations of Statistical Natural Language Processing," MIT Press, July 1999. 3. Dan Jurafsky, James H Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition, Pearson Education India, 2nd edition (2013).		
WEB REFERENCES		
5. https://huggingface.co/learn/nlp-course/chapter1/3?fw=pt 6. https://www.tutorialspoint.com/natural_language_processing/index.htm		

PERMISSIONED BLOCK CHAIN - HYPER LEDGER**(PROFESSIONAL ELECTIVE - II)****III B.TECH - I SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS30	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To understand the basic concepts of block chain technology and to explore the concepts in hyper ledger and hedera.
2. Integrate ideas from block chain technology into their own projects.
3. To understand and explore the different methods of Decentralization.
4. To describe business block chain and hyper ledger technologies.
5. To understand Blockchain and Hyper Ledger Technologies.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Demonstrate the foundation of the Block chain technology and understand the processes in payment and funding.
2. Identify the risks involved in building Block chain applications& Review of legal implications using smart contracts.
3. Understand the types, benefits and limitation of block chain and Explore the block chain decentralization and hyper ledger concepts.
4. Identify the use cases of hyper ledger technology.
5. Evaluate alternative blockchain and their applicability.

UNIT – I	INTRODUCTION	CLASSES: 12
Introduction, Scenarios, Challenges Articulated, Blockchain, Blockchain Characteristics, Opportunities Using Blockchain, History of Blockchain. Evolution of Blockchain : Evolution of Computer Applications, Centralized Applications, Decentralized Applications, Stages in Blockchain Evolution, Consortia, Forks, Public Blockchain Environments, Type of Players in Blockchain Ecosystem, Players in Market.		
UNIT – II	BLOCKCHAIN CONCEPTS	CLASSES: 14
Blockchain Concepts: Changing of Blocks, Hashing, Merkle-Tree, Consensus, Mining and Finalizing Blocks, Currency aka tokens, security on blockchain, data storage on blockchain, wallets, coding on blockchain: smart contracts, peer-to-peer network, types of blockchain nodes, risk associated with blockchain solutions, life cycle of blockchain transaction.		
UNIT – III	ARCHITECTING BLOCKCHAIN SOLUTIONS	CLASSES-12
Use of Blockchain, Blockchain relevance Framework, Blockchain Reference Architecture, Types of Blockchain Applications, Architecture Considerations, Platform for Architecture, Approaches for Designing Blockchain Platforms		

UNIT – IV	ETHEREUM BLOCKCHAIN IMPLEMENTATIONS	CLASSES: 12
Ethereum Eco system- Ethereum Development- Ethereum Tool Stack- Ethereum Virtual Machine-Smart Contract Programming – Ethereum IDE and Types – Ethereum Account – Ethereum Wallets -Decentralized Applications – Metamask – Tuna Fish Use case Implementation.		
UNIT – V	HYPER LEDGE IMPLEMENTATION	CLASSES: 10
Use cases – Car Owner Tracking – Calling Chain Code Implementation – Inter Planetary File System (IPFS) – Data Storage and Retrieval System – Zero-Knowledge Proof- Blockchain with IoT – Blockchain with AI/ML		
TEXT BOOKS		
1. Ambadas, Arshad Sarfarz Ariff, Sham “Blockchain for Enterprise Application Developers”, Wiley, 2020, Andreas M. Antonopoulos, “Mastering Bitcoin: Programming the Open Blockchain” , O’Reilly, 2. Mastering Blockchain - Distributed ledgers, decentralization and smart contracts explained, AuthorImran Bashir, Packt Publishing Ltd, Second Edition, ISBN 978-1- 78712-544-5, 2017		
REFERENCE BOOKS		
1. Blockchain: A Practical Guide to Developing Business, Law, and Technology Solutions, Joseph Bambara, Paul R. Allen, Mc Graw Hill. 2. Blockchain: Blueprint for a New Economy, Melanie Swan, O’Reilly. 3. Blockchain Basics: A Non-Technical Introduction in 25 Steps 1st Edition, by Daniel Drescher. 4. Bitcoin and Cryptocurrency Technologies: Author- Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder, Princeton University, 2016. 5. Wattenhofer, R. Distributed Ledger Technology: The Science of the Blockchain, 2017, 1st edition, CreateSpace Independent Publishing Platform, United States.		
WEB LINKS		
1. https://www.simplilearn.com/blockchain-certification-training-course?utm_source=google&utm_medium=cpc&utm_term=learn%20blockchain%20technology&utm_content=12655265231-140511553858-666605259310&utm_device=c&utm_campaign=Search-TechCluster-PG-SD-CBlockchain-Blockchain-IITK-IN-Main-AllDevice-adgroup-Blockchain-Technology-New-Exact&gclid=Cj0KCQjwi7GnBhDXARIsAFLvH4Ipls1bJLOXezVWNiLuHqc9Z2bwBX23YJ2LPYHrhEnnJpPCuACdBzQaArvHEALw_wcB 2. https://www.udemy.com/course/build-your-blockchain-az/ 3. https://www.udemy.com/course/ethereum-and-solidity-the-complete-developers-guide/ 4. https://www.udemy.com/course/complete-dapp-solidity-react-blockchain-development/		

ETHICAL HACKING

PROFESSIONAL ELECTIVE - 2

III B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CY10	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
Contact Classes:63	Tutorial Classes:	Practical Classes:				Total Classes:63		

COURSE OBJECTIVES

The course will enable the students to:

1. The aim of the course is to introduce the methodologies and framework of hacking for enhancing security.
2. The course includes-Impacts of Hacking; Types of Hackers
3. Information Security Models;
4. Information Security Program; Business Perspective; Planning a Controlled Attack;
5. Framework of Steps (Reconnaissance, Enumeration, Vulnerability Analysis, Exploitation, Deliverable and Integration)

COURSE OUTCOMES

At the end of the course, students are able to:

1. Gain the knowledge of the use and availability of tools to support an ethical hack
2. Gain the knowledge of interpreting the results of a controlled attack
3. Understand the role of politics, inherent and imposed limitations
4. Understand the role of metrics for planning of a test
5. Comprehend the dangers associated with penetration testing

UNIT - I	INTRODUCTION TO HACKING	CLASSES: 15
Introduction: Hacking Impacts, The Hacker Framework: Planning the test, Sound Operations, Reconnaissance, Enumeration, Vulnerability Analysis, Exploitation, Final Analysis, Deliverable, and Integration. Information Security Models: Computer Security, Network Security, Service Security, Application Security, Security Architecture Information Security Program: The Process of Information Security, Component Parts of Information Security Program, Risk Analysis and Ethical Hacking.		
UNIT – II	THE BUSINESS PERSPECTIVE	CLASSES: 14
The Business Perspective: Business Objectives, Security Policy, Previous Test Results, Business Challenges. Planning for a Controlled Attack: Inherent Limitations, Imposed Limitations, timing is Everything, Attack Type, Source Point, Required Knowledge, Multi-Phased Attacks, Teaming and Attack Structure, Engagement Planner, The Right Security Consultant, The Tester, Logistics, Intermediates, Law Enforcement.		
UNIT – III	PREPARING FOR A HACK	CLASSES: 10
Preparing for a Hack: Technical Preparation, Managing the Engagement. Reconnaissance: Social Engineering, Physical Security, Internet Reconnaissance.		

UNIT – IV	ENUMERATION TECHNIQUES	CLASSES: 14
Enumeration: Enumeration Techniques, Soft Objective, Looking Around or Attack, Elements of Enumeration, Preparing for the Next Phase. Exploitation: Intuitive Testing, Evasion, Threads and Groups, Operating Systems, Password Crackers, RootKits, applications, Wardialing, Network, Services and Areas of Concern.		
UNIT - V	DELIVERABLE	CLASSES: 10
Deliverable: The Deliverable, The Document, Overall Structure, Aligning Findings, Presentation. Integration: Integrating the Results, Integration Summary, Mitigation, Defense Planning, Incident Management, Security Policy, Conclusion.		
TEXT BOOKS		
1. James S. Tiller, "The Ethical Hack: A Framework for Business Value Penetration Testing", Auerbach Publications, CRC Press.		
REFERENCE BOOKS		
1. EC-Council, "Ethical Hacking and Countermeasures Attack Phases", Cengage Learning. 2. Michael Simpson, Kent Backman, James Corley, "Hands-On Ethical Hacking and Network Defense", Cengage Learning		

SEMANTIC WEB**(PROFESSIONAL ELECTIVE - II)****III B.TECH - I SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS37	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the semantic web and its importance.
2. Understand the attribute languages and how to use in semantic web.
3. Understand the Rule based languages.
4. Understand the different methods to development in internet.
5. Learn different types of semantic web tools.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Classify the ontologies towards to the internet.
2. Apply the attribute languages on internet.
3. Apply the rule based languages for improving the usage of semantic web.
4. Analyze the methods to development of the internet.
5. Use semantic web tools for efficient development and usage of web.

UNIT - I	THE FUTURE OF THE INTERNET	CLASSES: 12
Introduction, The Syntactic Web, The Semantic Web, How the Semantic Web Will Work. Ontology in Computer Science: Defining the Term Ontology, Differences Among Taxonomies, Thesauri, and Ontologies, Classifying Ontologies, Web Ontologies, Web Ontology Description Languages, Ontology, Categories, and Intelligence.		
UNIT - II	KNOWLEDGE REPRESENTATION IN DESCRIPTION LOGIC	CLASSES: 12
Introduction, An Informal Example, The Family of Attributive Languages, Inference Problems. RDF and RDF Schema: Introduction, XML Essentials, RDF, RDF Schema, A Summary of the RDF/RDF Schema Vocabulary. OWL: Introduction, Requirements for Web Ontology Description Languages, Header Information, Versioning, and Annotation Properties, Properties, Classes, Individuals, Data types, A Summary of the OWL Vocabulary.		
UNIT - III	RULE LANGUAGES	CLASSES: 14
Introduction, Usage Scenarios for Rule Languages, Datalog, RuleML, SWRL, TRIPLE. Semantic Web Services: Introduction, Web Service Essentials, OWL-S Service Ontology, An OWL-S Example.		
UNIT - IV	METHODS FOR ONTOLOGY DEVELOPMENT	CLASSES: 15
Introduction, Uschold and King Ontology Development Method, Toronto Virtual Enterprise Method, Methontology, KACTUS Project Ontology Development Method, Lexicon-Based Ontology Development Method, Simplified Methods. Ontology Sources: Introduction, Metadata, Upper Ontologies, Other Ontologies of Interest, Ontology Libraries.		

UNIT - V	SEMANTIC WEB SOFTWARE TOOLS & SOFTWARE AGENTS	CLASSES: 12
Introduction, Metadata and Ontology Editors, Reasoners, Other tools. Introduction, Agent Forms, Agent Architecture, Agents in the Semantic web Context. Semantic Desktop: Introduction, Semantic Desktop Metadata, Semantic Desktop Ontologies, Semantic Desktop Architecture, Semantic Desktop Related Applications. Ontology Application in Art: Introduction, Ontologies for the Description of Works of Art, Metadata Schemas for The Description of Works of Art, Semantic Annotation of Art Images.		
TEXT BOOKS		
1. Semantic Web- Concepts, Technologies and applications, 2007th edition (24 January 2007) K.Breitman, Marco Antonio Casanova and Walter Truszkowski, Springer..		
REFERENCE BOOKS		
1. InformationSharingontheSemantingWeb,HeinerStuckenschmidt, Frankvan Harmelen, Springer. 2. Semantic Web Primer, Grigoris Antoniou, FrankVan 3. Semantic Web Services: Concepts, Technologies and Applications, Rudi Studer, Stephan Grimm, Andrees Abeker, Springer 4. Towards the Semantic Web: Ontology Driven Knowledge Management, John Davis, Dieter Fensal, Frank Van Harmelen, J.Wiley.		
WEB LINKS		
1. https://www.techtarget.com/searchcio/definition/Semantic-Web 2. https://www.w3.org/RDF/Metalog/docs/sw-easy MOOC:1. https://www.coursera.org/learn/web-data EBOOK:1. https://open.hpi.de/courses/semanticweb 3. http://prof.mau.ac.ir/images/Uploaded_files/A%20Semautic%20Web%20Primer-The%20MIT%20Press%20(2012)%5B7460174%5D.PDF		

COMPUTER NETWORKS LAB

III B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS19	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Able to understand the practical approach to network communication protocols.
2. To understand network layers, structure/format and role of each network layer.
3. To know how to implement various network application such as data transmission between client and server, file transfer, real-time multimedia transmission.
4. To get the knowledge on the various Routing Protocols/Algorithms and Internetworking.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Understand the practical approach to network communication protocols.
2. Understand network layers, structure/format and role of each network layer.
3. Able to design and implement various network application such as data transmission between client and server, file transfer, real-time multimedia transmission.
4. Understand the various Routing Protocols/Algorithms and Internetworking.

LIST OF EXPERIMENTS

WEEK-1	Networking Commands
1. To Study the commands ARP, HOSTNAME, IPCONFIG, IPCONFIG/ALL, IPCONFIG/RENEW, IPCONFIG/RELEASE, NBTSTAT, NETDIAG, NETSTAT, NSLOOKUP, PING, ROUTE, TRACERT,	
WEEK-2	Socket Program for Echo/Ping/Talk commands.
2. a) To write a socket program for implementation of echo. b) To write a client-server application for chat using TCP	
WEEK-3	File transfer
3. To Perform File Transfer in Client & Server Using TCP/IP	
WEEK-4	Remote Command Execution
4. IMPLEMENTATION OF REMOTE COMMAND EXECUTION	
WEEK-5	Create a socket (UDP)
5. To write a program to implement simple client-server application using UDP.	
WEEK-6	Web page downloading
6. To download a webpage using Java	

WEEK-7	TCP Module Implementation
7. To write a socket program for implementation of TCP Protocol using client server communications.	
WEEK-8	Implementation of RMI
8. IMPLEMENTATION OF REMOTE METHOD INVOCATION	
WEEK-9	Implementation of Client in Server communication in Java
9. To write a socket program for implementation of client and server program in java language.	
WEEK-10	Program to display all the cookie information using this session lifetime.
10. Program to display all the cookie information using this session lifetime. connect the cables..	
WEEK-11	Create a simple program to send mail using SMTP Protocol.
11. Create a simple program to send mail using java mail.	
WEEK-12	A Simple Program using JSP.
12. A Simple Program using JSP.	
TEXT BOOKS	
1. "Computer Networks: A Systems Approach" by Larry L. Peterson and Bruce S. Davie	
2. "Data Communication and Networking" by Behrouz A Forouzan	

CLOUD AND DEVOPS LAB

III B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT16	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100
COURSE OBJECTIVES: The course should enable the students to: 1. Apply the concepts of AWS and its cloud services. 2. Deploy and use virtual instances 3. Implement continuous integration using AWS CodeBuild. 4. Implement end-to-end continuous integration and continuous deployment (CI/CD) using AWS CodePipeline								
COURSE OUTCOMES: At the end of the course, student will be able to: 1. Deploy secured virtual instances in Amazon AWS. 2. Deploy various cloud services like S3, Databases 3. Proficiency in basic Git and GitHub commands. 4. Deploy automated applications								
LIST OF EXPERIMENTS								
WEEK-1								
Create Amazon AWS EC2 Linux instance with conceptual understanding of SSH client software protocol and keys.								
WEEK-2								
Create Amazon AWS EC2 Windows server instance with conceptual understanding of RDP (Remote Desktop Protocol).								
WEEK-3								
Create cloud storage Bucket using Amazon Simple Storage Service (S3). Perform the following operations: A. Create a folder within a S3 Bucket B. Upload content to S3 C. Change permissions to allow public access of contents. D. Set Meta Data on an S3 Bucket E. Delete an S3 Bucket and its content								
WEEK-4								
Launch and connect to an Amazon Relational Data Base (RDS) Service, Use any one from MySQL, Oracle, SQL Server and postgre SQLDataBase engines.								
WEEK-5								
Study and practice Git and GitHub commands								

WEEK-6
Install and configure Jenkins to build and deploy Java or Web Applications
WEEK-7
Install and configure Docker tool and perform Docker commands for content management.
WEEK-8
Create a Docker image and push Docker image into AWS ECR
WEEK-9
Create Repositories, Cloning, and Pushing Code Changes using AWS CodeCommit.
WEEK-10
Implement continuous integration using AWS Codebuild
WEEK-11
Automate application deployment into EC2 using AWS Code Deploy
WEEK-12
Implement CI/CD using AWS Code pipeline that automates the build, test, and deployment phases
TEXT BOOKS:
1. Thomas Eri, "Cloud Computing: Concepts, Technology & Architecture", Edition: 1st or later 2. Douglas E. Comer, "The Cloud Computing Book", Edition: 1st, 2023 3. Stephen Baron, "AWS: The Complete Beginners Guide", Kindle Edition 2020
REFERENCE BOOKS:
1. Hands-On Azure Devops: Cidc Implementation For Mobile, Hybrid, And Web Applications using Azure Devops and Microsoft Azure: CICD Implementation for ... Devops and Microsoft Azure (English Edition) Paperback – 1 January 2020. byMiteshSoni 2. Jeff Geerling, "Ansible for Devops: Server and configuration management for humans", First Edition, 2015. 3. David Johnson, "Ansible for Devops: Everything You Need to Know to Use Ansible for Devops", Second Edition, 2016. 4. MariotTsitoara, "Ansible 6. Beginning Git and GitHub: A Copenhensive Guide to Version Control, Project Management, and Teamwork for the New Developer", Second Edition, 2019. 5. https://www.jenkins.io/user-handbook.pdf 6. https://maven.apache.org/guides/getting-started/

ADVANCED COMMUNICATION SKILLS LAB

III B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS03	HSMC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	40	60	100
COURSE OBJECTIVES								
This Lab focuses on using multi-media instruction for language development to meet the following targets:								
1. To improve the students' fluency in English, through a well-developed vocabulary and enable them to listen English spoken at normal conversational speed by educated English speakers and respond appropriately in different socio-cultural and professional contexts. 2. Further, they would be required to communicate their ideas relevantly and coherently in writing. 3. To prepare all the students for their placements.								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Accomplishment of sound vocabulary and its proper use contextually. 2. Flair in Writing and felicity in written expression. 3. Enhanced job prospects. 4. Effective Speaking Abilities.								
UNIT-I								
Activities on Fundamentals of Inter-personal Communication and Building Vocabulary - Starting a conversation-responding appropriately and relevantly-using the right body language - Role Play in different situations & Discourse Skills- using visuals - Synonyms and antonyms, word roots, one-word substitutes, prefixes and suffixes, study of word origin, business vocabulary, analogy, idioms and phrases, collocations & usage of vocabulary.								
UNIT-II								
Activities on Reading Comprehension -General Vs. Local Comprehension, Reading for facts, guessing meanings from context, scanning, skimming, inferring meaning, critical reading.								
UNIT-III								
Activities on Writing Skills -Structure and presentation of different types of writing - letter writing/Resume writing/e-correspondence/Technical report writing/planning for writing - improving one's writing.								
UNIT-IV								
Activities on Presentation Skills - Oral presentations (individual and group) through JAM sessions / seminars / PPTs and written presentations through posters/projects/reports/e-mails/assignments etc.								
UNIT-V								
Activities on Group Discussion and Interview Skills -Dynamics of group discussion, intervention, summarizing, modulation of voice, body language, relevance, fluency and organization of ideas and rubrics for evaluation-Concept and process, pre-interview planning, opening strategies, answering strategies, interview through tele-conference & video-conferencing and Mock Interviews.								

REFERENCE BOOKS

1. Raman, M & Sharma, S. (2009). Technical Communication. Oxford University Press.
2. Rani.S.(2011). Advanced Communication Skills Laboratory Manual. Pearson Education.
3. Anderson,V.(2007). Technical Communication. Cengage Learning pvt. Ltd.
4. Kelly M. Quintanilla & Shawn T. Wahl. (2011). Business and Professional Communication: Keys for Work place Excellence. Sage South Asia Edition. Sage Publications.
5. Stev.D & David T.McMahan. (2012). The Basics of Communication: A Relational Perspective. Sage South Asia Edition. Sage Publications.
6. McMurrey.D & Buckley.J.(2012). Handbook for Technical Communication Cengage Learning.
7. Sen.L.(2009). Communication Skills. PHI Learning Pvt Ltd.
8. Vishvamohan,A.(2009). English for Technical Communication for Engineering Students. Tata Mc Graw Hill.
9. Book son TOFEL/GRE/GMAT/CAT/IELTS by Barron's/DELTA/Cambridge University Press.
10. Tomalin, B& Thomas, B.(2009).International English for Call Centers. Macmillan Publishers.

HUMAN VALUES AND PROFESSIONAL ETHICS

III B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS10	MC	L	T	P	C	CIE	SEE	Total
		2	-	-	0	50	0	50

COURSE OBJECTIVES

The course will enable the students to:

1. To help the students appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2. To facilitate the development of a Holistic perspective among students towards life, profession and happiness, based on a correct understanding of the Human reality and the rest of Existence. Such a holistic perspective forms the basis of Value based living in a natural way.
3. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually satisfying human behavior and mutually enriching interaction with Nature.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. It ensures students sustained happiness through identifying the essentials of human values and skills.
2. It facilitates a correct understanding between profession and happiness
3. It helps students understand practically the importance of trust, mutually satisfying human behaviour and enriching interaction with nature.
4. Ability to develop appropriate technologies and management patterns to create harmony in professional and personal life.

UNIT-I	INTRODUCTION	CLASSES: 12
Need, basic Guidelines, Content and Process for Value Education: Understanding the need, basic guidelines, content and process for Value Education. Self-Exploration - Its content and process; 'Natural Acceptance' and Experiential Validation - as the mechanism for self-exploration. Continuous Happiness and Prosperity - A look at basic Human Aspirations. Right understanding, Relationship and Physical Facilities - the basic requirements for fulfillment of aspirations of every human being with their correct priority. Understanding Happiness and Prosperity correctly - A critical appraisal of the current scenario. Method to fulfill the above human aspirations: understanding and living in harmony at various levels.		
UNIT-II	UNDERSTANDING HARMONY IN THE HUMAN BEING - HARMONY IN MYSELF!	CLASSES: 12
Understanding human being as a co-existence of the sentient 'I' and the material 'Body'. Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha. Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer). Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, meaning of Prosperity in detail. Programs to ensure Sanyam and Swasthya.		
UNIT-III	UNDERSTANDING HARMONY IN THE FAMILY AND SOCIETY - HARMONY IN HUMAN - HUMAN RELATIONSHIP	CLASSES: 12
Understanding harmony in the Family the basic unit of human interaction. Understanding values in human - human relationship; meaning of Nyaya and program for its fulfillment to ensure Ubhay-tripti; Trust (Vishwas) and Respect (Samman) as the foundational values of relationship. Understanding the meaning of Vishwas; Difference between intention and competence. Understanding the meaning of Samman, Difference between respect and differentiation; the other salient values in relationship. Understanding the harmony in the society (society being an extension of family): Samadhan, Samridhi, Abhay, Sah-astiva as comprehensive Human Goals. Visualizing a universal harmonious order in society - Undivided Society (AkhandSamaj), Universal Order (Sarvabhaum Vyavastha) - from family to world family!		

UNIT-IV	UNDERSTANDING HARMONY IN THE NATURE AND EXISTENCE - WHOLE EXISTENCE AS CO-EXISTENCE	CLASSES: 12
Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature - recyclability and self-regulation in nature. Understanding Existence as Co-existence (Sah-astiva) of mutually interacting units in all-pervasive space. Holistic perception of harmony at all levels of existence.		
UNIT-V	IMPLICATIONS OF THE ABOVE HOLISTIC UNDERSTANDING OF HARMONY ON PROFESSIONAL ETHICS	CLASSES: 12
Natural acceptance of human values, Definitiveness of Ethical Human Conduct, Basic for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics: 1. Ability to utilize the professional competence for augmenting universal human order, 2. Ability to identify the scope and characteristics of people-friendly and eco-friendly production systems, 3. Ability to identify and develop appropriate technologies and management patterns for above production systems. 4. Case studies of typical holistic technologies, management models and production systems. Strategy for transition from the present state to Universal Human Order. 5. At the level of individual: as socially and ecologically responsible engineers, technologists and managers 6. At the level of society: as mutually enriching institutions and organizations.		
TEXT BOOKS		
1. R. R. Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics. 2. Prof. K. V. Subba Raju, 2013, Success Secrets for Engineering Students, Smart Student Publications, 3rd Edition.		

III B.TECH II SEMESTER SYLLABUS

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

III B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT39	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course should enable the students to: 1. State Space Searching techniques to solve a real-world problem 2. Heuristic Searching techniques to optimize a problem iteratively 3. Randomized Search and Emergent Systems to determine the best solution for the developed model 4. Optimal Path algorithms to determine shortest paths in different circuits 5. Problem Decomposition to decompose a problem into manageable sub-components is a necessity in complex problem-solving tasks								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Apply appropriate State Space Searching techniques to solve a real-world problem. 2. Apply appropriate Heuristic Searching techniques to solve a real-world problem. 3. Analyze the problem and infer new knowledge using Randomized Search and Emergent Systems. 4. Apply Optimal Paths Algorithms to real world problems. 5. Analyze the Problem Decomposition Methods and their test cases.								
UNIT - I	INTRODUCTION: ARTIFICIAL INTELLIGENCE						CLASSES: 9	
Introduction: Artificial Intelligence, Historical Backdrop, What is Artificial Intelligence, AI Agents –types of Agents, Applications of AI. State Space Search: Generate and Test, Simple Search, Depth First Search (DFS), Breadth First Search (BFS), Comparison of BFS and DFS, Depth Bounded DFS (DBDFS), Depth First Iterative Deepening (DFID).								
UNIT - II	HEURISTIC SEARCH						CLASSES: 10	
Heuristic Search: Heuristic Functions, Best First Search, Hill Climbing, Local Maxima, Solution Space Search, Variable Neighbourhood Descent, Beam Search, Tabu Search, Peak to Peak Methods.								
UNIT - III	RANDOMIZED SEARCH AND EMERGENT SYSTEMS						CLASSES: 10	
Randomized Search and Emergent Systems: Iterated Hill Climbing, Simulated Annealing, Genetic Algorithms, the Travelling Salesman Problem, Neural Networks, Emergent Systems.								
UNIT - IV	FINDING OPTIMAL PATHS						CLASSES: 10	
Finding Optimal Paths: Brute Force, Branch and Bound, Refinement Search, Dijkstra’s Algorithm, Algorithm A*, Admissibility of A*, Iterative Deepening A* (IDA*), Recursive Best First Search (RBFS), Pruning the Closed List, Pruning the Open List, Divide and Conquer Beam Stack Search.								
UNIT - V	PROBLEM DECOMPOSITION AND CASE STUDIES						CLASSES: 9	
Problem Decomposition: SAINT, Dendral, Goal Trees, Rule based Systems, XCON, Rule Based Expert Systems. Case Studies: Autonomous Vehicle, Medical Image Analysis.								

TEXT BOOKS
1. A First Course in Artificial Intelligence, Deepak Khemani, Tata Mc-Graw Hill Education, 2013.
REFERENCE BOOKS
1. AI – A Modern Approach, Stuart Russel, Peter Norvig, 4th Edition, Pearson Education, 2010. 2. Artificial Intelligence, Kevin Knight, Elaine Rich, Shivshankar B. Nair, 3rd Edition, Tata Mc-Graw Hill Education, 2019.

AUTOMATA AND COMPILER DESIGN

III B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT11	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the fundamental concepts of formal languages and regular expressions.
2. Identify the need of compiler and stages of compiler.
3. Enumerating top down and bottom up parsing techniques used in compilation process.
4. Understand Syntax directed translation scheme and different ways of representing Intermediate code.
5. Issues in design of Code generation and different code optimization techniques.

COURSE OUTCOMES

The course should enable the students to:

1. Able to employ finite state machines and regular expressions for modelling and solving computing problems.
2. Classify compilation phases and construction of Derivation trees for the grammar.
3. Construct the parsing tables using different types of parsing approaches.
4. Able to apply Syntax directed translation for the given expression and different ways of representing Intermediate code.
5. Analyze the issues in code generation and applying different code optimization techniques.

UNIT - I	INTRODUCTION TO FINITE AUTOMATA	CLASSES: 13
Introduction to Finite Automata: Central Concepts of Finite Automata – Alphabets, Strings, Languages. Deterministic Finite Automata - Formal Definition, Design of DFA, Nondeterministic Finite Automata - Formal Definition, Design of NFA, Conversion from NFA with ϵ -transitions to NFA without ϵ -transitions. Conversion from NFA to DFA.		
Regular Expressions: Introduction to Regular Expressions, Identity rules of Regular Expression, Algebraic Laws for Regular Expressions, Conversion of Finite Automata to Regular Expressions.		
UNIT - II	INTRODUCTION TO COMPILERS	CLASSES: 12
Introduction to Compilers: Language Processing system, Phases of Compiler, Pass and Phases of Translation, Boot Strapping.		
Lexical Analysis: Role of Lexical analyzer, Recognition of Tokens, Context Free Grammar, Derivations- Leftmost and Rightmost Derivations, Parse Trees, Ambiguity, Elimination of Left Recursion, Elimination of Left Factoring.		
UNIT - III	TOP DOWN PARSING	CLASSES: 13
Top Down Parsing: Types of Parsers, Calculation of First and Follow, Construction of LL(1) Parsing table, Recursive Descent Parser.		
Bottom up Parsing: Introduction, Classification of Bottom up parsing- LR Parser, Operator precedence Parser. Shift Reduce parser, Construction of parsing tables- LR(0), SLR(1), CLR(1), LALR(1).		

UNIT - IV	SYNTAX-DIRECTED TRANSLATION	CLASSES: 10
Syntax-Directed Translation: Syntax-Directed Definition, Types of Attributes, Annotated parse tree, Syntax Directed Translation Scheme, SDT for Infix to Postfix. Intermediate-Code Generation: Introduction, Types of Intermediate code, Implementation of Three Address Code.		
UNIT-V	RUN-TIME ENVIRONMENTS	CLASSES: 12
Run-Time Environments: Source Language Issues, Storage Allocation Techniques, Activation Record. Code Optimization: Principle sources of Code optimization, Basic Blocks, Optimization of Basic Blocks, Loop optimization, DAG representation of Basic Block. Code Generation: Issues in design of Code Generation, Object code forms, Peephole Optimization.		
TEXT BOOKS		
1. Introduction to Automata Theory, Languages, and Computation, 3rd Edition, John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, Pearson Education. 2. Compilers: Principles, Techniques and Tools By Aho, Lam, Sethi, and Ullman, Second Edition, Pearson, 2014. 3. Aho, Sethi & Ullman, "Compilers: Principles, Techniques and Tools", Pearson Education		
REFERENCE BOOKS		
1. Compilers: Principles, Techniques and Tools By Aho, Sethi, and Ullman, Addison-Wesley, 1986 2. Introduction to Formal languages Automata Theory and Computation, Kamala Krithivasan, Rama R, Pearson. 3. V Raghvan, "Principles of Compiler Design", McGraw-Hill.		

DATA MINING AND MACHINE LEARNING

III B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS16	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. Learn Data Mining concepts, preprocessing techniques 2. Understand associate rules for frequent pattern mining and learn concept of supervised and un supervised learning techniques 3. Discuss various classification algorithms. 4. Befamiliarwithbasicmachinelearningalgorithmswithclassificationandclustering. 5. Gain experience of doing in dependent study and research.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. Perform the preprocessing of data and apply mining techniques on it. 2. Recognize the characteristics of machine learning that makes it useful to solve real-world problems. 3. Design and implement machine learning solutions of classification, regression problems. 4. Choose an appropriate clustering technique to solve real world problems. 5. Choose a suitable machine learning model, implement and examine the performance of the chosen model for a given real world problems.								
UNIT - I	INTRODUCTION						CLASSES: 12	
INTRODUCTION TO DATA MINING: Introduction, What is Data Mining, Definition, Knowledge Discovery from Data (KDD), What Kinds of Data can be Mined, Data Mining Tasks, Integration of Data Mining System With A Database or Data Warehouse System, Types of Data Sets and Attribute Values. PREPROCESSING: Data Quality, Major Tasks in Data Pre-processing, Data Cleaning and Data Integration, Data Reduction, Data Transformation and Data Discretization, Measures of Similarity and Dissimilarity-Basics								
UNIT - II	ASSOCIATION RULES						CLASSES: 12	
INTRODUCTION TO MACHINE LEARNING: Introduction to machine learning, learning problems and scenarios, need for machine learning, types of learning, standard learning tasks. ASSOCIATION RULES: Problem Definition, Frequent Item Set Generation, The APRIORI Principle, Support and Confidence Measures, Association Rule Generation; APRIORI Algorithm, Compact Representation of Frequent Item Set- Maximal Frequent Item Set								
UNIT - III	SUPERVISED LEARNING						CLASSES: 14	
Learning a Class from Examples, Linear, Non-linear, Multi-class and Multi-label classification, Perceptron: – multilayer neural networks – back propagation - learning neural networks structures –support vector machines:–soft margin SVM–going beyond linearity generalization and over fitting– regularization– validation. Decision Trees: ID3, Classification and Regression Trees (CART), Regression: Linear Regression, Multiple Linear Regression, Logistic Regression.								

UNIT - IV	UNSUPERVISED LEARNING	CLASSES: 14
Introduction to clustering, Hierarchical: AGNES, DIANA, Partitional: K-means clustering, K-Mode Clustering, Self-Organizing Map, Expectation Maximization, Gaussian Mixture Models, Principal Component Analysis (PCA), Locally Linear Embedding (LLE), Factor Analysis, Neural Networks: Introduction, Perceptron, Multilayer Perceptron, Support vector machines: Linear and Non Linear, Kernel Functions, K-Nearest Neighbors.		
UNIT - V	ENSEMBLE AND PROBABILISTIC LEARNING	CLASSES: 12
Ensemble Learning Model Combination Schemes, Voting, Error-Correcting Output Codes, Bagging: Random Forest Trees, Boosting: Adaboost, Gradient Boosting, Xg boost, Stacking, Bayesian Learning, Bayes Optimal Classifier, Naïve Bayes Classifier, Bayesian Belief Networks, Mining Frequent Patterns		
TEXT BOOKS		
1. Data Mining: Concepts and Techniques – Jiawei Han, Micheline Kamber, Jian Pei (2012), 3rd edition, Elsevier, United States of America. 2. Giuseppe Bonaccorso — Machine Learning Algorithms II, 2nd Edition, Packt, 2018.		
REFERENCE BOOKS		
1. Data Mining Techniques, Arun K Pujari, 3rd Edition, Universities Press. 2. Data Mining Principles & Applications—T.V S veresh Kumar, B. Esware Reddy, Jagadish S Kalimani, Elsevier. 3. Ethem Al paydin,—Introduction to Machine Learning II, PHI, 2004 4. Stephen Marshland, —Machine Learning: An Algorithmic Perspective II, CRC Press Taylor & Francis, 2nd Edition, 2015		

BIG DATA TECHNOLOGIES

(PROFESSIONAL ELECTIVE – III)

III B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6DS06	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Contact Classes:61	Tutorial Classes:	Practical Classes:			Total Classes:61			

COURSE OBJECTIVES

The course will enable the students to:

1. Understand what big data is and how Big Data Analytics can help organizations achieve a competitive advantage.
2. Provide an overview of Apache Hadoop and its ecosystem components.
3. Understand Map Reduce Jobs
4. Processing of Big Data with advanced architectures like Spark.
5. To understand practical machine learning scalable and easy.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Understand fundamentals of Big Data analytics.
2. Investigate Hadoop framework and Hadoop Distributed File system.
3. Demonstrate the MapReduce programming model to process the big data along with Hadoop tools.
4. Implement Big Data code in Apache Spark (in PySpark).
5. Run Supervised and Unsupervised machine learning on Large-Scale Data.

UNIT-I	INTRODUCTION TO BIG DATA ANALYTICS	CLASSES :12
Introduction to Big Data Analytics: Big Data, Scalability and Parallel Processing, Designing Data Architecture, Data Sources, Quality, Pre-Processing and Storing, Data Storage and Analysis, Big Data Analytics Applications and Case Studies.		
UNIT-II	INTRODUCTION TO HADOOP	CLASSES-12
Introduction to Hadoop: Introduction, Hadoop and its Ecosystem, Hadoop Distributed File System, MapReduce Framework and Programming Model, Hadoop Yarn, Hadoop Ecosystem Tools. Hadoop Distributed File System Basics: HDFS Design Features, Components, HDFS User Commands. Hadoop Ecosystem Components: Using Apache Pig, Hive, Sqoop, Flume, Oozie, HBase.		
UNIT-III	MAPREDUCE, HIVE AND PIG	CLASSES:12
MapReduce, Hive and Pig: Introduction, MapReduce Map Tasks, Reduce Tasks and MapReduce Execution, Composing MapReduce for Calculations and Algorithms, Hive, HiveQL, Pig.		

UNIT-IV	LARGE-SCALE DATA PROCESSING WITH PYSARK	CLASSES:12
Apache Spark, Spark programming. (Python and PySpark), RDDs, Data Frames, Spark SQL, PySpark, NumPy, SciPy, Code Optimization, Cluster Configurations, Linear Algebra Computation in Large Scale, Distributed File Storage Systems.		
UNIT-V	LARGE SCALE MACHINE LEARNING WITH SPARK	CLASSES:13
Basic statistics, Data sources, Pipelines, Extracting, transforming and selecting features, Classification and Regression, Clustering, Collaborative filtering, Frequent Pattern Mining, Model selection and tuning.		
TEXT BOOKS		
1. Raj Kama! and Preeti Saxena, "Big Data Analytics Introduction to Hadoop, Spark, and Machine-Learning", McGraw Hill Education, 2018 ISBN: 9789353164966, 9353164966 2. Douglas Eadline, "Hadoop 2 Quick-Start Guide: Learn the Essentials of Big Data Computing in the Apache Hadoop 2 Ecosystem", 1st Edition, Pearson Education, 2016. ISBN13: 978-9332570351		
REFERENCE BOOKS		
1. Tom White, "Hadoop: The Definitive Guide", 4th Edition, O'Reilly Media, 2015. ISBN-13: 978-9352130672 2. Perrin, J. (2020). Spark in action (2nd Ed.). (Covers Apache Spark 3 with examples in Java, Python, and Scala) O'Reilly Media Inc. 3. Arshdeep Bahga, Vijay Madiseti, "Big Data Analytics: A Hands-On Approach", 1st Edition, VPT Publications, 2018. ISBN-13: 978-0996025577 4. Damji, J., Wenig, B., Das, T., Lee, D. (2020). Learning spark (2nd Ed.) O'Reilly Media Inc. 5. Nudurupati, S. (2021). Essential PySpark for scalable data analytics: A beginner's guide to harnessing the power and ease of PySpark 3 Packet Publishing		

DEEP LEARNING

(PROFESSIONAL ELECTIVE – III)

III B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AI17	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
Contact Classes: 60	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 60			

COURSE OBJECTIVES

The course should enable the students to:

1. To learn Deep learning techniques and their applications.
2. To acquire the knowledge of neural network architectures, Deep learning methods and algorithms.
3. To understand CNN and RNN algorithms and their applications.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Understand various learning models.
2. Design and develop various Neural Network Architectures.
3. Understand approximate reasoning using Convolution Neural Networks.
4. Analyze and design Deep learning algorithms in different applications.
5. Ability to apply CNN and RNN techniques to solve different applications.

UNIT-I	BASICS	CLASSES: 12
Introduction: Historical Trends in Deep Learning, McCulloch Pitts Neuron, Thresholding Logic, Perceptrons, Perceptron Learning Algorithm. Representation Power of MLPs, Sigmoid Neurons, Gradient Descent, Feed forward Neural Networks, Representation Power of Feed forward Neural Networks		
UNIT-II	FEED FORWARD NEURAL NETWORKS	CLASSES: 12
Feed Forward Neural Networks: Back propagation, Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMS Prop, Adam, Eigenvalues and eigenvectors, Eigenvalue Decomposition, Basis Principal Component Analysis and its interpretations, Singular Value Decomposition		
UNIT-III	AUTO ENCODERS & REGULARIZATION	CLASSES: 12
Auto encoders: Dealing with human-generated data, defining big data, architectural foundation, analytical data warehouses, Hadoop, data in motion and streaming data, integration of big data with traditional data Regularization: Bias Variance Tradeoff, L2 regularization, Early stopping, Dataset augmentation, Parameter sharing and tying, Injecting noise at input, Ensemble methods, Dropout, Greedy Layer wise Pre-training, Better activation functions, Better weight initialization methods, Batch Normalization		
UNIT-IV	CONVOLUTION NEURAL NETWORKS	CLASSES: 12
Convolutional Neural Network: The Convolution Operation, Motivation, Pooling, Convolution and Pooling as an Innately Strong Prior, Variants of the Basic Convolution Function, Structured Outputs, Data Types. LeNet, AlexNet, ZF-Net, VGGNet, GoogLeNet, ResNet, Visualizing Convolutional Neural Networks, Guided Back propagation, Deep Dream, Deep Art, Fooling Convolutional Neural Networks		

UNIT-V	RECURRENT NEURAL NETWORKS	CLASSES: 12
Building a cognitive health care application: Recurrent Neural Networks, Back propagation through time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, GRU, LSTMs, Encoder Decoder Models, Attention Mechanism, Attention over images		
TEXT BOOKS		
1. Goodfellow. I., Bengio. Y. and Courville. A., “ Deep Learning”, MIT Press, 2016.		
REFERENCE BOOKS		
1. Tom M. Mitchell, “Machine Learning”, MacGraw Hill, 1997. 2. Stephen Marsland, “Machine Learning - An Algorithmic Perspective “, CRC Press, 2009. 3. LiMin Fu, “Neural Networks in Computer Intelligence”, McGraw-Hill edition, 1994.		

SMART CONTRACT DEVELOPMENT AND AUDIT

(PROFESSIONAL ELECTIVE - III)

III B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS31	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the fundamentals of blockchain technology and its relevance to smart contracts.
2. To develop skills in creating secure and efficient smart contracts.
3. To explore the process of auditing smart contracts for security vulnerabilities.
4. To equip students with the knowledge to identify and mitigate risks in smart contract development.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Explain the principles of blockchain technology and its role in enabling smart contracts.
2. Develop functional smart contracts using programming languages like Solidity.
3. Identify potential security vulnerabilities in smart contracts and apply best practices to address them.
4. Perform comprehensive audits on smart contracts to identify security flaws.
5. Analyze and enhance real world applications and case studies.

UNIT - I	INTRODUCTION	CLASSES:14
Introduction to blockchain and decentralized systems, Smart contracts and their applications. Basics of Ethereum and Solidity programming.		
UNIT - II	SMART CONTRACT DEVELOPMENT	CLASSES:14
Programming constructs in Solidity, Datatypes, Functions, Events, and Modifiers. Creating and deploying basic smart contracts on Ethereum blockchain.		
UNIT - III	SECURITY IN SMART CONTRACTS	CLASSES:10
Common vulner abilities in smart contracts: reentrancy, overflow, underflow, etc. Best practices for writing secure smart contracts, code review, and testing.		
UNIT - IV	SMART CONTRACT AUDIT	CLASSES:10
Smart contract auditing process, tools, and techniques. Identifying and mitigating security risks in smart contracts.		
UNIT - V	REAL WORLD APPLICATIONS AND CASESTUDIES	CLASSES:12
Practical use cases of smart contracts: Decentralized finance (DeFi), Non-fungible tokens (NFTs), Supply chain, Case studies of successful and failed smart contract implementations.		
TEXT BOOKS		
1. "Mastering Ethereum: Building Smart Contracts and DApps" by Andreas M. Antonopoulos and Gavin Wood 2018 edition. 2. "Decentralized Finance (DeFi)" by Craig Calcaterra, WulfA. Kaal, 2021 edition		

CRIME INVESTIGATION & DIGITAL FORENSICS

(PROFESSIONAL - III)

III B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CY16	PEC	L	T	P	C	CIA	SEE	TOTAL
		3	-	-	3	40	60	100
Contact Classes:60		Tutorial Classes:		Practical Classes:		Total Classes:60		

COURSE OBJECTIVES

The course will enable the students to:

1. Provides an in-depth study of the rapidly changing and fascinating field of computer forensics.
2. Combines both the technical expertise and the knowledge required to investigate, detect and prevent digital crimes.
3. Knowledge on digital forensics legislations, digital crime, forensics processes and procedures, data acquisition and validation, e-discovery tools
4. E-evidence collection and preservation, investigating operating systems and file systems, network forensics, art of steganography and mobile device forensics

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Understand the fundamentals of cybercrime and issues.
2. Understand different investigation tools for cybercrime.
3. Understand basics of Forensic Technology and Practices.
4. Analyze different laws, ethics and evidence handling procedures.

UNIT – I	INTRODUCTION	CLASSES:10
Introduction: Introduction and Overview of Cyber Crime, Nature and Scope of Cyber Crime, Types of Cyber Crime: Social Engineering, Categories of Cyber Crime, Property Cyber Crime.		
UNIT – II	CYBER CRIME ISSUES	CLASSES:10
Cyber Crime Issues: Unauthorized Access to Computers, Computer Intrusions, White collar Crimes, Viruses and Malicious Code, Internet Hacking and Cracking, Virus Attacks, Pornography, Software Piracy, Intellectual Property, Mail Bombs, Exploitation, Stalking and Obscenity in Internet, Digital laws and legislation, Law Enforcement Roles and Responses.		
UNIT - III	INVESTIGATION	CLASSES:08
Investigation: Introduction to Cyber Crime Investigation, Investigation Tools, eDiscovery, Digital Evidence Collection, Evidence Preservation, E-Mail Investigation, E-Mail Tracking, IP Tracking, E-Mail Recovery, Hands on Case Studies. Encryption and Decryption Methods, Search and Seizure of Computers, Recovering Deleted Evidences, Password Cracking.		
UNIT – IV	DIGITAL FORENSICS	CLASSES:12
Digital Forensics: Introduction to Digital Forensics, Forensic Software and Hardware, Analysis and Advanced Tools, Forensic Technology and Practices, Forensic Ballistics and Photography, Face, Iris and Fingerprint Recognition, Audio Video Analysis, Windows System Forensics, Linux System Forensics, Network Forensics.		

UNIT - V	LAWS & ACTS	CLASSES:10
Laws and Acts: Laws and Ethics, Digital Evidence Controls, Evidence Handling Procedures, Basics of Indian Evidence ACT IPC and CrPC, Electronic Communication Privacy ACT, Legal Policies.		
TEXT BOOKS		
1. Nelson Phillips and Enfinger Steuart, "Computer Forensics and Investigations", Cengage Learning, New Delhi, 2009. 2. Kevin Mandia, Chris Prosise, Matt Pepe, "Incident Response and Computer Forensics ", Tata McGraw - Hill, New Delhi, 2006.		
REFERENCE BOOKS		
1. Robert M Slade," Software Forensics", Tata McGraw - Hill, New Delhi, 2005. 2. Bernadette H Schell, Clemens Martin, "Cybercrime", ABC – CLIO Inc, California, 2004. 3. "Understanding Forensics in IT ", NIIT Ltd, 2005		

IMAGE PROCESSING TECHNIQUES

(PROFESSIONAL ELECTIVE-III)

III B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS39	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The students are able to:

1. To become familiar with digital image fundamentals
2. To get exposed to image enhancement techniques in Spatial and Frequency domain.
3. To get knowledge on various image transforms.
4. To understand the image restoration and segmentation techniques.
5. To become familiar with image compression methods

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Know and understand the basics and fundamentals of digital image processing.
2. Operate on images using the techniques of smoothing, sharpening and enhancement.
3. Analyze the image in the frequency domain using various transforms.
4. Learn the basics and applications of restoration and segmentation.
5. Apply the image compression techniques in real time applications.

UNIT-I	INTRODUCTION OF IMAGE PROCESSING	CLASSES: 12
Digital Image Fundamentals: Origins of Digital image processing, Fundamental steps in Digital image processing, Components of an image Processing System, Elements of visual perception, image sensing and acquisition, image sampling and quantization, Some basic relationships between pixels, distance measures.		
UNIT-II	IMAGE ENHANCEMENT	CLASSES: 12
Enhancement in Spatial Domain : Gray level transformations, Histogram processing, Basics of Spatial Filtering– Smoothing and Sharpening Filters. Enhancement in Frequency Domain : Introduction to Fourier Transform– Smoothing and Sharpening Frequency domain filters – Ideal, Butterworth and Gaussian filters, Homomorphic filtering.		
UNIT-III	IMAGE TRANSFORMS	CLASSES: 12
2D-Discrete Fourier Transform and its properties, Discrete Cosine Transform (DCT) and properties, Haar Transform, Hadamard Transform, Hotelling Transform and Slanttrans form.		
UNIT-IV	IMAGE RESTORATION AND SEGMENTATION	CLASSES: 12
Image Restoration- Model of the Image Degradation/Restoration Process, Noise Models, Inverse Filtering, Minimum Mean Square Error (Wiener) Filtering. Image Segmentation-Detection of discontinuities, Edge Detection, Edge linking and boundary detection, Thresholding: Global and Optimal Thresholding, Region based segmentation, Water Shed Segmentation algorithm.		

UNIT-V	IMAGE COMPRESSION	CLASSES: 12
Image Compression Models, Huffman and Arithmetic Coding, Error Free Compression, Lossy Compression, Lossy and Lossless Predictive Coding, Transform Based Compression, JPEG 2000 Standards.		
TEXT BOOKS		
1. R.C. Gonzalez and R.E. Woods, Digital Image Processing, Second Edition, Pearson Education 3rd edition 2008 2. Anil Kumar Jain, Fundamentals of Digital Image Processing, Prentice Hall of India. 4TH edition		
REFERENCE BOOKS		
1. Digital Image Processing- S Jayaraman, S Esakkirajan, T Veerakumar- Mc Graw Hill Edn, 2010. 2. Murat Tekalp, Digital Video Processing" Prentice Hall, 2nd edition 2015. 3. S. Sridhar , Digital Image Processing, Oxford University Press, 2nd Ed,		

BIG DATA TECHNOLOGIES LAB**(PROFESSIONAL ELECTIVE – III LAB)****III B.TECH - II SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6DS07	PEC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Get familiar with Hadoop distributions, configuring Hadoop and performing File management tasks.
2. Understand the storage of data in a distributed file system.
3. Introduce the basics required to develop map reduce programs.
4. Introduce programming tools PIG & HIVE in the Hadoop ecosystem.
5. Demonstrate the usage of PySpark to implement ML concepts.

COURSE OUTCOMES

At the end of the course, students will be able to:

1. Connect to hadoop cluster, experiment with various Linux and HDFS commands to store data.
2. Apply the knowledge of MapReduce programming to process the stored data in HDFS.
3. Write scripts using Pig and retrieve data from Hadoop using HiveQL.
4. Create data processing pipelines with Spark
5. Build and tune machine learning models with Spark ML

LIST OF EXPERIMENTS

WEEK-I		
i) Perform setting up and installing Vmware for Hadoop and Linux. ii) Basic Linux Commands iii) Run basic HDFS shell commands		
WEEK-II		
Implement the following file management tasks in Hadoop: i) Adding files and directories ii) Retrieving files iii) Deleting files and directories.		
WEEK-III		
i) Develop a MapReduce program to calculate the frequency of a given word in a given file. ii) Develop a MapReduce program to find the maximum temperature in each year.		
WEEK-IV		
Design MapReduce algorithms to take a very large file of integers and produce as output: i) The largest integer ii) The average of all the integers. iii) The same set of integers, but with each integer appearing only once. * iv) The count of the number of distinct integers in the input.*		

WEEK-V		
Implement Matrix Multiplication on Hadoop Using Map Reduce .		
WEEK-VI		
i) Run Pig and perform basic PIG commands. ii) Write Pig Latin scripts to sort, group, join, project, and filter your data.		
WEEK-VII		
i) Practice Basic HiveQL Commands, read data from various File Formats and create Data Definition Statements and Data Manipulation Statements. ii) Write Queries using select.		
WEEK-VIII		
i) Interactive Analysis with the Spark Shell ii) Writing and running Spark program		
WEEK-IX		
Implement the following algorithms for classification using PySpark. i) Logistic Regression ii) Decision Tree Classifier iii) Naive Bayes		
WEEK-X		
Implement the following algorithms for clustering using PySpark. i) K-Means ii) Latent Dirichlet Allocation (LDA) iii) Gaussian Mixture Model (GMM)		
WEEK-XI		
Implement collaborative filtering using spark ML library.		
WEEK-XII		
Implement FP-Growth using Spark ML Library.		
TEXT BOOKS		
1. Hadoop: The Definitive Guide, 4th Edition - O'Reilly Media 2. Singh, Pramod. Machine Learning with PySpark: With Natural Language Processing and Recommender Systems. Apress, 2018. 3. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015.		

DEEP LEARNING LAB

III B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AI19	PEC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. The objective of this lab is to get an overview of the various Deep learning techniques and can able to demonstrate those using python.
2. Implement classification of linearly separable Data using Perceptron
3. Implement Back Propagation methods.
4. Implement a Recurrent Neural Network for IMDB movie review classification problem

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Implement deep neural networks to solve real world problems
2. Choose appropriate pre-trained model to solve real time problem
3. Interpret the results of two different deep learning models

WEEK-I	SINGLE LAYER PERCEPTRON	CLASSES: 03
Implement Single Layer Perceptron in Python		
WEEK-II	MULTILAYER PERCEPTRON	CLASSES: 03
Implement Multi-Layer Perceptron in Python		
WEEK-III	ACTIVATION FUNCTIONS	CLASSES: 03
Demonstrate normalization of input data, basic activation functions such as the softmax, sigmoid, dsigmoid, etc		
WEEK-IV	NEURAL NETWORK	CLASSES: 03
Build a neural network for logistic regression to minimize the cost function and update the parameters.		
WEEK-V	BACKWORD PROPAGATION	CLASSES: 03
Implement backward propagation neural network for a two class classification with a single hidden layer, non-linear activation function like tanh and compute the cross entropy loss.		
WEEK-VI	DEEP NEURAL NETWORK-ReLU	CLASSES: 03
Build a deep neural network with more than one hidden layer, non-linear functions like ReLU.		
WEEK-VII	DEEP NEURAL NETWORK	CLASSES: 03
Build deep neural network to any classification problem and compare its accuracy to logistic regression.		

WEEK-VIII	REGULARIZATION	CLASSES: 03
Apply Regularization techniques in deep learning model with backward propagation		
UNIT-IX	MINI-BATCH	CLASSES: 03
Implement mini batch optimization technique to improve the performance of deep learning model.		
UNIT-X	CONVOLUTION NEURAL NETWORKS	CLASSES: 03
Implement Convolution Neural Networks in Python		
UNIT-XI	RECURRENT NEURAL NETWORKS	CLASSES: 03
Implement a Recurrent Neural Network for IMDB movie review classification problem		
UNIT-XII	LONG SHORT TERM MEMORY NETWORKS	CLASSES: 03
Implement LST Min Python		
Software Packages required: • Keras • Tensorflow • PyTorch		
TEXT BOOKS		
1. Reza Zadeh and Bharath Ramsundar, "Tensorflow for Deep Learning", O' Reilly publishers, 2018.		
REFERENCES		
1. https://github.com/fchollet/deep-learning-with-python-notebooks		

SMART CONTRACT DEVELOPMENT AND AUDIT LAB

(PROFESSIONAL ELECTIVE-III LAB)

III B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS32	PEC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To develop and deploy smart contracts on local Block chain.
2. To deploy the smart contract on test networks.
3. To deploy and publish smart contracts on Ethereum test network.
4. To deploy chain code on permissioned Block chain
5. To design and develop a Full-fledged DApp using Ethereum/Hyper ledger.

COURSE OUTCOMES

At the end of the course the students are able to:

1. Develop and test smart contract on local Block chain.
2. Develop and test smart contract on Ethereum test networks.
3. Write and deploy smart contract using Remix IDE and Metamask
4. Write and deploy chain code in Hyperledger Fabric.
5. Develop and test a Full-fledged DApp using Ethereum/Hyperledger

WEEK – 1 & 2 INTRODUCTION: LOCAL BLOCK CHAIN

Introduction to Truffle, establishing local Block chain using Truffle

Mini Project: Allocation of the groups

WEEK-3 & 4 SMART CONTRACTS AND CHAIN CODE

Solidity programming language, chain code (Java/JavaScript/Go), deployment on Truffle local Block chain.

Mini Project: Topic selection

WEEK- 5 & 6 DEPLOYMENT AND PUBLISHING SMART CONTRACTS ON ETHEREUM TEST NETWORK

Ethereum Test networks (Ropsten/Gorelli/Rinkeby), deployment on test networks, Web3.js/Web3.py for interaction with Ethereum smart contract

Mini Project: Topic validation and finalizing software requirements

WEEK -7 & 8 CHAIN CODE DEPLOYMENT IN HYPERLEDGER FABRIC

Chain code deployment in Hyperledger Fabric

Mini project: Study required front end tools

WEEK -9 &10 MINI-PROJECT ON DESIGN AND DEVELOPMENT OF A DAPPS USING ETHEREUM/HYPERLEDGER FABRIC

Mini-project on Design and Development of a DApps using Ethereum/Hyperledger Fabric Implementation of Mini Project

1. Design, configure and testing of mini project
2. Report submission as per guidelines

WEEK-11&12	CASE STUDIES
Implementation of Successful and failed smart contracts – DeFi, NFTs	
TEXT BOOKS	
1. Ethereum Smart Contract Development, Mayukh Mukhopadhyay, Packt publication. 2. Hands-on Smart Contract Development with Hyperledger Fabric V2, Matt Zand, Xun Wu and Mark Anthony Morris, O'Reilly	
REFERENCE BOOKS	
1. Mastering Block chain, Imran Bashir, Packt Publishing 2. Introducing Ethereum and Solidity, Chris Dannen, APress. 3. Hands-on Block chain with Hyperledger, Nitin Gaur, Packt Publishing.	
WEB REFERENCES	
1. https://trufflesuite.com/ 2. https://metamask.io/ 3. https://remix.ethereum.org/ 4. https://www.hyperledger.org/use/fabric	

CRIME INVESTIGATION & DIGITAL FORENSICS LAB**(PROFESSIONAL ELECTIVE-II LAB)****III B.TECH - II SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CY17	PEC	L	T	P	C	CIA	SEE	TOTAL
		-	-	3	1.5	40	60	100
Contact Classes:60		Tutorial Classes:		Practical Classes:		Total Classes:60		

COURSE OBJECTIVES

The course will enable the students to:

1. To provide students with a comprehensive overview of collecting, investigating, preserving, and presenting evidence of cybercrime left in digital storage devices, emails, browsers, mobile devices using different Forensics tools.
2. To Understand file system basics and where hidden files may lie on the disk, as well as how to extract the data and preserve it for analysis.
3. Understand some of the tools of e-discovery.
4. To understand the network analysis, Registry analysis and analyze attacks using different forensics tools.

COURSE OUTCOMES

At the end of the course, student will be able to

1. Learn the importance of a systematic procedure for investigation of data found on digital storage media that might provide evidence of wrong-doing.
2. To learn the file system storage mechanisms and retrieve files in hidden format.
3. Learn the use of computer forensics tools used in data analysis.
4. Learn how to find data that may be clear or hidden on a computer disk, find out the open ports for the attackers through network analysis, Registry analysis.

LIST OF EXPERIMENTS

WEEK – 1	EMAIL ANALYSIS
1. Perform email analysis using the tools like Exchange EDB viewer, MBOX viewer and View user mailboxes and public folders, Filter the mailbox data based on various criteria, Search for particular items in user mailboxes and public folders	
WEEK – 2	BROWSER HISTORY ANALYSIS
2. Perform Browser history analysis and get the downloaded content, history, saved logins, searches, websites visited etc. using Foxton Forensics tool, Dumpzilla.	
WEEK – 3	MOBILE ANALYSIS
3. Perform mobile analysis in the form of retrieving call logs, SMS log, all contacts list using the forensics tool like SAFT	
WEEK – 4	REGISTRY ANALYSIS
4. Perform Registry analysis and get boot time logging using process monitor tool	

WEEK – 5	DISK IMAGING AND CLONING
5. Perform Disk imaging and cloning the using the X-way Forensics tools	
WEEK – 6	DATA ANALYSIS
6. Perform Data Analysis i.e. History about open file and folder, and view folder actions using Last view activity tool	
WEEK – 7	NETWORK ANALYSIS
7. Perform Network analysis using the Network Miner tool.	
WEEK – 8	INFORMATION FOR INCIDENT RESPONSE
8. Perform information for incident response using the crowd Response tool	
WEEK – 9	FILE TYPE DETECTION
9. Perform File type detection using Autopsy tool	
WEEK – 10	MEMORY CAPTURE AND ANALYSIS
10. Perform Memory capture and analysis using the Live RAM capture or any forensic tool	
TEXT BOOKS	
1. Real Digital Forensics for Handheld Devices, E. P. Dorothy, Auerback Publications, 2013. 2. The Basics of Digital Forensics: The Primer for Getting Started in Digital Forensics, J. Sammons, Syngress Publishing, 2012.	
REFERENCE BOOKS	
1. Handbook of Digital Forensics and Investigation, E. Casey, Academic Press, 2010. 2. Malware Forensics Field Guide for Windows Systems: Digital Forensics Field Guides, C. H. Malin, E. Casey and J. M. Aquilina, Syngress, 2012. 3. The Best Damn Cybercrime and Digital Forensics Book Period, J. Wiles and A.Reyes, Syngress, 2007	

IMAGE PROCESSING TECHNIQUES LAB**(PROFESSIONAL ELECTIVE-III LAB)****III B.TECH - II SEMESTER**

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS40	PEC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the acquisition and storage of different types of images
2. Able to Analyze and Apply different image transforms and their properties
3. Gain knowledge on different Image Smoothing & Sharpening algorithms in time and frequency domain
4. Know about different algorithms for image restoration, segmentation and compression

COURSE OUTCOMES

On the successful completion of the course, students will be able to

1. Understand the acquisition and storage of different types of images
2. Understand Analyze and Apply different image transforms and their properties
3. Apply different Image Smoothing & Sharpening algorithms in time and frequency domain
4. Apply different algorithms for image restoration
5. Apply different techniques for image segmentation
6. Apply different image compression techniques

LIST OF EXPERIMENTS

WEEK-1	INTRODUCTION TO IMAGES
a) To acquire an image, store in different formats and display the properties of the images b) Simulation and Display of an Image, Negative of an Image(Binary & Gray Scale)	
WEEK - 2	TRANSFORMATION TECHNIQUES
a) Implementation of Relationships between Pixels b) Implementation of Transformations of an Image	
WEEK - 3	ANALYSIS OF HISTOGRAMS
a) Display of bit planes of an Image b) Analyze Contrast stretching of a low contrast image, Histogram, and Histogram Equalization	
WEEK - 4	IMAGE SMOOTHENING
Implementation of Image Smoothing Filters(Mean and Median filtering of an Image)	
WEEK - 5	IMAGE SHARPENING
Implementation of image sharpening filters and Edge Detection using Gradient Filters	

WEEK - 6	FOURIER TRANSFORM AND PROPERTIES
The program is to a) Find the Fourier Transform of an image and also the inverse fourier transform. b) Verify the rotation property of Fourier Transform c) Verify the convolution property of the Fourier Transform	
WEEK - 7	DCT AND HADAMARD TRANSFORM
a) The program is to find the Discrete Cosine Transform of an image and also the inverse discrete cosine transform. b) To generate Hadamard matrix for N=4, find the Hadamard transform for a matrix of size 4 x 4. To generate a binary figure of size (256 x 256 and find its Hadamard transform as well as verify the inverse Hadamard.	
WEEK - 8	INVERSE FILTERING
To study the effect of inverse filtering in restoring degraded images.	
WEEK - 9	MORPHOLOGICAL OPERATIONS
a) Program for morphological operation: erosion and dilation. b) Program for opening and closing of the image. c) To fill the region of interest for the image.	
WEEK - 10	SEGMENTATION
a) Region based Segmentation. b) Segmentation using watershed transformation	
WEEK - 11	COLOUR IMAGE PROCESSING
Conversion between colour spaces. a) Display of colour images b) Conversion between colour spaces	
WEEK - 12	Image Compression Techniques
Implementation of Image Compression techniques. Study and compare the efficiency of any two schemes.	
TEXT BOOKS	
1. R1 Gonzalez and woods; Digital Image Processing with MatLab, TM	

DATA MINING AND MACHINE LEARNING LAB

III B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS17	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Make use of Data sets in implementing the machine learning algorithms
2. Implement the machine learning concepts and algorithms in any suitable language of choice.
3. To impart knowledge on the basic concepts underlying machine learning.
4. To acquaint with the process of selecting features for model construction.
5. To familiarize different types of machine learning techniques.

COURSE OUTCOMES

Upon successful completion of the course, the student is able to:

1. Design and implement machine learning solutions to classification, regression problems.
2. Analyze the complexity of Machine Learning algorithms and their limitations.
3. Apply appropriate data sets to the Machine Learning algorithms.
4. Identify and apply Machine Learning algorithms to solve real world problems.
5. Apply supervised and unsupervised techniques on various data sets.

LIST OF EXPERIMENTS

WEEK-1 BASICS

Write a program to demonstrate the following

- a) Operation of data types in Python.
- b) Different Arithmetic Operations on numbers in Python.
- c) Create, concatenate and print a string and access substring from a given string.
- d) Append, and remove lists in python.
- e) Demonstrate working with tuples in python.
- f) Demonstrate working with dictionaries in python.

WEEK-2 STATISTICAL OPERATIONS

Using python write a NumPy program to compute the

- a) Expected Value
- b) Mean
- c) Standard deviation
- d) Variance
- e) Covariance
- f) Covariance Matrix of two given arrays.

WEEK-3	DATA PREPROCESSING – CONTINUOUS/DISCRETE DATA
For a given set of training data examples stored in a .CSV file, demonstrate Data Preprocessing in Machine learning with the following steps - Getting the dataset. - Importing libraries. - Importing datasets. - Finding Missing Data. - Finding Outliers - Splitting dataset into training and test set. - Feature scaling.	
WEEK-4	DATA PREPROCESSING – CATEGORICAL DATA
For a given set of training data examples stored in a .CSV file, demonstrate Data Preprocessing in Machine learning with the following steps - Getting the dataset. - Importing libraries. - Importing datasets. - Finding Missing Data. - Encoding Categorical Data. - Splitting dataset into training and test set. - Feature scaling.	
WEEK-5	DECISION TREE
Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample	
WEEK-6	LINEAR REGRESSION
Build a linear regression model using python for a particular data set by - Splitting Training data and Test data. - Evaluate the model (intercept and slope). - Visualize the training set and testing set - predicting the test set result - compare actual output values with predicted values	
WEEK-7	MULTIPLE LINEAR REGRESSION
Build a multiple linear regression model using python for a particular data set by - Splitting Training data and Test data. - Evaluate the model (intercept and slope). - Visualize the training set and testing set - predicting the test set result - compare actual output values with predicted values	

WEEK-8**LOGISTIC REGRESSION**

The dataset contains information of users from a company's database. It contains information about UserID, Gender, Age, EstimatedSalary, and Purchased. Use this dataset for predicting that a user will purchase the company's newly launched product or not by Logistic Regression model.

User ID	Gender	Age	EstimatedSalary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	0
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	0
15728773	Male	27	58000	0
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	0
15727311	Female	35	65000	0
15570769	Female	26	80000	0
15606274	Female	26	52000	0
15746139	Male	20	86000	0
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1
15617482	Male	45	26000	1
15704583	Male	46	28000	1

WEEK-9**CLUSTERING**

A python program to implement K-Means, Hierarchical Clustering and PCA.

WEEK-10**KNN & SVM**

Write a Python program to implement KNN and SVM

WEEK-11**NAIVE BAYES**

Write a program to implement the naïve Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.

WEEK-12**RANDOM FOREST**

Implement Random Forest Algorithm using Python.

TEXT BOOKS

1. Aurélien Géron - Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd Edition. September 21019, O'Reilly Media, Inc., ISBN: 9781492032649.
2. Tom Mitchel "Machine Learning", Tata McGraw Hill, 2017.

WEB LINKS

1. <https://www.geeksforgeeks.org/machine-learning-with-python/>

ENVIRONMENTAL SCIENCE (only for Lateral Entry Students)

III B.TECH - II SEMESTER								
Course Code:	Category	Hours/Week			Credits	Maximum Marks		
A6BS11	ES	L	T	P	C	CIE	SEE	Total
		2	-	-	-	50	-	50
Contact Classes: 30	Tutorial Classes: 0	Practical Classes: 0			Total Classes:30			
COURSE OBJECTIVES								
The course will enable the students to:								
1. To identify the consequences of human actions on the web of life, global economy and quality of human life. 2. To know the strategies for scientific, social, economic and legal environmental protection. 3. To know the reasons for environmental pollution.								
COURSE OUTCOMES								
On Successful completion of this course, Students will be able to:								
1. Identify the consequences of human actions on the web of life, global economy and quality of human life. 2. Evaluate the strategies for scientific, social, economic and legal environmental protection. 3. Study the impact of conservation of biodiversity. 4. Analyze the reasons for environmental pollution. 5. Assess the environmental impact of air, water, biological and socio-economical aspects and risk assessment towards sustainable future.								
UNIT - I	ECOSYSTEMS					CLASSES:7		
Ecosystems: Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food web and ecological pyramids. Flow of energy, Biogeochemical cycles, Bio accumulation, Bio magnification, eco system value, services and carrying capacity.								
UNIT - II	NATURAL RESOURCES					CLASSES: 8		
Natural Resources: Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems-case studies. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources. Land resources: Forest resources. Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy source, case studies.								
UNIT - III	BIODIVERSITY AND BIOTIC RESOURCES					CLASSES: 5		
Bio diversity and Biotic Resources: Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values. India as a mega diversity nation: Hot spots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of bio diversity: In-Situ and Ex-situ conservation. National Biodiversity Act-Case studies.								

UNIT - IV	ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES	CLASSES:5
Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, Air Pollution, Water Pollution drinking water quality standards. Soil Pollution Impacts of modern agriculture, Noise Pollution Health hazards, standards. Concepts of bioremediation. Global Environmental Problems and Global Efforts: Ozone depletion and Ozone depleting substances (ODS) Concepts of Bioremediation International conventions/Protocols: Earth summit, Kyoto protocol and Montréal Protocol; NAPCC-GoI Initiatives, COP 24, COP25.		
UNIT - V	ENVIRONMENTAL POLICY, LEGISLATION & EIA	CLASSES:5
Environmental Policy, Legislation & Environmental Impact Assessment (EIA): Environmental Protection act, Legal aspects Air Act1981, Water Act, Forest Act, Wildlife Act, Municipal solid waste management, biomedical waste management hazardous waste management and handling rules. Environmental Impact Assessment: EIA structure, methods of baseline data acquisition. Strategies for risk assessment, Towards Sustainable Future: Concept of Sustainable Development, Urban Sprawl, Concept of Green Building.		
TEXT BOOKS		
1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission. 2. Environmental Studies by R.Rajagopalan, Oxford University Press.		

IV B.TECH I SEMESTER SYLLABUS

DISTRIBUTED COMPUTING

IV B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS22	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Understand the foundations of distributed systems.
2. Learn issues related to clock Synchronization and the need for global state in distributed systems
3. Learn distributed mutual exclusion and deadlock detection algorithms
4. Understand the significance of agreement, fault tolerance and recovery protocols in Distributed Systems
5. Learn the characteristics of peer-to-peer and distributed shared memory systems

COURSE OUTCOMES

By the end of the course, the student will able to:

1. Elucidate the foundations and issues of distributed systems
2. Illustrate various synchronization issues and global state for distributed systems.
3. Illustrate Mutual Exclusion and Deadlock detection algorithms in distributed systems
4. Describe the agreement protocols and fault tolerance mechanisms in distributed systems
5. Describe the features of peer-to-peer and distributed shared memory systems.

UNIT - I	A MODEL OF DISTRIBUTED COMPUTATIONS	CLASSES: 12
Definition, Relation to computer system components, Motivation, Relation to parallel systems, Message-passing systems versus shared memory systems, Primitives for distributed communication, Synchronous versus asynchronous executions, Design issues and challenges. A model of distributed computations: A distributed program, A model of distributed executions, Models of communication networks, Global state, Cuts, Past and future cones of an event, Models of process communications. Logical Time: A framework for a system of logical clocks, Scalar time, Vector time, Physical clock synchronization: NTP.		
UNIT - II	MESSAGE ORDERING & SNAPSHOTS	CLASSES: 12
Global state and snapshot recording algorithms: Introduction, System model and definitions, Snapshot algorithms for FIFO channels, Snapshot algorithms for non-FIFO channels, Message Ordering & Snapshots: Message ordering and group communication: Message ordering paradigms, Asynchronous execution with synchronous communication, Synchronous program order on an asynchronous system, Group communication, Causal order (CO), Total order.		
UNIT - III	DISTRIBUTED MUTEX& DEADLOCK	CLASSES: 12
Distributed Mutex& Deadlock: Distributed mutual exclusion algorithms: Introduction, Preliminaries, Lamport's algorithm, Ricart-Agrawala algorithm, Maekawa's algorithm, Suzuki-Kasami's broadcast algorithm. Deadlock detection in distributed systems: Introduction, System model, Preliminaries, Models of deadlocks, Knapp's classification, Algorithms for the single resource model, the AND model and the OR model.		

UNIT - IV	RECOVERY & CONSENSUS	CLASSES: 12
Recovery & Consensus: Check pointing and rollback recovery: Introduction, Background and definitions, Issues in failure recovery, Checkpoint-based recovery, Log-based rollback recovery, Coordinated check pointing algorithm, Algorithm for asynchronous check point in grand recovery. Consensus and agreement algorithms: Problem definition, Overview of results, Agreement in a failure, free system, Agreement in synchronous systems with failures.		
UNIT - V	PEER-TO-PEER COMPUTING AND OVERLAY GRAPHS	CLASSES: 12
Peer-to-peer computing and overlay graphs: Introduction, Data indexing and overlays, Chord–Content addressable networks, Tapestry. Distributed shared memory: Abstraction and advantages, Memory consistency models, Shared memory Mutual Exclusion.		
TEXT BOOKS		
1. Distributed computing: Principles, algorithms, and systems, Ajay Kshemkalyani and MukeshSinghal, Cambridge University Press, 2011. 2. Distributed Systems Concepts and Design, George Coulouris, Jean Dollimore and Tim Kindberg, Fifth Edition, Pearson Education, 2012.		
REFERENCE BOOKS		
1. Distributed Operating Systems: Concepts and Design, Pradeep K Sinha, Prentice Hall of India, 2007. 2. Advanced concepts in operating systems. MukeshSinghal and Niranjana G. Shivaratri, McGraw-Hill, 1994. 3. Java in Distributed Systems, Concurrency, Distribution and Persistence, Marko Boger, 2001. 4. Distributed Systems: Principles and Paradigms, Tanenbaum A.S., Van Steen M., Pearson Education, 2007. 5. Kshemkalyani Ajay D, Mukesh Singhal, Distributed Computing: Principles, Algorithms and Systems, Cambridge Press, 2011 6. Mukesh Singhal, Niranjana G Shivaratri, Advanced Concepts in Operating systems, McGraw Hill Publishers, 1994.		
WEB LINKS		
1. https://onlinecourses.nptel.ac.in/noc21_cs15/preview 2. https://www.distributedsystemscourse.com/		

CRYPTOGRAPHY AND NETWORK SECURITY

IV B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CY06	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Contact Classes:54	Tutorial Classes:	Practical Classes:			Total Classes:54			
COURSE OBJECTIVES The course will enable the students to: 1. To understand the objectives of information security. 2. To inculcate confidentiality, integrity, authentication and availability. 3. To Introduce various cryptographic algorithms and public-key cryptosystem 4. Demonstrate a PGP key pair and encrypted e- mail message. 5. To give a clear idea on fundamental ideas of public-key cryptography, web security and Firewalls.								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Ability to understand basic cryptographic algorithms. 2. Ability to understand message and web authentication and security issues. 3. Ability to identify information system requirements for both of them such as client and server. 4. Understand the current legal issues towards information security. 5. Understand Intrusions and intrusion detection.								
UNIT-I	CRYPTOGRAPHY CONCEPTS					CLASSES :12		
What is Introduction, The need for security, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security. Cryptography Concepts and Techniques: Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, steganography, key range and key size, possible types of attacks.								
UNIT-II	SYMMETRIC KEY CIPHERS AND ASYMMETRIC KEY CIPHERS					CLASSES:12		
Symmetric key Ciphers: Block Cipher principles, DES, AES, Blowfish, RC5, IDEA, Block cipher operation, Stream ciphers, RC4. Asymmetric key Ciphers: Principles of public key cryptosystems, RSA algorithm, Elgamal Cryptography, Diffie-Hellman Key Exchange, and Knapsack Algorithm.								
UNIT-III	CRYPTOGRAPHIC HASH FUNCTIONS					CLASSES:14		
Message Authentication, Secure Hash Algorithm (SHA-512), Message authentication codes: Authentication requirements, HMAC, CMAC, Digital signatures, Elgamal Digital Signature Scheme. Key Management and Distribution: Symmetric Key Distribution Using Symmetric & Asymmetric Encryption, Distribution of Public Keys, Kerberos, X.509 Authentication Service, Public – Key Infrastructure.								

UNIT-IV	TRANSPORT-LEVEL SECURITY	CLASSES:10
Web security considerations, Secure Socket Layer and Transport Layer Security, HTTPS, Secure Shell (SSH). Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN, IEEE 802.11i Wireless LAN Security.		
UNIT-V	E-MAIL SECURITY	CLASSES:10
Pretty Good Privacy, S/MIME IP Security: IP Security overview, IP Security architecture, Authentication Header, Encapsulating security payload, Combining security associations, Internet Key Exchange.		
TEXT BOOKS		
1. Cryptography and Network Security - Principles and Practice: William Stallings, Pearson Education, 6th Edition. 2. Cryptography and Network Security: Atul Kahate, McGraw Hill, 3rd Edition.		
REFERENCE BOOKS		
1. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition. 2. Cryptography and Network Security: Forouzan Mukhopadhyay, McGraw Hill, 3rd Edition. 3. Information Security, Principles, and Practice: Mark Stamp, Wiley India. 4. Principles of Computer Security: WM. Arthur Conklin, Greg White, TMH. 5. Introduction to Network Security: Neal Krawetz, CENGAGE Learning. 6. Network Security and Cryptography: Bernard Menezes, CENGAGE Learning.		

KNOWLEDGE REPRESENTATION AND REASONING

IV B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AI18	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Contact Classes: 44	Tutorial Classes: NIL	Practical Classes: NIL			Total Classes: 44			

COURSE OBJECTIVES

The course should enable the students to:

1. To investigate the key concepts of knowledge representation (KR) techniques and different notations.
2. To integrate the KR view as a knowledge engineering approach to model organizational knowledge.
3. To introduce the study of ontologies as a KR paradigm and applications of ontologies.
4. To understand various KR techniques.
5. To understand process, knowledge acquisition and sharing of ontology.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Analyze and design knowledge based systems intended for computer implementation.
2. Acquire theoretical knowledge about principles for logic-based representation and reasoning.
3. Ability to understand knowledge-engineering process.
4. Ability to implement production systems, frames, inheritance systems and approaches to handle uncertain or incomplete knowledge.

UNIT-I	INTRODUCTION TO KNOWLEDGE REPRESENTATION	CLASSES: 10
The Key Concepts: Knowledge, Representation, and Reasoning, Why Knowledge Representation and Reasoning, Knowledge-Based Systems, Why knowledge Representation, Why Reasoning, The Role of Logic, Propositional Logic basics, Soundness & Completeness, Resolution Proof, Semantic Tableaux, Binary Decision Diagrams .		
UNIT-II	THE LANGUAGE OF FIRST-ORDER LOGIC	CLASSES: 08
Introduction, The Syntax, The Semantics, Interpretations, Denotation, Satisfaction and Models, Logical Consequence Why We Care, Explicit and Implicit Belief, Knowledge-Based Systems. Expressing Knowledge. Knowledge Engineering Vocabulary, Basic Facts, Complex Fact, Terminological Fact, Entailments, Abstract Individuals, Other Sorts of Facts.		
UNIT-III	INTRODUCTION OF RESOLUTION	CLASSES: 10
The Propositional Case, Resolution Derivations, An Entailment Procedure, Handling Variables and Quantifiers, First-Order Resolution, Answer Extraction., Skolemization, Equality, Dealing with Computational Intractability, The First-Order Case, The Herbrand Theorem, The Propositional Case , The Implications , SAT Solvers, Most General Unifiers, Other Refinements .		
UNIT-IV	REASONING WITH HORN CLAUSES	CLASSES: 8
Reasoning with Horn Clauses: Horn Clauses, Resolution Derivations with Horn Clauses, SLD Resolution, Goal Trees, Computing SLD Derivations, Backward Chaining, Forward Chaining, The First-Order Case.		

UNIT-V	PROCEDURAL CONTROL OF REASONING	CLASSES: 08
Procedural Control of Reasoning: Facts and Rules, Rule Formation and Search Strategy, Algorithm Design, Specifying Goal Order, Committing to Proof Methods, Controlling Backtracking, Negation as Failure Dynamic Databases, The PLANNER Approach.		
TEXT BOOKS		
1. Language, Proof and Logic, Jon Barwise & John Etchemendy, CSLI Publications (1999); ch 9-11. 2. Knowledge representation and Reasoning, Ronald J. Brachman & Hector J. Levesque, Elsevier (2004); ch 2-6, 9, 11, 14, 15. 3. The Description Logic Handbook: Theory, implementation, and applications, Franz Baader, Deborah L. McGuinness, Daniele Nardi and Peter F. Patel-Schneider, Cambridge University Press (2010); ch 2, 5-6.		
REFERENCE BOOKS		
1. Rajendra A. Akerkar and Priti S. Sajja, Knowledge-Based Systems; Jones & Bartlett Publishers, 1st Edition, 2009. You can also check http://isbn.nu/http://isbn.nu/9780763776473 for prices of new and used copies at online bookstores like Amazon, Abebooks.com, Half.com and a few others. 2. J. Giarratano and G. Riley, `` Expert Systems -- Principles and Programming, 4th Edition, Thomson/PWS Publishing Company, 2004. 3. Ernest Friedman-Hill, "Jess in Action" Manning Publications, 2003. □ 4. A. Gonzalez and D. Dankel, "The Engineering of Knowledge-Based Systems"; Second Edition, Prentice Hall, 2004. VEER MADHO SINGH BHANDARI UTTARAKHAND TECHNICAL UNIVERSITY, DEHRADUN Syllabus Knowledge Based System (AIT-316) L: T: P: C: 3:0:0:3 Credits-4 5. Introduction to Expert Systems. By Peter Jackson. Addison Wesley Longman, 1999. ISBN 0-201-87686-8. 6. C. Nikolopoulos, "Expert Systems" Marcel Dekker Inc. 1997. ISBN 0 8247 9927 the CPE/CSC 480 textbook, Artificial Intelligence: A Modern Approach (2nd ed.) by Stuart Russell and Peter Norvig.		
WEB REFERENCES		
1. Knowledge representation and reasoning/Ronald J. Brachman, Hector J. Levesque. p. cm. includes bibliographical references and index. ISBN: 1-55860-932-6 1. Knowledge representation (Information theory) 2. Reasoning. I. Levesque, Hector J., 1951- II. Title		
E -TEXT BOOKS		
1. https://www.cin.ufpe.br/~mtcfa/files/in1122/Knowledge%20Representation%20and%20Reasoning.pdf		
MOOCS COURSE		
1. https://web.stanford.edu/class/cs227/ 2. https://www.emse.fr/~zimmermann/Teaching/KRR/		

COGNITIVE COMPUTING

IV B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AI21	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Contact Classes: 60	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 60			
COURSE OBJECTIVES The course should enable the students to: 1. Use the Innovation Canvas to justify potentially successful products. 2. Discuss about natural language technologies to business problems. 3. Understand the hadoop and analytical database. 4. Explain various descriptive models. 5. Learning from patterns in healthcare data.								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Understand applications in Cognitive Computing. 2. Understand Natural language processor role in Cognitive computing. 3. Apply various types of databases. 4. Evaluate the process of taking a product to market. 5. Use smarter transportation infrastructure using analytics.								
UNIT-I	FOUNDATION OF COGNITIVE COMPUTING						CLASSES: 13	
Foundation of Cognitive Computing: cognitive computing as a new generation, the uses of cognitive systems, system cognitive, gaining insights from data, Artificial Intelligence as the foundation of cognitive computing, understanding cognition Design Principles for Cognitive Systems: Components of a cognitive system, building the corpus, bringing data into cognitive system, machine learning, hypotheses generation and scoring, presentation and visualization services								
UNIT-II	NATURAL LANGUAGE PROCESSING IN SUPPORT OF A COGNITIVE SYSTEM						CLASSES: 13	
Natural Language Processing in support of a Cognitive System: Role of NLP in a cognitive system, semantic web, Applying Natural language technologies to Business problems Representing knowledge in Taxonomies and Ontologies: Representing knowledge, Defining Taxonomies and Ontologies, knowledge representation, models for knowledge representation, implementation considerations								
UNIT-III	RELATIONSHIP BETWEEN BIG DATA AND COGNITIVE COMPUTING						CLASSES: 12	
Relationship between Big Data and Cognitive Computing: Dealing with human-generated data, defining big data, architectural foundation, analytical data warehouses, Hadoop, data in motion and streaming data, integration of big data with traditional data Applying Advanced Analytics to cognitive computing: Advanced analytics is on a path to cognitive computing, Key capabilities in advanced analytics, Using advanced analytics to create value, Impact of open source tools on advanced analytics								

UNIT-IV	THE BUSINESS IMPLICATIONS OF COGNITIVE COMPUTING	CLASSES: 12
The Business Implications of Cognitive Computing: Preparing for change, advantages of new disruptive models, knowledge meaning to business, difference with a cognitive systems approach, meshing data together differently, using business knowledge to plan for the future answering business questions in new ways , building business specific solutions, making cognitive computing a reality, cognitive application changing the market.		
UNIT-V	BUILDING A COGNITIVE HEALTH CARE APPLICATION	CLASSES: 10
Building a cognitive health care application: Foundations of cognitive computing for healthcare, constituents in healthcare ecosystem, learning from patterns in healthcare Data, using cognitive applications to improve health and wellness. Smarter cities-Cognitive Computing in Government: Characteristics of smart city, rise of open data movement with fuel cognitive cities, internet of everything and smarter cities, understanding the owner ship and value of data, building a smarter transportation infrastructure using analytics to close workforce skills gap.		
TEXT BOOKS		
1. Judith H Hurwitz, Marcia Kaufman, Adrian Bowles , “Cognitive computing and Big Data Analytics” , Wiley		
REFERENCE BOOKS		
1. Vijay V Raghavan, Venkat N. Gudivada, Venu Govindaraju, Cognitive Computing: Theory and Applications: Volume 35 (Handbook of Statistics), North Holland.		

BLOCKCHAIN TAXONOMY & CRYPTO CURRENCIES

(PROFESSIONAL ELECTIVE – IV)

IV B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS33	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To understand the basic concepts of block chain technology and to explore the driving force behind the crypto currency Bitcoin.
2. Integrate ideas from block chain technology into their own projects.
3. To understand about the different methods of Decentralization using Block Chain and different Bitcoins and Alternative Coins.
4. Explore the block chain decentralization and cryptography concepts.
5. Enumerate the Bitcoin features and its alternative options.
6. To explore the driving force behind the crypto currency Bitcoin.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Demonstrate the foundation of the Block chain technology and understand the evolution of blockchain technology.
2. Understand the types, benefits and limitation of block chain and Explore the block chain decentralization and cryptography concepts.
3. Identify the risks involved in building Block chain applications& Review of legal implications using smart contracts.
4. Enumerate the Bit coin features and its alternative options.
5. Describe and deploy the crypto currency concepts and its applications.

UNIT - I	INTRODUCTION	CLASSES: 11
Introduction, Scenarios, Challenges Articulated, Blockchain, Blockchain Characteristics, Opportunities Using Blockchain, History of Blockchain. Evolution of Blockchain: Evolution of Computer Applications, Centralized Applications, Decentralized Applications, Stages in Blockchain Evolution, Consortia, Forks, Public Blockchain Environments, Type of Players in Blockchain Ecosystem, Players in Market.		
UNIT - II	BLOCKCHAIN CONCEPTS	CLASSES: 11
Blockchain Concepts: Changing of Blocks, Hashing, Merkle-Tree, Consensus, Mining and Finalizing Blocks, Currency aka tokens, security on blockchain, data storage on blockchain, wallets, coding on blockchain: smart contracts, peer-to-peer network, types of blockchain nodes, risk associated with blockchain solutions, life cycle of blockchain transaction.		
UNIT - III	BLOCKCHAIN APPLICATIONS, BITCOIN AND CRYPTOCURRENCY	CLASSES: 14
Blockchain-Outside of Currencies: Internet of Things, Government, Health, Finance, Media, Secure Voting and Digital Identity, Real Estate, Education. BitCoin and Cryptocurrency: What is Bitcoin, The Bitcoin Network, The Bitcoin Mining Process, Mining Developments, Bitcoin Wallets, Decentralization and Hard Forks, Ethereum Virtual Machine (EVM), Merkle Tree		

UNIT - IV	BITCOIN - ALTERNATIVE COINS	CLASSES: 9
Bitcoin, Transactions, Bitcoin payments, Bitcoin properties – Transaction life cycle – creation of coin –sending payments – double spending using blockchain – bitcoin anonymity – Ether: Ethereum properties, Alternative Coins, Bitcoin limitations, Namecoin, Litecoin, Primecoin, Zcash		
UNIT - V	CRYPTOCURRENCY -REGULATIONS	CLASSES: 14
Cryptocurrency: History, Distributed Ledger, Bitcoin protocols - Mining strategy and rewards, Ethereum - Construction, DAO, Smart Contract, GHOST, Vulnerability, Attacks, Sidechain, Namecoin. Cryptocurrency Regulation: Stakeholders, Roots of Bit coin, Legal Aspects-Crypto currency Exchange, Black Market and Global Economy. Applications: Internet of Things, Medical Record Management System, Domain Name Service and future of Blockchain. Decentralization and Cryptography.		
TEXT BOOKS		
1. Mastering Blockchain - Distributed ledgers, decentralization and smart contracts explained, AuthorImran Bashir, Packt Publishing Ltd, Second Edition, ISBN 978-1- 78712-544-5, 2017 2. 2. Bitcoin and Cryptocurrency Technologies, Author- Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder, Princeton University, 2016		
REFERENCE BOOKS		
1. Blockchain: A Practical Guide to Developing Business, Law, and Technology Solutions, Joseph Bambara, Paul R. Allen, Mc Graw Hill. 2. Blockchain: Blueprint for a New Economy, Melanie Swan, O'Reilly. 3. Blockchain Basics: A Non-Technical Introduction in 25 Steps 1st Edition, by Daniel Drescher. 4. Bitcoin and Cryptocurrency Technologies: Author- Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, Steven Goldfeder, Princeton University, 2016.		
WEB LINKS		
1. https://www.tutorialspoint.com/blockchain/index.htm 2. https://www.javatpoint.com/blockchain-tutorial 3. https://www.javatpoint.com/bitcoin 4. https://www.geeksforgeeks.org/comparison-of-different-cryptocurrencies/		

VULNERABILITY ASSESSMENT & PENETRATION TESTING

(PROFESSIONAL ELECTIVE – IV)

IV B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CY13	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Explain the basic principles and techniques of how attackers can enter the Computer System.
2. Put Acquired Knowledge into practice by performing ethical penetration tests and hiding the intrusion.
3. Perform analyses of data breaches and audits of information technology security.
4. Evaluate the strengths and weaknesses of various information technology solutions regarding data security.
5. Independently present and perform demonstrations of pen tests for educational purposes.
6. Evaluate the societal role of hacking from a social, ethical and economic standpoint.

COURSE OUTCOMES

At the end of the course, students will be able to:

1. Understand the penetration testing.
2. Understand to vulnerability assessment.
3. Evaluates if the system is susceptible to any known vulnerabilities, assigns severity levels to those vulnerabilities, and recommends remediation or mitigation, if and whenever needed.
4. Reverse engineer compiled applications to discover exploitable weaknesses.
5. Write new exploits to test various types of vulnerabilities on clients, against servers, and to escalate privileges.

UNIT-I	OWASP	CLASSES:10
Introduction to web applications security, threats and OWASP principles, introduction to secure design, web server: introduction a secure setup of apache, firewalling a server Browser: general concepts, functionalities, browsers war, configuration (HTTP-cookies, contents, scripting etc. attack to browsers, and users tracking/profiling (third party cookies, super cookies, XSS, CSFR, Command Injection), browser security (add-ons, plugins, same-origin policy etc.) & secure browsing.		
UNIT-II	OWASP PRIVACY PRESERVING	CLASSES:10
Attacks to privacy, (spyware & backdoors, browser, email etc.) Tracking techniques: (HTTP cookies, third party cookies, browser fingerprinting, CSP) Advanced browser configuration, anonymity and onion routing (Tor). Internet E-mail: Architecture and infrastructure, functions, agents and standards, MIME & PGP, phishing, spamming & spoofing, DKIM, SPF, introduction to email forensics.		
UNIT-III	INTRODUCTION TO ETHICAL HACKING TERMINOLOGY	CLASSES:10
Five stages of hacking-Vulnerability Research-Legal implication of hacking Impact of hacking. Foot printing & Social engineering.		

UNIT-IV	INFORMATION GATHERING METHODOLOGIES	CLASSES:10
Competitive Intelligence- DNS Enumerations- Social Engineering attacks. Scanning & Enumeration Port Scanning- Network Scanning- Vulnerability Scanning- NMAP scanning tool- OS Finger printing Enumeration. System Hacking Password.		
UNIT-V	SNIFFERS & SQL INJECTION ACTIVE & PASSIVE SNIFFING	CLASSES:10
ARP Poisoning- Session Hijacking- DNS Spoofing- Conduct SQL Injection attack – Counter measures. Cracking techniques- Key loggers- Escalating privileges- Hiding Files-Steganography technologies- Countermeasures.		
TEXT BOOKS		
1. Kimberly Graves, “CEH: Official Certified Ethical Hacker Review Guide”, Wiley Publishing Inc., 2007. ISBN: 978-0-7821-4437-6.		
REFERENCE BOOKS		
1. Shakeel Ali & Tedi Heriyanto, “Backtrack -4: Assuring security by penetration testing”, PACKT Publishing., 2011. ISBN: 978-1-849513-94-4.		

SPEECH AND VIDEO PROCESSING

(PROFESSIONAL ELECTIVE – IV)

IV B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS41	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Understand the underlying mechanisms of speech processing concepts to analyse and manipulate speech signals.
2. Explore the theoretical foundations of speech recognition and Gaussian Mixture models for effective speech recognition.
3. Gain proficiency in the principles of video formation, perception, and representation
4. Master motion estimation criteria, ranging from optical flow to pixel-based and gradient-based methods.
5. Develop expertise in advanced video analysis techniques and object tracking, while also understanding the concepts of compensation for enhanced video processing.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Describe the mechanisms of human speech production systems and methods for speech feature extraction.
2. Understand basic algorithms of speech analysis and speech recognition.
3. Explain basic techniques in digital video processing, including imaging characteristics and sensors.
4. Apply motion estimation and object tracking algorithms on video sequence.
5. Employ diverse techniques in object tracking and segmentation to contribute to real-world video analysis and applications.

UNIT-I	SPEECH PROCESSING CONCEPTS	CLASSES:12
The speech production mechanism, Discrete time speech signals, Pole-Zero modeling of speech, relevant properties of the fast Fourier transform for speech recognition, convolution, linear and nonlinear filter banks, spectral estimation of speech using DFT. Linear Prediction analysis of speech.		
UNIT-II	SPEECH RECOGNITION	CLASSES:14
Real and Complex Cepstrum, application of cepstral analysis to speech signal, feature extraction for speech, static and dynamic feature for speech recognition, robustness issues, discrimination in the feature space, feature selection, MFCC, LPCC, Distance measures, vector quantization models. Gaussian Mixture model, HMM.		
UNIT-III	BASICS OF VIDEO PROCESSING	CLASSES:10
Video formation, perception and representation: Principle of color video, video cameras, video display, pinhole model, CAHV model, Camera motion, Shape model, motion model, Scene model, two-dimensional motion models. Three-Dimensional Rigid Motion, Approximation of projective mapping.		

UNIT-IV	MOTION ESTIMATION TECHNIQUES	CLASSES:12
Optical flow, motion representation, motion estimation criteria, optimization methods, pixel-based motion estimation, Block matching algorithm, gradient Based, Intensity matching, feature matching, frequency domain motion estimation, Depth from motion. Motion analysis applications: Video Summarization, video surveillance		
UNIT-V	OBJECT TRACKING AND SEGMENTATION	CLASSES:12
2D and 3D video tracking, blob tracking, kernel based counter tracking, feature matching, filtering Mosaicing, video segmentation, mean shift based, active shape model, video shot boundary detection. Interframe compression, Motion compensation		
TEXT BOOKS		
1. Fundamentals of Speech recognition – L. Rabiner and B. Juang, Prentice Hall signal processing series. 2. Digital Video processing, A Murat Tekalp, Prentice Hall.		
REFERENCE BOOKS		
1. "Speech and Audio Signal Processing", B.Gold and N. Morgan, Wiley. 2. "Digital image sequence processing, Compression, and analysis", Todd R. Reed, CRC Press 3. "Handbook of Image and Video processing", Al Bovik, Academic press, second Edition		
WEB LINKS		
1. https://cmusphinx.github.io/wiki/tutorialconcepts/ 2. https://www.coursera.org/learn/digital 3. https://users.cs.cf.ac.uk/Dave.Marshall/Multimedia/node259.html 4. https://manipulation.csail.mit.edu/segmentation.html		

WEB AND SOCIAL MEDIA ANALYTICS

(PROFESSIONAL ELECTIVE – V)

IV B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS46	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES:

The course will enable the students to:

1. Exposure to various web and social media analytic techniques.
2. To understand natural language processing concepts on text analytics.
3. To understand sentiment analysis.
4. To gain knowledge on search engine optimization and web analytics.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Knowledge on decision support systems.
2. Apply natural language processing concepts on text analytics.
3. Understand sentiment analysis.
4. Knowledge on search engine optimization and web analytics.
5. Understand the concepts of Social Analytics and Social Network Analysis and Prescriptive Analytics.

UNIT – I	AN OVERVIEW OF BUSINESS INTELLIGENCE	CLASSES: 12
An Overview of Business Intelligence, Analytics, and Decision Support: Analytics to Manage a Vaccine Supply Chain Effectively and Safely, Changing Business Environments and Computerized Decision Support, Information Systems Support for Decision Making, The Concept of Decision Support Systems (DSS), Business Analytics Overview, Brief Introduction to Big Data Analytics.		
UNIT – II	TEXT ANALYTICS AND TEXT MINING	CLASSES: 12
Text Analytics and Text Mining: Machine Versus Men on Jeopardy, The Story of Watson, Text Analytics and Text Mining Concepts and Definitions, Natural Language Processing, Text Mining Applications, Text Mining Process, Text Mining Tools.		
UNIT - III	SENTIMENT ANALYSIS	CLASSES: 10
Sentiment Analysis: Sentiment Analysis Overview, Sentiment Analysis Applications, Sentiment Analysis Process, Sentiment Analysis and Speech Analytics.		
UNIT – IV	WEB ANALYTICS	CLASSES: 10
Web Analytics, Web Mining: Security First Insurance Deepens Connection with Policyholders, Web Mining Overview, Web Content and Web Structure Mining, Search Engines, Search Engine Optimization, Web Usage Mining (Web Analytics), Web Analytics Maturity Model and Web Analytics, Tools.		
UNIT – V	SOCIAL ANALYTICS AND SOCIAL NETWORK ANALYSIS	CLASSES: 12
Social Analytics and Social Network Analysis: Social Analytics and Social Network Analysis, Social Media Definitions and Concepts, Social Media Analytics. Prescriptive Analytics - Optimization and Multi-Criteria Systems: Multiple Goals, Sensitivity Analysis, What-If Analysis, and Goal Seeking.		

TEXT BOOKS

1. Ramesh Sharda, Dursun Delen, Efraim Turban, BUSINESS INTELLIGENCE AND ANALYTICS: SYSTEMS FOR DECISION SUPPORT, Pearson Education.

REFERENCE BOOKS

1. Rajiv Sabherwal, Irma Becerra-Fernandez, "Business Intelligence – Practice, Technologies and Management", John Wiley 2011.
2. Lariss T. Moss, Shaku Atre, "Business Intelligence Roadmap", Addison-Wesley It Service.
3. Yuli Vasiliev, "Oracle Business Intelligence: The Condensed Guide to Analysis and Reporting", SPD Shroff, 2012.

WEB REFERENCE

1. <https://www.searchenginejournal.com/social-media-analytics/463340/#close>

PERCEPTION AND COMPUTER VISION

(PROFESSIONAL ELECTIVE – V)

IV B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS55	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the Fundamental Concepts related to sources, shadows and shading.
2. To understand the Geometry of Multiple Views.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Implement fundamental image processing techniques required for computer vision.
2. Implement boundary tracking techniques.
3. Apply chain codes and other region descriptors, Hough Transform for line, circle, and ellipse detections.
4. Apply 3D vision techniques and Implement motion related techniques.
5. Develop applications using computer vision techniques.

UNIT - I CAMERAS & INTERREFLECTIONS

CLASSES:14

CAMERAS: Pinhole Cameras.

Radiometry – Measuring Light: Light in Space, Light Surfaces, Important Special Cases.

Sources, Shadows, And Shading: Qualitative Radiometry, Sources and Their Effects, Local Shading Models, Application: Photometric Stereo, Interreflections: Global Shading Models.

Color: The Physics of Color, Human Color Perception, Representing Color, A Model for Image Color, Surface Color from Image Color.

UNIT - II LINEAR FILTERS & TEXTURE

CLASSES: 12

Linear Filters: Linear Filters and Convolution, Shift Invariant Linear Systems, Spatial Frequency and Fourier Transforms, Sampling and Aliasing, Filters as Templates.

Edge Detection: Noise, Estimating Derivatives, Detecting Edges.

Texture: Representing Texture, Analysis (and Synthesis) Using Oriented Pyramids, Application: Synthesis by Sampling Local Models, Shape from Texture.

UNIT - III THE GEOMETRY OF MULTIPLE VIEWS

CLASSES: 10

The Geometry of Multiple Views: Two Views

Stereopsis: Reconstruction, Human Stereopsis, Binocular Fusion, Using More Cameras

Segmentation by Clustering: What Is Segmentation? Human Vision: Grouping and Gestalt,

Applications: Shot Boundary Detection and Background Subtraction, Image Segmentation by Clustering Pixels, Segmentation by Graph-Theoretic Clustering,

UNIT - IV SEGMENTATION BY FITTING A MODEL

CLASSES:10

Segmentation by Fitting a Model: The Hough Transform, Fitting Lines, Fitting Curves, Fitting as a Probabilistic Inference Problem, Robustness

Segmentation and Fitting Using Probabilistic Methods: Missing Data Problems, Fitting, and Segmentation, The EM Algorithm in Practice.

Tracking With Linear Dynamic Models: Tracking as an Abstract Inference Problem, Linear Dynamic Models, Kalman Filtering, Data Association, Applications and Examples

UNIT - V	GEOMETRIC CAMERA MODELS & MODEL-BASED VISION	CLASSES: 12
Geometric Camera Models: Elements of Analytical Euclidean Geometry, Camera Parameters and the Perspective Projection, Affine Cameras and Affine Projection Equations. Geometric Camera Calibration: Least-Squares Parameter Estimation, A Linear Approach to Camera Calibration, Taking Radial Distortion into Account, Analytical Photogrammetry, An Application: Mobile Model-Based Vision: Initial Assumptions, Obtaining Hypotheses by Pose Consistency, Obtainings Hypotheses by pose Clustering, Obtaining Hypotheses Using Invariants, Verification, Application: Registration In Medical Imaging Systems, Curved Surfaces and Alignment.		
TEXT BOOKS		
1. David A. Forsyth and Jean Ponce: Computer Vision – A Modern Approach, PHI Learning (Indian Edition), 2009.		
REFERENCE BOOKS		
1. E. R. Davies: Computer and Machine Vision – Theory, Algorithms and Practicalities, Elsevier (Academic Press), 4th edition, 2013. 2. R. C. Gonzalez and R. E. Woods “Digital Image Processing” Addison Wesley 2008. 3. Richard Szeliski “Computer Vision: Algorithms and Applications” Springer-Verlag London Limited 2011.		

BLOCKCHAIN FOR FINTECH AND NON FINTECH

(PROFESSIONAL ELECTIVE – V)

IV B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS34	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To Introduce block chain technology and Cryptocurrency
2. explore the driving force behind the cryptocurrency Bitcoin. Along with the Decentralization, Cryptography, Bitcoins with its alternative coins, Smart contracts and outside of currencies.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Understand the basics principles of Cryptography and Block Chain
2. Understand the bit coin network and Extensibility of Block Chain Concepts
3. Developing Decentralized apps on Ethereum Networks
4. Explore Corda and Block Chain Testing
5. Applying Block Chain Principles in FinTech and Non FinTech.

UNIT - I	FIRST PRINCIPLES AND BLOCK CHAIN STRUCTURE	CLASSES: 12
Basic Crypto graphic techniques, Digital Signatures, Distributed Consensus, Crypto in Distributed Consensus Blockchain - The Structure, Hashcash, Blockchain Data Structure, Blockchain Network, Consensus in Blockchain Network		
UNIT - II	CRYPTOCURRENCY AND BITCOIN	CLASSES: 14
Crypto currency and Bitcoin, What is Crypto currency, Bitcoin Network, Bitcoin Mining Extensibility of Blockchain concepts, Digital Identity verification, Block chain Neutrality, Digital art, Blockchain Environment		
UNIT - III	SMART CONTRACT AND ETHEREUM	CLASSES: 16
Smart Contract and Ethereum, Smart Contract - What it is and Why it is for?, Ether and Ethereum Network, Solidity, Building your first Dapps		
UNIT - IV	CORDA DLT	CLASSES: 16
Corda DLT, Hello, World!, Writing and testing a contract, Transaction workflows in Corda		
UNIT - V	FINTECH	CLASSES: 16
Various Fintech Applications using Blockchain – WeTrade, Marcopolo, Identity management (KYC) using Blockchain, Cross-border trading, Verification and Auditing		

TEXT BOOKS

1. Ambadas, Arshad Sarfarz Ariff, Sham "Blockchain for Enterprise Application Developers", Wiley
2. Andreas M. Antonopoulos, "Mastering Bitcoin: Programming the Open Blockchain" , O'Reilly

REFERENCE BOOKS

1. Blockchain: A Practical Guide to Developing Business, Law, and Technology Solutions, Joseph Bambara, Paul R. Allen, Mc Graw Hill.
2. Blockchain: Blueprint for a New Economy, Melanie Swan, O'Reilly
3. Blockchain Basics: A Non-Technical Introduction in 25 Steps 1st Edition, by Daniel Drescher

WEB AND DATABASE SECURITY (PROFESSIONAL ELECTIVE - V)

IV B.TECH - I SEMESTER

Course Code:	Category	Hours/Week			Credits	Maximum Marks		
A6CY30	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Contact Classes:	Tutorial Classes:	Practical Classes:			Total Classes:			

COURSE OBJECTIVES

The course will enable the students to:

1. Give an Overview of information security
2. Give an overview of Access control of relational databases

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Understand the Web architecture and applications
2. Understand client side and server-side programming
3. Understand how common mistakes can be bypassed and exploit the application
4. Identify common application vulnerabilities

UNIT-I	INTRODUCTION TO WEB SECURITY	CLASSES: 08
The Web Security, The Web Security Problem ,Risk Analysis and Best Practices Cryptography and the Web: Cryptography and Web Security, Working Cryptographic Systems and Protocols, Legal Restrictions on Cryptography, Digital Identification		
UNIT-II	PRIVACY PROTECTING TECHNIQUES	CLASSES: 09
The Web's War on Your Privacy, Privacy-Protecting Techniques, Backups and Antitheft, Web Server Security, Physical Security for Servers, Host Security for Servers, Securing Web Applications		
UNIT-III	DATABASE SECURITY	CLASSES: 08
Database Security: Recent Advances in Access Control, Access Control Models for XML, Database Issues in Trust Management and Trust Negotiation, Security in Data Warehouses and OLAP Systems		
UNIT-IV	REENGINEERING FOR DATABASES	CLASSES: 08
Security Re-engineering for Databases: Concepts and Techniques, Database Watermarking for Copyright Protection, Trustworthy Records Retention, Damage Quarantine and Recovery in Data Processing Systems, Hippocratic Databases: Current Capabilities		
UNIT-V	FUTURE TRENDS ON PRIVACY IN DATABASE	CLASSES: 08
Future Trends Privacy in Database Publishing: A Bayesian Perspective, Privacy-enhanced Location based Access Control, Efficiently Enforcing the Security and Privacy Policies in a Mobile Environment		

TEXT BOOKS

1. Web Security, Privacy and Commerce Simson Garfinkel, Gene Spafford, O'Reilly.
2. Handbook on Database security applications and trends Michael Gertz, Sushil Jajodia

REFERENCE BOOKS

1. Andrew Hoffman, Web Application Security: Exploitation and Countermeasures for Modern Web Applications, O'reilly
2. Jonathan LeBlanc Tim Messerschmidt, Identity and Data Security for Web Development - Best practices, O'reilly
3. McDonald Malcolm, Web Security for Developers, No Starch Press, US

AUGMENTED REALITY AND VIRTUAL REALITY

(PROFESSIONAL ELECTIVE-V)

IV B.TECH - I SEMESTER

Course Code:	Category	Hours/Week			Credits	Maximum Marks		
A6AI22	PEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Contact Classes: 54	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 54			

COURSE OBJECTIVES

The course should enable the students to:

1. Familiarize the students with the fundamentals of Virtual Reality.
2. Impart the knowledge of 3D orientation for understanding the behaviour of VR system with the environment.
3. Deal with the Development Tools and Frameworks in Virtual Reality.
4. Introduce the applications of Virtual Reality Systems.
5. Introduce technology and features of augmented reality.

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Describe the basic concepts of Virtual Reality and 3D Computer Graphics.
2. Apply 3D manipulation techniques in Virtual Reality.
3. Analyze Development Tools and Frameworks in Virtual Reality.
4. Develop a Virtual Reality application.
5. Evaluate Augmented Reality Systems.

UNIT-I	INTRODUCTION TO VR AND AR	Classes: 10
Introduction to VR and AR: History of VR and AR, Technology and Features of Augmented Reality, Comparison of AR and VR, Challenges with AR, AR Systems and Functionality, Human factors, Human visual system, Perception of depth, color, contrast, resolution, Stereo Rendering, VR Hardware: Head-coupled displays etc. VR Software, Geometric Modelling: From 2D to 3D, 3D space curves, 3D boundary representation. The Graphics Pipeline and OpenGL, Overview and Transformations, Rotation, translation, scaling, mode view matrix, projection matrix, Lighting and Shading, OpenGL Shading Language (GLSL), GLSL vertex and fragment shaders.		
UNIT-II	VISUAL COMPUTATION IN VIRTUAL REALITY	Classes: 12
Visual computation in virtual reality: 3D Interaction Techniques: 3D Manipulation Techniques and Input Devices, 3D Travel Tasks, Travel Techniques, Theoretical Foundations of Wayfinding, Types of CentredWayfinding Support, Evaluating Wayfinding Aids, System Control, Classification, Graphical Menus, Voice Commands, Gestural Commands, Tools, Multi-modal System Control Techniques, Case Study: Mixing System Control Methods, Symbolic Input Tasks.		
UNIT-III	FRAMING USING 3D VIRTUAL REALITY	Classes: 10
Framing using 3D virtual reality: Development Tools and Frameworks in Virtual Reality: VR. X3D Standard; Vega, MultiGen, Virtools etc., World Space, World Coordinate, World Environment, Objects - Geometry, Position/Orientation, Hierarchy, Bounding Volume, Scripts and other attributes, VR Environment - VR Database, Tessellated Data, LODs, Graphical User Interface, Control Panel, 2D Controls.		

UNIT-IV	VR APPLICATIONS	Classes: 12
VR Applications: Pose Tracking I, Tracking with light house, Pose Tracking II, Advanced positional tracking, Panoramic Imaging and Cinematic, VR Spatial Sound and the Vestibular System, VR Engines and Other Aspects of VR, Latency, eye tracking, post-rendering warp. The Future: Virtual environment, modes of interaction Application of VR in Digital Entertainment: VR Technology in Film & TV Production. VR Technology in Physical Exercises and Games, Demonstration of Digital, Entertainment by VR.		
UNIT-V	MIXED REALITY APPLICATION DEVELOPMENT	Classes: 10
Augmented and Mixed Reality: Taxonomy, technology and features of augmented reality, difference between AR and VR, Challenges with AR, AR systems and functionality, Augmented reality methods, visualization techniques for augmented reality, wireless displays in educational augmented reality applications, mobile projection interfaces, marker-less tracking for augmented reality, enhancing interactivity in AR environments, evaluating AR systems.		
TEXT BOOKS		
1. LaValle "Virtual Reality", Cambridge University Press, 2016. 2. John Vince, —Virtual Reality Systems —, Pearson Education Asia, 2007.		
REFERENCE BOOKS		
1. Alan B Craig, William R Sherman and Jeffrey D Will, —Developing Virtual Reality Applications: Foundations of Effective Designll, Morgan Kaufmann, 2009. 2. Alan B. Craig, Understanding Augmented Reality, Concepts and Applications, Morgan Kaufmann, 2013.		

CRYPTOGRAPHY AND NETWORK SECURITY LAB

IV B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CY08	PCC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	40	60	100
Tutorial Classes: Nil		Practical Classes: 21			Total Classes: 21			
COURSE OBJECTIVES								
The course will enable the students to:								
1. Explain the objectives of information security.								
2. Explain the importance and application of confidentiality, integrity, authentication and availability.								
3. Understand various cryptographic algorithms.								
4. Understand Buffer OverFlow attacks, Intrusion Detection Systems and Honeypots.								
5. Understand basic cryptographic algorithms, message and web authentication and security issues.								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Identify information system requirements for both of them such as client and server.								
2. Understand the current legal issues towards information security.								
3. Understand the process of capturing Network traffic using tools(Ethereal, Wireshark, Tcpdump)								
4. Implement Cryptographic algorithms in C/C++/Java								
5. Create applications in Client Server architecture.								
6. Set up secure mail and web communication channels								
LIST OF EXPERIMENTS								
WEEK-1								
1. Write a C program that contains a string (char pointer) with a value 'Hello world'. The program should XOR each character in this string with 0 and displays the result.								
WEEK-2								
2. Write a C program that contains a string (char pointer) with a value 'Hello world'. The program should AND or and XOR each character in this string with 127 and display the result.								
WEEK-3								
3. Write a Java program to perform encryption and decryption using the following algorithms								
a. Caesar cipher b. Substitution cipher c. Hill Cipher								
WEEK-4								
4. Write a C/JAVA program to implement the DES algorithm logic.								
WEEK-5								
5. Write a C/JAVA program to implement the Blowfish algorithm logic.								

WEEK-6

6. Write a C/JAVA program to implement the Rijndael algorithm logic.

WEEK-7

7. Write the RC4 logic in Java Using Java cryptography; encrypt the text "Hello world" using Blowfish. Create your own key using Java key tool.

WEEK-8

8. Write a Java program to implement RSA algorithm.

WEEK-9

9. Implement the Diffie-Hellman Key Exchange mechanism using HTML and JavaScript.

WEEK-10

10. Calculate the message digest of a text using the SHA-1 algorithm in JAVA.

WEEK-11

11. Calculate the message digest of a text using the MD5 algorithm in JAVA.

TEXT BOOKS

1. Cryptography and Network Security - Principles and Practice: William Stallings, Pearson Education, 6th Edition.
2. Cryptography and Network Security: Atul Kahate, McGraw Hill, 3rd Edition.

REFERENCE BOOKS

1. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition.
2. Cryptography and Network Security: Forouzan Mukhopadhyay, McGraw Hill, 3rd Edition.
3. Information Security, Principles, and Practice: Mark Stamp, Wiley India.
4. Principles of Computer Security: WM. Arthur Conklin, Greg White, TMH.
5. Introduction to Network Security: Neal Krawetz, CENGAGE Learning.
6. Network Security and Cryptography: Bernard Menezes, CENGAGE Learning.

IV B.TECH II SEMESTER SYLLABUS

NATURE INSPIRED COMPUTING FOR DATA SCIENCE

(PROFESSIONAL ELECTIVE – VI)

IV B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS47	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To understand the fundamentals of nature inspired techniques which influence computing,
2. To understand the Swarm Intelligence Computing
3. To understand the Immuno computing techniques
4. To familiarize the DNA.
5. To understand the scope of the DNA Computing and Adleman's experiment

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. The basics Natural systems.
2. The concepts of Natural systems and its applications
3. Basic Natural systems functions(operations)
4. Natural design considerations.
5. Integration of Hardware and software in Natural applications

UNIT - I	INTRODUCTION	CLASSES: 12
Introduction: From Nature to Nature Computing, Philosophy, Three Branches: A brief overview, Individual, Entities, and agents – Parallelism and Distributivity interactivity, Adaptation – Feedback – Self – Organization - Complexity Emergence and Bottom-up Vs. Top Down – Determination, Chaos and Fractals.		
UNIT - II	COMPUTING INSPIRED BY NATURE	CLASSES: 12
Computing Inspired by Nature: Evolutionary Computing, Hill Climbing and Simulated Annealing, Darwin's Dangerous Idea, Genetic principles, Standard Evolutionary Algorithm-Genetic Algorithms, Reproduction – Crossover, Mutation, Genetic Programming		
UNIT - III	SWARM INTELLIGENCE	CLASSES: 12
Swarm Intelligence: Introduction- Ant Colonies, Ant Foraging Behavior, Ant Colony Optimization, SACO and scope of ACO algorithms, Ant Colony Algorithm(ACA), Swarm Robotics, Foraging for food, Social Adaptation of Knowledge		
UNIT - IV	IMMUNO COMPUTING	CLASSES: 12
IMMUNO COMPUTING : Introduction – Immune System, Physiology and main components, Pattern Recognition and Binding, Immune Network Theory-Danger theory, Evaluation Interaction-Immune algorithms, Introduction-Genetic algorithms, Bone Marrow Models, Forest Algorithms, Artificial Immune Networks		

UNIT - V	COMPUTING WITH NEW NATURAL MATERIALS	CLASSES: 12
COMPUTING WITH NEW NATURAL MATERIALS: DNA Computing: Motivation, DNA Molecule, Adleman's experiment, Test tube programming language, Universal DNA Computers, PAM Model, Splicing Systems, Liptons Solution to SAT Problem, Scope of DNA Computing, From Classical to DNA Computing		
TEXT BOOKS		
1. Leandro Nunes de Castro, " Fundamentals of Natural Computing, Basic Concepts, Algorithms and Applications", Chapman & Hall/CRC, Taylor and Francis Group, 2007.		
REFERENCE BOOKS		
1. Floreano D. and Mattiussi C., "Bio-Inspired Artificial Intelligence: Theories, Methods, and Technologies", MIT Press, Cambridge, MA, 2008. 2. Albert Y.Zomaya, "Handbook of Nature-Inspired and Innovative Computing", Springer, 2006. 3. Marco Dorigo, Thomas Stutzle," Ant Colony Optimization", PHI,2006		

HUMAN COMPUTER INTERACTION

(PROFESSIONAL ELECTIVE – VI)

IV B.TECH - I SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT38	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To learn the foundations of Human Computer Interaction
2. To become familiar with the design technologies for individuals and persons with disabilities.
3. To be aware of mobile HCI.
4. To learn the guidelines for user interface
5. To get an understanding of web interface design

COURSE OUTCOMES

Upon successful completion of the course, the student is able to learn:

1. Design effective dialog for HCI
2. Design effective HCI for individuals and persons with disabilities.
3. Assess the importance of user feedback.
4. Explain the HCI implications for designing multimedia/ecommerce/e-learning Web sites.
5. Develop meaningful user interface

UNIT-I	FOUNDATION OF HCI	CLASSES: 12
The Human: I/O channels, Memory, Reasoning and problem solving; The computer: Devices, Memory, processing and networks; Interaction: Models, frameworks, Ergonomics, styles, elements, interactivity- Paradigms.		
UNIT-II	DESIGN & SOFTWARE PROCESS	CLASSES: 12
Interactive Design: Basics, process, scenarios, navigation, screen design, Iteration and prototyping. HCI in software process: Software life cycle, usability engineering, Prototyping in practice, design rationale. Design rules: principles, standards, guidelines, rules. Evaluation Techniques, Universal Design.		
UNIT-III	MODELS AND THEORIES	CLASSES: 10
HCI Models: Cognitive models: Socio-Organizational issues and stakeholder requirements – Communication and collaboration models-Hypertext, Multimedia and WWW.		
UNIT-IV	MOBILE HCI	CLASSES: 12
Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools. - Case Studies.		
UNIT-V	WEB INTERFACE DESIGN	CLASSES: 09
Designing Web Interfaces: Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow - Case Studies		

TEXT BOOKS

1. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, —Human Computer InteractionII, 3rd Edition, Pearson Education, 2004 (UNIT I, II & III)
2. Brian Fling, —Mobile Design and DevelopmentII, First Edition, O'Reilly Media Inc., 2009 (UNIT – IV)

REFERENCE BOOKS

1. Bill Scott and Theresa Neil, —Designing Web InterfacesII, First Edition, O'Reilly, 2009. (UNIT-V)
2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech,
3. User Interface Design, Soren Lauesen , Pearson Education.

FINANCIAL RISK MANAGEMENT FOR CRYPTO CURRENCIES

(PROFESSIONAL ELECTIVE – VI)

IV B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS35	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To learn about the Blockchain
2. To learn about information on Crypto currencies and their types
3. To learn Crypto currencies with qualitative Risks
4. To learn Crypto currencies with quantitative Risks
5. Learn about Future options on Crypto currencies

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Apply Cryptographic algorithm
2. Apply the Inferential statistics and Probability
3. Understand the Regulatory issues of Cryptocurrencies
4. Understand the Descriptive analysis and Volatile Behaviour
5. Understand the Exchanges of Trading

UNIT - I	BLOCKCHAIN	CLASSES: 12
Blockchain in a Nutshell, Network and Nodes, Cryptographic Algorithm, Distributed Ledger, Consensus Algorithm-Proof of Work, Proof of Stake, Byzantine Fault Tolerance, Delegated Proof of Stake.		
UNIT - II	CRYPTOCURRENCIES AND THEIR TYPES	CLASSES: 12
General Information on Crypto currencies – Creation, Objectives, Users of Crypto currencies, Proof-of-Work Coins – Bitcoin, Ethereum, Litecoin, Monero, Byzantine Fault Tolerance Coins, Proof of Stake Coins		
UNIT - III	DEALING CRYPTOCURRENCIES WITH QUALITATIVE RISKS	CLASSES: 12
Crypto currency Market, Regulatory Issues- Regulation in place, Design Risk-Selfish mining-ICO, Operational Risk- Exchanges, Other Operational Risks.		
UNIT - IV	DEALING CRYPTOCURRENCIES WITH QUANTITATIVE RISKS	CLASSES: 11
Descriptive Analysis of Returns, Correlation and Diversification, Flash Crash, Pump and Dump, Distributional Behavior, Volatile Behaviour- ARMA-GARCH Model, Market Efficiency.		
UNIT - V	OPTIONS ON CRYPTOCURRENCIES	CLASSES: 10
Futures- Chicago Mercantile Exchange Futures, Cryptocurrency Exchanges Trading Futures, Options on Cryptocurrencies, Volatile Measures of Cryptocurrencies, Stochastic Volatility.		

TEXT BOOKS

1. Financial Risks Management for Cryptocurrencies by Eline Van der Auwera, Wim Schoutens, Springer Nature Switzerland AG 2020

REFERENCE BOOKS

1. <https://financialcrimeacademy.org/cryptocurrency-risk-management/>

CLOUD SECURITY

(PROFESSIONALELECTIVE – VI)

IV B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS38	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Impart fundamental concepts in the area of virtualization.
2. Impart knowledge in infrastructure as a service.
3. Explain monitoring, management and applications.
4. Understand the concept of design patterns.
5. Learn to design the trusted cloud computing system.

COURSE OUTCOMES

At the end of the course, students will be able to:

1. Ability to use the fundamentals of virtualization.
2. Analyze the cloud security challenges, information security concepts.
3. Analyze and apply the security algorithms on cloud data and applications.
4. Build a cloud security application.
5. Create a scenario for threat protection via authentication protocols, PKI protocols

UNIT-I	FUNDAMENTALS	CLASSES:10
FUNDAMENTALS: System Modeling, Clustering and Virtualization: distributed system models and enabling technologies, computer clusters for scalable parallel computing, virtual machines and virtualization of clusters and datacenters. Introduction to cloud computing, migrating in to cloud, enriching the integration of service paradigm for cloud era, the enterprise cloud computing paradigm. Three-Tier Architecture of the Cloud		
UNIT -II	CLOUD SECURITY CHALLENGES	CLASSES:10
Cloud Security Challenges, Cloud Information Security Objectives, Cloud Security Services, Secure Cloud Software Requirements, Cloud Security Policy Implementation, Infrastructure Security, Data Security and Storage, Privacy in Cloud.		
UNIT -III	SECURITY ALGORITHMS	CLASSES:10
Cryptography: Concepts and Techniques: Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, Steganography, key range and key size, possible types of attacks, Hashing Algorithms		
UNIT -IV	CLOUD DATA SECURITY	CLASSES:10
Threats and Vulnerabilities to Infrastructure, Data, and Access Control; Risk Management and Risk Assessment in Cloud, Cloud Service Provider Risks, Virtualization Security Management in the Cloud, Trusted Cloud Computing, Identity Management and Access Control.		

UNIT -V	IDENTITY AND ACCESS MANAGEMENT	CLASSES:10
Standards for Security: SAML OAuth, OpenID, SSL/TLS, Encrypting Data and Key Management, Creating a Cloud Security Strategy, The Future of Security in Cloud Computing		
TEXT BOOKS		
1. Ronald L. Krutz, Russell Dean Vines, "Cloud Security: A Comprehensive Guide to Secure Cloud Computing", Wiley Publishing, 2010. 2. Tim Mather, Subra Kumara swamy, and Shahed Latif, " Cloud Security and Privacy", Published by O'Reilly Media, Inc., 2009		
REFERENCE BOOKS		
1. Charlie Kaufman, Network Security: Private Communication in a Public World, 2nd edition, Prentice Hall. Atul Kahate 2008. 2. Cryptography and Network Security, 2nd edition, Tata McGrawhill, India. Robert Bragg, Mark Rhodes 2004.		
WEB LINKS		
1. https://www.udemy.com/course/du-cryptography/?kw=Cryptography&src=sac 2. https://www.udemy.com/course/learn-cryptography-basics-in-python/ 3. https://www.udemy.com/course/du-cryptography/		

IOT CLOUD PROCESSING AND ANALYTICS

(PROFESSIONALELECTIVE – VI)

IV B.TECH - II SEMESTER

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS42	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Overview of Internet of Things
2. The application areas of IOT
3. The revolution of Internet in Mobile Devices, Cloud & Sensor Networks
4. Building blocks of Internet of Things and characteristics

COURSE OUTCOMES

Upon successful completion of the course, the student is able to:

1. Review Internet of Things (IoT).
2. Demonstrate various business models relevant to IoT.
3. Construct designs for web connectivity
4. Organize sources of data acquisition related to IoT, integrate to enterprise systems.
5. Internet of Things with Cloud technologies.

UNIT-I	INTRODUCTION TO INTERNET OF THINGS	CLASSES-10
The Internet of Things- An Overview of Internet of things, Internet of Things Technology, behind IoTs Sources of the IoTs, Examples OF IoTs, Design Principles For Connected Devices, Internet connectivity, Application Layer Protocols- HTTP, HTTPS, FTP.		
UNIT-II	LAYER DESIGNING AND AFFORDABILITY	CLASSES-12
Business Models for Business Processes in the Internet of Things, IoT/M2M systems LAYERS AND designs standardizations, Modified OSI Stack for the IoT/M2M Systems ,ETSI M2M domains and High-level capabilities, Communication Technologies, Data Enrichment and Consolidation and Device Management Gateway Ease of designing and affordability.		
UNIT-III	PROTOCOLS	CLASSES-10
Design Principles for the Web Connectivity for connected-Devices, Web Communication protocols for Connected Devices, Message Communication protocols for Connected Devices, Web Connectivity for connected-Devices.		
UNIT-IV	DATA ANALYTICS	CLASSES-12
Data Acquiring, Organizing and Analytics in IoT/M2M, Applications/Services/Business Processes, IOT/M2M Data Acquiring and Storage, Business Models for Business Processes in the Internet Of Things, Organizing Data, Transactions, Business Processes, Integration and Enterprise Systems		
UNIT-V	DATA STORAGE AND APPLICATIONS	CLASSES-13
Data Collection, Storage and Computing Using a Cloud Platform for IoT/M2M Applications/Services, Data Collection, Storage and Computing Using cloud platform Everything as a service and Cloud Service Models, IOT cloud-based services using the Xively (Pachube/COSM), Nimbits and other platforms Sensor, Participatory Sensing, Actuator, Radio Frequency Identification, and Wireless, Sensor Network Technology, Sensors Technology, Sensing the World		

TEXT BOOKS

1. Internet of Things: Architecture, Design Principles And Applications, Rajkamal, McGraw Hill Higher Education
2. Internet of Things, A.Bahgya and V.Madisetti, University Press, 2015

REFERENCE BOOKS

1. Designing the Internet of Things, Adrian McEwen and Hakim Cassimally, Wiley
2. Getting Started with the Internet of Things, Cuno Pfister , Oreilly

WEB REFERENCES

1. <http://donionsixbjtiohce24abfgsffo2l4tk26qx464zylumgejukfq2vead.onion/onions.php>
2. <https://www.propub3r6espa33w.onion>
3. <http://nzh3fv6jc6jskki3.onion>
4. <http://rougmnvswfsmd4dq.onion>

ORGANIZATIONAL BEHAVIOUR

IV B.TECH - II SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS15	HSMC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. To provide the students with the conceptual framework and the theories 2. To understand the dynamics of Organizational Behavior.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. To understand the cognitive processes 2. Good understanding of the dynamics of Organizational Behavior.								
UNIT-I	INTRODUCTION TO ORGANIZATIONAL BEHAVIOR						CLASSES: 08	
Introduction to OB - Definition, Nature and Scope – Environmental and organizational context – Impact of IT, globalization, Diversity, Ethics, culture, reward systems and organizational design on Organizational Behaviour. Cognitive Processes-I: Perception and Attribution: Nature and importance of Perception – Perceptual selectivity and organization – Social perception – Attribution Theories – Locus of control – Attribution Errors –Impression Management.								
UNIT-II	COGNITIVE PROCESSES AND TECHNIQUES						CLASSES: 10	
Cognitive Processes-II: Personality and Attitudes – Personality as a continuum – Meaning of personality - Johari Window and Transactional Analysis - Nature and Dimension of Attitudes – Job satisfaction and organizational commitment-Motivational needs and processes- Work-Motivation Approaches Theories of Motivation- Motivation across cultures - Positive organizational behavior: Optimism – Emotional intelligence – Self-Efficacy.								
UNIT-III	DYNAMICS OF OB						CLASSES: 12	
Dynamics of OB-I: Communication – types – interactive communication in organizations – barriers to communication and strategies to improve the follow of communication - Decision Making: Participative decision-making techniques – creativity and group decision making. Dynamics of OB –II Stress and Conflict: Meaning and types of stress –Meaning and types of conflict - Effect of stress and intra- individual conflict - strategies to cope with stress and conflict.								
UNIT-IV	POWER AND POLITICS						CLASSES: 08	
Dynamics of OB –III Power and Politics: Meaning and types of power – empowerment - Groups Vs. Teams – Nature of groups – dynamics of informal groups – dysfunctions of groups and teams – teams in modern work place.								

UNIT-V	PERFORMANCE AND PRACTICES	CLASSES: 08
Leading High performance: Job design and Goal setting for High performance- Quality of Work Life- Socio technical Design and High-performance work practices - Behavioral performance management: reinforcement and punishment as principles of Learning –Process of Behavioral modification - Leadership theories - Styles, Activities and skills of Great leaders.		
REFERENCE BOOKS		
1. Luthans, Fred: Organizational Behaviour 10/e, McGraw-Hill, 2009 2. McShane: Organizational Behaviour, 3e, TMH, 2008 3. Nelson: Organizational Behaviour, 3/e, Thomson, 2008. 4. Newstrom W. John & Davis Keith, Organizational Behaviour-- Human Behaviour at Work, 12/e, TMH, New Delhi, 2009. 5. Pierce and Gardner: Management and Organizational Behaviour: An Integrated perspective, Thomson, 2009. 6. Robbins, P. Stephen, Timothy A. Judge: Organizational Behaviour, 12/e, PHI/Pearson, New Delhi, 2009. 7. Pareek Udai: Behavioural Process at Work: Oxford & IBH, New Delhi, 2009. 8. Schermerhorn: Organizational Behaviour 9/e, Wiley, 2008. 9. Hitt: Organizational Behaviour, Wiley, 2008Aswathappa: Organizational Behaviour, 7/e, Himalaya, 2009 10. Mullins: Management and organizational Behaviour, Pearson, 2008. 11. McShane, Glinow: organizational Behaviour--Essentials, TMH, 2009. 12. Ivancevich: Organisational Behaviour and Management, 7/e, TMH, 2008.		

OPEN ELECTIVES OFFERED BY AERONAUTICAL DEPARTMENT

OE1		OE2	
A6AE19	Fundamentals of Avionics	A6AE25	Introduction to jets and rockets
A6AE20	Introduction to Aerospace Technology	A6AE26	Non-Destructive Testing Methods
OE3			
A6AE40	Introduction to Aircraft Industry		
A6AE41	Unmanned Aerial Vehicles		

FUNDAMENTALS OF AVIONICS

V SEMESTER - OPEN ELECTIVE - I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE19	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The purpose of this subject is to provide the students with the theoretical background and engineering applications. 1. Overview on Aviation using Electronics 2. Basic understanding about major electronics systems used for communication 3. Basic understanding about major devices, display and flight controls used in aircraft								
COURSE OUTCOMES At the end of the course the students are able to: 1. To explain the instrumentation used in avionics. 2. To classify various ranges of the communication techniques used in aircraft. 3. To distinguish between network systems, controlling parts & surfaces 4. To compare various principles of navigation systems 5. To build phenomena of auto pilot control system								
UNIT-I	BASICS & FLIGHT DECK AND DISPLAY SYSTEMS							
BASICS: Basic principles of Avionics, Typical avionics sub system in civil/military aircraft and space vehicles. FLIGHT DECK AND DISPLAY SYSTEMS: Flight deck display technologies, CRT, LED, LCD, Touch screen, Head up display, electronic instrumentation systems.								
UNIT-II	COMMUNICATION SYSTEMS							
AUDIO AND COMMUNICATION SYSTEMS: Aircraft audio systems, basic audio transmitter and receiver principles, VHF communication system, UHF communication systems.								
UNIT-III	FREQUENCY RANGING SYSTEM							
RANGING AND LANDING SYSTEMS: VHF Omnitrange, VOR receiver principles, distance maturity equipment, principles of operation, Instrument landing system, and localizer and glide slope. POSITIONING SYSTEM: Global positioning system principles, triangulation, position accuracy, applications in aviation								
UNIT-IV	NAVIGATION SYSTEM							
INERTIAL NAVIGATION SYSTEM: Principle of Operation of INS, navigation over earth, components of inertial Navigation systems, accelerometers, gyros and stabilized platform. SURVELLIENCE SYSTEM: ATC surveillance systems principles and operation interrogation and replay standards, Collision avoidance system, ground proximity warning system								
UNIT-V	AUTO FLIGHT SYSTEM							
AUTO FLIGHT SYSTEM: Automatic flight control systems fly by wire and fly by light technologies, flight director systems, and flight management systems.								

TEXT BOOKS

1. N. S. Nagaraja (1996), Elements of electronic navigation, 2nd edition, Tata McGraw Hill, New Delhi.
2. Janes W. Wasson, Jeppesen Sandersen (1994), Avionic systems Operation and maintenance,

REFERENCE BOOKS

1. Albert Hel Frick (2010), Principle of Avionics, 6th edition, Avionics Communications Inc, India.
2. H. J. Pallet (2010), Aircraft Instrumentation and Integrated systems, Pearson Education, New Delhi.
3. J. Powell (1998), Aircraft Radio Systems, Pitman publishers, London

INTRODUCTION TO AEROSPACE TECHNOLOGY

V SEMESTER: OPEN ELECTIVE - I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE20	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES								
The course will enable the students to:								
1. To introduce the basic concepts of Aerospace technology and the current developments in the field. 2. To provide knowledge on the basic principles on which the development of aerodynamics, Structures, propulsion and satellite systems.								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Compare the atmosphere conditions of different altitudes for spacecraft system 2. Analyze how lift, drag and thrust are generated and understand which components constitute them 3. Analyze the flight performance parameters with respective stability condition 4. Distinguish the different systems used in a satellite mission 5. Design lighter than air vehicle using Catia software								
UNIT-I	HISTORY OF FLIGHT- THE AEROSPACE ENVIRONMENT							
Balloons and dirigibles, heavier than air aircraft, commercial air transport, introduction of jet aircraft, helicopters, missiles, conquest of space, commercial use of space, exploring solar system and beyond. Earth's atmosphere, the temperature extremes of space, laws of gravitation, low earth orbit, microgravity, benefits of microgravity. The near earth radioactive environment. The magnetosphere. Environmental impact on spacecraft. Meteoroids and micrometeoroids, space debris. Planetary environments								
UNIT-II	AERODYNAMICS AND FLIGHT VEHICLE PROPULSION							
Anatomy of the airplane, helicopter, launch vehicles and missiles, space vehicles. Static forces and moments on the vehicle. Understanding engineering models. Aerodynamics of wings and bodies. Generation of lift. Sources of drag. Force and moment coefficients, center of pressure. Thrust for flight, the propeller, the jet engine, rocket engines- description, principles of operation. Governing equations.								
UNIT-III	FLIGHT VEHICLE PERFORMANCE AND STABILITY							
Performance parameters. Performance in steady flight, cruise, climb, range, endurance; accelerated Flight- symmetric maneuvers, turns, sideslips, takeoff and landing. Flight vehicle stability- longitudinal, lateral and directional- static, dynamic; trim, control. Handling qualities of airplanes								
UNIT-IV	SATELLITE SYSTEMS ENGINEERING- HUMAN SPACE EXPLORATION							
Satellite missions, an operational satellite system, elements of satellite, satellite subsystems. Satellite structures, mechanisms and materials. Power systems. Communication and telemetry. Thermal control. Attitude determination and control. Propulsion and station keeping. Space missions. Mission objectives. Case studies. Human space flight missions- goals, historical background. The Soviet and US missions. The Mercury, Gemini, Apollo (manned flight to the moon), Skylab, Apollo-Soyuz, Space Shuttle. International Space Station, extravehicular activity								

UNIT-V	INTRODUCTION TO ENGINEERING DESIGN, AIR TRANSPORTATION
Design as a critical component of engineering education- as a skill- the process, design thinking, design drawing. Design for mission, performance and safety requirements. Concurrent engineering. Computer aided engineering, design project. Example: the lighter-than – air vehicle student design project at MIT. Air Transportation Systems- civil, military- objectives- principal constituents- the vehicle, the ground facilities, the organization- role. Regulation- national and international. Indian effort- civil and military- in the field of Aerospace Engineering.	
TEXT BOOKS	
1. Newman, D., Interactive Aerospace Engineering and Design, (with software and reference material on CD), McGraw-Hill, 2002, ISBN 0-07-112254-0. 2. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082	
REFERENCE BOOKS	
1. Russell Mikel, Aerospace and Aeronautical Engineering, Willford press, 2017. 2. Ajoykumar Kundu, Mark A Price and David Riordan, Conceptual Design: An Industrial Approach, Wiley-Blackwell, 2019.	

INTRODUCTION TO JETS AND ROCKETS

VI Semester: OPEN ELECTIVE -II								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE25	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The course is intended to serve as an introduction to air breathing propulsion systems and Rocket Propulsion Systems. 1. Illustrate an overview of aerospace propulsion system. 2. Identify the foundation in fundamentals of thermodynamics. 3. Compare the ideal components and characteristics of jet engine 4. Interpret the performance of nozzles 5. Simplify the ideal performance analysis of rocket engines. 6. Select appropriate fuel for aerospace application.								
COURSE OUTCOMES At the end of the course the students are able to: 1. Explain the complexity in working of various engines 2. Interpret the elementary principles of thermodynamic cycles as applied to propulsion analysis 3. Analyze the process involved in individual components 4. Compare the nozzles with various operating conditions. 5. Determine Equations of motion in free space, Tsiokovsky's equation. 6. Classify the types of fuel in aviation and aerospace engineering.								
UNIT-I	INTRODUCTION TO AEROSPACE PROPULSION							
Propulsion system, Propulsive Systems – Evolution, Development, Growth and Challenges. Fundamentals of Thermodynamics – Variables, Thermodynamic Process, Introduction to IC Engines and Reciprocating Engines, Propellers and Working of Propellers.								
UNIT-II	PRINCIPLES OF JET PROPULSION							
Fundamentals of jet propulsion, Working Principle, Analysis of Ideal Jet Engine cycle, Engine components-merit- significance- ideal component characteristics, Classification – turbo jet, turbo fan, turbo prop and Ramjet engines. Basic Problems based on Engine Cycle.								
UNIT-III	RAMJET, SCRAMJET ENGINES AND NOZZLES							
Speed limitations of gas turbines, Basics of Ramjets, Combustors for liquid fuel ramjet engines, Combustion Instability and its Suppression, Solid fuel Ramjet Engines, SCRAM jet engines, Applications of RAM Jet and SCRAM Jet Engines to Missiles with Examples, Nozzles- Types of Nozzles, Converging-Diverging Nozzle, Variable Nozzle and Effects of Pressure Ratios on Engine Performance.								
UNIT-IV	ROCKET THEORY							
Applications of Rockets, Types of Rockets, Basics of Thermal Rocket Engine-Thermodynamics and Ideal Performance Analysis, Equations of motion-Rocket Motion in free space, Tsiokovsky's equation, Rocket Parameters, Burnout range, Burnout Velocity. Practical Problems								

UNIT-V	PROPELLANT ROCKETS
Solid Propulsion-Solid Propellant Rockets, Basic Configuration and Performance, Propellant Grain and Configuration, Propellant Characteristics Combustion Chamber, Ignition Process Liquid Propulsion - Design consideration of liquid rocket combustion chamber, injector, and propellant feed lines, valves, propellant tank outlet and helium pressurized and turbine feed systems-.BIO Fuels and Impact on the Atmosphere, Aviation turbine fuels - Requirements of aviation fuels of kerosene type.	
TEXT BOOKS	
1. Mechanics and Thermodynamics of Propulsion – Philip G Hill & Carl R Peterson , Pearson Publication – 2ndEdt 2. Rocket Propulsion Elements, Sutton, G.P., John Wiley, 1993.	
REFERENCE BOOKS	
1. The Jet Engine – Rolls Royce 2. Gas Turbines and Jet and Rocket Propulsion, M. L. Mathur, R. P. Sharma, Standard Publishers Distributors.	

NON-DESTRUCTIVE TESTING METHODS

VI SEMESTER - OPEN ELECTIVE - II								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE26	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: - To impart knowledge about the non-destructive testing methods. - To provide knowledge on the selection of NDT methods for application in engineering industries. - Classify the various NDT methods for detecting defects in the structural components. - Judge defects basing on data representation of testing								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: - Recognize various non-destructive techniques for engineering industries. - Select appropriate non-destructive technique for defects detection in manufactured/operating parts. - Perform inspection using major non-destructive testing methods. - Understand the importance and application of NDT in Aerospace structural analysis - Determine the defects basing on the principal of radiography								
UNIT-I	SURFACE TECHNIQUES							
Introduction to non-destructive testing (NDT) - importance of NDT techniques - types of NDT techniques - visual testing (direct and remote visual inspection) - principle and types of liquid penetrant tests (LPT) - advantages and limitations of LPT - applications of LPT.								
UNIT-II	MAGNETIC PARTICLE TESTING							
Introduction to magnetic particle testing (MPT) – principle, magnetization methods - demagnetization – advantages, limitations and applications of MPT, eddy current testing (ECT) method- principle advantages, limitations and applications								
UNIT-III	ULTRASONIC TESTING							
Introduction to ultrasonic testing (UT) -- principle of UT – UTprobes - UT inspection methods (pulse echo, transmission and phased array techniques) - advantages, limitations and applications								
UNIT-IV	RADIOGRAPHY TESTING							
Introduction to radiography testing (RT) - sources of X-rays and Gamma rays - characteristics of X-rays and Gamma rays (absorption, scattering) - filters and screens - film radiography and digital radiography (shadow formation, exposure factors, film handling and storage) --exposure charts - penetrameters - safety issues.								
UNIT-V	SPECIAL TECHNIQUES							
Acoustic emission testing (AET) principle, advantages, limitations - instrumentation and application of AET - infra red thermography (IRT) - contact and non-contact inspection methods - acoustic holography.								

TEXT BOOKS

1. Baldev Raj, T. Jayakumar, M. Thavasimuthu, "Practical Non-Destructive Testing", Narosa Publishing, London, 2012.
2. Paul E. Mix, "Introduction to Non Destructive Testing", A Training Guide, Wiley- Interscience, New Jersey, USA, June 2005.

REFERENCE BOOKS

1. ASM Metals Handbook, V-17, "Non-Destructive Evaluation and Quality Control", American Society of Metals, Metals Park, Ohio, USA, 2001
2. W.T. McGonnagle, "Non-Destructive Testing", McGraw Hill Book Co., USA, 2013.

INTRODUCTION TO AIRCRAFT INDUSTRY

VII SEMESTER: OPEN ELECTIVE -III								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE40	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The aim is to introduce students the overview of the Aircraft Industry. The course covers basic design process, Aircraft Stability & control and different mechanical systems. 1. Familiarize students with major types of aircraft industry. 2. Illustrate the role of regulatory bodies and its business context. 3. Exemplify the organisational structure of airlines and airports 4. Illustrate the importance of air navigation services and its types. 5. Familiarize students with air safety and security as a means for ensuring passenger safety.								
COURSE OUTCOMES At the end of the course the students are able to: 1. Discuss the types of aerospace industry and evolution of flying machine to airline alliances. 2. Explain standardisation and the importance of regulatory bodies. 3. Outline the organisational structure of airlines and airports. 4. Demonstrate major types of air navigation services and air traffic control services. 5. Illustrate the major processes involved in air safety and security.								
UNIT-I	AIRCRAFT INDUSTRY OVERVIEW							
Flying machine to airline alliances, Types of Aerospace Industry, Introduction to ages of engineering, Aerospace Manufacturing, Introduction to the space environment & human space exploration.								
UNIT-II	REGULATORY AND BUSINESS CONTEXT							
Regulatory bodies, Introduction to standardization, Standards and recommended practices (SARPS), Freedoms of the air, Role of governments, Major agreements and treaties, Business side of the industry – Key performance indicators (revenue passenger kilometer, yield etc.), Industry characteristics of passenger airlines.								
UNIT-III	AIRLINES AND AIRPORTS							
Introduction to airlines, Organizational structure, Types of airline personnel, Flight crew and cabin crew, Role of training with airlines, Organizational culture of an airline Introduction to airports – Airport personnel, Processing passengers and freight, Security								
UNIT-IV	AIR NAVIGATION SERVICES							
Introduction to air navigation services, Types of navigation services, Air traffic control (ATC) services, Structure of controlled airspace, Air traffic and control services, Classifications of airspace, Flight information services and alerting services, Air navigation facilities and air navigation service (ANS) providers								

UNIT-V	AIR SAFETY AND SECURITY
Introduction to air safety and security, Manufacturer responsibilities, Maintenance procedures, Airside safety, Safety in the air, Issues in air safety, Accident and incident investigation, Security – Response of International Civil Aviation Organization (ICAO).	
TEXT BOOKS	
1. Peter Belobaba, Amedeo Odoni and Cynthia Barnhart (2015), The Global Airline Industry (Aerospace Series), John Willey & Sons, Ltd, United Kingdom. 2. Connor R Walsh, Airline Industry – Strategies, Operations and Safety, Nova Science Publishers, New York.	
REFERENCE BOOKS	
1. Andrew R Thomas (2011), Soft Landing: Airline Industry Strategy, Service and Safety, A press, New York 2. Ahmed Abdelghany, Khaled Abdelghany (2009), Modelling Applications in the Airline Industry, Ashgate Publishing Ltd, England.	

UNMANNED AERIAL VEHICLES

VII SEMESTER - OPEN ELECTIVE - III								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6AE41	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES								
The course will enable the students to:								
1. To understand the evolution and applications of unmanned aerial vehicles. 2. To illustrate the subsystems of UAVs. 3. To explain the process involved in design of UAVs.								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Classify the Unmanned Aerial Vehicles. 2. Calculate the basic performance parameters for aircraft. 3. Identify and illustrate various payloads and propulsive systems. 4. Explain the functioning of subsystems in UAVs. 5. Illustrate the design process for a UAV.								
UNIT-I	INTRODUCTION TO UAV							
Historical Development, Overview of UAV Systems and sub-systems, Classification, Informal Categories, The Tier System, Classification Change.								
UNIT-II	BASIC AERODYNAMICS AND PERFORMANCE of UAV							
Basic Aerodynamic Equations, Air foils, lift, drag, moments, Aircraft Polar, The Real Wing and Airplane, Induced Drag, Total Air-Vehicle Drag, Flapping Wings, Rotary wings. Climbing Flight, Range, Endurance, Gliding Flight								
UNIT-III	PROPULSIVE SYSTEMS, STRUCTURES AND LOADS, PAYLOAD							
Thrust Generation and basic thrust equation, Sources of Power, Loads, types of loads, Materials, construction techniques. Payloads-Reconnaissance/Surveillance Payloads, Weapon Payloads, Other payloads								
UNIT-IV	UAV SUBSYSTEMS							
Mission Planning and Control Station- Types, Physical Configuration, Planning and Navigation, MPCs Interfaces, Modes of control, piloting and controlling mission, Autopilot system Launch Systems- Basic Considerations, launch Methods for Fixed, rotary wing UAV Recovery Systems								
UNIT-V	BASICS DESIGN AND CASE							
Introduction to Design and Selection of the System - Conceptual Phase, Preliminary Design, Detail Design, Selection of the System Case study on Indian UAVs (Rustom, Lakshya, AURA) Case study on Israeli-Heron, US- MQ9 Reaper								
TEXT BOOKS								
1. Paul Fahlstrom, Thomas Gleason - Introduction to UAV Systems-Wiley (2012) 2. Reg Austin, "Unmanned Air Systems: UAV Design, Development and Deployment", First Edition, Wiley Publishers, 2015.								
REFERENCE BOOKS								
1. Mirosaw Adamski, "Power units and power supply systems in UAV", New Edition, Taylor and Francis Group publishers, 2014. 2. Skafidas. "Microcontroller Systems for a UAV". KTH. TRITA-FYS 2002:51 ISSN 0280-316X. 34. 2002.								

OPEN ELECTIVES OFFERED BY COMPUTER SCIENCE AND ENGINEERING

OE1		OE2	
A6CS48	Software Project Management Fundamentals	A6CS09	Database Management System
A6CS37	Semantic Web	A6CS43	Data Warehousing and Business Intelligence
OE3			
A6CS51	Artificial Intelligence and Applications		
A6CS36	Cyber Security & Cyber laws		

SOFTWARE PROJECT MANAGEMENT FUNDAMENTALS

III B.TECH - II SEM - (OPEN ELECTIVE-I)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS48	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVE The course will enable the students to: 1. Understanding the specific roles within a software organization as related to project and process management. 2. Understanding the basic infrastructures competences (e.g., process modelling and measurements.). 3. Describe the principles, techniques, methods & tools for model based management of software projects, assurance of product quality and process adherence (quality assurance), as well as experienced based creation and improvements of models (process management). 4. Understanding the basic steps of project planning, project management, quality assurance, and process management and their relationships. 5. To provide basic project management skills with a strong emphasis on issues and problems associated with delivering successful IT projects.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. To describe and determine the purpose and importance of project management from the perspectives of planning, tracking and completion of project. 2. To compare and differentiate organization structures and project structures. 3. To implement a project to manage project schedule, expenses and resources with the application of suitable project management tools. 4. Understand and practice the process of project management and its application in delivering successful IT projects. 5. Conduct Case Study on “The Command Center Processing and Display System-Replacement” (CCPDS-R)								
UNIT - I	CONVENTIONAL SOFTWARE MANAGEMENT						CLASSES: 12	
The waterfall model, conventional software Management performance. Evolution of Software Economics: Software Economics, pragmatic software cost estimation.								
UNIT - II	IMPROVING SOFTWARE ECONOMICS						CLASSES: 12	
Reducing Software product size, improving software processes, improving team effectiveness, improving automation, Achieving required quality, peer inspections. The old way and the new: The principles of conventional software engineering, principles of modern software management, transitioning to an iterative process.								
UNIT - III	LIFE CYCLE PHASES						CLASSES: 12	
Engineering and production stages, inception, Elaboration, construction, transition phases. Artefacts of the process: The artefact sets, Management artefacts, Engineering artefacts, programmatic artefacts. Model based software architectures: A management perspective and technical perspective.								

UNIT - IV	WORK FLOWS OF THE PROCESS	CLASSES: 14
Software Process Workflows, Inter Trans Workflows. Checkpoints Of The Process: Major Mile Stones, Minor Milestones, Periodic Status Assessments. Iterative Process Planning: Work Break Down Structures, Planning Guidelines, Cost And Schedule Estimating, Interaction Planning Process, Pragmatic Planning. Project Organizations And Responsibilities: Line-Of-Business Organizations, Project Organizations, Evolution Of Organizations. Process Automation: Automation Building Blocks, The Project Environment.		
UNIT - V	PROJECT CONTROL AND PROCESS INSTRUMENTATION	CLASSES: 15
The Server Core Metrics, Management Indicators, Quality Indicators, Life Cycle Expectations Pragmatic Software Metrics, Metrics Automation. Tailoring The Process: Process Discriminates, Example. Future Software Project Management: Modern Project Profiles, Next Generation Software Economics, Modern Process Transitions.		
TEXT BOOKS		
1. Software Project Management, Walker Royce, Pearson Education, 2012 2. Software Project Management, BobHughes & Mike Cotterell, Fifth edition, TataMc-GrawHill, 2006		
REFERENCE BOOKS		
1. Applied Software Project Management, Andrew Stellman & Jennifer Greene, O'Reilly, 2006 2. Head First PMP, Jennifer Greene & Andrew Stellman, O'Reilly, 2007 3. Software Engineering Project Management, Richard H. Thayer & Edward Yourdon, second edition, Wiley India, 2004. 4. Agile Project Management, Jim Highsmith, Pearson education, 2004 5. The art of Project management, Scott Berkun, O'Reilly, 2005. 6. Software Project Management in Practice, PankajJalote, Pearson Education, 2002		
WEB LINKS		
1. https://www.tutorialspoint.com/software_engineering/software_project_management.htm 2. https://ebooks.lpude.in/management/mba/term_4/DCAP304_DCAP515_SOFTWARE_PROJECT_MANAGEMENT.pdf		
MOOCS : https://onlinecourses.nptel.ac.in/noc19_cs70/preview		
EBOOK : https://www.edulearn.net.in/wp-content/uploads/2019/09/MG6088-Software-Project-Management.pdf		

SEMANTIC WEB

III B.TECH - II SEMESTER - (OPEN ELECTIVE - I)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS37	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES								
The course should enable the students to:								
1. Understand the sematic web and its importance. 2. Understand the attribute languages and how to use in semantic web 3. Understand the Rule based languages 4. Understand the different methods to development in internet 5. Learn different types of semantic web tools.								
COURSE OUTCOMES								
Upon successful completion of this course, student will be able to:								
1. Classify the ontologies towards to the internet. 2. Apply the attribute languages on internet. 3. Apply the rule based languages for improving the usage of semantic web. 4. Analyze the methods to development of the internet. 5. Use semantic web tools for efficient development and usage of web.								
UNIT - I	THE FUTURE OF THE INTERNET						CLASSES: 12	
Introduction, The Syntactic Web, The Semantic Web, How the Semantic Web Will Work. Ontology in Computer Science: Defining the Term Ontology, Differences Among Taxonomies, Thesauri, and Ontologies, Classifying Ontologies, Web Ontologies, Web Ontology Description Languages, Ontology, Categories, and Intelligence.								
UNIT - II	KNOWLEDGE REPRESENTATION IN DESCRIPTION LOGIC						CLASSES: 12	
Introduction, An Informal Example, The Family of Attributive Languages, Inference Problems. RDF and RDF Schema: Introduction, XML Essentials, RDF, RDF Schema, A Summary of the RDF/RDF Schema Vocabulary. OWL: Introduction, Requirements for Web Ontology Description Languages, Header Information, Versioning, and Annotation Properties, Properties, Classes, Individuals, Data types, A Summary of the OWL Vocabulary.								
UNIT - III	RULE LANGUAGES						CLASSES: 14	
Introduction, Usage Scenarios for Rule Languages, Datalog, RuleML, SWRL, TRIPLE. Semantic Web Services: Introduction, Web Service Essentials, OWL-S Service Ontology, An OWL-S Example.								
UNIT - IV	METHODS FOR ONTOLOGY DEVELOPMENT						CLASSES: 15	
Introduction, Uschold and King Ontology Development Method, Toronto Virtual Enterprise Method, Methontology, KACTUS Project Ontology Development Method, Lexicon-Based Ontology Development Method, Simplified Methods. Ontology Sources: Introduction, Metadata, Upper Ontologies, Other Ontologic of Interest, Ontology Libraries.								
UNIT - V	SEMANTIC WEB SOFTWARE TOOLS & SOFTWAREAGENTS						CLASSES: 12	
Introduction, Metadata and Ontology Editors, Reasoners, Other tools. Introduction, Agent Forms, Agent Architecture, Agents in the Semantic web Context. Semantic Desktop: Introduction, Semantic Desktop Metadata, Semantic Desktop Ontologies, Semantic Desktop Architecture, Semantic Desktop Related Applications. Ontology Application in Art: Introduction, Ontologies for the Description of Works of Art, Metadata Schemas for The Description of Works of Art, Semantic Annotation of Art Images.								

TEXT BOOKS

1. Semantic Web- Concepts, Technologies and applications, 2007th edition (24 January 2007) K.Breitman, Marco Antonio Casanova and Walter Truszkowski, Springer..

REFERENCE BOOKS

1. Information Sharing on the Semantic Web, Heiner Stuckenschmidt, Frankvan Harmelen, Springer.
2. Semantic Web Primer, Grigoris Antoniou, FrankVan
3. Semantic Web Services: Concepts, Technologies and Applications, Rudi Studer, Stephan Grimm, Andrees Abeker, Springer
4. Towards the Semantic Web: Ontology Driven Knowledge Management, John Davis, Dieter Fensal, Frank Van Harmelen, J.Wiley.

WEB LINKS

1. <https://www.techtarget.com/searchcio/definition/Semantic-Web>
2. <https://www.w3.org/RDF/Metalog/docs/sw-easy>
MOOC:1. <https://www.coursera.org/learn/web-data>
EBOOK:1. <https://open.hpi.de/courses/semanticweb>
3. [http://prof.mau.ac.ir/images/Uploaded_files/A%20Semantic%20Web%20Primer-The%20MIT%20Press%20\(2012\)%5B7460174%5D.PDF](http://prof.mau.ac.ir/images/Uploaded_files/A%20Semantic%20Web%20Primer-The%20MIT%20Press%20(2012)%5B7460174%5D.PDF)

DATABASE MANAGEMENT SYSTEMS

IV B.TECH - I SEMESTER - (OPEN ELECTIVE-II)

Course	Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS09	OEC		L	T	P	C	CIE	SEE	Total
			3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Discuss the basic database concepts, applications, data models, schemas and instances.
2. Design Entity Relationship model for a database.
3. Demonstrate the use of constraints and relational algebra operations.
4. Describe the basics of SQL and construct queries using SQL
5. Understand the importance of normalization in databases.
6. Demonstrate the basic concepts of transaction processing and concurrency control.
7. Understand the concepts of database storage structures and identify the access techniques.

COURSE OUTCOMES

At the end of the course the students are able to:

1. Use the basic concepts of Database Systems in Database design
2. Design a Database using ER Modeling
3. Apply relational algebra and calculus concepts AND SQL for a wide range of database operations
4. Apply SQL queries and PL/SQL queries to interact with Database
5. Grasp transaction management principles including transaction states and concurrency control techniques

UNIT - I	INTRODUCTION	CLASSES: 09
Introduction and applications of DBMS, Purpose of database, Data Independence, Database System architecture- Levels, Database users and DBA. DATABASE DESIGN: Database Design Process, ER Diagrams - Entities, Attributes, Relationships, Constraints, keys, extended ER features, Generalization, Specialization, Aggregation, Conceptual design with the E-R model.		
UNIT - II	RELATIONAL MODEL & SCHEMA REFINEMENT	CLASSES: 09
RELATIONAL MODEL: Introduction to the relational model, Integrity constraints over relations, Enforcing integrity Constraints, Querying relational data, Logical database design: E-R to relational, Introduction to views, Destroying/altering tables and views. SCHEMAREFINEMENT AND NORMAL FORMS: Introduction to schema refinement, functional dependencies, reasoning about FDs. Normal forms: 1NF, 2NF, 3NF, BCNF, properties of decompositions, normalization, schema refinement in database design.		
UNIT - III	RELATIONAL ALGEBRA CALCULUS & SQL	CLASSES: 11
RELATIONAL ALGEBRA AND CALCULUS: Relational algebra operators, relational calculus – Tuple and domain relational calculus. SQL: Basics of SQL, DDL,DML,DCL, structure - creation, alteration, defining constraints - Primary key, foreign key, unique, not null, check, IN operator, Functions - aggregate functions, Built-in functions — numeric, date, string functions, set operations, sub-queries, correlated sub-queries		

UNIT - IV	SQL and PL/SQL	CLASSES: 11
SQL: Use of group by, having, order by clauses, join and its types, Exist, Any, All classes. Transaction control commands - Commit, Rollback, Save point, PL/SOL: Environment, block structure, variables, operators, data types, control structures; cursors, stored procedures, Triggers.		
UNIT - V	TRANSACTION & CONCURRENCY CONTROL	CLASSES: 10
TRANSACTIONS MANAGEMENT: Transaction concept, transaction state, concurrent executions, Serializability, recoverability, testing for serializability. CONCURRENCY CONTROL AND RECOVERY SYSTEM: Concurrency control, lock based protocols, Time-stamp based protocols, validation based protocols, multiple granularity and deadlock handling. Recovery system - failure classification, storage structure, recovery and atomicity, log based recovery.		
TEXT BOOKS		
1. Abraham Silbers Ghatz, Henry F. Korth, S. Sudharshan, "Database System Concepts". Sixth Edition, Tata McGraw Hill, 2011 2. Raghurama Krishnan, Johannes Gehrke, "Database Management Systems", TATA McGraw Hill. 3rd Edition, 2007. 3. R.P. Mahapatra & Govind Verma, Database Management Systems, Khanna Publishing House, 2013. 4. Michael McLaughlin, Oracle Database 11g PL/SOL Programming, Oracle press.		
REFERENCE BOOKS		
1. Peter Rob, Carlos Coronet, Database Systems Design Implementation and Management, 7th edition, 2009. 2. Scott UnTian. Michael McLaughlin, Ron Hardman. "Oracle database 10g PL/SQL programming ". 6th edition, Tata McGraw Hill, 2010 3. S.K Singh, "Database Systems Concepts. Design and Applications", First edition, Pearson Education, 2006. 4. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems". Fourth Edition. Pearson/Addition wesley, 2007		
WEB REFERENCES		
1. https://www.learndb.com/databases/how-to-convert-er-diagram-to-relational-database 2. https://www.w3schools.com/sql/sql_create_table.asp 3. https://www.ibm.com/docs/en/zos-basic-skills?topic=zos-what-is-database-management-system&external_link=true 4. https://www.cs.ucdavis.edu/~green/courses/ecs165a-w11/		
E-TEXT BOOKS		
1. https://www.freebookcentre.net/Database/Free-Database-Systems-Books-Download.html 2. https://www.ddegjust.ac.in/studymaterial/mca-3/ms-11.pdf		
MOOC COURSE		
1. https://www.mooc-list.com/tags/dbms-extensions		

DATA WAREHOUSING AND BUSINESS INTELLIGENCE

IV B.TECH - I SEMESTER - (OPEN ELECTIVE-II)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS43	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. This course is concerned with extracting data from the information systems that deal with the day-to-day operations and transforming it into data that can be used by businesses to drive high-level decision making 2. Students will learn how to design and create a data warehouse, and how to utilize the process of extracting, transforming, and loading (ETL) data into data warehouses.								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. Understand architecture of data warehouse and OLAP operations. 2. Understand Fundamental concepts of BI and Analytics 3. Application of BI Key Performance indicators and Design of Dashboards, Implementation of Web Analytics 4. Understand Utilization of Advanced BI Tools and their Implementation. 5. Implementation of BI Techniques and BI Ethics.								
UNIT - I	DATA WAREHOUSE						CLASSES:12	
DATA WAREHOUSE: Data Warehouse-Data Warehouse Architecture- Multidimensional Data Model-Data cube and OLAP Technology-Data Warehouse Implementation -DBMS schemas for Decision support - Efficient methods for Data cube computation.								
UNIT - II	BUSINESS INTELLIGENCE						CLASSES: 10	
Business Intelligence: Introduction – Definition, Leveraging Data and Knowledge for BI, BI Components, BI Dimensions, Information Hierarchy, Business Intelligence and Business Analytics. BI Life Cycle. Data for BI - Data Issues and Data Quality for BI.								
UNIT - III	BI IMPLEMENTATION						CLASSES: 10	
BI Implementation - Key Drivers, Key Performance Indicators and Performance Metrics, BI Architecture/Framework, Best Practices, Business Decision Making, Styles of BI-vent-Driven alerts-A Cyclic process of Intelligence Creation. The value of Business intelligence -Value driven and Information use.								
UNIT - IV	ADVANCED BI						CLASSES:10	
Advanced BI – Big Data and BI, Social Networks, Mobile BI, emerging trends, Description of different BI-Tools (Pentaho, KNIME)								
UNIT - V	BI IMPLEMENTATION						CLASSES: 12	
Business intelligence implementation-Business Intelligence and integration implementation-connecting in BI systems- Issues of legality- Privacy and ethics- Social networking and BI.								

TEXT BOOKS

1. Data Mining – Concepts and Techniques - JIAWEI HAN & MICHELINE KAMBER, Elsevier.
2. Rajiv Sabherwal “Business Intelligence” Wiley Publications, 2012.

REFERENCE BOOKS

1. Efraim Turban, Ramesh Sharda, Jay Aronson, David King, Decision Support and Business Intelligence Systems, 9th Edition, Pearson Education, 2009.
2. David Loshin, Business Intelligence - The Savy Manager's Guide Getting Onboard with Emerging IT, Morgan Kaufmann Publishers, 2009
3. Philo Janus, Stacia Misner, Building Integrated Business Intelligence Solutions with SQL Server, 2008 R2 & Office 2010, TMH, 2011.
4. Business Intelligence Data Mining and Optimization for decision making [Author: Carlo-Verellis] [Publication: (Wiley)]
5. Data Warehousing, Data Mining & OLAP- Alex Berson and Stephen J. Smith- Tata McGrawHill Edition, Tenth reprint 2007
6. Building the Data Warehouse- W. H. Inmon, Wiley Dreamtech India Pvt. Ltd.
7. Data Mining Introductory and Advanced topics –MARGARET H DUNHAM, PEA.

WEB LINKS

1. <https://www.coursera.org/learn/data-warehousing-business-intelligence>
2. https://www.vssut.ac.in/lecture_notes/lecture1428550844.pdf

ARTIFICIAL INTELLIGENCE AND APPLICATIONS

IV B.TECH - II SEMESTER - (OPEN ELECTIVE-III)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS51	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. The difference between optimal reasoning Vs. human like reasoning
2. Understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities
3. Apply the Expert and rule based systems in knowledge representation techniques
4. Understand the applications of AI: namely Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning.
2. Formulate an efficient problem space for problem solving and Represent knowledge using the appropriate technique
3. Design and develop expert systems using Prolog and solve uncertainty problems.
4. Analyze real world problems and implement using ANN, CNN and AI techniques.
5. Implement the concepts of AI in Healthcare, Robotics and Agriculture.

UNIT - I	INTRODUCTION AND PROBLEM SOLVING	CLASSES: 14
Introduction: History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A*, Constraint Satisfaction, Game Playing, Alpha-Beta Pruning.		
UNIT - II	LOGIC AND KNOWLEDGE REPRESENTATION	CLASSES: 14
Logic Concepts and Logic Programming: Introduction, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming. Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.		
UNIT - III	EXPERT SYSTEM AND APPLICATIONS	CLASSES:14
Expert System and Applications: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs. Traditional Systems, rule based expert systems black board systems truth maintenance systems, Application of Expert Systems.		
UNIT - IV	ANN, CNN	CLASSES:12
Artificial Neural Networks: Introduction, Artificial Neural Networks, Single-Layer Feed Forward Networks, Multi-Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks, Recurrent Networks. Convolutional Neural Networks: Pooling, Padding, Layers used to build ConvNets, Working of CNN, Python Libraries used for AI: Tensor flow, Keras, Theano, Pytorch, Scikit Learn, Pandas, Image Classification using CNN with Python.		

UNIT - V	APPLICATIONS & CASE STUDIES OF AI	CLASSES: 11
Applications of AI in real world, Future of AI, Benefits and Risks of AI. Case Studies: AI in Healthcare: Medical Imaging and Diagnostic, AI in Agriculture: Autonomous Tractors, Controlling Pest Infestations, Soil and Crops Health Monitoring, AI in Robotics: Use of Artificial Intelligence in Robotics		
TEXT BOOKS		
1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011 2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004		
REFERENCE BOOKS		
1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009. 2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson. 3. Introduction to Artificial Intelligence and expert systems Dan W.Patterson. PHI. 4. Artificial Intelligence by George Flugerrearson fifth edition.		
WEB LINKS		
1. https://www.ibm.com/topics/artificial-intelligence#:~:text=At%20its%20simplest%20form%2C%20artificial,in%20conjunction%20with%20artificial%20intelligence2 2. MOOCS: https://www.mygreatlearning.com/great-lakes-artificial-intelligence-and-machine-learning? 3. E BOOK: https://www.freebookcentre.net/CompuScience/Free-Artificial-Intelligence-Books-Download.html		

CYBER SECURITY AND CYBER LAWS

IV B.TECH - II SEMESTER - (OPEN ELECTIVE-III)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CS36	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. To learn cybercrime cyber security and cyber law.
2. To understand the cyber-attacks.
3. To understand information gathering.
4. To learn how to detect and prevent a cyber-attack and tools for mitigating them.
5. Develop Competencies For Dealing With Frauds And Deceptions (Confidence Tricks, Scams) And Other Cyber Crimes.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Explain the basics of cyber security, cybercrime and cyber law.
2. Classify various types of attacks and learn the tools to launch the attacks.
3. Apply various tools to perform information gathering.
4. Understand Cybercrimes and cyber laws.
5. Explore The Legal And Policy Developments In Various Countries To Regulate Cyberspace.

UNIT - I	INTRODUCTION	CLASSES: 16
Cyber Security-definition, Need for Cyber Security, Protection, Security, risk, threat, vulnerability, exploit, attack, confidentiality, integrity, availability, non-repudiation, authentication, authorization, codes, plain text, encryption, decryption, cipher text, key, ciphers, Symmetric and asymmetric cryptography, Public key, private key, Crypt analysis, Cyber forensics.		
UNIT - II	CYBER ATTACKS	CLASSES:14
CYBER ATTACKS (definitions and examples): Denial-of-service attacks, Man-in-the middle attack, Phishing, spoofing and spam attacks, Drive by attack, Password attack, SQL injection attack, Cross-site scripting attack, Eavesdropping attack, Birthday attack, Malware attacks, Social Engineering attacks		
UNIT - III	HANDLING THE ATTACKS	CLASSES: 11
Firewalls, logging and intrusion detection systems, e-mail security, security issues in operating systems, ethics of hacking and cracking.		
UNIT - IV	CYBER CRIMES	CLASSES: 12
Cyber Crime and Information Security – History of Cyber Crime, classifications of Cyber Crimes – Tools and Methods –Password Cracking, Key loggers, Spywares, SQL Injection – Network Access Control – Cloud Security – Web Security – Wireless Security		
UNIT - V	CYBER SPACE AND CYBER LAWS	CLASSES: 16
Cyber Space- Fundamental definitions -Interface of Technology and Law – Jurisprudence and-Jurisdiction in Cyber Space - Indian Context of Jurisdiction - Enforcement agencies – Need for IT act - UNCITRAL – E-Commerce basics .Information Technology Act, 2000 - Aims and Objects — Overview of the Act – Jurisdiction Cyber Law, A Global Perspective on Cyber Crimes; Cyber Laws – The Indian IT Act – Cybercrime and Punishment.		

TEXT BOOKS

1. Anand Shinde, "Introduction to Cyber Security Guide to the World of Cyber Security", Notion Press, 2021.
2. Nina Godbole, Sunit Belapure, "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", Wiley Publishers, 2011.

REFERENCE BOOKS

1. David Kim, Michael G. Solomon, "Fundamentals of Information Systems Security", Jones & Bartlett Learning Publishers, 2013
2. Patrick Engebretson, "The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made easy", Elsevier, 2011
3. Justice Yatindra Singh: Cyber Laws, Universal Law Publishing Co., New Delhi
4. Farouq Ahmed, Cyber Law in India, New Era publications, New Delhi

OPEN ELECTIVES OFFERED BY DEPARTMENT OF ECE

OE1		OE2	
A6EC55	Microprocessors and Interfacing	A6EC57	Microcontrollers and Applications
A6EC56	Principles of Communications	A6EC58	Fundamentals of Image processing
OE3			
A6EC59	Introduction to Sensors and Actuators		
A6EC60	Introduction to Computer Vision		

MICRO PROCESSORS AND INTERFACING

OPEN ELECTIVE COURSE - I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC55	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the basic of 8, 16 bit microprocessor architectures and its functionalities.
2. Develop an assembly language programming skills of various processors.
3. Interface different peripheral devices with microprocessors and microcontrollers.
4. Interface memory devices to 8086 processor.
5. Analyze Serial communication schemes of microprocessor based systems.

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Understand the architecture of microprocessor.
2. Understand the programming model of microprocessors.
3. Interface different external peripherals with microprocessors.
4. Analyze a problem and formulate appropriate computing solution for processor based application.
5. Develop an assembly language program for specified application in communication.

UNIT-I	MICROPROCESSORS ARCHITECTURE	CLASSES: 09
Overview of 8085, 8086 architecture- functional diagram, Register organization, memory segmentation, Memory addresses, physical memory organization.		
UNIT-II	SIGNAL DESCRIPTION OF 8086	CLASSES: 09
Signal description of 8086, timing diagrams, Interrupt structure of 8086, Vector interrupt table, Interrupts of 8086 Processor, Interrupt service routine.		
UNIT-III	INSTRUCTION SET AND ASSEMBLY LANGUAGE PROGRAMMING OF 8086	CLASSES: 09
Programming model, Addressing modes, Instruction set, Assembler directives, Programs involving logical, branch and call instructions, Sorting, evaluating arithmetic expressions, and string manipulations.		
UNIT-IV	I/O INTERFACE	CLASSES: 09
Introduction to 8255 PPI, various modes of operation of 8255 PPI, interfacing 8255 to 8086, Stepper motor interfacing, D/A & A/D converter, Memory interfacing to 8086.		
UNIT-V	SERIAL COMMUNICATION INTERFACE	CLASSES: 09
Serial communication standards, serial data transfer schemes, 8251 USART architecture, Interfacing of 8251/USART to 8086 Processor.		

TEXT BOOKS

1. D.V.Hall, Microprocessors and Interfacing. TMGH, 2nd edition 2006.
2. Advanced microprocessors and peripherals-A.K ray and K.M.Bhurchandani, TMH, 2nd edition 2006.
3. R.S.Gaonkar, Microprocessor Architecture: Programming and Applications with the 8085/8080A, Penram International Publishing, 1996.

REFERENCE BOOKS

1. D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface. Morgan Kaufman Publishers.
2. Micro computer system 8086/8088 family architecture, programming and design- By Liu and GA Gibson, PHI, 2nd Ed.,

WEB REFERENCES

1. <http://www.freebookcentre.net/electronics-ebooks-download/Microprocessor-and-Microcontroller.html>
2. <http://coen.boisestate.edu/smlcoo/smlcoo-courses/ece-332-microprocessors-fall07/lecture-notes/>
3. <http://www.freebookcentre.net/electronics-ebooks-download/Introduction-to-Microcontrollers-Lecture-Notes.html>

E-TEXT BOOKS

1. <http://gen.lib.rus.ec/book/index.php?md5=67C5AC79DC8180A7F0641609D0C7800C>
2. <http://www.faadooengineers.com/threads/9039-8085-microprocessor-by-RAMESH-GANOKAR-ebook-pdf-download>
3. https://e.edim.co/123389964/The_8051_Microcontroller_Architecture_Programming_And_Applications.pdf
4. https://e.edim.co/123389964/A.K._Ray_and_K.M._BhurchandiAdvanced_Microprocessors_and_Peripherals_3e-Tata_Mcgraw_Hill.pdf

MOOC COURSE

1. <https://www.mooc-list.com/tags/microprocessors>
2. <https://www.coursera.org/courses?query=microprocessor>

PRINCIPLES OF COMMUNICATIONS

OPEN ELECTIVE COURSE-I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC56	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
PREREQUISITE: Knowledge on Electron devices and Electronic circuits								
COURSE OBJECTIVES The students are able to: 1. Describe the basic concepts of analog communication 2. Significance of digital & analog communication 3. Compare the digital modulation techniques 4. Learn the various multiple access techniques 5. Understanding the Satellite and Optical fiber concepts.								
COURSE OUTCOMES Upon completion of the course, students will be able to: 1. Understand the concepts of analog modulations 2. Illustrate the basic concepts of digital modulations. 3. Analyse the various digital modulation methodologies. 4. Distinguish the concepts of multiple access techniques. 5. Differentiate the various orbits and associated satellite launches								
UNIT-I	FUNDAMENTALS OF ANALOG COMMUNICATION						CLASSES: 09	
Modulation, Principles of Amplitude Modulation, AM modulator and Demodulator, Types of Amplitude modulation (DSBSC, SSBSC, VSB). Angle modulation FM and PM waveforms, phase deviation and modulation index, frequency deviation and percent modulation.								
UNIT-II	DIGITAL TRANSMISSION						CLASSES: 09	
Introduction, Sampling theorem, Types of pulse modulations, PCM, companding, differential pulse code modulation, delta modulation, adaptive delta modulation, Inter symbol interference.								
UNIT-III	DIGITAL MODULATION TECHNIQUES						CLASSES: 09	
Introductions, Shannon limit for information capacity, ASK FSK, BPSK QPSK and DPSK modulators and demodulators, comparison of digital modulation techniques.								
UNIT-IV	SPREAD SPECTRUM AND MULTIPLE ACCESS TECHNIQUES						CLASSES: 09	
Introduction, Pseudo noise sequence, DS spread spectrum with coherent binary PSK, FH spread spectrum, multiple access techniques-FDMA, TDMA and CDMA.								
UNIT-V	SATELLITE AND OPTICAL COMMUNICATION						CLASSES: 09	
Satellite Communication Systems, Keplers Law, LEO, MEO and GEO Orbits, Link model Optical Communication Systems-Fiber losses-scattering, absorption attenuation and bending losses, types of optical Sources and Detectors.								

TEXT BOOKS

1. Wayne Tomasi, —Advanced Electronic Communication Systems, 6/e, Pearson Education, 2007.
2. Simon Haykin, —Communication Systems, 4th Edition, JohnWiley & Sons., 2001.

REFERENCE BOOKS

1. H.Taub, D L Schilling, G Saha, |Principles of Communication |3/e, 2007.
2. B.P.Lathi, |Modern Analog And Digital Communication systems |, 3/e, Oxford University Press, 2007
3. Blake — Electronic Communication Systems, Thomson Delmar Publications, 2002.
4. Martin S.Roden — Analog and Digital Communication System, 3rd Edition, PHI, 2002.
5. B.Sklar, |Digital Communication Fundamentals and Applications |2/e Pearson Education 2007.

WEB REFERENCES

1. <https://personal.utdallas.edu/~torlak/courses/ee4367/lectures/FIBEROPTICS.pdf>
2. Bricker G (2012) 2-D bar codes, Journal of Computing Sciences in Colleges, **28**:1, (25-32), Online publication date: 1-Oct-2012.

E-TEXT BOOKS

1. https://books.google.co.in/books/about/Principles_Of_Communication.html?id=6Zunu4Acfg8C

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc18_ee26/preview

MICRO CONTROLLER & APPLICATIONS

OPEN ELECTIVE COURSE-II								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC57	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. To introduce the students about architectural features of microcontrollers and its registers 2. To introduce about the instruction set of 8051 3. To know about the I/O Ports and Interrupts of 8051 4. To know about Timers/Counters of 8051 5. To introduce Arduino programming and interfacing of sensors								
COURSE OUTCOMES Upon completion of the course, students will be able to: 1. Understand the architectural features of MCS-51 and select a suitable microcontroller to suit the application. 2. Develop programs for control applications using assembly language and embedded C 3. Use timers and counters for delay generation and event counting and Illustrate the use of interrupts and service routines 4. Write algorithms and develop programs for serial data communication applications. 5. Design microcontroller based-applications for simple real-world applications								
UNIT-I	8051 ARCHITECTURE						CLASSES: 09	
Introduction to microcontrollers, RISC, CISC, Harvard and Von Neumann architectures. Architecture of 8051, Signal descriptions of 8051, General purpose registers of 8051, register banks, Memory organization.								
UNIT-II	INSTRUCTION SET OF 8051						CLASSES: 09	
Assembly language Instruction format, 8051 Addressing modes, Instruction set of 8051: Classification, syntax and function of instructions, Simple programs.								
UNIT-III	I/O PORT AND INTERRUPT PROGRAMMING						CLASSES: 09	
I/O Ports of 8051,Features of I/O ports, configuring I/O ports, interface I/O devices such as LED, buzzer, push-button switch, example programs with assembly & C. Interrupts of 8051 and its priorities, IE and IP registers, Interrupt enabling/disabling and priority setting, example programs in assembly and C.								
UNIT-IV	TIMERS/COUNTERS AND SERIAL I/O						CLASSES: 09	
Bit structure and function of TMOD and TCON registers, Timer/Counter modes of operations, Timer/Counter programs in assembly and C. Bit structure and function of SCON, PCON registers, SBUF register, Serial Communication modes in 8051, programs on serial communication.								
UNIT-V	INTRODUCTION TO ARDUINO						CLASSES: 09	
Introduction to Arduino-uno board, Analog and Digital pins, programming structure of Arduino, introduction to sensors and actuators , Sensor interfacing, programming to sensors, Motor interfacing, LCD interfacing.								

TEXT BOOKS

1. The 8051 Microcontroller(3rd edition) -Kenneth J Ayala
2. The 8051 Microcontroller & Embedded systems using assembly and C (2ndEdition) –M.A.Mazidi , J.C. Mazidi & R.D.McKinlay ISBN: 81-317-1026-2

REFERENCE BOOKS

1. The 8051 Microcontroller(4th Edition)- MacKenzie , ISBN:81-317-2018-7
2. The 8051 Microcontroller(1st Edition) – Dr. Uma Rao & Andhe Paallavi, ISBN: 81-317-3252-5
3. Microcontrollers & applications, Ramani Kalpathi, & Ganesh Raja , ISBN: 81-888-4918-9
4. Programming Arduino: Getting Started with Sketches, Second Edition (Tab) 2nd Edition – Simon Monk

WEB REFERENCES

1. <https://www.the8051microcontroller.com/web-references>

E-TEXT BOOKS

1. <https://www.freebookcentre.net/Electronics/Microcontroller-Books.html>
2. <https://www.freebookcentre.net/Electronics/Microcontroller-Application-Books.html>

MOOC COURSE

1. <https://nptel.ac.in/courses/117/104/117104072/>

FUNDAMENTALS OF IMAGE PROCESSING

OPEN ELECTIVE COURSE - II								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC58	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The students are able to: 1. To become familiar with digital image fundamentals 2. To get exposed to image enhancement techniques in Spatial and Frequency domain. 3. To learn concepts of degradation function and restoration techniques. 4. To study the image segmentation and representation techniques. 5. To become familiar with image compression and recognition methods								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. Know and understand the basics and fundamentals of digital image processing. 2. Operate on images using the techniques of smoothing, sharpening and enhancement. 3. Apply the restoration concepts and filtering techniques on digital images. 4. Learn the basics of segmentation and features extraction. 5. Analyze the compression and recognition methods for color images.								
UNIT-I	INTRODUCTION OF IMAGE PROCESSING						CLASSES: 09	
Digital Image Fundamentals-Elements of visual perception, image sensing and acquisition, image sampling and quantization, basic relationships between pixels – neighbourhood, adjacency, connectivity, distance measures								
UNIT-II	IMAGE ENHANCEMENT						CLASSES: 09	
Spatial Domain: Gray level transformations – Histogram processing – Basics of Spatial Filtering–Smoothing and Sharpening Spatial Filtering, Image Enhancement in Frequency Domain: Low Pass and High Pass Filters in Frequency Domain, Homomorphic filtering								
UNIT-III	IMAGE TRANSFORMS						CLASSES: 09	
Image Transforms: 2 D- Discrete Fourier Transform and its properties, Discrete Cosine Transform (DCT), Haar Transform, Hadamard Transform, Hotelling Transform and Slant transform.								
UNIT-IV	IMAGE SEGMENTATION						CLASSES: 09	
Introduction-Edge detection, Edge linking and boundary detection – Thresholding – Region based segmentation – Region growing – Region splitting and merging- Watershed segmentation algorithm.								
UNIT-V	IMAGE COMPRESSION						CLASSES: 09	
Image Compression Models, Huffman and Arithmetic Coding, Error Free Compression, Lossy Compression, Lossy and Lossless Predictive Coding, Transform Based Compression, JPEG 2000 Standards.								
TEXT BOOKS 1. R.C. Gonzalez and R.E. Woods, Digital Image Processing, Second Edition, Pearson Education 3rd edition 2008 2. Digital Image Processing- S Jayaraman, S Esakkirajan, T Veerakumar- McGraw Hill Edn., 2010.								

REFERENCE BOOKS

1. Anil Kumar Jain, Fundamentals of Digital Image Processing, Prentice Hall of India. 2nd edition 2004
2. Murat Tekalp, Digital Video Processing" Prentice Hall, 2nd edition 2015.
3. S. Sridhar, Digital Image Processing, Oxford University Press, 2nd Ed, 2016.

WEB REFERENCES

1. <http://homepages.inf.ed.ac.uk/rbf/BOOKS/VERNON/Chap004.pdf>

E-TEXT BOOKS

1. https://books.google.co.in/books/about/Digital_Image_Processing.html?id=a62xQ2r_f8wC

MOOC COURSE

1. <https://nptel.ac.in/courses/117105079/>

INTRODUCTION TO SENSORS AND ACTUATORS

OPEN ELECTIVE COURSE-III								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC59	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understanding basic laws and phenomena on which operation of sensors and actuators - Transformation of energy
2. Create analytical design and development solutions for sensors and actuators
3. To know the basic laws of behaviour of sensors and actuators
4. To able to know about the Standards for Smart Sensor Interface
5. Analyse the development and application of sensors and actuators.

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Apply the fundamental physical and technical base of sensors and actuators
2. Analyse various premises, approaches, procedures and results related to sensors and actuators
3. Analyse basic laws and phenomena that define behaviour of sensors and actuators
4. Apply the Smart Sensor Interface in various applications
5. Develop the application of sensors and actuators.

UNIT-I	SENSORS & TRANSDUCERS	CLASSES: 09
Sensors/Transducers: Principles, Classification, Parameters, Characteristics, Environmental Parameters (EP), Characterization. Mechanical and Electromechanical Sensors: Introduction, Resistive Potentiometer, Resistance Strain Gauge, Semiconductor Strain Gauges, Capacitive Sensors, Electrostatic Transducer, Ultrasonic Sensors.		
UNIT-II	THERMAL SENSORS	CLASSES: 09
Thermal Sensors: Introduction, Gas thermometric Sensors, Thermal Expansion Type Thermometric Sensors, Acoustic Temperature Sensor, Dielectric Constant and Refractive Index Thermo-sensors, Nuclear Thermometer, Thermo-EMF Sensors, Thermal Radiation Sensors, Quartz Crystal Thermo-electric Sensors, Spectroscopic Thermometry, Noise Thermometry.		
UNIT-III	RADIATION SENSORS	CLASSES: 09
Introduction – Basic Characteristics – Types of Photosensistors– X-ray and Nuclear Radiation Sensors– Fiber Optic Sensors. Electro Analytical Sensors: Introduction – The Electrochemical Cell – The Cell Potential – Standard Hydrogen Electrode (SHE) – Liquid Junction and Other Potentials – Sensor Electrodes.		
UNIT-IV	SMART SENSORS	CLASSES: 09
Introduction, Primary Sensors, Excitation, Converters, Compensation, Information Coding/Processing, Data Communication, Standards for Smart Sensor Interface, the Automation. Sensors Applications: Introduction, On-board Automobile Sensors (Automotive Sensors), Home Appliance Sensors.		
UNIT-V	ACTUATORS	CLASSES: 09
Actuators: Pneumatic and Hydraulic Actuation Systems- Actuation systems, Pneumatic and hydraulic systems, Directional Control valves, Pressure control valves, Cylinders, Servo and proportional control valves.		

TEXT BOOKS
1. D. Patranabis, "Sensors and Transducers", PHI Learning Private Limited. 2. W. Bolton, "Mechatronics", Pearson Education Limited.
REFERENCE BOOKS
1. Patranabis, "Sensors and Actuators", 2nd Edition, PHI, 2013.
WEB REFERENCES
1. https://www.journals.elsevier.com/sensors-and-actuators
E-TEXT BOOKS
1. https://www.sciencedirect.com/handbook/handbook-of-sensors-and-actuators
MOOC COURSE
1. https://www.classcentral.com/course/swayam-sensors-and-actuators-14285

INTRODUCTION TO COMPUTER VISION

OPEN ELECTIVE COURSE-III								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC60	OEC	L	T	P	C	CIA	SEE	Total
		3		-	3	40	60	100
COURSE OBJECTIVES The course should enable the students to: 1. To review image processing techniques for computer vision. 2. To understand the concepts of Image Enhancement. 3. To understand the basics of segmentation and its applications. 4. To know about the feature extraction and Hough Transform. 5. To understand three-dimensional image analysis techniques.								
COURSE OUTCOMES Upon successful completion of the course, the student is able to 1. Implement fundamental image processing techniques required for computer vision. 2. Analyze the various spatial and frequency domain filtering. 3. Understand the idea about segmentation 4. Apply Hough Transform for line, circle, and ellipse detections. 5. Apply 3D vision techniques and motion related techniques.								
UNIT-I	IMAGE PROCESSING FOUNDATIONS						CLASSES: 09	
Fundamental steps in Digital Image Processing, Components of Digital Image Processing Image sensing and acquisition, Image formation model, Pixels, Basic relationship between pixels, Sampling and Quantization in Digital Image Processing								
UNIT-II	IMAGE ENHANCEMENT						CLASSES: 09	
Intensity transformations, contrast stretching, histogram equalization, Spatial filtering: Smoothing filters, sharpening filters, Frequency domain filtering, Homomorphic filtering,								
UNIT-III	BAICS OF SEGMENTATION						CLASSES: 09	
Edge detection, Thresholding, Region growing, Fuzzy clustering, Watershed algorithm, Active contour models, and Texture feature based segmentation, Graph based segmentation, Wavelet based Segmentation - Applications of image segmentation.								
UNIT-IV	FEATURE EXTRACTION						CLASSES: 09	
First and second order edge detection operators, Phase congruency, Localized feature extraction -detecting image curvature, shape features, Introduction to Hough transform, shape skeletonization, Boundary descriptors, Moments, Texture descriptors.								
UNIT-V	3D IMAGE VISUALIZATION						CLASSES: 09	
Sources of 3D Data sets, Slicing the Data set, Arbitrary section planes, Volumetric display, Stereo Viewing, Ray tracing, Reflection, Surfaces, Multiple connected surfaces, Image processing in 3D, Measurements on 3D images.								

BOOKS

1. Rafael C. Gonzalez, Richard E. Woods, Digital Image Processing', Pearson, Education, Inc., Second Edition, 2004.
2. Mark Nixon, Alberto Aguado, "Feature Extraction and Image Processing", Academic Press, 2008.
3. D. L. Baggio et al., "Mastering OpenCV with Practical Computer Vision Projects", Packt Publishing, 2012.

REFERENCE BOOKS

1. E. R. Davies, "Computer & Machine Vision", Fourth Edition, Academic Press, 2012.
2. Jan Erik Solem, "Programming Computer Vision with Python: Tools and algorithms for analyzing images", O'Reilly Media, 2012.
3. R. Szeliski, "Computer Vision: Algorithms and Applications", Springer 2011.
4. Simon J. D. Prince, "Computer Vision: Models, Learning, and Inference", Cambridge University Press, 2012.

WEB REFERENCES

1. <https://machinelearningmastery.com/what-is-computer-vision/>

E-TEXT BOOKS

1. <https://machinelearningmastery.com/computer-vision-books/>

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc21_ee23/preview

INTRODUCTION TO MOBILE COMMUNICATIONS

OPEN ELECTIVE COURSE-IV								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EC61	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES The course should enable the students to: - Understand mobile radio communication principles and to study the recent trends adopted in cellular systems and wireless standards. - To fill the skill gap in the domain of Cellular Technology. - Get hands on practice in the field of cellular technology and related disciplines. - To appreciate the contribution of wireless Communication networks to overall technological growth. - To provide an overview of wireless Communication networks and its applications in communication engineering.								
COURSE OUTCOMES Upon successful completion of the course, the student is able to - Explain the basics of mobile telecommunication system - Illustrate the generations of telecommunication systems in wireless network - Demonstrate knowledge on : cellular concepts like frequency reuse, fading, equalization, and hand off strategies - Determine the functionality of network layer and Identify a routing protocol for a given Adhoc networks - Implement the functionality of Transport and Application layer								
UNIT-I	INTRODUCTION TO MOBILE COMPUTING						CLASSES: 09	
Applications of Mobile Computing- Generations of Mobile Communication Technologies-MAC Protocols – SDMA-TDMA-FDMA-CDMA								
UNIT-II	MOBILE TELE COMMUNICATION SYSTEM						CLASSES: 09	
GSM–Architecture–Protocols–Connection Establishment–Frequency Allocation –Routing–Mobility Management – Security–GPRS-UMTS-Architecture								
UNIT-III	CELLULAR CONCEPT						CLASSES: 09	
Limitations of conventional mobile system, concept of frequency reuse, cluster size, cellular system Architecture, channel assignment strategies, call hand off strategies-hard hand off and soft hand off, prioritizing hand off.								
UNIT-IV	WIRELESS NETWORKS						CLASSES: 09	
Wireless networks – Advantages and applications of Wireless LAN, WLAN technology – RF and IR wireless LAN, diffuse, IEEE802.11architecture, Physical layer, MAC layer, Introduction to WI-FI, Bluetooth architecture.								
UNIT-V	MOBILE NETWORK AND TRANSPORT LAYER						CLASSES: 09	
Introduction to Mobile IP, requirements, IP packet delivery, Agent discovery, Registration, Tunneling and encapsulation, Optimization, Reverse tunneling; Mobile adhoc networks–Routing, Destination sequence distance vector, Dynamic source routing.								

TEXT BOOKS

1. Theodore S. Rappaport - Wireless Communications Principles and Practice, 2nd Edition, Pearson Education, 2003.
2. Andreas F. Molisch – Wireless Communications, JohnWiley, 2nd Edition, 2006.

REFERENCE BOOKS

1. Kamilo Feher-Wireless Digital Communications, PHI, 2003
2. W.C.Y. Lee - Mobile Cellular Communications, 2nd Edition, McGraw-Hill, 1995.
3. Yi Bing Lin - Wireless and Mobile Network Architectures, 2nd Edition, Wiley, 2008.

WEB REFERENCES

1. https://ptolemy.berkeley.edu/books/leeseshia/releases/LeeSeshia_DigitalV2_2.pdf

E-TEXT BOOKS

1. <https://ptolemy.berkeley.edu/books/leeseshia/>

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc20_ee98/preview

BASICS OF EMBEDDED SYSTEMS DESIGN

OPEN ELECTIVE COURSE-IV								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A5EC62	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Explains about the basic functions, structure, concepts and applications of embedded systems.
2. Describes the Core of the Embedded System
3. Explains about the tools used to develop in an embedded environment
4. Gives the knowledge about the development of embedded software using RTOS.
5. Gives the knowledge about the Advance Architectures

COURSE OUTCOMES

Upon successful completion of the course, the student is able to

1. Understand the selection procedure of Processors in the embedded domain.
2. Design Procedure for Embedded Firmware.
3. Implement the Real time Operating Systems applications in Embedded Systems
4. Analyze the Correlation between task synchronization and latency issues
5. Understand the concepts of Advance architecture

UNIT-I	INTRODUCTION TO EMBEDDED SYSTEMS	CLASSES:09
Definition of Embedded System, Embedded Systems Vs. General Computing Systems, History of Embedded Systems, Classification, Major Application Areas, Purpose of Embedded Systems, Characteristics and Quality Attributes of Embedded Systems		
UNIT-II	TYPICAL EMBEDDED SYSTEM	CLASSES:09
Core of the Embedded System: General Purpose and Domain Specific Processors, ASICs, PLDs, Commercial Off-The-Shelf Components (COTS), Sensors and Actuators.		
UNIT-III	EMBEDDED SOFTWARE DEVELOPMENT TOOLS	CLASSES:09
Host and target machines, linker/locators for embedded software, getting embedded software in to the target system, debugging techniques: Testing on host machine, using laboratory tools.		
UNIT-IV	RTOS BASED EMBEDDED SYSTEM DESIGN	CLASSES:09
Operating System Basics, Types of Operating Systems, Tasks, Process and Threads, Multiprocessing and Multitasking, Task Scheduling. Message queues, mailboxes and pipes.		
UNIT-V	INTRODUCTION TO ADVANCED ARCHITECTURES	CLASSES:09
ARM and SHARC, processor and memory organization and instruction level parallelism; networked embedded systems: bus protocols, I2Cbus and CANbus.		

TEXT BOOKS

1. Introduction to Embedded Systems-Shibu K.V, McGraw-Hill.

REFERENCE BOOKS

1. Embedded Systems-Raj Kamal, TMH.
2. Embedded System Design-Frank Vahid, Tony Givargis, John Wiley.
3. Embedded Systems –Lyla,Pearson,2013
4. An Embedded Software Primer-David E.Simon, Pearson Education.

WEB REFERENCES

1. <https://www.elprocus.com/embedded-system-design/>

E-TEXT BOOKS

1. <https://www.phindia.com/Books/BookDetail/9788120347304/embedded-system-design-chattopadhyay>

MOOC COURSE

1. https://onlinecourses.nptel.ac.in/noc20_ee98/preview

OPEN ELECTIVES OFFERED BY EEE

OE1		OE2	
A6EE55	Electrical Wiring and Safety Measures	A6EE57	Neural Networks and Fuzzy Logic
A6EE56	Electrical Materials	A6EE58	Energy Storage Systems
OE3			
A6EE59	Solar Energy and Applications		
A6EE65	Instrumentation and Control		

ELECTRICAL WIRING AND SAFETY MEASURES

III YEAR II SEMESTER - (OPEN ELECTIVE – I)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE55	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To Study the wiring diagram of residential.
2. To understand the Safety measures of Electrical wiring
3. Apply the Different types of safety rules in Electrical Systems.
4. Design the Different types of Residential Electrification Methods.
5. Identifying Different types of Substations.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. To Analyze the safety measures & state safety precautions.
2. To Understand the Different Methods of Earthlings.
3. Apply the Different types of safety rules in Electrical Systems.
4. Design the Different types of Residential Electrification Methods.
5. Identifying Different types of Substations.

SYLLABUS

UNIT-I	BASICS OF ELECTRICAL INSTALLATIONS	CLASSES: 12
Electric Supply System, Three phase four wire distribution system, Protection of Electric Installation against over load, short circuit and Earth fault, General requirements of electrical installations, testing of installations, Types of loads, Systems of wiring, Service connections, Service Mains, Sub-Circuits, Location of Outlets, Location of Control Switches, Location of Main Board and Distribution board, Guide lines for Installation of Fittings, Load Assessment, Permissible voltage drops and sizes of wires, estimating and costing of Electric installations.		
UNIT-II	EARTHING	CLASSES: 08
Introduction & importance, Factors affecting Earth Resistance, Methods of earthing Substation and Transmission tower earthing, Neutral and Earth wire, Transformer Neutral Earthing.		
UNIT-III	SAFETY & PREVENTION OF ACCIDENTS	CLASSES: 08
Definition of terminology used in safety, I.E. Act & statutory regulations for safety of persons & equipments working with electrical installation. Dos & don'ts for substation operators as listed in IS. Meaning & causes of electrical accidents factors on which severity of shock depends.		
UNIT-IV	RESIDENTIAL BUILDING ELECTRIFICATION	CLASSES: 10
General rules guidelines for wiring of Residential Installation and positioning of equipments. Principles of circuit design in lighting and power circuits. Procedures for designing the circuits and deciding the number of sub-circuits. Method of drawing single line diagram & wiring diagram.		



UNIT-V	RULES : SUBSTATION AND METERS	CLASSES: 12
Rule 28: Voltage level definitions. Rule 30: Service lines & apparatus on consumer premises. Rule 31: Cut-out on consumer's premises. Rule 46: Periodical inspection & testing of consumer's installation. Rule 47: Testing of consumer's installation. Rule 54: Declared voltage of supply to consumer. Rule 55: Declared frequency of supply to consumer. Rule 56: Sealing of meters & cut-outs. Rule 77: Clearances above ground of the lowest conductor. Rule 79: Clearances between conductors & trolley wires. Rule 87: Lines crossing or approaching each other. Rule 88: Guarding.		
TEXT BOOKS		
1. K.B. Raina, S.K.Bhattacharya Electrical Design; Estimating and costing New Age International (p) Limited, New Delhi Surjit Singh. 2. Electrical Estimating and costing Dhanpat Rai and company, New Delhi .J.B.Gupta. 3. A course in Electrical Installation, Estimating & costing, S.K.Kataria & sons, S.L. Uappal. 4. Electrical wiring Estimating and costing, Khanna Publication. ,A.K.Sawhney.		
REFERENCE BOOKS		
1. Electrical Machine Design Danpat Rai & co. 2. The Electricity Rule 2005 Universal Law Publishing Co. Pvt. Ltd. N. Alagapan, S. Ekambaram 3. Electrical Estimating and costing Tata Mc Graw Hill Publication, New Delhi ,Surjit Singh 4. TarlokSibgh Installation, Commissioning & Maintenance of Electrical Equipment, S.K.Kataria & Sons 5. B.V.S.Rao Operation & Maintenance of Electrical Machines Vol I & II, Media Promoters & Publisher Ltd. Mumbai		
WEB REFERENCES		
1. https://electrical-engineering-portal.com > Technical Articles 2. https://www.st-andrews.ac.uk/staff/policy/healthandsafety/publications/electricalsafety/ 3. https://www.cpwd.gov.in/Publication/Internal2013.pdf		
E-TEXT BOOKS		
1. https://books.google.co.in/books?isbn=0323170064 2. https://www.jove.com/science.../electrical-safety-precautions-and-basic-equipment		
MOOC COURSE		
1. https://nptel.ac.in/courses/103106071/5 2. https://nptel.ac.in/courses/108108099/28 3. https://nptel.ac.in/courses/124107001/		

ELECTRICAL MATERIALS

III Year II Semester - (OPEN ELECTIVE – I)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE56	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To analyze the various Conducting materials.
2. To Explain the various Semi conducting materials.
3. To Explain The Dielectrics and Insulators.
4. To Explain the Different Magnetic Materials.
5. To Describe the Optical properties of Fibers

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Analyze the various Conducting materials.
2. Explain the various Semi conducting materials.
3. Explain The Dielectrics and Insulators.
4. Explain the Different Magnetic Materials.
5. Describe the Optical properties of Fibers.

SYLLABUS

UNIT-I	CONDUCTORS	CLASSES: 10
Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.		
UNIT-II	SEMICONDUCTORS	CLASSES: 08
General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semi-conductors, extrinsic Semi-conductors, hall effect, drift, mobility, diffusion in Semiconductors, and their applications, superconductors.		
UNIT-III	DIELECTRICS AND INSULATORS	CLASSES: 12
Properties Of Gaseous, Liquid And Solid Dielectric, Dielectric As A Field Medium, Electric Conduction In Gaseous, Liquid And Solid Dielectric, Breakdown In Dielectric Materials, Mechanical And Electrical Properties Of Dielectric Materials, Effect Of Temperature On Dielectric Materials, Polarization, Loss Angle And Dielectric Loss, Petroleum Based Insulating Oils, Transformer Oil, Capacitor Oils, Properties, Solid Electrical Insulating Materials, Fibrous, Paper Boards, Yarns, Cloth Tapes, Sleeving Wood, Impregnation, Plastics, Filling And Bounding Materials, Fibrous, Film, Mica, Rubber, Mica Based Materials, Ceramic Materials, Classification Of Insulation (Solid) And Application In AC And DC Machines.		



UNIT-IV	MAGNETIC MATERIALS	CLASSES: 10
Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magnetostriction.		
UNIT-V	OPTICAL PROPERTIES OF SOLIDS	CLASSES: 10
Photo emission, photo emission materials, electro luminescence junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).		
TEXT BOOKS		
1. "Electrical Engineering Materials", Dekker, PHI Pbs. 2. "Electrical Engineering Materials", Indulkar, S.Chand		
REFERENCE BOOKS		
1. "Electrical Engineering Materials", Tareev 2. "Electrical Engineering Materials", Yu. Koritsky. 3. "Electrical Engineering Materials", R.K.Rajput, Laxmi Pbs		
WEB REFERENCES		
1. https://physics.info/dielectrics/ 2. https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265 3. web.mit.edu/course/6/6.732/www/6.732-pt2.pdf		
E-TEXT BOOKS		
1. https://easyengineering.net/electrical-engineering-materials-by-dekker/ 2. https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/		
MOOC COURSE		
1. https://nptel.ac.in/courses/108108076/ 2. https://nptel.ac.in/courses/112104203/3 3. https://onlinecourses.nptel.ac.in/noc18_ee14/		

NEURAL NETWORKS AND FUZZY LOGIC

IV YEAR I SEMESTER - (OPEN ELECTIVE – II)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE57	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the fundamental theory and concepts of neural networks, neuro-modeling, several neural network paradigms and its applications.
2. To understand the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic control and other machine intelligence applications of fuzzy logic.
3. To understand the basics of an evolutionary computing paradigm known as genetic algorithms and its application to engineering optimization problems.

COURSE OUTCOMES

The students should be able to

1. Comprehend the concepts of feed forward neural networks
2. Analyze the various feedback networks.
3. Understand the concept of fuzziness involved in various systems and fuzzy set theory.
4. Comprehend the fuzzy logic control and adaptive fuzzy logic and to design the fuzzy control using genetic algorithm.
5. Analyze the application of fuzzy logic control to real time systems.

SYLLABUS

UNIT-I	INTRODUCTION & ESSENTIALS TO NEURAL NETWORKS	CLASSES: 12
Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin-Huxley Neuron Model, Integrate-and-Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCullochPiUs Model, Historical Developments, Potential Applications of ANN. Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN — Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application		
UNIT-II	SINGLE & MULTI LAYER FEED FORWARD NEURAL NETWORKS	CLASSES: 10
Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training. Algorithms: Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications. Credit Assignment Problem, Generalized Delta Rule, and Derivation of Back-propagation (BP) Training, Summary of Back-propagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements.		
UNIT-III	ASSOCIATIVE MEMORIES-I	CLASSES: 08
Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory).		



UNIT-IV	ASSOCIATIVE MEMORIES-II	CLASSES: 10
Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem. Architecture of Hopfield Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hopfield Network Summary and Discussion of Instance/Memory Based Learning Algorithms, Applications		
UNIT-V	FUZZY LOGIC	CLASSES: 10
Classical & Fuzzy Sets: Introduction to classical sets – properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions. Fuzzy Logic System Components: Fuzzification, Membership value assignment, development of rule base and decision making system, De-fuzzification to crisp sets, De-fuzzification methods.		
TEXT BOOKS		
1. Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications, Rajasekharan and Pal, PHI. 2. Neural Networks and Fuzzy Logic, C. Naga Bhaskar, G. Vijay Kumar, BS Publications.		
REFERENCE BOOKS		
1. Artificial Neural Networks, B. Yegnanarayana, PHI. 2. Artificial Neural Networks, Zaruda, PHI. 3. Neural Networks and Fuzzy Logic System, Bail Kosko, PHI. 4. Fuzzy Logic and Neural Networks, M. Amirhavalili, Scitech Publications India Pvt. Ltd. 5. Neural Networks, James A Freeman and Davis Skapura, Pearson Education. 6. Neural networks by Satish Kumar, TIVIH, 2004 7. Neural Networks, Simon Hakens, Pearson Education. 8. Neural Engineering, C.Eliasmith and CH.Anderson, PHI.		
WEB REFERENCES		
1. users.monash.edu/~app/CSE5301/Lnts/LaD.pdf 2. https://engineering.purdue.edu/~tsoukala/rational.html 3. https://pdfs.semanticscholar.org/5e31/c55a00eb3945e3e483caa2e146a95c12f5aa.pdf		
E-TEXT BOOKS		
1. https://www.mheducation.co.in/computer.../neural-networks-fuzzy-systems/text-book 2. www.crectirupati.com/sites/default/files/lecture_notes/NNFL.pdf 3. www.vssut.ac.in/lecture_notes/lecture1423723637.pdf		
MOOCCOURSE		
1. https://nptel.ac.in/noc/individual_course.php?id=noc19-ge07 2. https://nptel.ac.in/courses/108104049/16 3. https://nptel.ac.in/courses/117105084/ 4. https://nptel.ac.in/syllabus/127105006/		

ENERGY STORAGE SYSTEMS

IV Year I Semester - (OPEN ELECTIVE – II)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE58	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To enable the student to understand the need for energy storage, devices and technologies available and their applications.
2. To Analyze the characteristics of energy from various sources and need for storage
3. To Classify various types of energy storage and various devices used for the purpose
4. To Explain the Pumped hydro Storage System.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Analyze the characteristics of energy from various sources and need for storage
2. Classify various types of energy storage and various devices used for the purpose
3. Explain the Pumped hydro Storage System.
4. Identify various real time applications.

SYLLABUS

UNIT-I	ELECTRICAL ENERGY STORAGE TECHNOLOGIES	CLASSES: 12
Characteristics of electricity, Electricity and the roles of EES, High generation cost during peak-demand periods, Need for continuous and flexible supply, Long distance between generation and consumption, Congestion in power grids, Transmission by cable.		
UNIT-II	NEEDS FOR ELECTRICAL ENERGY STORAGE	CLASSES: 08
Emerging needs for EES, More renewable energy, less fossil fuel, Smart Grid uses, The roles of electrical energy storage technologies, The roles from the viewpoint of a utility, The roles from the viewpoint of consumers, The roles from the viewpoint of generators of renewable energy.		
UNIT-III	FEATURES OF ENERGY STORAGE SYSTEMS	CLASSES: 08
Classification of EES systems , Mechanical storage systems, Pumped hydro storage (PHS), Compressed air energy storage (CAES), Flywheel energy storage (FES), Electrochemical storage systems, Secondary batteries, Flow batteries, Chemical energy storage, Hydrogen (H ₂), Synthetic natural gas (SNG).		
UNIT-IV	TYPES OF ELECTRICAL ENERGY STORAGE SYSTEMS	CLASSES: 10
Electrical storage systems, Double-layer capacitors (DLC), Superconducting magnetic energy storage (SMES), Thermal storage systems, Standards for EES, Technical comparison of EES technologies.		



UNIT-V	APPLICATIONS	CLASSES: 12
Present status of applications, Utility use (conventional power generation, grid operation & service) , Consumer use (uninterruptable power supply for large consumers), New trends in applications, Renewable energy generation, Smart Grid, Smart Micro grid, Smart House, Electric vehicles, Management and control hierarchy of storage systems, Internal configuration of battery storage systems, External connection of EES systems, Aggregating EES systems and distributed generation (Virtual Power Plant), Battery SCADA–aggregation of many dispersed batteries.		
TEXT BOOKS		
1. James M. Eyer, Joseph J. Iannucci and Garth P. Corey, “Energy Storage Benefits and Market Analysis”, Sandia National Laboratories, 2004. 2. The Electrical Energy Storage by IEC Market Strategy Board.		
REFERENCE BOOKS		
1. Jim Eyer, Garth Corey, Energy Storage for the Electricity Grid: Benefits and Market Potential Assessment Guide, Report, Sandia National Laboratories, Feb 2010.		
WEB REFERENCES		
1. https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118991978.hces212		
E-TEXT BOOKS		
1. https://www.pewtrusts.org/~media/.../energy_storage-backs_up_power_supply.pdf 2. https://energy.mit.edu/wp-content/uploads/2018/04/Energy-Storage-for-the-Grid.pdf 3. https://www.adb.org/sites/default/files/.../handbook-battery-energy-storage-system.pdf		
MOOC COURSE		
1. nptel.ac.in/courses/112105221/56 2. nptel.ac.in/courses/108108036/9 3. https://nptel.ac.in/courses/108102047/7		

SOLAR ENERGY AND APPLICATIONS

IV YEAR II SEMESTER - (OPEN ELECTIVE – III)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE59	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To Study about solar modules and PV system design and their applications
2. To Deal with grid connected PV systems
3. To Discuss about different energy storage systems

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Understanding of principles and technologies for solar thermal energy collection, conversion and utilization
2. Understanding of solar heating systems, liquid based solar heating systems for buildings.
3. Identify, formulate and solve simple to complex problems of solar thermal energy conversion and storage.
4. Identify and understand solar thermal systems 'components and their function.
5. Analyze hot water load and solar resource data and use this information to properly size a solar thermal system

SYLLABUS

UNIT-I	INTRODUCTION	CLASSES: 10
Characteristics of sunlight – semiconductors and P-N junctions –behavior of solar cells – cell properties – PV cell interconnection		
UNIT-II	STAND ALONE PV SYSTEM	CLASSES: 10
Solar modules – storage systems – power conditioning and regulation - MPPT- protection – standalone PV systems design – sizing		
UNIT-III	GRID CONNECTED PV SYSTEMS	CLASSES: 10
PV systems in buildings – design issues for central power stations – safety – Economic aspect – Efficiency and performance - International PV programs		
UNIT-IV	ENERGY STORAGE SYSTEMS	CLASSES: 10
Impact of intermittent generation – Battery energy storage – solar thermal energy storage – pumped hydroelectric energy storage		
UNIT-V	APPLICATIONS	CLASSES: 10
Water pumping – battery chargers – solar car – direct-drive applications –Space – Telecommunications.		



TEXT BOOKS
1. Solanki C.S., "Solar Photovoltaics: Fundamentals, Technologies And Applications", PHI Learning Pvt. Ltd.,2015. 2. Stuart R.Wenham, Martin A.Green, Muriel E. Watt and Richard Corkish, "Applied Photovoltaics", 2007,Earthscan, UK. Eduardo Lorenzo G. Araujo, "Solar electricity engineering of photovoltaic systems", Progenia,1994
REFERENCE BOOKS
1. Frank S. Barnes & Jonah G. Levine, "Large Energy storage Systems Handbook", CRC Press, 2011. 2. McNeils, Frenkel, Desai, "Solar & Wind Energy Technologies", Wiley Eastern, 1990 5 S.P. Sukhatme , "Solar Energy", Tata McGraw Hill,1987.
WEB REFERENCES
1. https://www.loc.gov/rr/scitech/tracer-bullets/solar-updatetb.html 2. https://www.oxfordreference.com/view/10.1093/oi/authority.20110803100516798 3. https://link.springer.com/journal/11949
E-TEXT BOOKS
1. https://courses.edx.org/c4x/DelftX/ET.3034TU/asset/solar_energy_v1.1.pdf 2. bookstore.teri.res.in/books/9788179935736
MOOC COURSE
1. https://nptel.ac.in/courses/112105051/ 2. https://nptel.ac.in/courses/121106014/18 3. https://nptel.ac.in/courses/112105050/ 4. https://nptel.ac.in/syllabus/112105051/

INSTRUMENTATION AND CONTROL

IV Year II Semester - (OPEN ELECTIVE – III)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6EE65	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To introduce the basic principles of all measuring instruments
2. To deal with the measurement of voltage, current measurements
3. To understand students how different types of meters work and their construction
4. To understand the different ways of system representations such as Transfer function representation and state space representations and to assess the system dynamic response
5. To assess the system performance using time domain analysis and methods for improving it

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Identify the instruments suitable for typical measurements.
2. Apply the knowledge about transducers and instrument transformers to use them effectively.
3. Improve the system performance by selecting a suitable controller for a specific application
4. Apply various time domain techniques to assess the system performance

SYLLABUS

UNIT-I	INTRODUCTION TO MEASURING INSTRUMENTS	CLASSES: 10
Classification —deflecting, control and damping torques — Ammeters and Voltmeters — PMMC, moving iron type instruments — expression for the deflecting torque and control torque — Errors and compensations, extension of range using shunts and series resistance. Electrostatic Voltmeters-electrometer type and attracted disc type — Extension of range of E.S. Voltmeters.		
UNIT-II	POTENTIOMETERS & INSTRUMENT TRANSFORMERS	CLASSES: 08
Principle and operation of D.C. Crompton's potentiometer—standardization — Measurement of unknown resistance, current, voltage. A.C. Potentiometers: polar and coordinate types standardization — applications. CT and PT — Ratio and phase angle errors.		
UNIT-III	TRANSDUCERS & OSCILLOSCOPES	CLASSES: 12
Definition of transducers, Classification of transducers, Advantages of Electrical transducers, Characteristics and choice of transducers; Principle operation of LVDT and capacitor transducers; LVDT Applications, Strain gauge and its principle of operation, gauge factor, Thermistors, Thermo couples, Piezo electric transducers, photovoltaic, photo conductive cells, photo diodes. CRO: Cathode ray oscilloscope-Cathode ray tube-time base generator- horizontal and vertical amplifiers-CRO probes-applications of CRO - Measurement of phase and frequency-lissajous patterns.		
UNIT-IV	CLASSIFICATION OF CONTROL SYSTEMS	CLASSES: 10
Concepts of Control Systems- Open Loop and closed loop control systems and their differences- Different examples of control systems- Classification of control systems Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations – Impulse Response and transfer functions. Block diagram algebra – Representation by Signal flow graph – Reduction using mason's gain formula.		



UNIT-V	TIME RESPONSE ANALYSIS	CLASSES: 10
Standard test signals – Time response of first order systems –Characteristic Equation of Feedback control systems, Transient response of second order systems – Time domain specifications .Effects of proportional derivative, proportional integral systems.		
TEXT BOOKS		
1. Electrical and Electronic Measurements and Instrumentation, R. K. Rajput, S. Chand & Company Ltd. 2. Electrical Measuring Instruments and Measurements, S. C. Bhargava, BS Publications. 3. I. J. Nagrath and M. Gopal", "Control Systems Engineering", New Age International (P) Limited, Publishers, 5th edition, 2009 4. "B. C. Kuo", "Automatic Control Systems", John wiley and sons, 8th edition, 2003.		
REFERENCE BOOKS		
1. Electrical & Electronic Measurement & Instruments, A.K.Sawhney Dhanpat Rai & Co. Publications. 2. Electrical and Electronic Measurements, G. K. Banerjee, PHI Learning Pvt. Ltd. 3. Electrical Measurements and Measuring Instruments, Golding and Widdis, Reem Publications. 4. N. K. Sinha", "Control Systems", New Age International (P) Limited Publishers, 3rdEdition, 1998. 5. "NISE", "Control Systems Engineering", John wiley, 6th Edition, 2011. 6. "Katsuhiko Ogata", "Modern Control Engineering", Prentice Hall of India Pvt. Ltd., 3rd edition, 1998.		
WEB REFERENCES		
1. home.mit.bme.hu/~virosztek/docs/mt.../Principles_of_electrical_measurement.pdf 2. https://www.mccdaq.com/handbook/chapt_4.aspx		
E-TEXT BOOKS		
1. www.vssut.ac.in/lecture_notes/lecture1423813026.pdf 2. www.vssut.ac.in/lecture_notes/lecture1423904331.pdf 3. www.ent.mrt.ac.lk/~rohan/teaching/EN5001/Reading/DORFCH1.pdf		
MOOC COURSE		
1. https://nptel.ac.in/syllabus/108106070/ 2. https://nptel.ac.in/courses/108105064/ 3. https://nptel.ac.in/courses/108101037/		



OPEN ELECTIVES OFFERED BY INFORMATION TECHNOLOGY

OE1		OE2	
A6IT28	Adhoc & Sensor Networks	A6IT30	Security Testing
A6IT35	Object Oriented Programming	A6IT36	Human Computer Interaction
OE3			
A6IT37	Introduction to Computer Networks		
A6IT31	High Performance Computing		

AD-HOC & SENSOR NETWORKS

III Year II Semester - (OPEN ELECTIVE – I)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT28	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the concepts of sensor networks
2. To understand the MAC and transport protocols for ad hoc networks
3. To understand the security of sensor networks
4. To understand the applications of adhoc and sensor networks

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Know the principles and characteristics of wireless sensor networks.
2. Use proactive routing protocols function and their implications on data transmission delay and bandwidth consumption.
3. Find issues in MAC Layer Protocols, transport layer protocols in MANETS and WSNs.
4. Find security issues and familiar with security key management mechanisms MANETs and WSNs.
5. Practice network simulations on various simulators.

SYLLABUS

UNIT-I	INTRODUCTIONS TO AD HOC NETWORKS	CLASSES: 10
Introduction to Ad Hoc Networks - Characteristics of MANETs, Applications of MANETs and Challenges of MANETs. Routing in MANETs - Criteria for classification, Taxonomy of MANET routing algorithms, Topology- based routing algorithms- Proactive : DSDV; Reactive : DSR, AODV; Hybrid: ZRP; Position-based routing algorithms- Location Services -DREAM, Quorum-based; Forwarding Strategies : Greedy Packet, Restricted Directional Flooding-DREAM, LAR.		
UNIT-II	BROADCASTING TECHNIQUES	CLASSES: 10
Data Transmission - Broadcast Storm Problem, Rebroadcasting Schemes -Simple-flooding, Probability-based Methods, Area-based Methods, Neighbor Knowledge-based: SBA, Multipoint Relaying, AHBP. Multicasting : Tree-based : AMRIS, MAODV; Mesh-based : ODMRP, CAMP; Hybrid : AMRoute, MCEDAR.		
UNIT-III	GEOCASTING AND PROTOCOLS	CLASSES: 10
Geocasting : Data-transmission Oriented-LBM; Route Creation Oriented-GeoTORA, MGR. TCP over Ad Hoc TCP protocol overview, TCP and MANETs, Solutions for TCP over Ad hoc		
UNIT-IV	BASICS OF WIRELESS SENSOR NETWORKS	CLASSES: 10
Basics of Wireless, Sensors and Lower Layer Issues : Applications, Classification of sensor networks, Architecture of sensor network, Physical layer, MAC layer, Link layer, Routing Layer.		

UNIT-V	ISSUES OF WIRELESS SENSOR NETWORKS	CLASSES: 10
Upper Layer Issues of WSN: Transport layer, High-level application layer support, Adapting to the inherent dynamic nature of WSNs, Sensor Networks and mobile robots.		
TEXT BOOKS		
1. Ad Hoc and Sensor Networks – Theory and Applications, Carlos Corderio Dharma P. Aggarwal, World Scientific Publications, March 2006, ISBN – 981–256–681–3. 2. Wireless Sensor Networks: An Information Processing Approach, Feng Zhao, Leonidas Guibas, Elsevier Science, ISBN – 978-1-55860-914-3 (Morgan Kauffman).		
REFERENCE BOOKS		
1. Advanced Selenium Web Accessibility Testing, Software Automation Testing Secrets Revealed, Narayanan Palani, 2019 2. Automated Software Testing A Complete Guide - 2020 Edition-by Gerardus Blokdyk (Author). 3. Software Automation Testing Secrets Revealed Part 2 Selenium Webdriver 4. Mastering Web Automated Testing -Use of Selenium with Practical Answers to Testing Interview Questions Narayanan Palani, 2016. 5. J F DiMarzio, "Beginning Android Programming with Android Studio", 4th Edition, Wiley India Pvt Ltd, 2016. ISBN-13: 978-8126565580		
WEB LINKS		
1. https://www.guru99.com/selenium-tutorial.html. 2. https://www.browserstack.com/guide/automation-testing-tutorial		



OBJECT ORIENTED PROGRAMMING

III Year II Semester - (OPEN ELECTIVE – I)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT35	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course should enable the students to:

1. Understand the basic object oriented programming concepts and apply them in problem solving.
2. Illustrate inheritance and polymorphism concepts for reusing the program.
3. Demonstrate on the exception handling mechanism
4. Use multithreading concepts to develop inter process communication
5. Understand the basics of java collection framework

COURSE OUTCOMES

At the end of the course, student will be able to

1. Understand the basic programming concepts and apply them in problem solving
2. Implement object-oriented principles, such as encapsulation, inheritance, polymorphism, and abstraction.
3. Demonstrate the user defined exceptions by exception handling keywords (try, catch, throw, throws and finally).
4. Develop multithreaded applications with synchronization.
5. Build real world applications using Collection framework

SYLLABUS

UNIT-I	JAVA BASICS	CLASSES: 12
JAVA BASICS: Review of Object oriented concepts, History of Java, Java buzzwords, JVM architecture, Data types, Variables, arrays, operators, control statements, type conversion and casting, constructors, methods, Static block, Static Method ,String Handling, Scanner class		
UNIT-II	INHERITANCE, POLYMORPHISM, PACKAGES AND INTERFACES	CLASSES: 11
INHERITANCE AND POLYMORPHISM: Basic concepts, Types of inheritance, Member access rules, Usage of this and Super key word, Polymorphism, Abstract classes, Encapsulation, getter and setter methods, Dynamic method dispatch, Usage of final keyword. PACKAGES AND INTERFACES: Defining package, Access protection, importing packages, Defining and Implementing interfaces, and Extending interfaces		
UNIT-III	EXCEPTION HANDLING AND FILES	CLASSES: 10
EXCEPTION HANDLING: Exception types, Usage of Try, Catch, Throw, Throws and Finally keywords, Built-in Exceptions, Creating own Exception classes. I/O STREAMS AND FILES: Concepts of streams, Stream classes- Byte and Character stream, Reading console Input and Writing Console output, IO/Serialization, File Handling.		
UNIT-IV	MULTITHREADING	CLASSES: 10
MULTI THREADING: Concepts of Thread, Thread life cycle, creating threads using Thread class and Runnable interface, Synchronization, Thread priorities, Inter Thread communication, Concurrency.		



UNIT-V	COLLECTION FRAMEWORK	CLASSES: 10
COLLECTION FRAMEWORK: Introduction to Java Collections, Overview of Java Collection frame work, Generics, Commonly used Collection classes- Array List, Vector, Hash table, Stack, Enumeration, Iterator.		
TEXT BOOKS		
1. Herbert Schildt and Dale Skrien, <i>Java Fundamentals – A comprehensive Introduction</i>, McGraw Hill, 1st Edition, 2013. 2. Herbert Schildt, <i>Java the complete referencell</i>, McGraw Hill, Osborne, 7th Edition, 2011. 3. T.Budd, <i>Understanding Object- Oriented Programming with Javall</i>, Pearson Education, Updated Edition (New Java 2 Coverage), 1999.		
REFERENCE BOOKS		
1. P.J.Dietel and H.M.Dietel, <i>Java How to program</i>, Prentice Hall, 6th Edition, 2005. 2. P.Radha Krishna, <i>Object Oriented programming through Javall</i>, CRC Press, 1st Edition, 2007. 3. S.Malhotra and S.Choudhary, <i>Programming in Javall</i>, Oxford University Press, 2nd Edition, 2014.		



SECURITY TESTING

IV Year I Semester - (OPEN ELECTIVE – II)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT30	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100

COURSE OUTCOMES

The course will enable the students to:

1. Understand the basic principles and techniques of how attackers can enter computer systems.
2. Able to analyse the data breaches and audits of information technology security.
3. Able to understand the nitty-gritties of pen tests for educational purposes.
4. Evaluate the societal role of hacking from a social, ethical and economic standpoint

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Explain the basic principles and techniques of how attackers can enter computer systems.
2. Perform analyses of data breaches and audits of information technology security.
3. Evaluate the strengths and weaknesses of various information technology solutions regarding data security.
4. Present and perform demonstrations of pen tests for educational purposes.
5. evaluate the societal role of hacking from a social, ethical and economic standpoint

SYLLABUS

UNIT-I	INTRODUCTION TO OWASP	CLASSES: 10
OWASP: Introduction to web applications security, threats and OWASP principles, introduction to secure design, web server: introduction a secure setup of apache, firewalling a server Browser: general concepts, functionalities, browsers war, configuration (HTTP-cookies, contents, scripting etc. attack to browsers, and users tracking/profiling (third party cookies, super cookies, XSS, CSFR, Command Injection), browser security (add-ons, plugins, same-origin policy etc.) & secure browsing		
UNIT-II	OWASP Privacy preserving	CLASSES: 08
OWASP Privacy preserving: attacks to privacy, (spyware & backdoors, browser, email etc.) Tracking techniques: (HTTP cookies, third party cookies, browser fingerprinting, CSP) Advanced browser configuration, anonymity and onion routing (Tor). Internet E-mail: Architecture and infrastructure, functions, agents and standards, MIME & PGP, phishing, spamming & spoofing, DKIM, SPF, introduction to email forensics.		
UNIT-III	Introduction to Ethical Hacking Terminology	CLASSES: 12
Introduction to Ethical Hacking Terminology -Five stages of hacking-Vulnerability Research-Legal implication of hacking Impact of hacking. Foot printing & Social engineering.		
UNIT-IV	Information gathering methodologies	CLASSES: 10
Information gathering methodologies - Competitive Intelligence- DNS Enumerations- Social Engineering attacks. Scanning & Enumeration Port Scanning-Network Scanning- Vulnerability Scanning- NMAP scanning tool- OS Finger printing Enumeration. System Hacking Password.		



UNIT-V	Sniffers & SQL Injection	CLASSES: 10
Sniffers & SQL Injection Active and passive sniffing - ARP Poisoning- Session Hijacking- DNS Spoofing- Conduct SQL Injection attack – Counter measures. Cracking techniques- Key loggers- Escalating privileges- Hiding Files-Steganography technologies- Countermeasures.		
TEXT BOOKS		
1. Kimberly Graves, "CEH: Official Certified Ethical Hacker Review Guide", Wiley Publishing Inc., 2007.ISBN: 978-0-7821-4437-6.		
REFERENCE BOOKS		
1. Shakeel Ali & Tedi Heriyanto, "Backtrack -4: Assuring security by penetration testing", PACKT Publishing., 2011. ISBN: 978-1-849513-94-4.		

HUMAN COMPUTER INTERACTION

IV Year I Semester - (OPEN ELECTIVE – II)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT36	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To learn the foundations of Human Computer Interaction
2. To become familiar with the design technologies for individuals and persons with disabilities.
3. To be aware of mobile HCI.
4. To learn the guidelines for user interface
5. To get an understanding of web interface design

COURSE OUTCOMES

Upon successful completion of the course, the student is able to learn

1. Design effective dialog for HCI
2. Design effective HCI for individuals and persons with disabilities.
3. Assess the importance of user feedback.
4. Explain the HCI implications for designing multimedia/ecommerce/e-learning Web sites.
5. Develop meaningful user interface

SYLLABUS

UNIT-I	FOUNDATION OF HCI	CLASSES: 12
The Human: I/O channels, Memory, Reasoning and problem solving; The computer: Devices, Memory, processing and networks; Interaction: Models, frameworks, Ergonomics, styles, elements, interactivity- Paradigms.		
UNIT-II	DESIGN & SOFTWARE PROCESS	CLASSES: 12
Interactive Design: Basics, process, scenarios, navigation, screen design, Iteration and prototyping. HCI in software process: Software life cycle, usability engineering, Prototyping in practice, design rationale. Design rules: principles, standards, guidelines, rules. Evaluation Techniques, Universal Design.		
UNIT-III	MODELS AND THEORIES	CLASSES: 10
HCI Models: Cognitive models: Socio-Organizational issues and stakeholder requirements – Communication and collaboration models-Hypertext, Multimedia and WWW.		
UNIT-IV	MOBILE HCI	CLASSES: 12
Mobile Ecosystem: Platforms, Application frameworks- Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools. - Case Studies.		
UNIT-V	WEB INTERFACE DESIGN	CLASSES: 09
Designing Web Interfaces: Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow - Case Studies		



TEXT BOOKS

1. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, —Human Computer InteractionII, 3rd Edition, Pearson Education, 2004 (UNIT I, II & III)
2. Brian Fling Mobile Design and DevelopmentII, First Edition, O'Reilly Media Inc., 2009 (UNIT – IV)

REFERENCE BOOKS

1. Bill Scott and Theresa Neil, —Designing Web InterfacesII, First Edition, O'Reilly, 2009. (UNIT-V)
2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech,
3. User Interface Design, Soren Lauesen, Pearson Education.

INTRODUCTION TO COMPUTER NETWORKS

IV Year II Semester - (OPEN ELECTIVE – III)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT37	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. Understand and apply different network commands.
2. Analyse different networking functions and features for implementing optimal solutions
Apply different networking concepts for implementing network solution
3. Implement different network protocols.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Apply the basics of Physical layer in real time applications
2. Apply data link layer concepts, design issues, and protocols
3. Apply Network layer routing protocols and IP addressing
4. Implement the functions of Application layer and Presentation layer paradigms and Protocols

SYLLABUS

UNIT-I	INTRODUCTION TO COMPUTER NETWORKS	CLASSES: 10
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Introduction:

Network Types, LAN, MAN, WAN, Network Topologies Reference models- The OSI Reference Model- the TCP/IP Reference Model - A Comparison of the OSI and TCP/IP Reference Models, OSI Vs. TCP/IP, Lack of OSI models success

Physical Layer –Introduction to Guided Media- Twisted-pair cable, Coaxial cable and Fiber optic cable and unguided media: Wireless-Radio waves, microwaves, infrared.

Modern Networks – History of Internet, Wi-Fi Networks, Blue-Tooth and Edge Computing

UNIT-II	DATA LINK LAYER	CLASSES: 10
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Data link layer: Design issues, Framing: fixed size framing, variable size framing, flow control, error control, error detection and correction codes, CRC, Checksum: idea, one's complement internet checksum, services provided to Network Layer

Elementary Data Link Layer protocols: simplex protocol, Simplex stop and wait, Simplex protocol for Noisy Channel.

Sliding window protocol: One bit, Go back N, Selective repeat-Stop and wait protocol, Data link layer in HDLC: configuration and transfer modes, frames, control field, point to point protocol (PPP): framing transition phase, multiplexing, multi link PPP.

UNIT-III	MULTIPLE ACCESS TECHNIQUES USED	CLASSES: 10
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Media Access Control: Random Access: ALOHA, Carrier sense multiple access (CSMA), CSMA with Collision Detection, CSMA with Collision Avoidance, Controlled Access: Reservation, Polling, Token Passing, **Channelization:** frequency division multiple Access (FDMA), time division multiple access (TDMA), code division multiple access (CDMA).

Wired LANs: Ethernet, Ethernet Protocol, Standard Ethernet, Fast Ethernet (100 Mbps), Gigabit Ethernet, 10 Gigabit Ethernet.



UNIT-IV	NETWORK LAYER DESIGN ISSUES	CLASSES: 10
The Network Layer Design Issues – Store and Forward Packet Switching-Services Provided to the Transport layer- Implementation of Connectionless Service-Implementation of Connection Oriented Service- Comparison of Virtual Circuit and Datagram Networks, Routing Algorithms-The Optimality principle-Shortest path, Flooding, Distance vector, Link state, Hierarchical, Congestion Control algorithms-General principles of congestion control, Congestion prevention policies, Approaches to Congestion Control-Traffic Aware Routing- Admission Control-Traffic Throttling-Load Shedding. Traffic Control Algorithm: Leaky bucket & Token bucket. Internet Working: How networks differ- How networks can be connected- Tunneling, internetwork routing-, Fragmentation, network layer in the internet – IP protocols-IP Version 4 protocol-IPV4 Header Format, IP addresses, Class full Addressing, CIDR, NAT-, Subnets-IP Version 6-The main IPV6 header, Transition from IPV4 to IPV6, Comparison of IPV4 & IPV6- Internet control protocols- ICMP-ARPDHCP		
UNIT-V	TRANSPORT LAYER	CLASSES: 10
The Transport Layer: Transport layer protocols: Introduction-services- port number-User data gram protocol- User datagram-UDP services-UDP applications-Transmission control protocol: TCP services- TCP features-Segment- A TCP connection- windows in TCP- flow control-Error control, Congestion control in TCP. Application Layer: World Wide Web: HTTP, Electronic mail-Architecture- web based mail- email security-TELENET-local versus remote Logging-Domain Name System: Name Space, DNS in Internet, Resolution- Caching- Resource Records- DNS messages- Registrars-security of DNS Name Servers, SNMP		
TEXT BOOKS		
1. Computer Networks — Andrew S Tanenbaum, Fifth Edition. Pearson Education/PHI 2. Data Communications and Networks – Behrouz A. Forouzan, Fifth Edition TMH.		
REFERENCE BOOKS		
1. Data Communications and Networks- Achut S Godbole, Atul Kahate 2. Computer Networks, Mayank Dave, CENGAGE		

HIGH PERFORMANCE COMPUTING

IV Year II Semester - (OPEN ELECTIVE – III)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6IT31	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To learn about Modern Processors and concepts
2. To understand the concepts of Optimizations
3. To learn about Parallel Computers and Programming
4. To study about Memory Parallel Programming using Open MP and MPI

COURSE OUTCOMES

At the end of the course, student will be able to:

1. Transform algorithms in the computational area to efficient programming code for modern computer architectures
2. Write, organise and handle programs for scientific computations.
3. Use tools for performance optimisation and debugging and Analyse code with respect to performance.
4. Suggest and implement performance improvements
5. Write, organise and handle programs for MIP

SYLLABUS

UNIT-I	MODERN PROCESSORS	CLASSES: 09
Stored-program computer architecture, General-purpose cache-based microprocessor architecture, Memory hierarchies, Multicore processors, Multi-threaded processors, Vector processors.		
UNIT-II	BASIC OPTIMIZATION TECHNIQUES FOR SERIAL CODE	CLASSES: 09
Scalar profiling, Common sense optimizations, Simple measures, large impact, The role of compilers, C++ optimizations		
UNIT-III	PARALLEL COMPUTERS	CLASSES: 09
Taxonomy of parallel computing paradigms, Shared-memory computers, Distributed-memory computers, Hierarchical (hybrid) systems, Networks, Need for parallelization, Parallelism, Parallel scalability		
UNIT-IV	SHARED MEMORY PARALLEL PROGRAMMING WITH OPENMP	CLASSES: 08
introduction to Open MP, Parallel execution, Data scoping, Open MP work sharing for loops, Synchronization, Reductions, Loop scheduling, Tasking, Miscellaneous, Open MP-parallel Jacobi algorithm, Open MP: Wave front parallelization.		
UNIT-V	DISTRIBUTED-MEMORY PARALLEL PROGRAMMING WITH MPI	CLASSES: 09
Message passing, introduction to MPI, A simple example, Messages and point-to-point communication, Collective communication, Nonblocking point-to-point communication, Virtual topologies, MPI parallelization of a Jacobi solver, MPI implementation, Performance properties		



TEXT BOOKS

1. Georg Hager, Gerhard Welle in, "Introduction to High Performance Computing for Scientists and Engineers", Chapman & Hall/CRC Computational Science series, 2011.

REFERENCE BOOKS

1. Charles Severance, Kevin Dowd, "High Performance Computing", O'Reilly Media, 2nd Edition, 1998.
2. Kai Hwang, Faye Alaye Briggs, "Computer Architecture and Parallel Processing", McGraw Hill, 1984

OPEN ELECTIVES OFFERED BY MECH

OE1		OE2	
A6ME55	Basics of Mechanical Engineering	A6ME65	Fundamentals of Operations Research
A6ME60	Fundamentals of Engineering Materials	A6ME70	Introduction to Industrial Engineering
OE3			
A6ME75	Renewable Energy Sources		
A6ME80	Elements of Electric and Hybrid Vehicles		

BASICS OF MECHANICAL ENGINEERING

III Year II Semester - (OPEN ELECTIVE – I)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME55	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the basic concepts of mechanics of rigid and deformable bodies.
2. To know the fundamentals of metallurgy and material science.
3. To be able to conceptualize the principles of thermodynamics.
4. To know the basic elements of fabrication processes and equipment.
5. To know the basic concepts of CAD/CAM.

COURSE OUTCOMES

At the end of the course, the student should be able to:

1. Demonstrate basic concepts of mechanics of rigid and deformable bodies.
2. Explain fundamentals of metallurgy and material science.
3. Discuss basic concepts and principles of thermodynamics.
4. Demonstrate the elements of fabrication processes and equipment.
5. Explain basic concepts of CAD/CAM.

UNIT - I	MECHANICS	CLASSES: 10
MECHANICS OF RIGID BODIES: Force, Force system, Classification of force systems, triangle law, parallelogram law–derivation and problems, polygon law, resolution of forces in coplanar concurrent force system–problems; Definition of friction, laws of friction, coefficient of friction, types of friction, angle of friction, angle of repose, Definition of centroid and center of gravity, problems on centroid of plane areas; Definition of area moment of inertia, parallel axis theorem, perpendicular axis theorem, simple problems on area moment of inertia. MECHANICS OF DEFORMABLE BODIES: Stress, Strain, Types of stresses and strains, Poisson's ratio, stress-strain curve of mild steel, Hooke's Law, factor of safety, mechanical properties of materials, bulk modulus, shear modulus, bars of uniform and varying sections-simple problems.		
UNIT - II	METALLURGY & MATERIAL SCIENCE	CLASSES: 08
Bonds in Solids, crystallization of metals, Classification of steels, Necessity of alloying, types of solid solutions, lever rule, phase rule, cooling curve of pure iron, Iron-Iron carbide equilibrium diagram. Heat treatment–Annealing, quenching, normalizing, Hardening, tempering; Ceramic materials-classification, properties and applications.		
UNIT - III	THERMO DYNAMICS	CLASSES: 12
System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic view points, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry. First law of Thermodynamics – Corollaries, First law applied to a Closed System, applied to a flow system, Steady Flow Energy Equation and Limitations of the First Law, Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements, Carnot cycle and its specialties.		



UNIT - IV	MANUFACTURING	CLASSES: 10
Casting-Types of casting, advantage of casting and its applications, defects in castings; Patterns - Pattern making, Types, Materials used for patterns, pattern allowances; Welding-classification, Heat affected zone in welding, welding defects-causes and remedies. Basic extrusion process and its characteristics, Forging operations and principles; Basic machining operations-drilling, milling, turning, step turning, threading, grinding and shaping; Limits, fits and tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system.		
UNIT - V	CAD/CAM	CLASSES: 10
Fundamentals of CAD/CAM – Benefits of CAD, Computer configuration for CAD applications, Computer peripherals for CAD; Wire frame modeling: Definition, advantages, disadvantages, wire frame entities– analytic entities and synthetic entities. Surface modeling: Definition, advantages, disadvantages, surface entities–analytic entities and synthetic entities; Solid Modeling: Definition, constructive solid geometry, advantages, modeling entities.		
TEXT BOOKS		
1. K. Vijay Kumar Reddy and J. Suresh Kumar, Singer's Engineering Mechanics, B.S Publications 2. Avner, Introduction to physical metallurgy, Tata McGraw-Hill Edition. 3. P.K Nag, Thermodynamics, McGraw Hill Education. 4. R.K Jain, Production Technology, Khanna Publishers. 5. Ibrahim Zeid & R.Sivasubramanian, CAD/CAM Theory & Practice, McGraw Hill Education.		
REFERENCE BOOKS		
1. R.K. Bansal, Engineering Mechanics, Laxmi Publications 2. O.P. Khanna, Material science and metallurgy, Dhanpat Ray Publications 3. Pakirappa, V.Naveen Kumar, V.Naresh, Engineering Thermodynamics, Durga Publishing House 4. P.C Sharma, Production Technology, S.CHAND. 5. P. Radha Krishnan, S. Subramanyan and V. Raju, CAD/CAM, New Age International Publishers. 6. Sadhu Singh, Strength of Materials, Khanna Publishers.		

FUNDAMENTALS OF ENGINEERING MATERIALS

III B. TECH.II SEM - (OPEN ELECTIVE – I)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME60	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand basic structure and crystal arrangement of materials
2. To understand the phase diagrams
3. To understand the advantages of heat treatment processes
4. To understand the application of composite materials.

COURSE OUTCOMES

At the end of the course, the student should be able to:

1. Explain basic concepts of crystal structure such as unit cells, crystal systems of metals etc.
2. Demonstrate the concept of alloying and formation of different types of phases in alloys.
3. Differentiate ferrous and non-ferrous alloys.
4. Explain various heat treatment processes.
5. Classify and explain polymers, ceramics and composites.

UNIT-I	CRYSTAL STRUCTURE	CLASSES: 10
Unit cells, crystal systems of metals, Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, determination of grain size, effect of grain size on the properties of alloys.		
UNIT-II	ALLOYS & PHASE DIAGRAMS	CLASSES: 09
Alloys- substitutional and interstitial solid solutions. Phase diagrams: Interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron Iron-carbide phase diagram and microstructure of ledeburite, austenite, ferrite and cementite.		
UNIT-III	FERROUS AND NON FERROUS ALLOYS	CLASSES: 10
Alloying of steel, properties of stainless steel and tool steels, maraging steels; cast irons - grey, white, malleable and spheroidal cast irons. Copper and copper alloys- brass, bronze and cupro-nickel; Aluminum and Aluminum alloys.		
UNIT-IV	HEAT TREATMENT OF STEEL	CLASSES: 08
Annealing, tempering, normalizing and spheroidising, austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.		



UNIT-V	POLYMERS, CERAMICS AND COMPOSITES	CLASSES: 12
Classification, properties and applications of polymers, ceramics, composites and nano materials.		
TEXT BOOKS		
1. V. Raghavan, "Material Science and Engineering", Prentice Hall of India Private Limited, 1999. 2. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011. 3. Sidney H. Avener, Introduction to Physical Metallurgy, 2nd edition, Tata McGraw hill education (P) Ltd, New Delhi, India, 2007.		
REFERENCE BOOKS		
1. W. D. Callister, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley India, 2006. 2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002. 3. V. D. Kodgire, Material Science and Metallurgy for engineers, 1st Edition, Everest, Pune, India, 2006.		

FUNDAMENTALS OF OPERATIONS RESEARCH

IV B. TECH.I SEM - (OPEN ELECTIVE-II)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME65	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To introduce the basic concepts of Operational Research
2. Understand about the types of models and linear programming, transportation and assignment problem
3. To understand the sequencing, game theory and inventory to the students.

COURSE OUTCOMES

At the end of the course, the student should be able to:

1. Describe types of models and solve linear programming problem.
2. Solve transportation and assignment problems.
3. Analyze sequencing and replacement models and apply them for optimization.
4. Apply gaming theory for optimal decision making.
5. Analyze inventory models to optimize the cost

UNIT-I	INTRODUCTION & ALLOCATION	CLASSES: 10
Development – Definition– Characteristics and Phases – Types of models – Operations Research models – applications. ALLOCATION: Linear Programming Problem – Formulation – Graphical solution – Simplex method –Artificial variables techniques: Two–phase method, Big-M method; Duality Principle.		
UNIT-II	TRANSPORTATION & ASSIGNMENT PROBLEMS	CLASSES: 10
TRANSPORTATION PROBLEM: Formulation – Optimal solution, unbalanced transportation problem – Degeneracy. ASSIGNMENT PROBLEM: Formulation – Optimal solution – Variants of Assignment Problem; Traveling Salesman problem.		
UNIT-III	SEQUENCING & REPLACEMENT	CLASSES: 12
SEQUENCING: Introduction – Flow –Shop sequencing – n jobs through two machines – n jobs through three machines. REPLACEMENT: Introduction – Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely.		
UNIT-IV	THEORY OF GAMES	CLASSES: 10
THEORY OF GAMES: Introduction –Terminology– Solution of games with saddle points and without saddle points– 2×2 games – $m \times 2$ & $2 \times n$ games – graphical method – $m \times n$ games - dominance principle.		
UNIT-V	INVENTORY	CLASSES: 08
INVENTORY: Introduction – Single item, Deterministic models – Types – Purchase inventory models with one price break and multiple price breaks –Stochastic models – demand discrete variable or continuous variable – Single Period model with no setup cost.		



TEXT BOOKS

1. J.K. Sharma, Operation Research: Theory and Applications, 5th Edition, MacMillian India Ltd., 2012.
2. A.C.S. Kumar, Operations Research: Quantitative Analysis for Business Decisions, 1st Edition, Yes Dee Publications, 2015.

REFERENCE BOOKS

1. A.M.Natarajan, P.Balasubramaniam, A. Tamilarasi, Operations Research, 2nd Edition, Pearson Education India, 2014.
2. Harvey M. Wagner, Principles of Operations Research, 2nd Edition, Prentice Hall India Learning Pvt. Ltd., 1980.
3. Frederick S. Hillier, Gerald J. Lieberman, Bodhibrata Nag, Preetam Basu, Introduction to Operations Research, 11th Edition, Tata McGraw Hill India, 2021.

INTRODUCTION TO INDUSTRIAL ENGINEERING

IV B. TECH.I SEM - (OPEN ELECTIVE-II)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME70	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the Industrial engineering principles, tools and concepts.
2. To know concepts like define, design, build, operate and improve production processes that convert resources to high quality products or services effectively, efficiently and safely.
3. To know the various types of plant layouts and their maintenance.

COURSE OUTCOMES

At the end of the course, the student should be able to:

1. Describe the tools and concepts of industrial engineering principles.
2. Differentiate various types of plant layouts and maintenance.
3. Apply the principles of method study and time study to analyze and optimize work processes for increased effectiveness.
4. Apply Statistical Methods for Quality Control and explain Total Quality Management principles.
5. Use the techniques like human resource management, industrial relations, wage incentive plans, Value engineering for the effective enterprise resource planning.

UNIT-I	INTRODUCTION	CLASSES: 10
Definition of industrial engineering (I.E), development, applications, role of an industrial engineer, differences between production management and industrial engineering, quantitative tools of IE and productivity measurement. concepts of management, importance, functions of management, scientific management, Taylor's principles, theory X and theory Y, Fayol's principles of management.		
UNIT-II	PLANT LAYOUT	CLASSES: 12
Factors governing plant location, types of production layouts, advantages and disadvantages of process layout and product layout, applications, quantitative techniques for optimal design of layouts, plant maintenance, preventive and breakdown maintenance.		
UNIT-III	WORK STUDY	CLASSES: 08
Importance, types of production, applications, work study, method study and time study, work sampling, PMTS. Micro-motion study, rating techniques, MTM, work factor system, principles of Ergonomics, flow process charts, string diagrams and Therbligs.		
UNIT-IV	STATISTICAL QUALITY CONTROL	CLASSES: 10
Quality control, Queuing assurance and its importance, SQC, attribute sampling inspection with single and double sampling, Control charts – X and R – charts X and S charts and their applications, numerical examples. TOTAL QUALITY MANAGEMENT: zero defect concept, quality circles, implementation, applications, ISO quality systems. Six sigma – definition, basic concepts.		



Concept of human resource management, personnel management and industrial relations, functions of personnel management, Job-evaluation, its importance and types, merit rating, quantitative methods, wage incentive plans, types.
VALUE ANALYSIS: Value engineering, implementation procedure, enterprise resource planning and supply chain management.

TEXT BOOKS

1. O.P Khanna, Industrial Engineering and management, 2nd Edition, Dhanpat Rai publications, 2018.
2. Martand Telsang, Industrial Engineering and Production Management, 3rd Edition, S.Chand & Company Ltd. New Delhi, 2013.

REFERENCE BOOKS

1. Bhattacharya D. K., Industrial Management, 2nd Edition, Vikas publishers, 2011.
2. J.G Monks, Operations Management, 3rd Edition, McGraw-Hill Publishers, 2007.
3. T.R. Banga, S.C.Sharma, N. K. Agarwal, Industrial Engineering and Management Science, 12th Edition, Khanna Publishers, 2008.

RENEWABLE ENERGY SOURCES

IV B. TECH.II SEM - (OPEN ELECTIVE-III)

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME75	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand the various energy sources and their availability.
2. To know the knowledge of solar radiation and its applications.
3. To analyse the performance of wind energy conversion systems.
4. To understand the Biomass conversion techniques and Geothermal sources and resources

COURSE OUTCOMES

At the end of the course, the student should be able to:

1. Demonstrate various energy sources and their availability.
2. Demonstrate the knowledge of solar radiation and its applications.
3. Analyse the performance of wind energy conversion systems.
4. Explain Biomass conversion techniques and Geothermal sources and resources
5. Explain various methods for harvesting the ocean energy sources.

UNIT-I	ENERGY SOURCES & THEIR AVAILABILITY	CLASSES: 10
Energy Sources and their Availability - Importance of Renewable Energy Resources – Present energy scenario – World energy status – Energy scenario in India – Classification of RES. Introduction to Solar, Wind, Geothermal, Bio-mass, Ocean Energy Sources, Comparison of these Energy Sources, Advantages and Limitations of RES.		
UNIT-II	INTRODUCTION TO SOLAR RADIATION & ITS APPLICATIONS	CLASSES: 10
Introduction to Solar radiation - beam and diffuse radiation, solar constant, measurement of solar radiation. Solar Energy Storage – Collectors: Flat plate and Concentrating collectors, Focusing and non-focusing type. Applications of Solar Energy: Solar water heater – Solar dryer – Solar distillation – Solar Photo-Voltaic Cell.		
UNIT-III	PRINCIPLE OF WIND ENERGY CONVERSION	CLASSES: 12
Principle of Wind Energy Conversion – Basic components of Wind Energy Conversion Systems, Wind mill Components, Various Types and their Constructional Features. Effect of Density, Frequency Variances, Angle of attack, and Wind Speed – Design Considerations of Horizontal and Vertical Axis Wind Machines.		
UNIT-IV	BIOMASS CONVERSION TECHNIQUES , GEOTHERMAL SOURCES AND RESOURCES	CLASSES: 10
Biomass conversion techniques – Biogas Generation – Types of biogas plants – Advantages and disadvantages of biogas plants – Types of Bio-digester – Advantages and disadvantages. Geothermal Sources like hydrothermal, geo-pressured hot dry rock, magma – Simple geothermal Power plant.		
UNIT-V	TIDAL ENERGY & OCEAN THERMAL ENERGY CONVERSION	CLASSES: 08
Tidal Energy – Principle of working, performance and limitations. Wave Energy – Principle of working. Ocean Thermal Energy Conversion – OTEC power plants – working principle and Limitations.		



TEXT BOOKS

1. G. D. Rai, Non-Conventional Energy Sources, 4th Edition, Khanna Publishers, 2000.
2. B H Khan, Non-Conventional Energy Resources, 2nd Edition, Tata McGraw Hill Education Pvt. Ltd, 2011

REFERENCE BOOKS

1. S. Hasan Saeed and D.K. Sharma, Non-Conventional Energy Resources, 3rd Edition, S.K. Kataria & Sons, 2012.
2. N.K. Bansal, M. Kleemann, M. Heliss, Renewable energy sources and conversion technology, Tata McGraw Hill, 1990.
3. S. P. Sukhatme, Solar Energy Principles and Application, Tata McGraw Hill Education Pvt. Ltd, 2009.

ELEMENTS OF ELECTRIC AND HYBRID VEHICLES

IV B. TECH. II SEM - (OPEN ELECTIVE-III)								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6ME80	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
COURSE OBJECTIVES The course will enable the students to: 1. To understand the impact of conventional vehicles on the society and different types of drive train topologies 2. To Evaluate the performance and energy consumption in electric vehicles 3. To understand the assessment, configuration and architecture in hybrid vehicles								
COURSE OUTCOMES At the end of the course, the student should be able to: 1. Analyze the impact of conventional vehicles on the society and different types of drive train topologies 2. Evaluate the performance and energy consumption in electric vehicles 3. Assess the configuration and architecture in hybrid vehicles 4. Choose the type of motor used in electric and hybrid electric vehicles 5. Design the drive train suitable for electric and hybrid vehicles								
UNIT-I	INTRODUCTION TO HYBRID ELECTRIC VEHICLES					CLASSES: 10		
History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies								
UNIT-II	ELECTRIC VEHICLES					CLASSES: 08		
Configurations of EVs, Performance of EVs, Tractive Effort in Normal Driving, Energy Consumption.								
UNIT-III	HYBRID ELECTRIC VEHICLES					CLASSES: 10		
Concept of Hybrid Electric Drive Trains. Architectures of Hybrid Electric Drive Trains.								
UNIT-IV	ELECTRIC PROPULSION SYSTEMS					CLASSES: 12		
Introduction, DC Motor Drives, Induction Motor Drives, Permanent Magnetic BLDC Motor Drives, SRM Drives								
UNIT-V	DESIGN PRINCIPLE OF SERIES AND PARALLEL HYBRID ELECTRIC DRIVE TRAIN					CLASSES: 10		
Design Principles of a Series (Electrical Coupling) Hybrid Drive Train: Operation Patterns, Control Strategies, Design Example. Design Principles of Parallel (Mechanically Coupled) Hybrid Electric Drive Train: Drive Train Configuration and Design Objectives, Control Strategies, Parametric Design of a Drive Train.								



TEXT BOOKS

1. Mehrdad Ehsani, Yimi Gao, Sebastian E. Gay, Ali Emadi, Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press, 2004.
2. Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.

REFERENCE BOOKS

1. James Larminie, John Lowry, Electric Vehicle Technology Explained, Wiley, 2003.
2. Allen Fuhs, Hybrid Vehicles and the future of personal transportation, CRC Press, 2011.

OPEN ELECTIVES OFFERED BY HSM

OE1		OE2	
A6HS08	Business Economics and Financial Analysis	A6HS11	Advanced Entrepreneurship
A6HS09	Basics of Entrepreneurship		
OE3			
A6HS12	Management Science		

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS

VI SEMESTER: OPEN ELECTIVE-I

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS08	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations
2. To understand the demand and supply, production function, cost analysis, markets, forms of business organizations, capital budgeting and financial accounting and financial analysis.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Understand the market dynamics namely, demand and supply, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures.
2. Gain an insight into how production function is carried out to achieve least cost combination of inputs and cost analysis.
3. Develop an understanding of Analyse how capital budgeting decisions are carried out.
4. Understanding the framework for both manual and computerised accounting process
5. Know how to analyse and interpret the financial statements through ratio analysis.

UNIT-I	INTRODUCTION & DEMAND ANALYSIS	CLASSES: 12
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Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.

UNIT-II	PRODUCTION & COST ANALYSIS	CLASSES: 12
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Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance.

UNIT-III	MARKETS & NEW ECONOMIC ENVIRONMENT	CLASSES: 12
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Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario.

UNIT-IV	CAPITAL BUDGETING	Classes: 12
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Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).

UNIT-V	INTRODUCTION TO FINANCIAL ACCOUNTING & FINANCIAL ANALYSIS	CLASSES: 12
Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, Trial Balance - Final Accounts (Tracing Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.		
TEXT BOOKS		
1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009. 2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013. 3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.		
REFERENCE BOOKS		
1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012. 2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012. 3. Lipsey & Chrystel, Economics, Oxford University Press, 2012. 4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012. 5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012. 6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012. 7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012. 8. Dwivedi: Managerial Economics, Vikas, 2012. 9. Shailaja & Usha: MEFA, University Press, 2012. 10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012. 11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011. 12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.		

BASIC OF ENTREPRENEURSHIP

VI SEMESTER: OPEN ELECTIVE-I								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS09	Foundation	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 50	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 50			
COURSE OBJECTIVES The course should enable the students to: 1. Understand and discover entrepreneurship 2. Build a strong foundation for students to start, build, and grow a viable and sustainable venture 3. Develop an entrepreneurial outlook and mindset , critical skills and knowledge 4. Mitigate three types of risks : Customer, Business Model, and Product/Technical								
COURSE OUTCOMES Upon successful completion of this course, student will be able to: 1. Entrepreneurship and Innovation minors will be able to sell themselves and their ideas. Students 2. Master oral and visual presentation skills and establish a foundation of confidence in the skills necessary to cause others to act. 3. The ability to imagine, recognize, and seize opportunities for innovation and new venture creation in the arts and culture sector 4. The know-how to bring capital to ideas in both commercial and nonprofit domains.								
UNIT-I	DISCOVER YOURSELF AND IDENTIFY PROBLEMS WORTH SOLVING						CLASSES: 10	
Discover Yourself: Find your flow, Effectuation, Case Study: Tristan Walker: The extroverted introvert, Identify your entrepreneurial style, Identify Problems Worth Solving: What is a business opportunity and how to identify it, Find problems around you that are worth solving, Methods for finding and understanding problems - (Observation, Questioning, DT, Jobs to be done (JTBD), How to run problem interviews to understand the customer's worldview, Introduction to Design Thinking – Process and Examples, Generate ideas that are potential solutions to the problem identified – DISRUPT, Class Presentation: Present the problem you "love"								
UNIT-II	CUSTOMER, BUSINESS MODEL, VALIDATION						CLASSES: 10	
CUSTOMER :Identify Your Customer Segments and Early Adopters - The difference between a consumer and a customer (decision maker), Market Types, Segmentation and Targeting, Defining the personas; Understanding Early Adopters and Customer Adoption Patterns, Identify the innovators and early adopters for your startup; Craft Your Value Proposition - Come up with creative solutions for the identified problems, Deep dive into Gains, Pains and “Jobs-To- Be-Done” (using Value Proposition Canvas, or VPC), Identify the UVP of your solution using the Value Proposition section of the VPC, Outcome-Driven Innovation, Class Presentation: Communicating the Value Proposition- 1 min Customer Pitch. BUSINESS MODEL: Get Started with Lean Canvas. VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.								
UNIT-III	MONEY AND TEAM						CLASSES: 12	
MONEY: Revenue Streams - Basics of how companies make money, Understand income, costs, gross and net margins, Identify primary and secondary revenue, streams ; Pricing and Costs - Value, price, and costs; Different pricing, Understand product costs and operations costs; Basics of unit costing								



strategies; Financing Your New Venture - How to finance business ideas, Various sources of funds available to an entrepreneur and pros and cons of each, What investors expect from you, Practice Pitching to Investors and Corporates.

TEAM: Team Building - Shared Leadership, Role of a good team in a venture's success; What to look for in a team; How do you ensure there is a good fit? Defining clear roles and responsibilities, How to pitch to candidates to join your startup, Explore collaboration tools and techniques - Brainstorming, Mind mapping, Kanban Board, Slack.

UNIT-IV	MARKETING & SALES	CLASSES: 10
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MARKETING & SALES: Positioning - Understand the difference between product and brand and the link between them, Define the positioning statement for your product/service and how it should translate into what your customers should see about that brand in the market place.

Channels & Strategy: Building Digital Presence and leveraging Social media, Creating your company profile page, Measuring the effectiveness of selected channels, Budgeting and planning.

UNIT-V	PLANNING & TRACKING	CLASSES: 08
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Sales Planning : Understanding why customers buy and how buying decisions are made; Listening, Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features, Follow-up and closing a sale; Asking for the sale

Planning & Tracking: Importance of project management to launch and track progress, Understanding time management, workflow, and delegation of tasks.

Business Regulation: Basics of business regulations of starting and operating a business; Importance of being compliant and keeping proper documentation, How to find help to get started.

TEXT BOOKS

1. S. R. Bhowmik & M. Bhowmik, Entrepreneurship, New Age International, 2007.
2. Steven Fisher, Ja-nae' Duane, The Startup Equation -A Visual Guidebook for Building Your Startup, Indian Edition, Mc Graw Hill Education India Pvt. Ltd, 2016.
3. D F Kuratko and T V Rao —Entrepreneurship-A South-Asian Perspective —Cengage Learning, 2012. (For PPT, Case Solutions Faculty may visit : login.cengage.com)

REFERENCE BOOKS

1. Vasant Desai — Small Scale industries and entrepreneurship|| Himalaya publishing 2012.
2. Rajeev Roy — Entrepreneurship|| 2e, Oxford, 2012.
3. B.Janakiram and M.Rizwana || Entrepreneurship Development: Text & Cases, Excel Books, 2011.
4. Stuart Read, Effectual Entrepreneurship, Routledge, 2013.
5. Nandan H, Fundamentals of Entrepreneurship, PHI, 2013

WEB REFERENCES

1. <http://freevideolectures.com/Course/3641/Entrepreneurship-Through-the-Lens-of-Venture-Capital>
2. <http://www.onlinevideolecture.com/?course=mba-programs&subject=entrepreneurship>
3. http://nptel.ac.in/courses/122106032/Pdf/7_4.pdf
4. <https://www.scribd.com/doc/21516826/Entrepreneurship-Notes>
5. <http://freevideolectures.com/Course/3514/Economics-/Management-/Entrepreneurship/50>
6. Journal of Entrepreneurship & Organization Management, Vikalpa, IIMA, IIMB Review, Decision, IIMC, Vision, HBR.

ADVANCED ENTREPRENEURSHIP

VII SEMESTER: OPEN ELECTIVE-II

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS11	Foundation	L	T	P	C	CIA	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 50	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 50			

COURSE OBJECTIVES

The course should enable the students to:

1. Understand and **discover** entrepreneurship
2. Build a strong foundation for students to **start, build, and grow** a viable and sustainable venture
3. Develop an entrepreneurial outlook and **mindset**, critical **skills** and **knowledge**
4. Mitigate three types of **risks**: Customer, Business Model, and Product/Technical

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. The knowledge and mastery of organizational and management techniques in the arts, including leadership, organizational design, human resources, legal issues, and finance
2. The knowledge of the legal and ethical environment impacting business organizations and exhibit an understanding and appreciation of the ethical implications of decisions.
3. An ability to engage in critical thinking by analyzing situations and constructing and selecting viable solutions to solve problems.

UNIT-I	ORIENTATION TO GROWTH	CLASSES: 10
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ORIENTATION TO GROWTH: Getting Ready for Growth-Why growth stage is different compared to startup phase, Why Product-Market fit is not enough, Case study, To assess readiness for growth, To chart a growth path.

UNIT-II	CUSTOMER, BUSINESS MODEL, VALIDATION	CLASSES: 10
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CUSTOMERS :Expanding Customer Base-Revisit your business model and develop few variants (more business model types),Identify additional customer segments that your solution can address, Evaluate business models for the new customer segments, Relook at the Problem Statement (can you expand the scope and scalability of your business by repositioning your problem statement?), Explore additional ways to monetize.

BUSINESS MODEL: Get Started with Lean Canvas.

VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.

UNIT-III	TRACTION AND TEAM	CLASSES: 12
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TRACTION : Scaling - How to gain traction beyond early customers, Defining traction (in quantifiable terms) and identifying the most important metrics to measure traction, Calculate cost of new customer acquisition, Estimate your customer lifetime value(LTV),Identifying waste in your operations and focusing your team on what is important for traction.

Channels and Strategy - The Bullseye framework, Identify Channels using Bulls Eye Frame work, Measuring the effectiveness of selected channels, Budgeting and planning.

UNIT-IV	MONEY & SALES	CLASSES: 10
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MONEY: Growing Revenues - Stabilizing key revenue streams, Developing additional revenue streams (licensing, franchising), Exploring new channels and partnerships, Sales Planning - Understanding why customers buy and how buying decisions are made; Listening skills, Sales planning, setting targets, Unique



Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features, Follow-up and closing a sale; Asking for the sale. Strengthening Sales - Follow-up and closing a sale; Asking for the sale, Building a professional sales team, Sales compensation and incentives, Sales planning, setting targets. Improving Margins - Testing price elasticity, Optimizing costs and operational expenses, Advanced concepts of unit costing. Financial Modeling - Financial modeling of your venture's growth, Analyzing competitor and peer's financial models.		
UNIT-V	SUPPORT	CLASSES: 08
SUPPORT: Legal - Overview of legal issues and their impact on entrepreneurs, Importance of getting professional help (legal and accounting), Importance of being compliant and keeping proper documentation, Patents and Intellectual property, Trademarks. Mentors, Advisors, and Experts - The importance of a Mentor and how to find one, Role of business advisors and experts for specific targets in your growth plan		
TEXT BOOKS		
1. Bygrave, W., & Zacharakis, A. (2017) Entrepreneurship, 4th Edition (3rd Edition is ok too) Wiley. 2. Carree, M. A., Thurik, A. R. "The impact of entrepreneurship on economic growth" In: Audretsch, D.B.Acs. 3. Handbook of Entrepreneurship Research. Berlin: Springer Verlag, 2010.		
REFERENCE BOOKS		
1. Eric, Reis (2017) The Startup Way: How Entrepreneurial Management Transforms Culture and Drives Growth 2. Thurik, A. R., Audretsch, D. B., Stam, E. "The rise of the entrepreneurial economy and the future of dynamic capitalism", Technovation. 3. Audretsch, D. B., Grilo, I., Thurik, A. R. (Eds). The Handbook of Research on Entrepreneurship Policy. Cheltenham, UK: Edward Elgar, 2007.		
WEB REFERENCES		
1. https://www.ediindia.org/ediilibrary.aspx 2. https://www.entrepreneurshipsecret.com/sources-of-information-for-entrepreneurship-development-contd		
KEY REFERENCE		
1. https://www.ediindia.org/EDILibrary.aspx 2. https://www.entrepreneurshipsecret.com/sources-of-information-for-entrepreneurship-development-contd		

MANAGEMENT SCIENCE

VIII SEMESTER: OPEN ELECTIVE-III

Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6HS12	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES

The course will enable the students to:

1. To familiarize & obtain Knowledge with the process of management and to provide basic insights into management practices
2. To understand the structure & Designing of an Organization
3. Familiarize with strategies used for analysis of an Organization
4. To understand the Contemporary Management Issues
5. To familiarize with the management skills which can be applied in the Organizational context to achieve Organizational goals.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Knowledge on management theories and practices
2. Understanding designing organizational structure
3. Understanding on the methods & charts used in operations management
4. Ability to understand the market and its environment
5. Understand the processes, functions etc. in Human Resources Management
6. Knowledge on Strategic alternatives
7. Familiar with the practices implemented in management
8. Understand the social responsibilities of Management.

UNIT-I	INTRODUCTION TO MANAGEMENT AND ORGANIZATION	CLASSES: 12
Concepts of Management and organization- nature, importance and Functions of Management, Systems Approach to Management-Taylor's Scientific Management Theory- Fayal's Principles of Management- Maslow's theory of Hierarchy of Human Needs- Douglas McGregor's Theory X and Theory Y - Herzberg Two Factor Theory of Motivation - Leadership Styles, Social responsibilities of Management, Designing Organizational Structures: Basic concepts related to Organization - Departmentation and Decentralization, Types and Evaluation of mechanistic and organic structures of organization and suitability.		
UNIT-II	OPERATIONS AND MARKETING MANAGEMENT	CLASSES: 12
Principles and Types of Plant Layout-Methods of Production(Job, batch and Mass Production), Work Study - Basic procedure involved in Method Study and Work Measurement - Business Process Reengineering(BPR) - Statistical Quality Control: control charts for Variables and Attributes (simple Problems) and Acceptance Sampling, TQM, Six Sigma, Deming's contribution to quality, Objectives of Inventory control, EOQ, ABC Analysis, Purchase Procedure, Stores Management and Store Records - JIT System, Supply Chain Management, Functions of Marketing, Marketing Mix, and Marketing Strategies based on Product Life Cycle, Channels of distribution.		
UNIT-III	HUMAN RESOURCES MANAGEMENT(HRM)	CLASSES: 12
Concepts of HRM, HRD and Personnel Management and Industrial Relations (PMIR), HRM vs. PMIR, Basic functions of HR Manager: Manpower planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Grievance Handling and Welfare Administration, Job Evaluation and Merit Rating - Capability Maturity Model (CMM) Levels - Performance Management System.		



UNIT-IV	PROJECT MANAGEMENT (PERT/CPM)	CLASSES: 12
Network Analysis, Program Evaluation and Review Technique (PERT), Critical Path Method (CPM), Identifying critical path, Probability of Completing the project within given time, Project Cost Analysis, Project Crashing (simple problems)		
UNIT-V	STRATEGIC MANAGEMENT AND CONTEMPORARY STRATEGIC ISSUES	CLASSES: 12
Mission, Goals, Objectives, Policy, Strategy, Programs, Elements of Corporate Planning Process, Environmental Scanning, Value Chain Analysis, SWOT Analysis, Steps in Strategy Formulation and Implementation, Generic Strategy alternatives. Bench Marking and Balanced Score Card as Contemporary Business Strategies.		
TEXT BOOKS		
1. A.R.Aryasri : Management Science, TMH, (Latest edition) 2. Stoner, Freeman, Gilbert, Management, 6th Ed, Pearson Education, New Delhi, 2004.		
REFERENCE BOOKS		
1. Kotler Philip and Keller Kevin Lane: Marketing Management, Pearson, 2012. 2. Koontz and Weihrich: Essentials of Management, McGraw Hill, 2012. 3. Thomas N. Duening and John M. Ivancevich Management - Principles and Guidelines, Biztantra, 2012. 4. Kanishka Bedi, Production and Operations Management, Oxford University Press, 2012. 5. Samuel C. Certo: Modern Management, 2012. 6. Schermerhorn, Capling, Poole and Wiesner: Management, Wiley, 2012. 7. Parnell: Strategic Management, Cengage, 2012. 8. Lawrence R Jauch, R. Gupta and William F. Glueck: Business Policy and Strategic Management Science, McGraw Hill, 2012.		
WEB REFERENCES		
1. https://www.smartzworld.com/notes/management-science-pdf-notes-ms-notes-pdf 2. https://www.alljntuworld.in/download/management-science-notes/		

VISION STATEMENT

To create and nurture competent Engineers and managers who would be enterprise leaders in all parts of the world with aims of reaching the skies and touching the stars and yet feet firmly planted on the ground – good human beings steeped in ethical and moral values.

MISSION STATEMENT

MLR Institute of Technology is committed to providing a positive, professional and conducive learning environment where all students are inspired to achieve their potential and strive for excellence in a global society as dignified professionals with the cooperation of all stakeholders.

QUALITY POLICY

We, at MLRIT, are committed to Educate, Enrich and Excel, in imparting Professional Education, by top-quality faculty; who endeavour to mentor the students as turn-key solution providers, while striving continually to improve through team work, innovation and research.

GOALS OF MLRIT

Goals of engineering education at undergraduate/graduate level:

- Equip students with industry – accepted career and life skills
- To create a knowledge warehouse for students
- To disseminate information on skills and competencies that is in use and in demand by the industry.
- To create learning environment where the campus culture acts as a catalyst to student fraternity to understand their core competencies, enhance their competencies and improve their career prospects.
- To provide base for lifelong learning and professional development in support of evolving career objectives, which include being informed, effective, and responsible participants within the engineering profession and in society.
- To prepare students for graduate study in Engineering and Technology.
- To prepare graduates to engineering practice by learning from professional engineering assignments.